Story of How POSTECH Defeats KAIST

Mathematics for Science Quiz

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology
(2026 Science Quiz Representative)
jkpxt14@postech.ac.kr

POSTECH–KAIST Science War

Pohang University of Science and Technology
vs
Korea Advanced Institute of Science and Technology

Preface

This book is devoted to the Mathematics section of Science Quiz in the POSTECH–KAIST Science War. It collects mathematical concepts, facts to memorize, past problems, and expected problems that are useful for competition preparation.

Science Quiz remains the primary target of this book. At the same time, selected results and techniques from mathematical olympiads, gifted-school and science-high-school entrance examinations, and other problem-solving competitions are included when they are concise, memorable, or broadly useful. The emphasis is therefore on meaningful results, effective formulas, and decisive ideas rather than on reproducing long proofs.

The main purpose of this text is not merely to record solutions, but to build a practical and reliable mathematical toolkit for Science Quiz. It includes standard theorems, important terminology, memorable facts about mathematicians and mathematical history, and problem-solving strategies that may be useful in an actual match.

Science Quiz requires competitors to move rapidly across broad areas of mathematics and solve each problem in roughly one or two minutes. Accordingly, this book combines the academic foundations of university mathematics with contest-style tools such as invariants, extremal arguments, coloring, symmetry, reverse engineering, fast transformations, and pattern recognition. The aim is not to reproduce long olympiad proofs, but to make the decisive idea or calculation available quickly when a short and creative problem appears.

I sincerely hope that this book contributes to POSTECH's victory. Good luck, and let the game begin.

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Contents

Notation and Conventions

The notation specified below is used throughout this book. When several typographic variants of the same letter exist, each variant is assigned a distinct role and the variants are not interchanged.

General Conventions

Scalars are written in ordinary italic letters, such as a, b, c, λ, t. Vectors are written in bold lowercase letters, such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Matrices are written in bold uppercase letters, such as $\mathbf{A}, \mathbf{B}, \mathbf{P}$. In particular, the identity matrix, a diagonal matrix, and a projection matrix are written as $\mathbf{I}_n$, $\mathbf{D}$, and $\mathbf{P}$, respectively. Unless otherwise stated, vectors are column vectors.

Round parentheses are used for ordered tuples and matrix delimiters. Square brackets are used for closed intervals and for area notation such as $[ABC]$. Braces are used for sets. In set-builder notation, the condition is separated by $|$, as in $\{x \in A \mid P(x)\}$. Determinants may be enclosed by vertical bars because a determinant is a scalar.

In symbolic statements, logical conjunction, disjunction, and negation are written as $\wedge$, $\vee$, and $\neg$. The words "and" and "or" are used in ordinary prose.

Notation	Meaning
$\mathbb{N}$	the set of positive integers, $\{1, 2, 3, \dots\}$
$\mathbb{N}_0$	the set of nonnegative integers, $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	the set of integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{R}$	the set of real numbers
$\mathbb{C}$	the set of complex numbers
$a \coloneqq b$	a is defined to be b
ε	a positive quantity in analysis, usually arbitrary
ϵ_{ijk}	the Levi–Civita symbol; the glyph ϵ is reserved for this use
$O(g(x))$	Big-O notation
$o(g(x))$	little-o notation
$\square$	the end of a proof or solution

Theorem-like Environments

Label	Usage
Definition	a precise meaning of a term or object
Theorem	a major mathematical result
Proposition	a useful result of more limited scope
Lemma	a supporting result or technical tool

Label	Usage
Corollary	a result that follows directly from another result
Formula	a computational identity or closed form
Algorithm	a step-by-step computational procedure
Fact	a verified mathematical or historical fact
Conjecture	a statement believed to be true but not proved in full generality
Hypothesis	a named assumption or proposed statement
Open Problem	a problem whose stated form remains unsolved
Idea	a concise account of a decisive mathematical idea
Strategy	a broadly useful problem-solving method
Tactic	a local, immediately applicable device
Remark	supplementary context or a caution

Set Theory and Logic

Uppercase letters such as A, B, X, Y usually denote sets, while lowercase letters such as a, b, x, y usually denote their elements unless the context indicates another meaning. The arrow $f : X \to Y$ specifies the domain and codomain of a function, while $x \mapsto f(x)$ specifies its action on an element.

Notation	Meaning
$x \in A$	x is an element of A
$x \notin A$	x is not an element of A
$A \subseteq B$	A is a subset of B
$A \subsetneq B$	A is a proper subset of B
$A \supseteq B$	A is a superset of B
$A \supsetneq B$	A is a proper superset of B
$A \cup B$	the union of A and B
$A \cap B$	the intersection of A and B
$A \setminus B$	the set difference of A and B
A^{c}	the complement of A in the understood ambient set
$\varnothing$	the empty set
$\mathcal{P}(A)$	the power set of A
$A \times B$	the Cartesian product of A and B
$A \sqcup B$	the disjoint union of A and B
$\lvert A \rvert$	the cardinality of a finite set A
$\{x \in A \mid P(x)\}$	the set of all $x \in A$ for which $P(x)$ holds
$x \sim y$	x and y are equivalent under an understood equivalence relation
$A/\sim$	the set of equivalence classes of A under $\sim$
$\forall x \in A$	for every $x \in A$
$\exists x \in A$	there exists $x \in A$
$\exists! x \in A$	there exists a unique $x \in A$
$\nexists x \in A$	there does not exist $x \in A$
$P \Rightarrow Q$	P implies Q
$P \Leftarrow Q$	Q implies P
$P \Leftrightarrow Q$	P if and only if Q
$P \wedge Q$	logical conjunction

Notation	Meaning
$P \vee Q$	logical disjunction
$\neg P$	logical negation
$f : X \to Y$	a function from X to Y
$x \mapsto f(x)$	the elementwise rule of a function
$\operatorname{dom} f$	the domain of f
$\operatorname{codom} f$	the codomain of f
id_X	the identity map on X
$g \circ f$	composition of functions
$f\vert_A$	restriction of f to A
$f : X \hookrightarrow Y$	an injective map
$f : X \twoheadrightarrow Y$	a surjective map
$f : X \xrightarrow{\sim} Y$	an isomorphism
$X \cong Y$	X and Y are isomorphic
$f(A)$	the image of A under f
$f^{-1}(B)$	the inverse image, or preimage, of B
$\mathbb{1}_A(x)$	the indicator function of A
$\aleph_0$	the cardinality of a countably infinite set
$\mathfrak{c}$	the cardinality of the continuum

The symbol $\mathbb{1}_A$ is blackboard bold and does not denote a vector. The all-ones vector is $\mathbf{1}$. Boldface is otherwise reserved for vectors and matrices.

Calculus and Analysis

Logarithms to bases e, 10, and 2 are written as $\ln x$, $\lg x$, and $\operatorname{lb} x$, respectively.

The coordinate conventions are

$$\begin{aligned} &\text{Cartesian:} && (x, y), \quad (x, y, z), \\ &\text{polar:} && (r, \theta), \\ &\text{cylindrical:} && (r, \theta, z), \\ &\text{spherical:} && (\rho, \theta, \phi). \end{aligned}$$

In polar, cylindrical, and spherical coordinates, θ is the azimuthal angle measured counterclockwise from the positive x-axis in the xy-plane. In spherical coordinates, ϕ is the polar angle measured from the positive z-axis. The glyph ϕ is reserved for this spherical polar angle. The glyph φ is used for functions, homomorphisms, Euler's totient function, and the golden ratio whenever one of these objects is introduced.

Notation	Meaning
$f'(a)$	the first derivative of f at a
$f^{(n)}(a)$	the nth derivative of f at a
$\dfrac{\mathrm{d}y}{\mathrm{d}x}$	the derivative of y with respect to x, with upright d
$\displaystyle\int_a^b f(x)\,\mathrm{d}x$	the definite integral of f over $[a, b]$
$\ln x$	the logarithm to base e
$\lg x$	the logarithm to base 10
$\operatorname{lb} x$	the logarithm to base 2

Notation	Meaning
$\log_a x$	the logarithm to the explicitly specified base a
$\arcsin x$	the principal inverse sine, with values in $[-\pi/2, \pi/2]$
$\arccos x$	the principal inverse cosine, with values in $[0, \pi]$
$\arctan x$	the principal inverse tangent, with values in $(-\pi/2, \pi/2)$
$C^k(I)$	the space of k times continuously differentiable functions on I
$C^\infty(I)$	the space of smooth functions on I
$\dfrac{\partial f}{\partial x_i}$	the partial derivative of f with respect to x_i
(x, y), (x, y, z)	Cartesian coordinates
(r, θ)	polar coordinates
(r, θ, z)	cylindrical coordinates
(ρ, θ, ϕ)	spherical coordinates
∇f	the gradient of f
$\nabla \cdot \mathbf{F}$	the divergence of $\mathbf{F}$
$\nabla \times \mathbf{F}$	the curl of $\mathbf{F}$
Δf	the Laplacian of f
$D\mathbf{F}(\mathbf{a})$	the total derivative, or Jacobian matrix, of $\mathbf{F}$ at $\mathbf{a}$
$\mathrm{Jac}(\mathbf{F})$	the Jacobian determinant of $\mathbf{F}$
$\mathrm{Hess}\, f$	the Hessian matrix of f
$\mathbf{r}(t)$	a parametrized curve
$\mathrm{d}s$	an element of arc length
$\displaystyle\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$	a line integral along C
$\displaystyle\iint_S \mathbf{F} \cdot \mathbf{n}\, \mathrm{d}S$	a flux integral across S
$B_r(x)$	the open ball of radius r centered at x
$\overline{B}_r(x)$	the closed ball of radius r centered at x
$x_n \to x$	convergence of a sequence
$a_n \nearrow L$	monotone increasing convergence to L
$a_n \searrow L$	monotone decreasing convergence to L
$\limsup a_n$	the limit superior of (a_n)
$\liminf a_n$	the limit inferior of (a_n)
$f_n \to f$	pointwise convergence when no other mode is specified
$f_n \rightrightarrows f$	uniform convergence
$f_n \xrightarrow{\text{a.e.}} f$	convergence almost everywhere
$\lVert f \rVert_\infty$	the supremum norm of f
$z = x + iy$	a complex number with real part x and imaginary part y
$\mathrm{Re}\, z$	the real part of z
$\mathrm{Im}\, z$	the imaginary part of z
$\overline{z}$	the complex conjugate of z
$\lvert z \rvert$	the modulus of z
$\arg z$	an argument of z
$\displaystyle\oint_C f(z)\, \mathrm{d}z$	a contour integral around a closed curve C
$\mathrm{Res}(f; a)$	the residue of f at a

Linear Algebra

Vectors and matrices are bold; scalars and linear maps are ordinary italic.

Notation	Meaning
$\mathbf{A} = (a_{ij})$	the matrix whose (i, j)-entry is a_{ij}
$\mathbf{A}^{\mathsf{T}}$	the transpose of $\mathbf{A}$
$\mathbf{A}^{\mathsf{H}}$	the conjugate transpose of $\mathbf{A}$
$\mathbf{A}^{-1}$	the inverse of $\mathbf{A}$
$\mathbf{I}_n$	the $n \times n$ identity matrix
$\mathbf{0}$	a zero vector or zero matrix, with dimension clear from context
$\mathbf{D}$	a diagonal matrix
$\mathbf{1}$	the all-ones vector
$\mathbf{J}_{m,n}$, $\mathbf{J}_n$	the $m \times n$ and $n \times n$ all-ones matrices
$\mathbf{P}_\sigma$	the permutation matrix associated with σ
$\det(\mathbf{A})$	the determinant of $\mathbf{A}$
$\mathrm{rk}(\mathbf{A})$	the rank of $\mathbf{A}$
$\mathrm{tr}(\mathbf{A})$	the trace of $\mathbf{A}$
$p_{\mathbf{A}}(\lambda)$	the characteristic polynomial of $\mathbf{A}$
$m_{\mathbf{A}}(\lambda)$	the minimal polynomial of $\mathbf{A}$
$\rho(\mathbf{A})$	the spectral radius of $\mathbf{A}$
$\sigma_i(\mathbf{A})$	the ith singular value of $\mathbf{A}$
M_{ij}	the minor of the entry a_{ij}
$C_{ij} = (-1)^{i+j} M_{ij}$	the cofactor of the entry a_{ij}
$\mathrm{adj}(\mathbf{A})$	the adjugate of $\mathbf{A}$
$\mathrm{Col}(\mathbf{A})$	the column space of $\mathbf{A}$
$\mathrm{Null}(\mathbf{A})$	the null space of $\mathbf{A}$
$\mathrm{Row}(\mathbf{A})$	the row space of $\mathbf{A}$
$\ker T$	the kernel of a linear map T
$\mathrm{im}\, T$	the image of a linear map T
$\mathbf{e}_i$	the ith standard basis vector
$\mathrm{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$	the span of the listed vectors
$U \oplus V$	the direct sum of subspaces U and V
$\langle \mathbf{u}, \mathbf{v} \rangle$	the inner product of $\mathbf{u}$ and $\mathbf{v}$
$\lVert \mathbf{v} \rVert$	the norm of $\mathbf{v}$
$\mathrm{proj}_{\mathbf{v}}\, \mathbf{x}$	the orthogonal projection of $\mathbf{x}$ onto $\mathrm{span}(\mathbf{v})$
$\mathbf{u} \perp \mathbf{v}$	$\mathbf{u}$ and $\mathbf{v}$ are orthogonal
$\mathbf{u} \times \mathbf{v}$	the cross product in $\mathbb{R}^3$

Differential Equations

Notation	Meaning
y'	the first derivative of y with respect to the independent variable
y''	the second derivative of y
$\dot{x}$	the first derivative of x with respect to time t
$\ddot{x}$	the second derivative of x with respect to time t
y_h	the homogeneous part of a solution
y_p	a particular solution
$c_1, c_2, \ldots$	arbitrary constants in a general solution
$W(y_1, y_2)(t)$	the Wronskian of y_1 and y_2

Notation	Meaning
$\mathcal{L}\{f(t)\}(s)$	the Laplace transform of $f(t)$
$\widehat{f}(\omega)$	the Fourier transform under the convention stated where it is used

Combinatorics and Discrete Mathematics

The symbol $\left(\!\binom{n}{k}\!\right)$ is reserved for combinations with repetition.

Notation	Meaning
$[n]$	the set $\{1, 2, \ldots, n\}$
$n!$	factorial
$\binom{n}{k}$	the number of k-element subsets of an n-element set
$\left(\!\binom{n}{k}\!\right)$	the number of size-k multisets chosen from an n-element set
$P(n, k)$	the number of k-permutations of an n-element set
$\left\{ {n \atop k} \right\}$	the Stirling number of the second kind
a_n	the nth term of a sequence
Δa_n	the forward difference $a_{n+1} - a_n$
$S_n = \displaystyle\sum_{k=1}^{n} a_k$	the nth partial sum
$A(x) = \displaystyle\sum_{n \geq 0} a_n x^n$	an ordinary generating function
$[x^n]A(x)$	the coefficient of x^n in $A(x)$
$\omega_n = e^{2\pi i/n}$	a primitive nth root of unity
$\sigma = (a_1\ a_2\ \cdots\ a_k)$	cycle notation for a permutation
$\mathrm{inv}(\sigma)$	the inversion number of σ
$\mathrm{Fix}(g)$	the set of objects fixed by g
$G = (V, E)$	a graph with vertex set V and edge set E
$\deg(v)$	the degree of the vertex v
K_n	the complete graph on n vertices
C_n	the cycle graph on n vertices
P_n	the path graph on n vertices

Probability and Statistics

Random variables are usually denoted by uppercase letters such as X, Y, Z; events are denoted by uppercase letters such as A, B, E.

Notation	Meaning
$\mathbb{P}(A)$	the probability of the event A
$\mathbb{P}(A \mid B)$	the conditional probability of A given B
$\mathbb{E}[X]$	the expectation of X
$\mathrm{Var}(X)$	the variance of X
$\mathrm{Cov}(X, Y)$	the covariance of X and Y
$X \sim \mathrm{Bin}(n, p)$	a binomial distribution
$X \sim \mathrm{Poi}(\lambda)$	a Poisson distribution
$X \sim \mathrm{N}(\mu, \sigma^2)$	a normal distribution with mean μ and variance σ^2

Notation	Meaning
$X_n \xrightarrow{\text{a.s.}} X$	almost-sure convergence
$X_n \xrightarrow{\mathbb{P}} X$	convergence in probability
$X_n \xrightarrow{L^p} X$	convergence in L^p
$X_n \xrightarrow{\text{d}} X$	convergence in distribution

Geometry

Uppercase letters such as A, B, C, D denote points. When the usual triangle notation is adopted, the corresponding lowercase letters a, b, c denote the side lengths opposite A, B, C, respectively. For a triangle, R, r, s, and Δ denote the circumradius, inradius, semiperimeter, and area.

Notation	Meaning
$\triangle ABC$	the triangle with vertices A, B, C
$\angle ABC$	the angle with vertex B
$\overline{AB}$	the segment joining A and B
AB	the length of $\overline{AB}$
$[ABC]$ or Δ	the area of $\triangle ABC$
$\operatorname{conv}(S)$	the convex hull of S
R	the circumradius of a triangle
r	the inradius of a triangle
s	the semiperimeter of a triangle
I	the number of interior lattice points in Pick's Theorem
B	the number of boundary lattice points in Pick's Theorem
$\parallel$	parallel
$\perp$	perpendicular

Number Theory

Unless otherwise stated, divisibility and congruence statements are made in $\mathbb{Z}$.

Notation	Meaning
$a \mid b$	a divides b
$a \nmid b$	a does not divide b
$\gcd(a, b)$	the greatest common divisor of a and b
$\operatorname{lcm}(a, b)$	the least common multiple of a and b
$a \equiv b \pmod{n}$	a is congruent to b modulo n
$[a]_n$	the residue class of a modulo n
$\mathbb{Z}/n\mathbb{Z}$	the ring of residue classes modulo n
$(\mathbb{Z}/n\mathbb{Z})^\times$	the multiplicative group of units modulo n
$a^{-1} \pmod{n}$	the multiplicative inverse of a modulo n
$\left(\frac{a}{p}\right)$	the Legendre symbol
$\varphi(n)$	Euler's totient function
$\operatorname{ord}_n(a)$	the multiplicative order of a modulo n
$\nu_p(n)$	the exponent of p in the prime factorization of n

Notation	Meaning
$\tau(n)$	the number of positive divisors of n
$\sigma(n)$	the sum of the positive divisors of n
$\mu(n)$	the Möbius function
p	usually a prime number in number-theoretic contexts
$\mathbb{F}_q$	the finite field with q elements

Algebra

Capital letters such as G, H denote groups, R, S denote rings, and F, K, L denote fields; lowercase letters denote their elements unless the context requires otherwise.

Notation	Meaning
G	usually a group
S_n	the symmetric group on n symbols
A_n	the alternating group on n symbols
D_{2n}	the dihedral group of order $2n$
e	the identity element of a group
$\|G\|$	the order of a finite group G
$\langle S\rangle$	the subgroup generated by S
$H \leq G$	H is a subgroup of G
$H \trianglelefteq G$	H is a normal subgroup of G
G/H	the quotient group of G by H
$\ker\varphi$	the kernel of a homomorphism φ
$\operatorname{im}\varphi$	the image of a homomorphism φ
$\operatorname{Aut}(G)$	the automorphism group of G
$G \cdot x$	the orbit of x under a group action
G_x	the stabilizer of x under a group action
$Z(G)$	the center of G
$C_G(x)$	the centralizer of x in G
$\mathrm{GL}_n(F)$	the general linear group over F
$\mathrm{SL}_n(F)$	the special linear group over F
R	usually a ring
$I \trianglelefteq R$	I is an ideal of R
R/I	the quotient ring of R by I
F or K	usually a field
$F[x]$	the polynomial ring over F
$\deg f$	the degree of a polynomial f
$\operatorname{Disc}(f)$	the discriminant of a polynomial f
$\operatorname{char} F$	the characteristic of F
$\operatorname{Gal}(L/K)$	the Galois group of the extension L/K

Topology

Notation	Meaning
$(X, \mathcal{T})$	a topological space
$\operatorname{int} A$	the interior of A
$\overline{A}$	the closure of A
∂A	the boundary of A
$X \cong Y$	homeomorphic spaces when the context is topological
$\pi_1(X, x_0)$	the fundamental group of X based at x_0
$\chi(X)$	the Euler characteristic of X

Miscellaneous Topics

Mathematical Constants

The table contains constants that occur frequently across several areas of mathematics. A defining expression is used whenever a single-letter symbol is not needed elsewhere in the book.

Symbol	Name	Meaning or definition	Exact value or approximation
i	Imaginary unit	The distinguished square root of -1 in $\mathbb{C}$.	$i^2 = -1$
$\sqrt{2}$	Pythagoras constant	The positive length of the diagonal of a unit square.	$1.41421356237\ldots$
e	Euler's number	The base of the natural logarithm: $e = \sum_{n=0}^{\infty} 1/n!$.	$2.71828182846\ldots$
π	Pi	The ratio of the circumference of a Euclidean circle to its diameter.	$3.14159265359\ldots$
φ	Golden ratio	The positive solution of $x^2 = x + 1$.	$\dfrac{1+\sqrt{5}}{2} \approx 1.61803398875\ldots$
γ	Euler–Mascheroni constant	$\gamma = \lim_{n\to\infty}\left(\sum_{k=1}^{n} \frac{1}{k} - \ln n\right)$.	$0.577215664902\ldots$
$\zeta(3)$	Apéry's constant	The value of the Riemann zeta function at 3: $\sum_{n=1}^{\infty} 1/n^3$.	$1.20205690316\ldots$
G	Catalan's constant	$G = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}$.	$0.915965594177\ldots$

Other Notation

Notation	Meaning
$\lfloor x \rfloor$	the floor of x
$\lceil x \rceil$	the ceiling of x
$\{x\}$	the fractional part $x - \lfloor x \rfloor$
RH	the Riemann Hypothesis, after it has been introduced
P vs NP	the P versus NP problem, after it has been introduced

Part I

Mathematical Concepts

Chapter 1

Set Theory and Logic

Related POSTECH Courses: (MATH202) Set Theory
Related Books: Introduction to Set Theory (Karel Hrbacek, Thomas Jech)

Definition 1.1. Set and Element
A set is a collection of objects, called its elements. We write $x \in A$ if x is an element of A, and $x \notin A$ otherwise. The relation $A \subseteq B$ means

$$\forall x, \qquad x \in A \Rightarrow x \in B.$$

Formula 1.2. Basic Set Operations
For sets A and B inside an ambient set U, we use

$$A \cup B = \{x \in U \mid (x \in A) \vee (x \in B)\},$$

$$A \cap B = \{x \in U \mid (x \in A) \wedge (x \in B)\},$$

and

$$A \setminus B = \{x \in A \mid x \notin B\}.$$

Formula 1.3. Set Algebra
For subsets $A, B, C \subseteq U$, the following identities are often useful:

$$A \cup \varnothing = A, \qquad A \cap U = A, \qquad A \cup A = A, \qquad A \cap A = A.$$

The distributive laws are

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

The absorption laws are

$$A \cup (A \cap B) = A, \qquad A \cap (A \cup B) = A.$$

Formula 1.4. Basic Logical Equivalences
The most useful logical equivalences are

$$(P \Rightarrow Q) \iff (\neg Q \Rightarrow \neg P),$$

$$\neg(P \Rightarrow Q) \iff P \wedge \neg Q,$$

$$\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q), \qquad \neg(P \vee Q) \iff (\neg P) \wedge (\neg Q),$$

and

$$\neg(\forall x \in A,\ P(x)) \iff \exists x \in A \text{ such that } \neg P(x),$$
$$\neg(\exists x \in A \text{ such that } P(x)) \iff \forall x \in A,\ \neg P(x).$$

Strategy. To prove $P \Rightarrow Q$, it is often enough to prove the contrapositive $\neg Q \Rightarrow \neg P$. To disprove a universal statement, find a counterexample.

Formula 1.5. De Morgan's Laws
For subsets $A, B \subseteq U$, we have

$$(A \cup B)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}}, \qquad (A \cap B)^{\mathsf{c}} = A^{\mathsf{c}} \cup B^{\mathsf{c}}.$$

More generally,

$$\left(\bigcup_{i\in I} A_i\right)^{\mathsf{c}} = \bigcap_{i\in I} A_i^{\mathsf{c}}, \qquad \left(\bigcap_{i\in I} A_i\right)^{\mathsf{c}} = \bigcup_{i\in I} A_i^{\mathsf{c}}.$$

Definition 1.6. Power Set
The power set of a set A is

$$\mathcal{P}(A) = \{B \mid B \subseteq A\}.$$

If $|A| = n < \infty$, then

$$|\mathcal{P}(A)| = 2^n.$$

Definition 1.7. Function
A function $f : X \to Y$ assigns to each element $x \in X$ a unique element $f(x) \in Y$. The set X is the domain, and Y is the codomain. The notation $x \mapsto f(x)$ describes the rule on elements.

Definition 1.8. Injection, Surjection, and Bijection
A function $f : X \to Y$ is injective if distinct elements of X have distinct images. Equivalently,

$$\forall x_1, x_2 \in X, \qquad f(x_1) = f(x_2) \implies x_1 = x_2.$$

Once injectivity has been established, one may write $f : X \hookrightarrow Y$.

The function is surjective if every element of Y has a preimage:

$$\forall y \in Y, \qquad \exists x \in X \text{ such that } f(x) = y.$$

Once surjectivity has been established, one may write $f : X \twoheadrightarrow Y$. The function is bijective if it is both injective and surjective.

Proposition 1.9. Finite Injective-Surjective Criterion
If X and Y are finite sets with $|X| = |Y|$, then for a function $f : X \to Y$, the following are equivalent:

$$f \textit{ is injective} \iff f \textit{ is surjective} \iff f \textit{ is bijective.}$$

Proposition 1.10. Inverse Image Preserves Set Operations
If $f : X \to Y$ and $A, B \subseteq Y$, then

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B),$$

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B),$$

and

$$f^{-1}(Y \setminus A) = X \setminus f^{-1}(A).$$

Definition 1.11. Relation and Equivalence Relation
A binary relation on a set A is a subset $R \subseteq A \times A$. An equivalence relation $\sim$ is reflexive, symmetric, and transitive; explicitly,

$$\forall x \in A, \qquad x \sim x,$$

$$\forall x, y \in A, \qquad x \sim y \implies y \sim x,$$

and

$$\forall x, y, z \in A, \qquad (x \sim y \land y \sim z) \implies x \sim z.$$

The equivalence class of $a \in A$ is

$$[a] = \{x \in A \mid x \sim a\}.$$

Proposition 1.12. Equivalence Relations and Partitions
Equivalence relations on a set A are the same structure as partitions of A: each equivalence relation partitions A into equivalence classes, and each partition of A determines an equivalence relation.

Definition 1.13. Countability
A set A is countable if it is finite or if there exists a bijection $f : A \to \mathbb{N}$. In the countably infinite case,

$$\exists f : A \xrightarrow{\sim} \mathbb{N}, \qquad |A| = |\mathbb{N}| = \aleph_0.$$

A set is uncountable if it is not countable.

Theorem 1.14. Cantor's Diagonal Argument
The set $\mathbb{R}$ of real numbers is uncountable. In particular,

$$|\mathbb{N}| < |\mathbb{R}|.$$

Theorem 1.15. Cantor's Theorem
For every set A, there is no surjection from A onto $\mathcal{P}(A)$. Equivalently,

$$|A| < |\mathcal{P}(A)|.$$

Theorem 1.16. Schröder–Bernstein Theorem; Cantor–Bernstein Theorem
If there exists an injection $f : A \hookrightarrow B$ and an injection $g : B \hookrightarrow A$, then there exists a bijection

$$h : A \xrightarrow{\sim} B.$$

Thus two sets have the same cardinality once each can be embedded into the other.

Algorithm 1.17. Mathematical Induction
To prove a statement $P(n)$ for all integers $n \geq n_0$:

1. Prove the base case $P(n_0)$.
2. Prove the induction step: for every $k \geq n_0$, show that $P(k)$ implies $P(k+1)$.

Then $P(n)$ holds for all integers $n \geq n_0$.

Theorem 1.18. Well-Ordering Principle
Every nonempty subset of $\mathbb{N}$ has a least element.

Fact 1.19. Russell's Paradox
The expression "the set of all sets that do not contain themselves" leads to a contradiction in naive set theory. This is one reason modern set theory is formulated axiomatically.

Fact 1.20. Axiom of Choice and Zorn's Lemma
The Axiom of Choice, Zorn's Lemma, and the Well-Ordering Theorem are equivalent over the usual Zermelo–Fraenkel axioms. They are standard foundational tools in advanced algebra, analysis, and topology.

Chapter 2

Calculus and Analysis

Related POSTECH Courses: (MATH101) Calculus I, (MATH102) Calculus II, (MATH210) Applied Complex Variables, (MATH311) Analysis I, (MATH312) Analysis II
Related Books: Calculus With Applications (Peter D. Lax, Maria Shea Terrell), Multivariable Calculus with Applications (Peter D. Lax, Maria Shea Terrell), Principles of Mathematical Analysis (Walter Rudin)

One-Variable Calculus

Formula 2.1. Arithmetic and Geometric Progressions
For an arithmetic progression with first term a and common difference d, we have

$$a_n = a + (n-1)d, \qquad \sum_{k=1}^{n} a_k = \frac{n}{2}(a_1 + a_n).$$

For a geometric progression with first term a and common ratio r, we have

$$a_n = ar^{n-1}, \qquad \sum_{k=0}^{n-1} ar^k = a\frac{1-r^n}{1-r} \quad (r \neq 1).$$

If $|r| < 1$, then

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}.$$

Formula 2.2. Sums of Powers; Faulhaber's Formulas
The first power sums are

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2},$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2,$$

and

$$\sum_{k=1}^{n} k^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}.$$

More generally, for each fixed nonnegative integer m,

$$\sum_{k=1}^{n} k^m$$

is a polynomial in n of degree $m+1$ with leading coefficient $1/(m+1)$.

Formula 2.3. Finite Differences; Newton's Forward Difference Formula
For a sequence (a_n), define the forward difference by

$$\Delta a_n = a_{n+1} - a_n.$$

If $a_n = P(n)$ for a polynomial P of degree d with leading coefficient c, then

$$\Delta^d a_n = d!c, \qquad \Delta^{d+1} a_n = 0.$$

Conversely, a sequence with constant dth differences is polynomial in n of degree at most d. For every polynomial P of degree at most d,

$$P(n) = \sum_{k=0}^{d} \binom{n}{k} \Delta^k P(0).$$

Strategy. A table of successive differences is one of the most efficient methods to detect a hidden polynomial sequence. Constant first, second, and third differences indicate linear, quadratic, and cubic behavior, respectively.

Formula 2.4. Linear Recurrences with Constant Coefficients
Consider

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_m a_{n-m}.$$

Its characteristic polynomial is

$$r^m - c_1 r^{m-1} - c_2 r^{m-2} - \cdots - c_m.$$

If the roots $r_1, \ldots, r_m$ are distinct, then

$$a_n = A_1 r_1^n + \cdots + A_m r_m^n.$$

A root r of multiplicity q contributes terms

$$r^n,\ nr^n,\ \ldots,\ n^{q-1} r^n.$$

The constants are determined from the initial values.

Formula 2.5. First-Order Linear Recurrence
If

$$a_n = r a_{n-1} + b_n \qquad (n \geq 1),$$

then iteration gives

$$a_n = r^n a_0 + \sum_{j=1}^{n} r^{n-j} b_j.$$

In particular, if $b_n = b$ and $r \neq 1$, then

$$a_n = r^n a_0 + b \frac{r^n - 1}{r - 1}.$$

Formula 2.6. Nonhomogeneous Linear Recurrences
For a linear recurrence with constant coefficients, the general solution is

$$a_n = a_n^{(h)} + a_n^{(p)},$$

where $a_n^{(h)}$ solves the associated homogeneous recurrence and $a_n^{(p)}$ is one particular solution. If the forcing term has the form

$$P(n)\alpha^n,$$

try a particular solution of the form

$$n^s Q(n)\alpha^n,$$

where $\deg Q = \deg P$ and s is the multiplicity of α as a root of the characteristic polynomial.

Strategy. For recurrences, first identify whether the quickest language is characteristic roots, generating functions, a transfer matrix, or telescoping after a suitable normalization.

Definition 2.7. Limit of a Function; Epsilon–Delta Definition
Let $D \subseteq \mathbb{R}$, let a be an accumulation point of D, and let $f : D \to \mathbb{R}$. The statement

$$\lim_{x\to a} f(x) = L$$

means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to, but different from, a. Formally,

$$\forall\varepsilon > 0, \quad \exists\delta > 0, \quad \forall x \in D, \qquad 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Definition 2.8. One-Sided Limits and Limits at Infinity
The right-hand limit $\lim_{x\to a^+} f(x) = L$ is defined by

$$\forall\varepsilon > 0, \quad \exists\delta > 0, \quad \forall x \in D, \qquad 0 < x - a < \delta \implies |f(x) - L| < \varepsilon.$$

The left-hand definition is analogous, with $0 < a - x < \delta$. Moreover,

$$\lim_{x\to\infty} f(x) = L$$

means

$$\forall\varepsilon > 0, \quad \exists M > 0, \quad \forall x \in D, \qquad x > M \implies |f(x) - L| < \varepsilon.$$

Finally, $\lim_{x\to a} f(x) = \infty$ means

$$\forall M > 0, \quad \exists\delta > 0, \quad \forall x \in D, \qquad 0 < |x - a| < \delta \implies f(x) > M.$$

Theorem 2.9. Squeeze Theorem; Sandwich Theorem
Suppose that $g(x) \le f(x) \le h(x)$ for all x sufficiently close to a, except possibly at a, and that

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L.$$

Then

$$\lim_{x\to a} f(x) = L.$$

Definition 2.10. Continuity at a Point
Let $f : D \to \mathbb{R}$ and $a \in D$. The function f is continuous at a if

$$\lim_{x\to a} f(x) = f(a).$$

Equivalently,

$$\forall\varepsilon > 0, \quad \exists\delta > 0, \quad \forall x \in D, \qquad |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

The function is continuous on D if it is continuous at every point of D.

Definition 2.11. Derivative
Let f be defined near a. The derivative of f at a is

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists as a finite real number. Thus $f'(a) = L$ if and only if

$$\forall\varepsilon > 0, \quad \exists\delta > 0, \quad \forall h \in \mathbb{R}, \qquad 0 < |h| < \delta \implies \left|\frac{f(a+h) - f(a)}{h} - L\right| < \varepsilon.$$

Formula 2.12. Standard Trigonometric and Exponential Limits
The following limits are fundamental:

$$\lim_{x\to 0}\frac{\sin x}{x}=1,\qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2},$$

$$\lim_{x\to 0}\frac{e^x-1}{x}=1,\qquad \lim_{x\to 0}\frac{\ln(1+x)}{x}=1,$$

and

$$\lim_{x\to 0}(1+x)^{1/x}=e.$$

Strategy. These limits are often faster than L'Hôpital's Rule. The expressions $\sin x$, $1-\cos x$, e^x-1, $\ln(1+x)$, and $(1+x)^{1/x}$ are strong signals for these standard patterns.

Formula 2.13. Elementary Function Reference Table
The following domains, ranges, symmetries, and periods are standard.

Function	Domain	Range	Symmetry	Period
e^x	$\mathbb{R}$	$(0,\infty)$	neither	none
$\ln x$	$(0,\infty)$	$\mathbb{R}$	neither	none
$\sin x$	$\mathbb{R}$	$[-1,1]$	odd	2π
$\cos x$	$\mathbb{R}$	$[-1,1]$	even	2π
$\tan x$	$\mathbb{R}\setminus\{\frac{\pi}{2}+k\pi \mid k\in\mathbb{Z}\}$	$\mathbb{R}$	odd	π
$\arcsin x$	$[-1,1]$	$[-\frac{\pi}{2},\frac{\pi}{2}]$	odd	none
$\arccos x$	$[-1,1]$	$[0,\pi]$	neither	none
$\arctan x$	$\mathbb{R}$	$(-\frac{\pi}{2},\frac{\pi}{2})$	odd	none
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$	odd	none
$\cosh x$	$\mathbb{R}$	$[1,\infty)$	even	none
$\tanh x$	$\mathbb{R}$	$(-1,1)$	odd	none

Formula 2.14. Exact Trigonometric Values Table
For the standard first-quadrant angles,

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$

Moreover,

$$\sin\frac{\pi}{2}=1,\qquad \cos\frac{\pi}{2}=0,$$

and the remaining exact values follow from parity, periodicity, and reference-angle identities.

Formula 2.15. Trigonometric Identity and Synthesis Table
The principal trigonometric identities are

Type	Identities
Reciprocal	$\sec x=\dfrac{1}{\cos x},\quad \csc x=\dfrac{1}{\sin x},\quad \cot x=\dfrac{1}{\tan x}$
Quotient	$\tan x=\dfrac{\sin x}{\cos x},\quad \cot x=\dfrac{\cos x}{\sin x}$
Pythagorean	$\sin^2 x+\cos^2 x=1$, $1+\tan^2 x=\sec^2 x$, $1+\cot^2 x=\csc^2 x$

Type	Identities
Angle addition	$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
Angle addition	$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
Angle addition	$\tan(x \pm y) = \dfrac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$
Double angle	$\sin 2x = 2 \sin x \cos x$
Double angle	$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$
Double angle	$\tan 2x = \dfrac{2 \tan x}{1 - \tan^2 x}$
Half angle	$\sin^2 \dfrac{x}{2} = \dfrac{1 - \cos x}{2}, \quad \cos^2 \dfrac{x}{2} = \dfrac{1 + \cos x}{2}$
Product-to-sum	$2 \sin x \sin y = \cos(x - y) - \cos(x + y)$
Product-to-sum	$2 \cos x \cos y = \cos(x - y) + \cos(x + y)$
Product-to-sum	$2 \sin x \cos y = \sin(x + y) + \sin(x - y)$
Sum-to-product	$\sin x + \sin y = 2 \sin \dfrac{x + y}{2} \cos \dfrac{x - y}{2}$
Sum-to-product	$\cos x + \cos y = 2 \cos \dfrac{x + y}{2} \cos \dfrac{x - y}{2}$
Sum-to-product	$\sin x - \sin y = 2 \cos \dfrac{x + y}{2} \sin \dfrac{x - y}{2}$
Sum-to-product	$\cos x - \cos y = -2 \sin \dfrac{x + y}{2} \sin \dfrac{x - y}{2}$
Synthesis	$a \cos x + b \sin x = R \cos(x - \alpha), \quad R = \sqrt{a^2 + b^2}, \quad \cos \alpha = \dfrac{a}{R}, \quad \sin \alpha = \dfrac{b}{R} \quad (R > 0)$

Theorem 2.16. L'Hôpital's Rule
Let f and g be differentiable on a deleted neighborhood of a, and suppose that $g'(x) \neq 0$ there. Assume either

$$f(x) \to 0, \qquad g(x) \to 0,$$

or

$$|f(x)| \to \infty, \qquad |g(x)| \to \infty,$$

as $x \to a$. If

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L$$

exists in the extended real numbers, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

Strategy. Apply L'Hôpital's Rule only after identifying a $0/0$ or ∞/∞ form. Rewrite $0 \cdot \infty$ and $\infty - \infty$ as a quotient first. For 0^0, 1^∞, or ∞^0, take logarithms before differentiating.

Formula 2.17. Derivative and Antiderivative Table
The following table records the basic one-variable forms. Affine changes of variable are handled by the chain rule and substitution. The constant of integration is omitted from the right column.

$f(x)$	$f'(x)$	$\int f(x)\,\mathrm{d}x$
c	0	cx
x	1	$\dfrac{x^2}{2}$
x^a	ax^{a-1}	$\dfrac{x^{a+1}}{a+1} \quad (a \neq -1)$

$f(x)$	$f'(x)$	$\int f(x)\,\mathrm{d}x$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\ln\lvert x\rvert$
e^x	e^x	e^x
a^x	$a^x \ln a$	$\frac{a^x}{\ln a} \quad (a > 0,\ a \neq 1)$
$\ln x$	$\frac{1}{x}$	$x \ln x - x$
$\lg x$	$\frac{1}{x \ln 10}$	$\frac{x \ln x - x}{\ln 10}$
$\operatorname{lb} x$	$\frac{1}{x \ln 2}$	$\frac{x \ln x - x}{\ln 2}$
$\log_a x$	$\frac{1}{x \ln a}$	$\frac{x \ln x - x}{\ln a} \quad (a > 0,\ a \neq 1)$
$x \ln x$	$\ln x + 1$	$\frac{x^2}{2} \ln x - \frac{x^2}{4}$
$\frac{\ln x}{x}$	$\frac{1 - \ln x}{x^2}$	$\frac{1}{2}(\ln x)^2$
$\frac{1}{x \ln x}$	$-\frac{\ln x + 1}{x^2 (\ln x)^2}$	$\ln\lvert \ln x\rvert$
$\sin x$	$\cos x$	$-\cos x$
$\cos x$	$-\sin x$	$\sin x$
$\tan x$	$\sec^2 x$	$-\ln\lvert \cos x\rvert$
$\cot x$	$-\csc^2 x$	$\ln\lvert \sin x\rvert$
$\sec x$	$\sec x \tan x$	$\ln\lvert \sec x + \tan x\rvert$
$\csc x$	$-\csc x \cot x$	$\ln\lvert \csc x - \cot x\rvert$
$\sec^2 x$	$2 \sec^2 x \tan x$	$\tan x$
$\csc^2 x$	$-2 \csc^2 x \cot x$	$-\cot x$
$\sec x \tan x$	$\sec x \tan^2 x + \sec^3 x$	$\sec x$
$\csc x \cot x$	$-\csc x \cot^2 x - \csc^3 x$	$-\csc x$
$\arcsin x$	$\frac{1}{\sqrt{1 - x^2}}$	$x \arcsin x + \sqrt{1 - x^2}$
$\arccos x$	$-\frac{1}{\sqrt{1 - x^2}}$	$x \arccos x - \sqrt{1 - x^2}$
$\arctan x$	$\frac{1}{1 + x^2}$	$x \arctan x - \frac{1}{2} \ln(1 + x^2)$
$\sinh x$	$\cosh x$	$\cosh x$
$\cosh x$	$\sinh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$	$\ln(\cosh x)$
$\coth x$	$-\operatorname{csch}^2 x$	$\ln\lvert \sinh x\rvert$
$\operatorname{sech} x$	$-\operatorname{sech} x \tanh x$	$\arctan(\sinh x)$
$\operatorname{csch} x$	$-\operatorname{csch} x \coth x$	$\ln\left\lvert \tanh \frac{x}{2}\right\rvert$
$\operatorname{arsinh} x$	$\frac{1}{\sqrt{x^2 + 1}}$	$x \operatorname{arsinh} x - \sqrt{x^2 + 1}$
$\operatorname{arcosh} x$	$\frac{1}{\sqrt{x - 1}\sqrt{x + 1}}$	$x \operatorname{arcosh} x - \sqrt{x^2 - 1}$
$\operatorname{artanh} x$	$\frac{1}{1 - x^2}$	$x \operatorname{artanh} x + \frac{1}{2} \ln(1 - x^2)$
$\frac{1}{x^2 + a^2}$	$-\frac{2x}{(x^2 + a^2)^2}$	$\frac{1}{a} \arctan \frac{x}{a} \quad (a > 0)$
$\frac{1}{x^2 - a^2}$	$-\frac{2x}{(x^2 - a^2)^2}$	$\frac{1}{2a} \ln \left\lvert \frac{x - a}{x + a} \right\rvert \quad (a > 0)$
$\frac{1}{a^2 - x^2}$	$\frac{2x}{(a^2 - x^2)^2}$	$\frac{1}{2a} \ln \left\lvert \frac{a + x}{a - x} \right\rvert \quad (a > 0)$
$\frac{1}{\sqrt{a^2 - x^2}}$	$\frac{x}{(a^2 - x^2)^{3/2}}$	$\arcsin \frac{x}{a} \quad (a > 0)$
$\frac{1}{\sqrt{x^2 + a^2}}$	$-\frac{x}{(x^2 + a^2)^{3/2}}$	$\ln\left\lvert x + \sqrt{x^2 + a^2}\right\rvert$

$f(x)$	$f'(x)$	$\int f(x)\,\mathrm{d}x$
$\dfrac{1}{\sqrt{x^2-a^2}}$	$-\dfrac{x}{(x^2-a^2)^{3/2}}$	$\ln\left\|x+\sqrt{x^2-a^2}\right\| \quad (a>0)$
$\dfrac{x}{\sqrt{x^2+a^2}}$	$\dfrac{a^2}{(x^2+a^2)^{3/2}}$	$\sqrt{x^2+a^2}$
$\dfrac{x}{\sqrt{a^2-x^2}}$	$\dfrac{a^2}{(a^2-x^2)^{3/2}}$	$-\sqrt{a^2-x^2}$
$\sqrt{a^2-x^2}$	$-\dfrac{x}{\sqrt{a^2-x^2}}$	$\dfrac{x}{2}\sqrt{a^2-x^2}+\dfrac{a^2}{2}\arcsin\dfrac{x}{a}$
$\sqrt{x^2+a^2}$	$\dfrac{x}{\sqrt{x^2+a^2}}$	$\dfrac{x}{2}\sqrt{x^2+a^2}+\dfrac{a^2}{2}\ln\left\|x+\sqrt{x^2+a^2}\right\|$
$\sqrt{x^2-a^2}$	$\dfrac{x}{\sqrt{x^2-a^2}}$	$\dfrac{x}{2}\sqrt{x^2-a^2}-\dfrac{a^2}{2}\ln\left\|x+\sqrt{x^2-a^2}\right\|$

Tactic. An antiderivative is often exposed by a simple substitution, symmetry, completing the square, or integration by parts. Try those before expanding into a long calculation.

Theorem 2.18. Intermediate Value Theorem
Let f be continuous on $[a,b]$. If N is any number between $f(a)$ and $f(b)$, then there exists $c\in[a,b]$ such that
$$f(c)=N.$$

Strategy. The Intermediate Value Theorem is the theorem behind many existence arguments. To show that an equation has a solution, define a continuous function and show that its values cross the desired level.

Theorem 2.19. Extreme Value Theorem; Weierstrass Extreme Value Theorem
Let f be continuous on $[a,b]$. Then f attains both a maximum and a minimum on $[a,b]$.

Remark. The closed interval condition is important. A continuous function on an open interval need not attain a maximum or a minimum.

Theorem 2.20. Rolle's Theorem
Let f be continuous on $[a,b]$ and differentiable on (a,b). If $f(a)=f(b)$, then there exists $c\in(a,b)$ such that
$$f'(c)=0.$$

Theorem 2.21. Mean Value Theorem; Lagrange Mean Value Theorem
Let f be continuous on $[a,b]$ and differentiable on (a,b). Then there exists $c\in(a,b)$ such that
$$f'(c)=\frac{f(b)-f(a)}{b-a}.$$

Idea. The Mean Value Theorem says that somewhere on the interval, the instantaneous rate of change equals the average rate of change.

Theorem 2.22. Cauchy's Mean Value Theorem; Extended Mean Value Theorem
Let f and g be continuous on $[a,b]$ and differentiable on (a,b). Then there exists $c\in(a,b)$ such that
$$(f(b)-f(a))g'(c)=(g(b)-g(a))f'(c).$$
If $g'(x)\neq 0$ for every $x\in(a,b)$, then $g(b)\neq g(a)$ and the result may be written as
$$\frac{f'(c)}{g'(c)}=\frac{f(b)-f(a)}{g(b)-g(a)}.$$

Theorem 2.23. Derivative of an Inverse Function
Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be one-to-one, and suppose that f is differentiable at an interior point $x_0 \in I$ with $f'(x_0) \neq 0$. If the inverse $g = f^{-1}$ is continuous at $y_0 = f(x_0)$, then g is differentiable at y_0 and

$$g'(y_0) = \frac{1}{f'(x_0)}.$$

In particular, the continuity hypothesis is automatic when f is continuous and strictly monotone on an interval.

Theorem 2.24. Fundamental Theorem of Calculus
If f is continuous on $[a, b]$ and

$$F(x) = \int_a^x f(t)\,\mathrm{d}t,$$

then $F'(x) = f(x)$. Conversely, if $F' = f$ on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Theorem 2.25. Taylor's Theorem
Let f be $(n + 1)$ times differentiable on an open interval containing the closed interval with endpoints a and x. Then there exists a point ξ strictly between a and x such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k + \frac{f^{(n+1)}(\xi)}{(n + 1)!}(x - a)^{n+1}.$$

The final term is the Lagrange remainder.

Corollary 2.26. Taylor Error Bound
If

$$|f^{(n+1)}(t)| \leq M$$

for every t between a and x, then the Taylor remainder satisfies

$$|R_n(x)| \leq \frac{M}{(n + 1)!}|x - a|^{n+1}.$$

Strategy. Taylor expansion is one of the fastest tools for limits, approximations, and hidden series evaluations. Expressions involving e^x, $\sin x$, $\cos x$, $\ln(1+x)$, or $\arctan x$ often invite Taylor expansion.

Formula 2.27. Maclaurin Series
The most important Maclaurin series are

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n + 1)!}, \qquad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

$$\frac{1}{1 - x} = \sum_{n=0}^{\infty} x^n \qquad (|x| < 1),$$

$$\ln(1 + x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \qquad (-1 < x \leq 1),$$

and

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \qquad (|x| \le 1).$$

Also,

$$-\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n} \qquad (-1 \le x < 1),$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \qquad \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!},$$

and

$$\operatorname{artanh} x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} \qquad (|x| < 1).$$

Formula 2.28. Binomial Series
For any real or complex number β,

$$(1+x)^\beta = \sum_{n=0}^{\infty} \binom{\beta}{n} x^n \qquad (|x| < 1),$$

where

$$\binom{\beta}{n} = \frac{\beta(\beta-1)\cdots(\beta-n+1)}{n!}.$$

Formula 2.29. Chebyshev Polynomials
The Chebyshev polynomials of the first kind are characterized by

$$T_n(\cos\theta) = \cos(n\theta).$$

They satisfy

$$T_0(x) = 1, \qquad T_1(x) = x, \qquad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

Hence multiple-angle trigonometric expressions can be converted into polynomial calculations almost immediately; for example,

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta.$$

Formula 2.30. Classical Series Values
The following standard series values are useful in computation:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90},$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \ln 2, \qquad \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \frac{\pi}{4}.$$

More generally, the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$.

Formula 2.31. Geometric-Series Differentiation
Starting from

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \qquad (|x| < 1),$$

term-by-term differentiation gives

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1} \qquad (|x| < 1),$$

and hence

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2} \qquad (|x| < 1).$$

Formula 2.32. Telescoping Series
If

$$a_n = b_n - b_{n+1},$$

then the partial sums satisfy

$$\sum_{n=1}^{N} a_n = b_1 - b_{N+1}.$$

Hence, if $b_n \to L$, then

$$\sum_{n=1}^{\infty} a_n = b_1 - L.$$

Formula 2.33. Abel's Summation Formula; Summation by Parts
Let

$$A_k = \sum_{j=m}^{k} a_j.$$

Then

$$\sum_{k=m}^{n} a_k b_k = A_n b_n + \sum_{k=m}^{n-1} A_k (b_k - b_{k+1}).$$

This is the discrete analogue of integration by parts and is useful when one factor has simple partial sums while the other changes slowly.

Formula 2.34. Euler–Maclaurin Summation Formula
Let $p \geq 1$ and let $f \in C^{2p}([m,n])$, where $m < n$ are integers. Then

$$\sum_{k=m}^{n} f(k) = \int_m^n f(x)\,\mathrm{d}x + \frac{f(m)+f(n)}{2} + \sum_{r=1}^{p-1} \frac{B_{2r}}{(2r)!}\Big(f^{(2r-1)}(n) - f^{(2r-1)}(m)\Big) + R_p,$$

where B_{2r} are the Bernoulli numbers and

$$|R_p| \leq \frac{2\zeta(2p)}{(2\pi)^{2p}} \int_m^n |f^{(2p)}(x)|\,\mathrm{d}x.$$

The case $p = 1$ gives the trapezoidal approximation together with a rigorous error bound.

Formula 2.35. Asymptotic Growth Hierarchy
For fixed $p, q > 0$ and $c > 1$, as $n \to \infty$,

$$(\ln n)^p = o(n^q), \qquad n^q = o(c^n), \qquad c^n = o(n!), \qquad n! = o(n^n).$$

Strategy. For fast series calculations, first try to reduce the expression to a geometric series, a differentiated geometric series, a Maclaurin series, or a known special value such as $\pi^2/6$, $\pi^4/90$, $\ln 2$, or $\pi/4$.

Formula 2.36. Integration by Parts
If u and v are differentiable functions, then

$$\int u \, \mathrm{d}v = uv - \int v \, \mathrm{d}u.$$

Equivalently,

$$\int_a^b u(x)v'(x) \, \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, \mathrm{d}x.$$

Theorem 2.37. Leibniz Integral Rule; Differentiation under the Integral Sign
Let I be an open interval, let $a, b : I \to \mathbb{R}$ be continuously differentiable, and suppose that f and $\partial f/\partial t$ are continuous on

$$\{(x,t) \mid t \in I, \ a(t) \le x \le b(t)\}.$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{a(t)}^{b(t)} f(x,t) \, \mathrm{d}x = f(b(t),t)b'(t) - f(a(t),t)a'(t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x,t) \, \mathrm{d}x.$$

In particular, if the endpoints are constant, only the integral of the partial derivative remains.

Tactic. Introducing a parameter and differentiating under the integral sign is often called Feynman's trick. It is especially effective when direct integration is difficult but the parameter derivative is elementary.

Formula 2.38. Frullani's Integral
Let $f : [0, \infty) \to \mathbb{R}$ be continuous, suppose that the finite limits $f(0)$ and $f(\infty) \coloneqq \lim_{x\to\infty} f(x)$ exist, and assume that the improper integral below converges. For $a, b > 0$,

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} \, \mathrm{d}x = (f(0) - f(\infty)) \ln \frac{b}{a}.$$

A standard sufficient hypothesis is that f is continuously differentiable and $f' \in L^1(0, \infty)$.

Formula 2.39. Arc Length Formula
The length of the graph $y = f(x)$ on $[a, b]$ is

$$L = \int_a^b \sqrt{1 + (f'(x))^2} \, \mathrm{d}x.$$

More generally, if a curve is parametrized by $\mathbf{r}(t) = (x(t), y(t), z(t))$, then

$$L = \int_a^b \|\mathbf{r}'(t)\| \, \mathrm{d}t.$$

Formula 2.40. Viète's Infinite Product
For every real x,

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right),$$

where the value at $x = 0$ is interpreted by continuity. Setting $x = \pi/2$ gives Viète's classical product for $2/\pi$.

Idea. Viète's formula is a product formula for π. The key identity is the double-angle formula

$$\sin(2x) = 2 \sin x \cos x.$$

Formula 2.41. Euler's Formula
For every real number x,

$$e^{ix} = \cos x + i \sin x.$$

In particular,

$$e^{i\pi} + 1 = 0.$$

Remark. The identity $e^{i\pi} + 1 = 0$ is called Euler's identity. It connects the constants e, i, π, 1, and 0 in a single formula.

Formula 2.42. Gaussian Integral
The Gaussian integral is

$$\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}.$$

Equivalently,

$$\int_{0}^{\infty} e^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}.$$

Remark. The Gaussian integral is one of the most famous integrals in calculus. It is closely connected to the normal distribution in probability.

Formula 2.43. Fresnel Integrals
The Fresnel integrals satisfy

$$\int_{0}^{\infty} \cos(x^2)\,\mathrm{d}x = \int_{0}^{\infty} \sin(x^2)\,\mathrm{d}x = \sqrt{\frac{\pi}{8}}.$$

Remark. Fresnel integrals appear in wave optics, diffraction, and the Cornu spiral. They are memorable examples of non-elementary definite integrals with clean values.

Formula 2.44. Sine Integral; Dirichlet Integral
The sine integral is

$$\mathrm{Si}(x) = \int_{0}^{x} \frac{\sin t}{t}\,\mathrm{d}t.$$

The classical Dirichlet integral is

$$\int_{0}^{\infty} \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}.$$

Formula 2.45. Arctangent Integral
For $a > 0$,

$$\int_{-\infty}^{\infty} \frac{1}{x^2 + a^2}\,\mathrm{d}x = \frac{\pi}{a}, \qquad \int_{0}^{\infty} \frac{1}{x^2 + a^2}\,\mathrm{d}x = \frac{\pi}{2a}.$$

More generally,

$$\int_{-\infty}^{\infty} \frac{c}{(x-b)^2 + a^2}\,\mathrm{d}x = \frac{c\pi}{a}.$$

Formula 2.46. Gamma-Type Exponential Integrals
For $a > 0$ and every nonnegative integer n,

$$\int_{0}^{\infty} x^n e^{-ax}\,\mathrm{d}x = \frac{n!}{a^{n+1}}.$$

More generally, for $\mathrm{Re}(s) > 0$,

$$\int_{0}^{\infty} x^{s-1} e^{-ax}\,\mathrm{d}x = \frac{\Gamma(s)}{a^s}.$$

Formula 2.47. Damped Sine and Cosine Integrals
For $a > 0$,

$$\int_{0}^{\infty} e^{-ax} \cos bx\,\mathrm{d}x = \frac{a}{a^2 + b^2}, \qquad \int_{0}^{\infty} e^{-ax} \sin bx\,\mathrm{d}x = \frac{b}{a^2 + b^2}.$$

Strategy. Many fast integral problems reduce to this formula after completing a square or translating the variable. A shift such as $u = x - b$ does not change the infinite limits.

Formula 2.48. Symmetry Trick for Definite Integrals
If f is integrable on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(a+b-x)\,\mathrm{d}x.$$

Hence, if

$$f(x) + f(a+b-x) = C$$

is constant, then

$$\int_a^b f(x)\,\mathrm{d}x = \frac{C(b-a)}{2}.$$

Strategy. This is one of the most useful Integration-Bee-style tricks. Typical signals are intervals such as $[0, \pi/2]$, expressions involving $\tan x$ and $\cot x$, or a denominator that becomes complementary after the substitution $x \mapsto a+b-x$.

Formula 2.49. Parity and Periodicity Shortcuts for Integrals
If f is integrable on $[-a, a]$, then

$$f(-x) = -f(x) \quad \Longrightarrow \quad \int_{-a}^{a} f(x)\,\mathrm{d}x = 0,$$

and

$$f(-x) = f(x) \quad \Longrightarrow \quad \int_{-a}^{a} f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x.$$

If f has period T, then for every c,

$$\int_c^{c+T} f(x)\,\mathrm{d}x = \int_0^T f(x)\,\mathrm{d}x.$$

Formula 2.50. Reciprocal-Substitution Identity
If the improper integrals converge, the substitution $x = 1/t$ gives

$$\int_0^\infty f(x)\,\mathrm{d}x = \int_0^\infty \frac{f(1/x)}{x^2}\,\mathrm{d}x.$$

Therefore,

$$\int_0^\infty f(x)\,\mathrm{d}x = \frac{1}{2}\int_0^\infty \left(f(x) + \frac{f(1/x)}{x^2}\right)\mathrm{d}x.$$

This often reveals a cancellation or symmetry hidden in an integral on $(0, \infty)$.

Formula 2.51. Weierstrass Substitution; Tangent Half-Angle Substitution
The substitution

$$t = \tan\frac{x}{2}$$

converts rational expressions in $\sin x$ and $\cos x$ into rational functions of t:

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1+t^2}.$$

Algorithm 2.52. Fast Integral Recognition Checklist
Before expanding or integrating directly, check in this order:

1. parity, periodicity, and interval symmetry;
2. the substitutions $x \mapsto a-x$, $x \mapsto 1/x$, or $x \mapsto a/x$;
3. a hidden derivative for substitution or integration by parts;

4. a parameter that permits differentiation under the integral sign;
5. Beta–Gamma, Gaussian, Wallis, or a standard trigonometric integral;
6. the tangent half-angle substitution for rational trigonometric expressions.

Formula 2.53. Wallis Product
The Wallis product is

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots .$$

Definition 2.54. Gamma Function
For $x > 0$, the Gamma function is defined by

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} \, \mathrm{d}t.$$

Proposition 2.55. Gamma Function and Factorials
The Gamma function satisfies

$$\Gamma(x+1) = x\Gamma(x).$$

In particular, for every positive integer n,

$$\Gamma(n) = (n-1)!.$$

Also,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Definition 2.56. Beta Function; Euler Beta Integral
For $x, y > 0$, the Beta function is defined by

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1} \, \mathrm{d}t.$$

Formula 2.57. Beta–Gamma Identity
The Beta and Gamma functions are related by

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

In particular, for nonnegative integers m, n,

$$\int_0^1 x^m (1-x)^n \, \mathrm{d}x = B(m+1, n+1) = \frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Formula 2.58. Stirling's Formula
As $n \to \infty$,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n .$$

Remark. Stirling's formula is the standard approximation for large factorials. It often appears in counting, probability, and asymptotic estimation.

Formula 2.59. Trapezoidal Rule
The trapezoidal approximation is

$$\int_a^b f(x) \, \mathrm{d}x \approx \frac{b-a}{2} \big(f(a) + f(b)\big).$$

It has degree of precision 1.

Formula 2.60. Simpson's Rule
Simpson's rule is

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right).$$

It has degree of precision 3.

Remark. Simpson's rule uses quadratic interpolation, but the resulting formula is exact for every polynomial of degree at most 3.

Theorem 2.61. p-Series Test
The series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$.

Theorem 2.62. Direct Comparison Test
Let $0 \le a_n \le b_n$ *for all sufficiently large* n. *If* $\sum b_n$ *converges, then* $\sum a_n$ *converges. If* $\sum a_n$ *diverges, then* $\sum b_n$ *diverges.*

Proposition 2.63. Absolute Convergence Implies Convergence
If

$$\sum_{n=1}^{\infty} |a_n|$$

converges, then $\sum_{n=1}^{\infty} a_n$ *converges.*

Theorem 2.64. Integral Test
Suppose f *is positive, continuous, and decreasing on* $[1, \infty)$, *and let* $a_n = f(n)$. *Then*

$$\sum_{n=1}^{\infty} a_n$$

and

$$\int_1^{\infty} f(x)\,\mathrm{d}x$$

either both converge or both diverge.

Theorem 2.65. Limit Comparison Test
Let $a_n, b_n > 0$. *If*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

for some finite number $c > 0$, *then*

$$\sum_{n=1}^{\infty} a_n \qquad \text{and} \qquad \sum_{n=1}^{\infty} b_n$$

converge or diverge together.

Theorem 2.66. Ratio Test; d'Alembert's Ratio Test
Let $a_n > 0$ *and suppose*

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$$

exists. If $L < 1$, *then* $\sum a_n$ *converges. If* $L > 1$, *then* $\sum a_n$ *diverges. If* $L = 1$, *the test is inconclusive.*

Theorem 2.67. Root Test; Cauchy's Root Test
Let $a_n \geq 0$ and suppose

$$L = \lim_{n\to\infty} \sqrt[n]{a_n}$$

exists. If $L < 1$, then $\sum a_n$ converges. If $L > 1$, then $\sum a_n$ diverges. If $L = 1$, the test is inconclusive.

Theorem 2.68. Cauchy–Hadamard Theorem
For a power series

$$\sum_{n=0}^{\infty} a_n(x-a)^n,$$

the radius of convergence R is given by

$$\frac{1}{R} = \limsup_{n\to\infty} \sqrt[n]{|a_n|}.$$

Equivalently, the series converges absolutely for $|x-a| < R$ and diverges for $|x-a| > R$.

Remark. This is the standard theorem behind the phrase "radius of convergence" for a power series. It is also the source of the name Cauchy–Hadamard.

Theorem 2.69. Termwise Differentiation and Integration of Power Series
Suppose

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$$

has radius of convergence $R > 0$. For $|x-a| < R$,

$$f'(x) = \sum_{n=1}^{\infty} na_n(x-a)^{n-1},$$

and

$$\int_a^x f(t)\,\mathrm{d}t = \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}.$$

The differentiated and integrated series have the same radius of convergence R. Convergence at the endpoints must be checked separately.

Theorem 2.70. Alternating Series Test; Leibniz Test
If

$$a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$$

and $a_n \to 0$, then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converges.

Formula 2.71. Alternating Series Error Estimate
Under the hypotheses of the Alternating Series Test, if

$$S = \sum_{n=1}^{\infty} (-1)^{n+1} a_n, \qquad S_N = \sum_{n=1}^{N} (-1)^{n+1} a_n,$$

then

$$|S - S_N| \leq a_{N+1}.$$

Strategy. Factorials and powers often suggest the Ratio Test, nth powers suggest the Root Test, and alternating signs with decreasing terms suggest the Alternating Series Test.

Theorem 2.72. Divergence Test; Term Test
If the series

$$\sum_{n=1}^{\infty} a_n$$

converges, then

$$\lim_{n\to\infty} a_n = 0.$$

Equivalently, if $\lim_{n\to\infty} a_n$ *does not exist or is not* 0*, then* $\sum_{n=1}^{\infty} a_n$ *diverges.*

Strategy. Before trying a sophisticated convergence test, first check whether the terms go to 0. If the terms do not tend to 0, the series diverges immediately.

Algorithm 2.73. Fast Series Checklist
For a convergence or summation problem:

1. Check the term limit first.
2. Look for telescoping, a geometric series, or a known power series.
3. For positive terms, compare only the dominant asymptotic factors.
4. Use the Ratio Test for factorials and exponentials, and the Root Test for expressions raised to the nth power.
5. For alternating terms, check monotonicity and use the first omitted term to control the error.

Multivariable Calculus

Definition 2.74. Gradient, Divergence, Curl, and Laplacian
For a scalar-valued function $f(x, y, z)$ and a vector field $\mathbf{F} = (P, Q, R)$, we write

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right),$$

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z},$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix},$$

and

$$\Delta f = \nabla \cdot \nabla f = f_{xx} + f_{yy} + f_{zz}.$$

Idea. The gradient points in the direction of steepest increase. Divergence measures source or sink behavior. Curl measures local rotation. The Laplacian measures how a value differs from its surrounding average.

Theorem 2.75. Clairaut's Theorem; Mixed Partial Derivative Theorem
If f has continuous second partial derivatives near a point, then the mixed partial derivatives agree at that point:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Definition 2.76. Differentiability; Total Derivative; Fréchet Derivative
Let $\mathbf{F} : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$, where U is open, and let $\mathbf{a} \in U$. The map $\mathbf{F}$ is differentiable at $\mathbf{a}$ if there exists a linear map $D\mathbf{F}(\mathbf{a}) : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\mathbf{F}(\mathbf{a}+\mathbf{h}) = \mathbf{F}(\mathbf{a}) + D\mathbf{F}(\mathbf{a})\mathbf{h} + \mathbf{r}(\mathbf{h}),$$

where

$$\lim_{\mathbf{h}\to\mathbf{0}} \frac{\|\mathbf{r}(\mathbf{h})\|}{\|\mathbf{h}\|} = 0.$$

Equivalently,

$$\begin{aligned}
&\forall\varepsilon > 0, \quad \exists\delta > 0, \quad \forall\mathbf{h} \in \mathbb{R}^n,\\
&(\mathbf{a}+\mathbf{h} \in U) \wedge \big(0 < \|\mathbf{h}\| < \delta\big) \implies\\
&\qquad \frac{\|\mathbf{F}(\mathbf{a}+\mathbf{h}) - \mathbf{F}(\mathbf{a}) - D\mathbf{F}(\mathbf{a})\mathbf{h}\|}{\|\mathbf{h}\|} < \varepsilon.
\end{aligned}$$

The linear map $D\mathbf{F}(\mathbf{a})$ is the total derivative; in standard coordinates, its matrix is the Jacobian matrix.

Theorem 2.77. Multivariable Chain Rule
Let $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ and $\mathbf{G} : \mathbb{R}^m \to \mathbb{R}^k$ be differentiable. Then

$$D(\mathbf{G} \circ \mathbf{F})(\mathbf{x}) = D\mathbf{G}(\mathbf{F}(\mathbf{x}))D\mathbf{F}(\mathbf{x}).$$

Theorem 2.78. Fubini's Theorem
Let $R \subseteq \mathbb{R}^2$ be a bounded region of the form

$$R = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$

where g_1 and g_2 are continuous and $g_1 \le g_2$. If f is continuous on R, then

$$\iint_R f(x, y)\,\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,\mathrm{d}y\,\mathrm{d}x.$$

If the same region is also described as

$$R = \{(x, y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\},$$

then the integral may equally be evaluated in the reverse order. More generally, for an integrable function on a product measure space, the two iterated integrals agree with the product-space integral.

Definition 2.79. Jacobian
For a map

$$\mathbf{F}(u_1, \ldots, u_n) = (x_1(u_1, \ldots, u_n), \ldots, x_n(u_1, \ldots, u_n)),$$

the Jacobian matrix is

$$D\mathbf{F}(\mathbf{u}) = \left(\frac{\partial x_i}{\partial u_j}\right)_{i,j=1}^{n}.$$

Its determinant

$$\mathrm{Jac}(\mathbf{F}) = \det(D\mathbf{F})$$

is called the Jacobian determinant.

Theorem 2.80. Change of Variables Formula
Let $U, V \subseteq \mathbb{R}^n$ be open, and let $\mathbf{F} : U \to V$ be a C^1 diffeomorphism. If $D \subseteq U$ is measurable and f is integrable on $\mathbf{F}(D)$, then

$$\int_{\mathbf{F}(D)} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \int_D f(\mathbf{F}(\mathbf{u}))\,|\det D\mathbf{F}(\mathbf{u})|\,\mathrm{d}\mathbf{u}.$$

The absolute value is essential because the integral measures unsigned volume, independently of orientation.

Remark. The Jacobian determinant is the local area or volume scaling factor of a change of variables.

Formula 2.81. Polar Coordinates
For polar coordinates (r, θ),

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r \geq 0, \qquad 0 \leq \theta < 2\pi.$$

The area element is

$$\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta.$$

The area enclosed by a polar curve $r = r(\theta)$ is

$$A = \frac{1}{2}\int_\alpha^\beta r(\theta)^2\,\mathrm{d}\theta.$$

Its arc length is

$$L = \int_\alpha^\beta \sqrt{r(\theta)^2 + (r'(\theta))^2}\,\mathrm{d}\theta.$$

Formula 2.82. Cylindrical and Spherical Coordinates
Cylindrical coordinates (r, θ, z) satisfy

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z,$$

with volume element

$$\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z.$$

Spherical coordinates (ρ, θ, ϕ) satisfy

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi,$$

where

$$\rho \geq 0, \qquad 0 \leq \theta < 2\pi, \qquad 0 \leq \phi \leq \pi.$$

The volume element is

$$\mathrm{d}V = \rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\phi.$$

Here θ is the azimuthal angle measured from the positive x-axis in the xy-plane, and ϕ is the polar angle measured from the positive z-axis.

Formula 2.83. Differential Operators in Standard Coordinates
Let

$$\mathbf{F} = F_r\mathbf{e}_r + F_\theta\mathbf{e}_\theta + F_z\mathbf{e}_z$$

in cylindrical coordinates. Then

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{\partial f}{\partial z}\mathbf{e}_z,$$

$$\nabla \cdot \mathbf{F} = \frac{1}{r}\frac{\partial}{\partial r}(rF_r) + \frac{1}{r}\frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z},$$

and

$$\Delta f = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}.$$

In spherical coordinates (ρ, θ, ϕ), where θ is the azimuthal angle and ϕ is the polar angle, write

$$\mathbf{F} = F_\rho \mathbf{e}_\rho + F_\theta \mathbf{e}_\theta + F_\phi \mathbf{e}_\phi.$$

Then

$$\nabla f = \frac{\partial f}{\partial \rho}\mathbf{e}_\rho + \frac{1}{\rho \sin\phi}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{1}{\rho}\frac{\partial f}{\partial \phi}\mathbf{e}_\phi,$$

$$\nabla \cdot \mathbf{F} = \frac{1}{\rho^2}\frac{\partial}{\partial \rho}(\rho^2 F_\rho) + \frac{1}{\rho \sin\phi}\frac{\partial F_\theta}{\partial \theta} + \frac{1}{\rho \sin\phi}\frac{\partial}{\partial \phi}(\sin\phi\, F_\phi),$$

and

$$\Delta f = \frac{1}{\rho^2}\frac{\partial}{\partial \rho}\left(\rho^2\frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2 \sin^2\phi}\frac{\partial^2 f}{\partial \theta^2} + \frac{1}{\rho^2 \sin\phi}\frac{\partial}{\partial \phi}\left(\sin\phi\frac{\partial f}{\partial \phi}\right).$$

Theorem 2.84. Inverse Function Theorem
Let $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable near $\mathbf{a}$. If

$$\det D\mathbf{F}(\mathbf{a}) \neq 0,$$

then $\mathbf{F}$ has a continuously differentiable local inverse near $\mathbf{a}$.

Theorem 2.85. Implicit Function Theorem
Let $F(x, y)$ be continuously differentiable near (a, b), with

$$F(a, b) = 0, \qquad \frac{\partial F}{\partial y}(a, b) \neq 0.$$

Then, near (a, b), the equation $F(x, y) = 0$ determines y as a differentiable function of x, and

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{F_x}{F_y}.$$

Formula 2.86. Second Derivative Test and Hessian
At a critical point (a, b) of a twice differentiable function $f(x, y)$, let

$$\Delta_H = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

If $\Delta_H > 0$ and $f_{xx}(a, b) > 0$, the point is a local minimum. If $\Delta_H > 0$ and $f_{xx}(a, b) < 0$, it is a local maximum. If $\Delta_H < 0$, it is a saddle point. If $\Delta_H = 0$, the test is inconclusive.

Theorem 2.87. Lagrange Multipliers
Suppose that f has a local maximum or minimum subject to the constraint

$$g(x_1, \ldots, x_n) = 0$$

at a point where $\nabla g \neq \mathbf{0}$. Then there exists a scalar λ such that

$$\nabla f = \lambda \nabla g.$$

Idea. At a constrained optimum, the level surface of f is tangent to the constraint surface. Thus their normal vectors are parallel.

Theorem 2.88. Euler's Homogeneous Function Theorem
Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable and homogeneous of degree k, meaning

$$f(t\mathbf{x}) = t^k f(\mathbf{x})$$

for $t > 0$. Then

$$\mathbf{x} \cdot \nabla f(\mathbf{x}) = k f(\mathbf{x}).$$

In coordinates,

$$\sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i} = k f.$$

Formula 2.89. Parametrized Curves
Let $\mathbf{r}(t)$ be a twice differentiable curve. Its velocity, speed, and acceleration are

$$\mathbf{v}(t) = \mathbf{r}'(t), \qquad \|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\|, \qquad \mathbf{a}(t) = \mathbf{r}''(t).$$

If $\mathbf{r}'(t_0) \neq \mathbf{0}$, the tangent line at t_0 is

$$\mathbf{r}(t_0) + s\mathbf{r}'(t_0), \qquad s \in \mathbb{R}.$$

The speed is constant if and only if

$$\mathbf{r}'(t) \cdot \mathbf{r}''(t) = 0.$$

Definition 2.90. Scalar Line Integral
If a curve C is parametrized by $\mathbf{r}(t)$ for $a \leq t \leq b$, then

$$\int_C f \, \mathrm{d}s = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, \mathrm{d}t.$$

The value is unchanged by any regular reparametrization.

Definition 2.91. Line Integral
If a curve C is parametrized by $\mathbf{r}(t)$ for $a \leq t \leq b$, then the line integral of a vector field $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, \mathrm{d}t.$$

Theorem 2.92. Fundamental Theorem for Line Integrals; Gradient Theorem
If $\mathbf{F} = \nabla f$ and C is a smooth curve from $\mathbf{a}$ to $\mathbf{b}$, then

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a}).$$

Hence the integral is path independent.

Proposition 2.93. Conservative-Field Test
Let $U \subseteq \mathbb{R}^n$ be open and simply connected, and let $\mathbf{F} : U \to \mathbb{R}^n$ be continuously differentiable. Then $\mathbf{F}$ is conservative if and only if its Jacobian matrix is symmetric:

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i} \qquad (1 \leq i, j \leq n).$$

In $\mathbb{R}^3$, this is equivalent to $\operatorname{curl} \mathbf{F} = \mathbf{0}$. Simple connectedness cannot in general be omitted.

Theorem 2.94. Green's Theorem
Let $D \subseteq \mathbb{R}^2$ be a bounded region whose positively oriented boundary $C = \partial D$ is piecewise smooth, and let P, Q have continuous first partial derivatives on an open set containing D. Then

$$\oint_C P \, \mathrm{d}x + Q \, \mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathrm{d}A.$$

Idea. Green's Theorem converts circulation along a boundary curve into curl over the region inside. It is the two-dimensional ancestor of Stokes' Theorem.

Theorem 2.95. Stokes' Theorem
Let S be an oriented piecewise smooth surface with positively oriented piecewise smooth boundary ∂S, and let $\mathbf{F}$ be continuously differentiable on an open set containing S. Then

$$\oint_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} \, \mathrm{d}S.$$

The orientation of ∂S is determined from that of S by the right-hand rule.

Idea. Stokes' Theorem says that circulation around the boundary equals total curl through the surface. It turns a boundary integral into an interior integral.

Theorem 2.96. Divergence Theorem; Gauss's Theorem
Let $V \subseteq \mathbb{R}^3$ be a bounded solid whose boundary $S = \partial V$ is piecewise smooth and oriented outward. If $\mathbf{F}$ is continuously differentiable on an open set containing V, then

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = \iiint_V \operatorname{div} \mathbf{F} \, \mathrm{d}V.$$

Idea. The Divergence Theorem says that outward flux through the boundary equals total source strength inside the region.

Remark. Green's Theorem, Stokes' Theorem, and the Divergence Theorem share the same philosophy: a boundary integral is equal to an interior integral. This is the geometric heart of vector calculus.

Real Analysis

Definition 2.97. Metric Space and Open Ball
A metric space is a pair (X, d) in which $d : X \times X \to \mathbb{R}$ satisfies, for all $x, y, z \in X$,

$$d(x,y) \geq 0, \qquad d(x,y) = 0 \iff x = y,$$

$$d(x,y) = d(y,x), \qquad d(x,z) \leq d(x,y) + d(y,z).$$

The open ball of radius $r > 0$ centered at x is

$$B_r(x) = \{y \in X \mid d(x,y) < r\}.$$

Definition 2.98. Open and Closed Sets in a Metric Space
A subset $U \subseteq X$ is open if

$$\forall x \in U, \quad \exists r > 0, \qquad B_r(x) \subseteq U.$$

A subset $F \subseteq X$ is closed if $X \setminus F$ is open.

Definition 2.99. Limit of a Map Between Metric Spaces
Let (X, d_X) and (Y, d_Y) be metric spaces, let $D \subseteq X$, let a be an accumulation point of D, and let $f : D \to Y$. The statement

$$\lim_{x \to a} f(x) = L$$

means

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \qquad 0 < d_X(x,a) < \delta \implies d_Y(f(x), L) < \varepsilon.$$

Definition 2.100. Convergence of a Sequence
A sequence (x_n) in a metric space (X, d) converges to $x \in X$, written $x_n \to x$, if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \qquad d(x_n, x) < \varepsilon.$$

Definition 2.101. Cauchy Sequence and Complete Metric Space
A sequence (x_n) in (X, d) is Cauchy if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \qquad d(x_m, x_n) < \varepsilon.$$

The metric space is complete if every Cauchy sequence in X converges to a point of X.

Definition 2.102. Continuity in Metric Spaces
Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is continuous at $x_0 \in X$ if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in X, \qquad d_X(x, x_0) < \delta \implies d_Y\big(f(x), f(x_0)\big) < \varepsilon.$$

It is continuous on X if it is continuous at every $x_0 \in X$.

Theorem 2.103. Sequential Criterion for Continuity
A map $f : X \to Y$ between metric spaces is continuous at $x \in X$ if and only if

$$x_n \to x \implies f(x_n) \to f(x)$$

for every sequence (x_n) in X.

Definition 2.104. Uniform Continuity
A map $f : (X, d_X) \to (Y, d_Y)$ is uniformly continuous if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x, y \in X, \qquad d_X(x, y) < \delta \implies d_Y\big(f(x), f(y)\big) < \varepsilon.$$

Unlike ordinary continuity, the same δ must work for every pair $x, y \in X$.

Definition 2.105. Pointwise and Uniform Convergence
Let $f_n : E \to Y$ and $f : E \to Y$, where (Y, d) is a metric space. The sequence (f_n) converges pointwise to f, written $f_n \xrightarrow{\text{ptw}} f$, if

$$\forall x \in E, \quad \forall \varepsilon > 0, \quad \exists N = N(x, \varepsilon) \in \mathbb{N}, \quad \forall n \geq N, \qquad d\big(f_n(x), f(x)\big) < \varepsilon.$$

It converges uniformly to f, written $f_n \rightrightarrows f$, if

$$\forall \varepsilon > 0, \quad \exists N = N(\varepsilon) \in \mathbb{N}, \quad \forall n \geq N, \quad \forall x \in E, \qquad d\big(f_n(x), f(x)\big) < \varepsilon.$$

For real- or complex-valued functions, this is equivalent to

$$\|f_n - f\|_\infty = \sup_{x \in E} |f_n(x) - f(x)| \longrightarrow 0.$$

Definition 2.106. Equicontinuity and Uniform Boundedness
A family $\mathcal{F}$ of maps from (X, d_X) to (Y, d_Y) is equicontinuous at $x_0 \in X$ if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall f \in \mathcal{F}, \quad \forall x \in X, \qquad d_X(x, x_0) < \delta \implies d_Y\big(f(x), f(x_0)\big) < \varepsilon.$$

A sequence of real-valued functions (f_n) is uniformly bounded if

$$\exists M > 0, \quad \forall n \in \mathbb{N}, \quad \forall x \in X, \qquad |f_n(x)| \leq M.$$

Theorem 2.107. Least Upper Bound Property; Completeness Axiom of $\mathbb{R}$
Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$. Equivalently, every nonempty subset bounded below has a greatest lower bound.

Definition 2.108. Limit Superior and Limit Inferior
For a real sequence (a_n), define

$$\limsup_{n \to \infty} a_n = \lim_{n \to \infty} \sup_{k \geq n} a_k, \qquad \liminf_{n \to \infty} a_n = \lim_{n \to \infty} \inf_{k \geq n} a_k,$$

whenever these extended real limits are allowed.

Theorem 2.109. Monotone Convergence Theorem for Sequences
Every bounded monotone real sequence converges. More precisely, every increasing sequence bounded above converges to its supremum, and every decreasing sequence bounded below converges to its infimum.

Theorem 2.110. Bolzano–Weierstrass Theorem
Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Theorem 2.111. Heine–Borel Theorem
A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Theorem 2.112. Heine–Cantor Theorem; Uniform Continuity on Compact Sets
If $K \subseteq \mathbb{R}^n$ is compact and $f : K \to \mathbb{R}^m$ is continuous, then f is uniformly continuous on K.

Strategy. In analysis questions, compactness usually means that one may extract a convergent subsequence or that a continuous function attains extrema. In $\mathbb{R}^n$, the fast recognition phrase is "closed and bounded."

Theorem 2.113. Cauchy Criterion for Convergence
In a complete metric space, a sequence converges if and only if it is Cauchy. In particular, for (a_n) in $\mathbb{R}$ or $\mathbb{C}$,

$$(a_n) \text{ converges} \iff \forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall m, n \geq N,\ |a_m - a_n| < \varepsilon.$$

Theorem 2.114. Uniform Cauchy Criterion
Let (Y, d) be complete and let $f_n : E \to Y$. Then (f_n) converges uniformly on E if and only if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad \forall x \in E, \qquad d\big(f_m(x), f_n(x)\big) < \varepsilon.$$

Theorem 2.115. Weierstrass M-Test
Let (f_n) be a sequence of functions on a set E. If

$$|f_n(x)| \leq M_n \quad \text{for all } x \in E$$

and

$$\sum_{n=1}^{\infty} M_n$$

converges, then the series

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly on E.

Theorem 2.116. Uniform Limit Theorem
If (f_n) is a sequence of continuous functions on a set E and $f_n \rightrightarrows f$ on E, then f is continuous on E.

Theorem 2.117. Term-by-Term Differentiation
Suppose (f_n) is a sequence of continuously differentiable functions on a closed bounded interval $[a, b]$, and suppose $(f_n(x_0))$ converges for some $x_0 \in [a, b]$. If $f_n' \rightrightarrows g$ on $[a, b]$, then $f_n \rightrightarrows f$ for a differentiable function f, and

$$f' = g = \lim_{n\to\infty} f_n'.$$

Theorem 2.118. Arzelà–Ascoli Theorem
Every uniformly bounded and equicontinuous sequence of real-valued continuous functions on a compact metric space has a uniformly convergent subsequence.

Theorem 2.119. Stone–Weierstrass Theorem
A subalgebra of $C(K,\mathbb{R})$ that contains the constants and separates points is dense in $C(K,\mathbb{R})$ under the supremum norm, provided K is compact.

Theorem 2.120. Banach Fixed-Point Theorem; Contraction Mapping Theorem
Let (X,d) be a nonempty complete metric space, and suppose that $T : X \to X$ is a contraction: there exists $q \in [0,1)$ such that

$$\forall x, y \in X, \qquad d\big(T(x), T(y)\big) \le q d(x,y).$$

Then T has a unique fixed point $x^* \in X$, and the iteration $x_{n+1} = T(x_n)$ converges to x^* for every initial point $x_0 \in X$.

Theorem 2.121. Baire Category Theorem
A nonempty complete metric space cannot be written as a countable union of nowhere dense closed sets. Equivalently, a countable intersection of open dense subsets of a complete metric space is dense.

Theorem 2.122. Tonelli's Theorem
Let (X,μ) and (Y,ν) be σ-finite measure spaces, and let $f : X \times Y \to [0,\infty]$ be measurable. Then the iterated integrals and the product-space integral agree, possibly with value $+\infty$:

$$\int_{X\times Y} f \,\mathrm{d}(\mu \times \nu) = \int_X \left(\int_Y f(x,y)\,\mathrm{d}\nu(y)\right) \mathrm{d}\mu(x) = \int_Y \left(\int_X f(x,y)\,\mathrm{d}\mu(x)\right) \mathrm{d}\nu(y).$$

In particular, a nonnegative series may be integrated term by term.

Theorem 2.123. Fatou's Lemma
If $f_n \ge 0$ are measurable, then

$$\int \liminf_{n\to\infty} f_n \,\mathrm{d}\mu \le \liminf_{n\to\infty} \int f_n \,\mathrm{d}\mu.$$

Theorem 2.124. Lebesgue Monotone Convergence Theorem; Beppo Levi Theorem
If $0 \le f_1 \le f_2 \le \cdots$ and $f_n \xrightarrow{\text{a.e.}} f$, then

$$\lim_{n\to\infty} \int f_n \,\mathrm{d}\mu = \int f \,\mathrm{d}\mu.$$

Theorem 2.125. Lebesgue Dominated Convergence Theorem
If $f_n \xrightarrow{\text{a.e.}} f$ and there exists an integrable function g such that $|f_n| \le g$ for all n, then

$$\lim_{n\to\infty} \int f_n \,\mathrm{d}\mu = \int f \,\mathrm{d}\mu.$$

Theorem 2.126. Density of Irrational Rotations
If $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, then the sequence of fractional parts

$$\{n\alpha\}, \qquad n = 0, 1, 2, \ldots,$$

is dense in $[0,1]$. Equivalently, the orbit of any point under rotation of the circle through an irrational multiple of 2π is dense in the circle.

Idea. Divide $[0,1)$ into N equal intervals and apply the pigeonhole principle to $0, \{\alpha\}, \ldots, \{N\alpha\}$. Two points differ by less than $1/N$ modulo 1, producing an arbitrarily small positive rotation step whose multiples approach every point of the circle.

Complex Analysis

Definition 2.127. Complex Differentiability; Holomorphic and Entire Functions
Let $U \subseteq \mathbb{C}$ be open and let $f : U \to \mathbb{C}$. The complex derivative of f at $z_0 \in U$ is

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0},$$

provided the limit exists independently of the direction of approach. The function is holomorphic on U if it is complex differentiable at every point of U, and entire if it is holomorphic on all of $\mathbb{C}$.

Theorem 2.128. Cauchy–Riemann Equations
Write

$$f(z) = u(x, y) + iv(x, y), \qquad z = x + iy.$$

If f is complex differentiable at $z_0 = (x_0, y_0)$ and the first partial derivatives exist there, then

$$\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0), \qquad \frac{\partial u}{\partial y}(x_0, y_0) = -\frac{\partial v}{\partial x}(x_0, y_0).$$

Conversely, if these partial derivatives are continuous near z_0 and satisfy the equations there, then f is complex differentiable at z_0.

Strategy. To test whether a given $f(z) = u(x, y) + iv(x, y)$ can be holomorphic, compute the four first partial derivatives and check the Cauchy–Riemann equations. Failure at one point immediately rules out holomorphicity there.

Proposition 2.129. Holomorphic Functions Have Harmonic Components
If $f = u + iv$ is holomorphic and u, v have continuous second partial derivatives, then

$$\Delta u = 0, \qquad \Delta v = 0.$$

Thus the real and imaginary parts of a holomorphic function are harmonic conjugates locally.

Theorem 2.130. Cauchy–Goursat Integral Theorem
If f is holomorphic on a simply connected domain $U \subseteq \mathbb{C}$, then for every closed piecewise smooth curve γ in U,

$$\oint_\gamma f(z)\, \mathrm{d}z = 0.$$

Equivalently, contour integrals of f in U are path-independent.

Theorem 2.131. Cauchy Integral Formula
Let f be holomorphic on an open set containing a positively oriented simple closed contour γ and its interior. If a lies inside γ, then

$$f(a) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z - a}\, \mathrm{d}z.$$

More generally, for every integer $n \geq 0$,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(z)}{(z - a)^{n+1}}\, \mathrm{d}z.$$

Strategy. When a contour integral contains $(z - a)^{-m}$ and the remaining factor is holomorphic, compare it directly with Cauchy's integral formula before attempting a parametrization of the contour.

Theorem 2.132. Maximum Modulus Principle
A nonconstant holomorphic function on a connected open set cannot attain a maximum of $|f|$ at an interior point. Consequently, if f is holomorphic on a bounded domain and continuous on its closure, then the maximum of $|f|$ occurs on the boundary.

Theorem 2.133. Liouville's Theorem
Every bounded entire function is constant.

Remark. Liouville's Theorem gives a short proof of the Fundamental Theorem of Algebra: if a nonconstant complex polynomial had no zero, then its reciprocal would be a bounded entire function.

Theorem 2.134. Taylor Series for Holomorphic Functions
If f is holomorphic on the open disk $|z-a|<R$, then

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(z-a)^n, \qquad |z-a|<R.$$

Thus a holomorphic function is analytic: complex differentiability implies local representability by a convergent power series.

Theorem 2.135. Laurent Series
If f is holomorphic on an annulus

$$r < |z-a| < R,$$

then it has a unique Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n$$

converging throughout the annulus. The coefficient c_{-1} is the residue of f at a.

Definition 2.136. Isolated Singularities and Residues
Let a be an isolated singularity of f. In the Laurent expansion about a, the residue is

$$\operatorname{Res}(f;a) = c_{-1}.$$

The singularity is removable if every negative-power coefficient is zero, a pole if only finitely many negative-power coefficients are nonzero, and essential if infinitely many are nonzero.

Formula 2.137. Residues at Poles
If a is a simple pole of f, then

$$\operatorname{Res}(f;a) = \lim_{z\to a}(z-a)f(z).$$

More generally, if a is a pole of order m, then

$$\operatorname{Res}(f;a) = \frac{1}{(m-1)!}\lim_{z\to a}\frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}\left((z-a)^m f(z)\right).$$

If $f(z) = g(z)/h(z)$, where g and h are holomorphic, $h(a)=0$, and $h'(a)\neq 0$, then

$$\operatorname{Res}(f;a) = \frac{g(a)}{h'(a)}.$$

Theorem 2.138. Residue Theorem
Let f be holomorphic inside and on a positively oriented simple closed contour γ, except for isolated singularities $a_1,\ldots,a_m$ inside γ. Then

$$\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i \sum_{k=1}^{m} \operatorname{Res}(f;a_k).$$

Strategy. Real Integrals by Residues. For a rational integral over $\mathbb{R}$, extend the integrand to a complex rational function, choose a semicircular contour, sum the residues in the appropriate half-plane, and verify that the contribution from the large arc vanishes. Trigonometric integrals over $[0, 2\pi]$ often become contour integrals under the substitution $z = e^{i\theta}$ and

$$\mathrm{d}\theta = \frac{\mathrm{d}z}{iz}.$$

Theorem 2.139. Rouché's Theorem
Let f and g be holomorphic inside and on a simple closed contour γ. If

$$|g(z)| < |f(z)| \qquad \text{for every } z \in \gamma,$$

then f and $f + g$ have the same number of zeros inside γ, counted with multiplicity.

Definition 2.140. Conformal Map
A holomorphic function f is conformal at z_0 if $f'(z_0) \neq 0$. At such a point, f preserves oriented angles and infinitesimal shapes, although it may change scale.

Definition 2.141. Möbius Transformation; Linear Fractional Transformation
A Möbius transformation is a map of the extended complex plane of the form

$$T(z) = \frac{az + b}{cz + d}, \qquad ad - bc \neq 0.$$

It is conformal wherever its derivative is finite and nonzero, and it maps generalized circles—circles and lines—to generalized circles.

Formula 2.142. Standard Conformal Maps
The following transformations are useful to recognize quickly:

$$z \mapsto az + b \quad \text{(rotation, dilation, and translation)},$$

$$z \mapsto \frac{1}{z} \quad \text{(inversion)}, \qquad z \mapsto z^n \quad \text{(angle multiplication)},$$

and

$$z \mapsto \frac{z - i}{z + i},$$

which maps the upper half-plane conformally onto the unit disk.

Chapter 3

Linear Algebra

Related POSTECH Courses: (MATH203) Applied Linear Algebra
Related Books: Linear Algebra and Its Applications (Gilbert Strang)

Definition 3.1. Linear Combination, Span, and Linear Independence
A vector of the form

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$$

is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$. Their span is

$$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k) = \{c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k : c_1, \ldots, c_k \in F\}.$$

The vectors are linearly independent if

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

implies $c_1 = \cdots = c_k = 0$.

Definition 3.2. Linear Map; Linear Transformation
Let V and W be vector spaces over the same field F. A map $T : V \to W$ is linear if

$$\forall \alpha, \beta \in F, \quad \forall \mathbf{u}, \mathbf{v} \in V, \qquad T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}).$$

Equivalently, T preserves vector addition and scalar multiplication. Its kernel and image are

$$\ker T = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}, \qquad \operatorname{im} T = \{T(\mathbf{v}) \mid \mathbf{v} \in V\}.$$

Fact 3.3. Translation Is Affine, Not Linear
For a fixed vector $\mathbf{v} \in \mathbb{R}^n$, the translation

$$T_{\mathbf{v}}(\mathbf{x}) = \mathbf{x} + \mathbf{v}$$

is bijective, with inverse $T_{-\mathbf{v}}$. It is a linear map if and only if $\mathbf{v} = \mathbf{0}$.

Formula 3.4. Inner Product and Adjoint Transfer
For real column vectors,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^{\mathsf{T}}\mathbf{v}.$$

For a real matrix $\mathbf{A}$ of compatible size,

$$\langle \mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}^{\mathsf{T}}\mathbf{x}, \mathbf{y} \rangle.$$

Hence, if $\mathbf{A}$ is symmetric,

$$\langle \mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle.$$

Over $\mathbb{C}$, use

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^{\mathsf{H}}\mathbf{v}$$

and replace $\mathbf{A}^{\mathsf{T}}$ by $\mathbf{A}^{\mathsf{H}}$.

Theorem 3.5. Rank–Nullity Theorem
Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

For a matrix $\mathbf{A} \in F^{m\times n}$, this becomes

$$n = \operatorname{rk}(\mathbf{A}) + \dim \operatorname{Null}(\mathbf{A}).$$

Strategy. Rank–Nullity connects variables, pivots, and free variables. It is the linear algebra version of accounting: nothing disappears; dimensions are redistributed.

Theorem 3.6. Row Rank Equals Column Rank
For every matrix $\mathbf{A}$,

$$\dim \operatorname{Row}(\mathbf{A}) = \dim \operatorname{Col}(\mathbf{A}).$$

This common dimension is called the rank of $\mathbf{A}$ and is denoted by $\operatorname{rk}(\mathbf{A})$.

Theorem 3.7. Invertible Matrix Theorem
For a square matrix $\mathbf{A} \in F^{n\times n}$, the following statements are equivalent:

$$\mathbf{A} \textit{ is invertible}, \qquad \det(\mathbf{A}) \neq 0, \qquad \operatorname{rk}(\mathbf{A}) = n,$$

$$\operatorname{Null}(\mathbf{A}) = \{\mathbf{0}\},$$

and

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

has a unique solution for every $\mathbf{b} \in F^n$.

Remark. This theorem is a dictionary. It translates between determinants, rank, null spaces, and solvability of linear systems.

Proposition 3.8. One-Sided Inverse for Square Matrices
If $\mathbf{A}, \mathbf{B} \in F^{n\times n}$ satisfy

$$\mathbf{A}\mathbf{B} = \mathbf{I}_n,$$

then both matrices are invertible and

$$\mathbf{B}\mathbf{A} = \mathbf{I}_n.$$

Thus, for square matrices, a right inverse is automatically a left inverse, and conversely.

Formula 3.9. Inverse of a 2×2 Matrix
If

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and $ad - bc \neq 0$, then

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Theorem 3.10. Cramer's Rule
Let $\mathbf{A} \in F^{n\times n}$ be invertible. The unique solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is given by

$$x_i = \frac{\det(\mathbf{A}_i)}{\det(\mathbf{A})},$$

where $\mathbf{A}_i$ is obtained from $\mathbf{A}$ by replacing its ith column with $\mathbf{b}$.

Formula 3.11. Adjugate Formula for the Inverse
For every square matrix $\mathbf{A}$,

$$\mathbf{A}\,\mathrm{adj}(\mathbf{A}) = \mathrm{adj}(\mathbf{A})\mathbf{A} = \det(\mathbf{A})\mathbf{I}_n.$$

Hence, if $\det(\mathbf{A}) \neq 0$, then

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\,\mathrm{adj}(\mathbf{A}).$$

Definition 3.12. Determinant
The determinant is a scalar associated to a square matrix $\mathbf{A} \in F^{n\times n}$, denoted by $\det(\mathbf{A})$. Geometrically, for real matrices, $|\det(\mathbf{A})|$ is the volume scaling factor of the linear transformation $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$.

Formula 3.13. Determinant as Volume
If $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{R}^n$, then the n-dimensional volume of the parallelepiped spanned by these vectors is

$$\left|\det\begin{pmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n\end{pmatrix}\right|.$$

In $\mathbb{R}^3$, this is the absolute value of the scalar triple product

$$|(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|.$$

Formula 3.14. Scalar and Vector Triple Products
For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$,

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \det(\mathbf{a}, \mathbf{b}, \mathbf{c}).$$

The vector triple-product identities are

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c},$$

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}.$$

Formula 3.15. Cross-Product Determinant Identity
For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$,

$$\det(\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}) = \det(\mathbf{a}, \mathbf{b}, \mathbf{c})^2.$$

Therefore, if $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are linearly independent, then

$$\mathbf{a} \times \mathbf{b}, \qquad \mathbf{b} \times \mathbf{c}, \qquad \mathbf{c} \times \mathbf{a}$$

are also linearly independent.

Proposition 3.16. Basic Properties of Determinants
For square matrices $\mathbf{A}, \mathbf{B} \in F^{n\times n}$, we have

$$\det(\mathbf{A}\mathbf{B}) = \det(\mathbf{A})\det(\mathbf{B}),$$

$$\det(\mathbf{A}^\mathsf{T}) = \det(\mathbf{A}),$$

and, if $\mathbf{A}$ is invertible,

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

Formula 3.17. Determinant of a Triangular Matrix
If $\mathbf{A}$ is upper triangular or lower triangular with diagonal entries $a_{11}, a_{22}, \dots, a_{nn}$, then

$$\det(\mathbf{A}) = a_{11}a_{22}\cdots a_{nn}.$$

Formula 3.18. Determinant of a Block Triangular Matrix
If the diagonal blocks are square, then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A}) \det(\mathbf{D}),$$

and the same formula holds for a lower block triangular matrix.

Proposition 3.19. Multilinearity and Alternation of the Determinant
The determinant is linear in each row separately and also linear in each column separately. If two rows or two columns are interchanged, the sign of the determinant changes. In particular, if the rows or columns of $\mathbf{A}$ *are linearly dependent, then*

$$\det(\mathbf{A}) = 0.$$

Strategy. In short determinant problems, first look for dependent rows or columns, row or column sums, repeated patterns, triangular form, and easy row operations. Direct expansion should usually be the last resort, except for 2×2 and 3×3 matrices.

Proposition 3.20. Characterization of the Determinant
Let

$$f : (F^n)^n \to F$$

be alternating and multilinear. Then

$$f(\mathbf{v}_1, \dots, \mathbf{v}_n) = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \det \begin{pmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{pmatrix}.$$

In particular, the determinant is the unique alternating multilinear function satisfying

$$\det(\mathbf{I}_n) = 1.$$

Formula 3.21. Laplace Expansion; Cofactor Expansion
Let $\mathbf{A} = (a_{ij}) \in F^{n \times n}$, and let C_{ij} be the cofactor of a_{ij}. Expanding along row i gives

$$\det(\mathbf{A}) = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

Expanding along column j gives

$$\det(\mathbf{A}) = \sum_{i=1}^{n} a_{ij} C_{ij}.$$

Formula 3.22. Sarrus' Rule
For a 3×3 matrix,

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Formula 3.23. Determinant and Elementary Row Operations
The determinant changes under elementary row operations as follows.

- Swapping two rows multiplies the determinant by -1.
- Multiplying one row by c multiplies the determinant by c.
- Adding a multiple of one row to another row does not change the determinant.

The same rules hold for columns.

Formula 3.24. Matrix Determinant Lemma
If $\mathbf{A} \in F^{n\times n}$ is invertible and $\mathbf{u}, \mathbf{v} \in F^n$, then

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\mathsf{T}) = \det(\mathbf{A}) \left(1 + \mathbf{v}^\mathsf{T}\mathbf{A}^{-1}\mathbf{u}\right).$$

In particular,

$$\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\mathsf{T}) = 1 + \mathbf{v}^\mathsf{T}\mathbf{u}.$$

Theorem 3.25. Sylvester's Determinant Theorem
If $\mathbf{A} \in F^{m\times n}$ and $\mathbf{B} \in F^{n\times m}$, then

$$\det(\mathbf{I}_m + \mathbf{A}\mathbf{B}) = \det(\mathbf{I}_n + \mathbf{B}\mathbf{A}).$$

This allows a determinant involving a large product to be replaced by a determinant of the smaller dimension.

Formula 3.26. Sherman–Morrison Formula
If $\mathbf{A}$ is invertible and $1 + \mathbf{v}^\mathsf{T}\mathbf{A}^{-1}\mathbf{u} \neq 0$, then

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^\mathsf{T})^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}^\mathsf{T}\mathbf{A}^{-1}}{1 + \mathbf{v}^\mathsf{T}\mathbf{A}^{-1}\mathbf{u}}.$$

It is the inverse counterpart of the Matrix Determinant Lemma.

Formula 3.27. Schur Complement Determinant Formula
If $\mathbf{A}$ is invertible, then

$$\det\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A})\det(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}).$$

The matrix $\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$ is the Schur complement of $\mathbf{A}$.

Formula 3.28. Special Matrix Recognition Table
For real or complex square matrices, the following implications permit rapid determinant, inverse, and eigenvalue calculations.

Matrix type	Defining relation	Immediate consequences
Diagonal	$\mathbf{D} = \operatorname{diag}(d_1, \ldots, d_n)$	eigenvalues d_i; $\det(\mathbf{D}) = \prod_i d_i$
Triangular	$a_{ij} = 0$ for $i > j$ or for $i < j$	eigenvalues are diagonal entries; determinant is their product
Orthogonal	$\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_n$	$\mathbf{Q}^{-1} = \mathbf{Q}^\mathsf{T}$; $\lvert\det(\mathbf{Q})\rvert = 1$
Unitary	$\mathbf{U}^\mathsf{H}\mathbf{U} = \mathbf{I}_n$	$\mathbf{U}^{-1} = \mathbf{U}^\mathsf{H}$; every eigenvalue has modulus 1
Idempotent	$\mathbf{P}^2 = \mathbf{P}$	eigenvalues belong to $\{0, 1\}$; $\operatorname{rk}(\mathbf{P}) = \operatorname{tr}(\mathbf{P})$
Involutory	$\mathbf{A}^2 = \mathbf{I}_n$	$\mathbf{A}^{-1} = \mathbf{A}$; eigenvalues belong to $\{-1, 1\}$
Nilpotent	$\mathbf{N}^k = \mathbf{0}$ for some k	every eigenvalue is 0; $\det(\mathbf{N}) = 0$
Permutation	one 1 in each row and column	$\mathbf{P}^{-1} = \mathbf{P}^\mathsf{T}$; determinant is ± 1
Orthogonal projection	$\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^\mathsf{T}$	$\mathbb{R}^n = \operatorname{Col}(\mathbf{P}) \oplus \operatorname{Null}(\mathbf{P})$

Formula 3.29. Vandermonde Determinant
For scalars $x_1, x_2, \ldots, x_n$,

$$\det\begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{1\leq i<j\leq n} (x_j - x_i).$$

Fact 3.30. Pascal Matrices
Define the lower triangular Pascal matrix by

$$\mathbf{P}_n = (p_{ij})_{0\le i,j\le n}, \qquad p_{ij} = \begin{cases} \binom{i}{j}, & j \le i, \\ 0, & j > i. \end{cases}$$

Since all diagonal entries are equal to 1, we have

$$\det(\mathbf{P}_n) = 1.$$

Define the symmetric Pascal matrix by

$$\mathbf{S}_n = (s_{ij})_{0\le i,j\le n}, \qquad s_{ij} = \binom{i+j}{i}.$$

This matrix also has determinant 1.

Remark. Pascal matrices have combinatorial entries while their determinants admit simple closed forms.

Formula 3.31. Cauchy Determinant
Let

$$c_{ij} = \frac{1}{x_i + y_j},$$

where every denominator is nonzero and the x_i and y_j are pairwise distinct within their respective families. Then

$$\det(c_{ij})_{1\le i,j\le n} = \frac{\displaystyle\prod_{1\le i<j\le n} (x_j - x_i)(y_j - y_i)}{\displaystyle\prod_{i=1}^{n}\prod_{j=1}^{n}(x_i + y_j)}.$$

The Hilbert matrix is a famous special case of a Cauchy matrix.

Formula 3.32. Hilbert Matrix Determinant
The $n \times n$ Hilbert matrix is

$$\mathbf{H}_n = \left(\frac{1}{i+j-1}\right)_{1\le i,j\le n}.$$

As a special Cauchy matrix, it satisfies

$$\det(\mathbf{H}_n) = \frac{\displaystyle\prod_{k=1}^{n-1}(k!)^4}{\displaystyle\prod_{k=1}^{2n-1} k!}.$$

Definition 3.33. Hadamard Matrix
A Hadamard matrix of order n is an $n \times n$ matrix $\mathbf{H}$ whose entries are all $+1$ or -1 and whose rows are mutually orthogonal. Equivalently,

$$\mathbf{H}\mathbf{H}^{\mathsf{T}} = n\mathbf{I}_n.$$

Formula 3.34. Hadamard Determinant
If $\mathbf{H}$ is a Hadamard matrix of order n, then

$$|\det(\mathbf{H})| = n^{n/2}.$$

Formula 3.35. Permutation Matrices
A permutation matrix $\mathbf{P}_\sigma$ has exactly one entry equal to 1 in each row and each column. It satisfies

$$\mathbf{P}_\sigma^{-1} = \mathbf{P}_\sigma^{\mathsf{T}}, \qquad \det(\mathbf{P}_\sigma) = \operatorname{sgn}(\sigma).$$

Its powers are determined by the cycle structure of σ.

Formula 3.36. Idempotent, Involutory, and Nilpotent Matrices
If $\mathbf{P}^2 = \mathbf{P}$, then $\mathbf{P}$ is idempotent and its eigenvalues belong to $\{0, 1\}$. If $\mathbf{A}^2 = \mathbf{I}_n$, then $\mathbf{A}$ is involutory and its eigenvalues belong to $\{-1, 1\}$. If $\mathbf{N}^k = \mathbf{0}$ for some k, then $\mathbf{N}$ is nilpotent and every eigenvalue is 0.

Formula 3.37. Eigenvalues of a Circulant Matrix
Let $\mathbf{C}$ be the circulant matrix whose first row is

$$(c_0, c_1, \ldots, c_{n-1}).$$

With $\omega_n = e^{-2\pi i/n}$, its eigenvalues are

$$\lambda_j = \sum_{k=0}^{n-1} c_k \omega_n^{jk}, \qquad j = 0, 1, \ldots, n-1.$$

Hence $\det(\mathbf{C}) = \prod_{j=0}^{n-1} \lambda_j$.

Formula 3.38. Discrete Fourier Matrix
Let $\omega_n = e^{2\pi i/n}$. The normalized discrete Fourier matrix is

$$\mathbf{F}_n = \frac{1}{\sqrt{n}} \left(\omega_n^{-jk}\right)_{0 \le j,k \le n-1}.$$

It is unitary:

$$\mathbf{F}_n^{\mathsf{H}} \mathbf{F}_n = \mathbf{I}_n.$$

Every circulant matrix is diagonalized by $\mathbf{F}_n$.

Formula 3.39. Symmetric Tridiagonal Toeplitz Matrix
For

$$\mathbf{T}_n = \begin{pmatrix} a & b & 0 & \cdots & 0 \\ b & a & b & \ddots & \vdots \\ 0 & b & a & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b \\ 0 & \cdots & 0 & b & a \end{pmatrix},$$

the eigenvalues are

$$a + 2b\cos\frac{k\pi}{n+1}, \qquad k = 1, 2, \ldots, n.$$

Formula 3.40. Continuant Recurrence for Tridiagonal Determinants
Let

$$\mathbf{T}_n = \begin{pmatrix} a_1 & b_1 & 0 & \cdots & 0 \\ c_1 & a_2 & b_2 & \ddots & \vdots \\ 0 & c_2 & a_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b_{n-1} \\ 0 & \cdots & 0 & c_{n-1} & a_n \end{pmatrix}.$$

If $D_n = \det(\mathbf{T}_n)$, then

$$D_0 = 1, \qquad D_1 = a_1, \qquad D_n = a_n D_{n-1} - b_{n-1} c_{n-1} D_{n-2}.$$

Formula 3.41. Householder Reflection
For a nonzero vector $\mathbf{v}$,

$$\mathbf{H} = \mathbf{I}_n - 2\frac{\mathbf{v}\mathbf{v}^\mathsf{T}}{\mathbf{v}^\mathsf{T}\mathbf{v}}$$

is symmetric, orthogonal, and involutory:

$$\mathbf{H}^\mathsf{T} = \mathbf{H}, \qquad \mathbf{H}^\mathsf{T}\mathbf{H} = \mathbf{I}_n, \qquad \mathbf{H}^2 = \mathbf{I}_n.$$

It reflects vectors across the hyperplane perpendicular to $\mathbf{v}$.

Formula 3.42. Rotation and Reflection Matrices
Rotation by angle θ in the plane is represented by

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Reflection across the line making angle θ with the positive x-axis is represented by

$$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

Formula 3.43. Eigenvalues of a Three-Dimensional Rotation Matrix
A rotation of $\mathbb{R}^3$ by angle θ around an axis through the origin has eigenvalues

$$1, \qquad e^{i\theta}, \qquad e^{-i\theta}.$$

The eigenvalue 1 corresponds to the rotation axis. Therefore, if $\theta \not\equiv 0, \pi \pmod{2\pi}$, the rotation matrix has exactly one real eigenvalue.

Proposition 3.44. Trace, Determinant, and Eigenvalues
Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity. Then

$$\operatorname{tr}(\mathbf{A}) = \lambda_1 + \cdots + \lambda_n$$

and

$$\det(\mathbf{A}) = \lambda_1\lambda_2\cdots\lambda_n.$$

Proposition 3.45. Cyclic Property of the Trace
Whenever the products are defined and square,

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA}).$$

More generally,

$$\operatorname{tr}(\mathbf{ABC}) = \operatorname{tr}(\mathbf{BCA}) = \operatorname{tr}(\mathbf{CAB}).$$

The factors may be cyclically permuted, but they may not in general be rearranged arbitrarily.

Definition 3.46. Eigenvalue and Characteristic Polynomial
Let $\mathbf{A} \in F^{n\times n}$. A scalar $\lambda \in F$ is an eigenvalue of $\mathbf{A}$ if there exists a nonzero vector $\mathbf{v} \in F^n$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

The characteristic polynomial of $\mathbf{A}$ is

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A}).$$

Proposition 3.47. Eigenvalues and the Characteristic Polynomial
A scalar λ is an eigenvalue of $\mathbf{A}$ if and only if

$$\det(\lambda \mathbf{I}_n - \mathbf{A}) = 0.$$

Equivalently, the eigenvalues of $\mathbf{A}$ are the roots of the characteristic polynomial $p_{\mathbf{A}}(\lambda)$.

Formula 3.48. All-Ones Matrix
Let $\mathbf{J}_n$ be the $n \times n$ matrix whose entries are all 1. Then

$$\mathbf{J}_n \mathbf{1} = n\mathbf{1}, \qquad \mathbf{J}_n \mathbf{v} = \mathbf{0} \quad \text{if} \quad v_1 + \cdots + v_n = 0.$$

Hence the eigenvalues of $\mathbf{J}_n$ are n with multiplicity 1 and 0 with multiplicity $n-1$. The matrix $\mathbf{J}_n - \mathbf{I}_n$ has eigenvalues $n-1$ with multiplicity 1 and -1 with multiplicity $n-1$.

Formula 3.49. Constant Diagonal and Off-Diagonal Matrix
Let $\mathbf{A} \in F^{n\times n}$ have diagonal entries a and off-diagonal entries b. Then

$$\mathbf{A} = (a-b)\mathbf{I}_n + b\mathbf{J}_n.$$

Counting algebraic multiplicity, its eigenvalues consist of

$$a + (n-1)b$$

once and

$$a - b$$

$n-1$ times. If these two values coincide, they together give the single eigenvalue of multiplicity n.

Formula 3.50. Rank-One Outer Product
Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be nonzero and set

$$\mathbf{A} = \mathbf{v}\mathbf{w}^\mathsf{T}.$$

Then $\mathrm{rk}(\mathbf{A}) = 1$ and

$$\mathbf{A}\mathbf{v} = (\mathbf{w}^\mathsf{T}\mathbf{v})\mathbf{v}.$$

The characteristic polynomial is

$$\lambda^{n-1}(\lambda - \mathbf{w}^\mathsf{T}\mathbf{v}).$$

Thus the eigenvalues, counted with algebraic multiplicity, are $\mathbf{w}^\mathsf{T}\mathbf{v}$ and 0 repeated $n-1$ times. If $\mathbf{w}^\mathsf{T}\mathbf{v} = 0$, every eigenvalue is 0 and $\mathbf{A}^2 = \mathbf{0}$.

Theorem 3.51. Cayley–Hamilton Theorem
Every square matrix satisfies its own characteristic polynomial. That is, if

$$p_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A}),$$

then

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}.$$

Remark. Cayley–Hamilton is a famous named theorem. It is especially useful for reducing high powers of a matrix to lower powers.

Formula 3.52. Companion Matrix and Linear Recurrences
For

$$p(x) = x^m - c_1 x^{m-1} - \cdots - c_m,$$

the companion matrix

$$\mathbf{C} = \begin{pmatrix} c_1 & c_2 & \cdots & c_{m-1} & c_m \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

has characteristic polynomial p. The recurrence

$$a_n = c_1 a_{n-1} + \cdots + c_m a_{n-m}$$

is encoded by multiplication by $\mathbf{C}$, so fast matrix exponentiation computes distant terms efficiently.

Formula 3.53. Fibonacci Q-Matrix
For

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

we have

$$\mathbf{Q}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \qquad (n \geq 1).$$

Taking determinants gives Cassini's identity

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n.$$

Formula 3.54. Matrix Geometric Series
Let $\mathbf{A}$ be an $n \times n$ matrix. If $\mathbf{I}_n - \mathbf{A}$ is invertible, then

$$\sum_{k=0}^{N-1} \mathbf{A}^k = (\mathbf{I}_n - \mathbf{A}^N)(\mathbf{I}_n - \mathbf{A})^{-1}.$$

If the spectral radius of $\mathbf{A}$ is less than 1, then

$$\sum_{k=0}^{\infty} \mathbf{A}^k = (\mathbf{I}_n - \mathbf{A})^{-1}.$$

Strategy. Fast Matrix Powers. To compute a large power $\mathbf{A}^N$, first look for one of the following: diagonalization, a short minimal polynomial, idempotence, involution, nilpotence, rank one, a circulant structure, or a decomposition into $a\mathbf{I}_n + b\mathbf{J}_n$. Cayley–Hamilton then reduces every sufficiently high power to a linear combination of lower powers.

Theorem 3.55. Diagonalization Criterion
A matrix $\mathbf{A} \in F^{n\times n}$ is diagonalizable over F if and only if F^n has a basis consisting of eigenvectors of $\mathbf{A}$.

Corollary 3.56. Distinct Eigenvalues Imply Diagonalizability
If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues in F, then $\mathbf{A}$ is diagonalizable over F.

Proposition 3.57. Polynomials of a Diagonalizable Matrix
Let $\mathbf{A} \in F^{n\times n}$ be diagonalizable over F, and suppose

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}, \qquad \mathbf{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

For a polynomial $p(x) \in F[x]$, we have

$$p(\mathbf{A}) = \mathbf{S}p(\mathbf{D})\mathbf{S}^{-1},$$

where

$$p(\mathbf{D}) = \operatorname{diag}(p(\lambda_1), \ldots, p(\lambda_n)).$$

In particular,

$$\det(p(\mathbf{A})) = p(\lambda_1)p(\lambda_2)\cdots p(\lambda_n).$$

Idea. Similar matrices represent the same linear transformation in different bases. Therefore, applying a polynomial to a diagonalizable matrix is the same as applying that polynomial to each diagonal entry after diagonalization.

Strategy. Distinct eigenvalues are an easy sufficient condition for diagonalizability. They are not necessary: a matrix can be diagonalizable even when some eigenvalues are repeated.

Theorem 3.58. Gershgorin Circle Theorem
Let $\mathbf{A} = (a_{ij}) \in \mathbb{C}^{n\times n}$. Every eigenvalue of $\mathbf{A}$ lies in at least one disk

$$\left\{ z \in \mathbb{C} \mid |z - a_{ii}| \le \sum_{j\neq i} |a_{ij}| \right\}.$$

Theorem 3.59. Perron–Frobenius Theorem, Positive-Matrix Form
If every entry of a real square matrix $\mathbf{A}$ is positive, then $\rho(\mathbf{A})$ is a positive simple eigenvalue with a positive eigenvector, and every other eigenvalue λ satisfies

$$|\lambda| < \rho(\mathbf{A}).$$

Theorem 3.60. Spectral Theorem for Real Symmetric Matrices
If $\mathbf{A} \in \mathbb{R}^{n\times n}$ is symmetric, then all eigenvalues of $\mathbf{A}$ are real, and there exists an orthogonal matrix $\mathbf{Q}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}.$$

Equivalently, $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$.

Remark. A real symmetric matrix has real eigenvalues and admits orthogonal diagonalization.

Definition 3.61. Conjugate Transpose; Hermitian Transpose
For a complex matrix $\mathbf{A} = (a_{ij})$, its conjugate transpose, or Hermitian transpose, is

$$\mathbf{A}^{\mathsf{H}} = \overline{\mathbf{A}}^{\mathsf{T}} = (\overline{a_{ji}}).$$

A complex square matrix is Hermitian if

$$\mathbf{A}^{\mathsf{H}} = \mathbf{A},$$

and unitary if

$$\mathbf{A}^{\mathsf{H}}\mathbf{A} = \mathbf{A}\mathbf{A}^{\mathsf{H}} = \mathbf{I}_n.$$

Proposition 3.62. Basic Rules for the Hermitian Transpose
For complex matrices of compatible sizes,

$$(\mathbf{A}+\mathbf{B})^{\mathsf{H}} = \mathbf{A}^{\mathsf{H}} + \mathbf{B}^{\mathsf{H}}, \qquad (c\mathbf{A})^{\mathsf{H}} = \bar{c}\,\mathbf{A}^{\mathsf{H}},$$

$$(\mathbf{A}\mathbf{B})^{\mathsf{H}} = \mathbf{B}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}}, \qquad (\mathbf{A}^{\mathsf{H}})^{\mathsf{H}} = \mathbf{A}.$$

If $\mathbf{A}$ is real, then $\mathbf{A}^{\mathsf{H}} = \mathbf{A}^{\mathsf{T}}$.

Proposition 3.63. Hermitian Matrix Quick Facts
If $\mathbf{A}$ *is Hermitian, then*

$$\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x} \in \mathbb{R}$$

for every $\mathbf{x} \in \mathbb{C}^n$*. All eigenvalues of* $\mathbf{A}$ *are real, and eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Theorem 3.64. Spectral Theorem for Hermitian Matrices
If $\mathbf{A} \in \mathbb{C}^{n\times n}$ *is Hermitian, then there exists a unitary matrix* $\mathbf{U}$ *and a real diagonal matrix* $\mathbf{D}$ *such that*

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathsf{H}}.$$

Equivalently, $\mathbb{C}^n$ *has an orthonormal basis consisting of eigenvectors of* $\mathbf{A}$*.*

Remark. The real analogue of a Hermitian matrix is a real symmetric matrix. The real analogue of a unitary matrix is an orthogonal matrix.

Definition 3.65. Markov Matrix; Stochastic Matrix
A column-stochastic matrix is a square matrix with nonnegative entries whose columns each sum to 1. Thus

$$\mathbf{1}^{\mathsf{T}}\mathbf{A} = \mathbf{1}^{\mathsf{T}}.$$

Consequently, 1 is an eigenvalue of $\mathbf{A}^{\mathsf{T}}$ and therefore also of $\mathbf{A}$. A probability column vector evolves by

$$\mathbf{p}_{k+1} = \mathbf{A}\mathbf{p}_k.$$

A stationary distribution is a probability vector satisfying $\mathbf{A}\mathbf{p} = \mathbf{p}$.

Remark. Markov matrices are linear algebra objects behind finite-state Markov chains. A stationary distribution is an eigenvector corresponding to the eigenvalue 1.

Theorem 3.66. Cauchy–Schwarz Inequality
For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$*, we have*

$$|\langle \mathbf{u}, \mathbf{v}\rangle| \le \|\mathbf{u}\|\,\|\mathbf{v}\|.$$

Equivalently, for real numbers $a_1, \ldots, a_n$ *and* $b_1, \ldots, b_n$*,*

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right).$$

Strategy. The expression $\sum a_i b_i$ is a strong signal for Cauchy–Schwarz. The vector form is often the most efficient method to recognize the inequality.

Corollary 3.67. Triangle Inequality
For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$*, we have*

$$\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Proposition 3.68. Properties of Orthogonal Matrices
A real square matrix $\mathbf{Q}$ *is orthogonal if*

$$\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_n.$$

If $\mathbf{Q}$ *is orthogonal, then*

$$\|\mathbf{Q}\mathbf{v}\| = \|\mathbf{v}\|, \qquad \langle \mathbf{Q}\mathbf{u}, \mathbf{Q}\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle, \qquad \det(\mathbf{Q}) = \pm 1.$$

Algorithm 3.69. Gaussian Elimination
Gaussian elimination uses elementary row operations to transform a matrix into an upper triangular or row echelon form. It is the standard computational method for solving linear systems, finding ranks, and computing inverses.

Theorem 3.70. LU Decomposition
When Gaussian elimination can be performed without row exchanges, a square matrix $\mathbf{A}$ *can be written as*

$$\mathbf{A} = \mathbf{L}\mathbf{U},$$

where $\mathbf{L}$ *is lower triangular and* $\mathbf{U}$ *is upper triangular.*

Theorem 3.71. Gram–Schmidt Process; QR Decomposition
From any linearly independent list

$$\mathbf{v}_1, \dots, \mathbf{v}_k$$

in an inner product space, one can construct an orthonormal list

$$\mathbf{q}_1, \dots, \mathbf{q}_k$$

with the same span. In matrix form, this leads to the QR decomposition

$$\mathbf{A} = \mathbf{Q}\mathbf{R},$$

where the columns of $\mathbf{Q}$ *are orthonormal and* $\mathbf{R}$ *is upper triangular.*

Remark. Gram–Schmidt is the standard procedure for turning an arbitrary basis into an orthonormal basis.

Formula 3.72. Orthogonal Projection Matrices
For a nonzero vector $\mathbf{v} \in \mathbb{R}^n$,

$$\operatorname{proj}_{\mathbf{v}} \mathbf{x} = \frac{\mathbf{v}^\mathsf{T}\mathbf{x}}{\mathbf{v}^\mathsf{T}\mathbf{v}}\mathbf{v},$$

and the projection matrix onto $\operatorname{span}(\mathbf{v})$ is

$$\mathbf{P}_{\mathbf{v}} = \frac{\mathbf{v}\mathbf{v}^\mathsf{T}}{\mathbf{v}^\mathsf{T}\mathbf{v}}.$$

If $\|\mathbf{v}\| = 1$, then

$$\mathbf{P}_{\mathbf{v}} = \mathbf{v}\mathbf{v}^\mathsf{T}, \qquad \mathbf{I}_n - \mathbf{v}\mathbf{v}^\mathsf{T}$$

projects onto the hyperplane $\mathbf{v}^\perp$.

If the columns of $\mathbf{Q} \in \mathbb{R}^{n\times k}$ are orthonormal, then the orthogonal projection matrix onto $\operatorname{Col}(\mathbf{Q})$ is

$$\mathbf{P} = \mathbf{Q}\mathbf{Q}^\mathsf{T}.$$

More generally, if the columns of $\mathbf{A} \in \mathbb{R}^{n\times k}$ are linearly independent, then

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}.$$

Over $\mathbb{C}$, replace every transpose by the Hermitian transpose.

Proposition 3.73. Projection Matrix Criterion
A matrix $\mathbf{P}$ *is a projection matrix if*

$$\mathbf{P}^2 = \mathbf{P}.$$

Over $\mathbb{R}$*, it is an orthogonal projection matrix if, in addition,*

$$\mathbf{P}^\mathsf{T} = \mathbf{P}.$$

Over $\mathbb{C}$*, the corresponding condition is*

$$\mathbf{P}^\mathsf{H} = \mathbf{P}.$$

Definition 3.74. Positive Definite Matrix
A real symmetric matrix $\mathbf{A}$ is positive definite if

$$\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$$

for every nonzero vector $\mathbf{x}$.

Theorem 3.75. Sylvester's Criterion
A real symmetric matrix is positive definite if and only if all of its leading principal minors are positive.

Definition 3.76. Singular Values
Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. The singular values of $\mathbf{A}$ are the nonnegative square roots of the eigenvalues of

$$\mathbf{A}^{\mathsf{T}}\mathbf{A}.$$

Theorem 3.77. Singular Value Decomposition; SVD
Every real matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ can be written as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}},$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices and $\mathbf{\Sigma}$ is a rectangular diagonal matrix whose diagonal entries are the singular values of $\mathbf{A}$.

Remark. Singular Value Decomposition applies to every real matrix, not only to square or diagonalizable matrices. This is why SVD is one of the most useful decompositions in applied linear algebra.

Theorem 3.78. Jordan Canonical Form
Over the complex numbers, every square matrix is similar to a Jordan matrix. That is, for every $\mathbf{A} \in \mathbb{C}^{n\times n}$, there exists an invertible matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

is in Jordan canonical form.

Remark. Jordan canonical form describes the structure that remains when a matrix cannot be diagonalized.

Chapter 4

Differential Equations

Related POSTECH Courses: (MATH200) Differential Equations
Related Books: Elementary Differential Equations and Boundary Value Problems (William E. Boyce, Richard C. DiPrima, Douglas B. Meade)

Theorem 4.1. Picard–Lindelöf Existence and Uniqueness Theorem
Consider the initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

If f and $\partial f/\partial y$ are continuous near (t_0, y_0), then there exists a unique solution near $t = t_0$.

Remark. The decisive conclusion is existence and uniqueness. It tells us when an initial condition determines exactly one solution.

Theorem 4.2. Superposition Principle
If $y_1, \dots, y_k$ are solutions of a homogeneous linear differential equation, then any linear combination

$$c_1 y_1 + \cdots + c_k y_k$$

is also a solution.

Remark. The Superposition Principle is the differential-equation version of linear algebra. It is one of the main reasons linear equations are much easier to handle than nonlinear equations.

Proposition 4.3. General Solution of a Nonhomogeneous Linear Equation
For a nonhomogeneous linear differential equation, the general solution has the form

$$y = y_h + y_p,$$

where y_h is the general solution of the associated homogeneous equation and y_p is one particular solution of the nonhomogeneous equation.

Formula 4.4. Separable First-Order Equation
A first-order equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(y)$$

is separable. When $h(y) \neq 0$, rewrite it as

$$\frac{1}{h(y)}\,\mathrm{d}y = g(x)\,\mathrm{d}x$$

and integrate both sides.

Formula 4.5. First-Order Linear Equation; Integrating Factor
The equation
$$y' + p(t)y = q(t)$$
has integrating factor
$$\mu(t) = e^{\int p(t)\,\mathrm{d}t}.$$
Then
$$(\mu y)' = \mu q,$$
so
$$y(t) = \frac{1}{\mu(t)}\left(\int \mu(t)q(t)\,\mathrm{d}t + C\right).$$

Formula 4.6. Exact Differential Equation
An equation
$$M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$$
is exact if
$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$
Then there is a potential function $F(x,y)$ such that
$$F_x = M, \qquad F_y = N,$$
and the solution is given implicitly by
$$F(x,y) = C.$$

Formula 4.7. Bernoulli Equation
The Bernoulli equation
$$y' + p(t)y = q(t)y^n \qquad (n \neq 0, 1)$$
becomes a first-order linear equation after the substitution
$$u = y^{1-n}.$$

Tactic. For a first-order ODE, first check whether it is separable, linear, exact, or Bernoulli. These four patterns cover many quick undergraduate differential-equation questions.

Formula 4.8. Second-Order Constant-Coefficient Equation
Consider the homogeneous equation
$$ay'' + by' + cy = 0, \qquad a \neq 0.$$
Its characteristic equation is
$$ar^2 + br + c = 0.$$
If the roots are distinct real numbers r_1, r_2, then
$$y_h = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$
If the equation has a repeated real root r, then
$$y_h = (c_1 + c_2 t)e^{rt}.$$
If the roots are $\alpha \pm i\beta$ with $\beta \neq 0$, then
$$y_h = e^{\alpha t}\left(c_1 \cos \beta t + c_2 \sin \beta t\right).$$

Remark. The characteristic equation converts a constant-coefficient linear ODE into an algebraic equation. This is the key trick.

Definition 4.9. Wronskian
For two differentiable functions y_1 and y_2, their Wronskian is

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t).$$

Proposition 4.10. Wronskian and Linear Independence
If

$$W(y_1, y_2)(t_0) \neq 0$$

for some t_0, then y_1 and y_2 are linearly independent.

Formula 4.11. Abel's Identity
For

$$y'' + p(t)y' + q(t)y = 0,$$

the Wronskian of two solutions satisfies

$$W(t) = Ce^{-\int p(t)\,\mathrm{d}t}.$$

In particular, the Wronskian is either always zero or never zero on an interval where p is continuous.

Formula 4.12. Euler–Cauchy Equation
The equation

$$ax^2y'' + bxy' + cy = 0$$

suggests the trial solution $y = x^r$. This gives the indicial equation

$$ar(r-1) + br + c = 0.$$

Algorithm 4.13. Undetermined Coefficients
For a constant-coefficient equation with forcing term made from polynomials, exponentials, sines, or cosines, guess a particular solution of the same type. If the guess overlaps with y_h, multiply the guess by the smallest power of t that removes the overlap.

Formula 4.14. Variation of Parameters
For

$$y'' + p(t)y' + q(t)y = g(t)$$

with fundamental solutions y_1, y_2 of the homogeneous equation, a particular solution is

$$y_p = -y_1 \int \frac{y_2 g}{W}\,\mathrm{d}t + y_2 \int \frac{y_1 g}{W}\,\mathrm{d}t,$$

where

$$W = y_1y_2' - y_1'y_2.$$

Formula 4.15. Exponential Growth and Decay
The equation

$$y' = ky$$

has solution

$$y(t) = Ce^{kt}.$$

If $k > 0$, this models exponential growth. If $k < 0$, this models exponential decay.

Formula 4.16. Logistic Equation
The logistic equation is

$$y' = ky\left(1 - \frac{y}{K}\right),$$

where $k > 0$ is the growth rate and $K > 0$ is the carrying capacity. Its equilibrium solutions are

$$y = 0 \qquad \text{and} \qquad y = K.$$

Remark. The logistic equation is the standard model for population growth with limited resources. The parameter K represents the limiting population.

Formula 4.17. Simple Harmonic Oscillator
The equation

$$x'' + \omega^2 x = 0$$

has general solution

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t.$$

Equivalently,

$$x(t) = A \cos(\omega t - \delta)$$

for suitable constants A and δ.

Remark. This is the equation behind ideal mass–spring motion and many small oscillation models in physics.

Formula 4.18. Damped Harmonic Oscillator
The equation

$$mx'' + cx' + kx = 0$$

models a damped harmonic oscillator. Its characteristic equation is

$$mr^2 + cr + k = 0.$$

The motion is underdamped, critically damped, or overdamped according as

$$c^2 - 4mk < 0, \qquad c^2 - 4mk = 0, \qquad c^2 - 4mk > 0.$$

Definition 4.19. Laplace Transform
The Laplace transform of a function $f(t)$ is

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty e^{-st} f(t)\, \mathrm{d}t.$$

Formula 4.20. Laplace Transform Table
Let $F(s) = \mathcal{L}\{f(t)\}(s)$. The basic transforms are

$f(t)$	$\mathcal{L}\{f(t)\}(s)$	Conditions
1	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
t^n	$\frac{n!}{s^{n+1}}$	$n \in \mathbb{N}_0$
e^{at}	$\frac{1}{s-a}$	$\mathrm{Re}(s) > a$
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$	$n \in \mathbb{N}_0,\ s > a$
$\sin at$	$\frac{a}{s^2 + a^2}$	$\mathrm{Re}(s) > 0$

$f(t)$	$\mathcal{L}\{f(t)\}(s)$	Conditions
$\cos at$	$\dfrac{s}{s^2+a^2}$	$\mathrm{Re}(s) > 0$
$e^{at}\sin bt$	$\dfrac{b}{(s-a)^2+b^2}$	$\mathrm{Re}(s) > a$
$e^{at}\cos bt$	$\dfrac{s-a}{(s-a)^2+b^2}$	$\mathrm{Re}(s) > a$
$\sinh at$	$\dfrac{a}{s^2-a^2}$	$\mathrm{Re}(s) > \|a\|$
$\cosh at$	$\dfrac{s}{s^2-a^2}$	$\mathrm{Re}(s) > \|a\|$
$f'(t)$	$sF(s) - f(0)$	
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$	
$tf(t)$	$-F'(s)$	
$\int_0^t f(\tau)\,\mathrm{d}\tau$	$\dfrac{F(s)}{s}$	
$u(t-a)f(t-a)$	$e^{-as}F(s)$	$a > 0$
$f(t+T) = f(t)$	$\dfrac{\int_0^T e^{-st}f(t)\,\mathrm{d}t}{1-e^{-sT}}$	$T > 0,\ s > 0$

Proposition 4.21. Laplace Shift and Convolution
If $F(s) = \mathcal{L}\{f(t)\}(s)$*, then*

$$\mathcal{L}\{e^{at}f(t)\}(s) = F(s-a).$$

If

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,\mathrm{d}\tau,$$

then

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f\}\mathcal{L}\{g\}.$$

Formula 4.22. Fourier Series
For a piecewise smooth function f on $[-L, L]$,

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right),$$

where

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,\mathrm{d}x, \qquad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,\mathrm{d}x.$$

If f is even, then $b_n = 0$; if f is odd, then $a_n = 0$. At a jump discontinuity, the series converges to the midpoint of the left and right limits.

Formula 4.23. Classical Fourier Series
For $-\pi < x < \pi$,

$$x = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin(nx),$$

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(nx),$$

and, for $0 < x < \pi$,

$$1 = \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\sin((2k+1)x)}{2k+1}.$$

Definition 4.24. Fourier Transform
For an integrable function $f : \mathbb{R} \to \mathbb{C}$, define

$$\widehat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,\mathrm{d}t.$$

Under suitable hypotheses, the inversion formula is

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(\omega)e^{i\omega t}\,\mathrm{d}\omega.$$

Formula 4.25. Fourier Transform Table
With the convention above, the standard identities are

Function or operation	Fourier transform
$f'(t)$	$i\omega\widehat{f}(\omega)$
$f(t-a)$	$e^{-i\omega a}\widehat{f}(\omega)$
$e^{iat}f(t)$	$\widehat{f}(\omega - a)$
$f(at)$, $a \neq 0$	$\frac{1}{\lvert a\rvert}\widehat{f}\left(\frac{\omega}{a}\right)$
$tf(t)$	$i\frac{\mathrm{d}}{\mathrm{d}\omega}\widehat{f}(\omega)$
$(f * g)(t)$	$\widehat{f}(\omega)\widehat{g}(\omega)$
e^{-at^2}, $a > 0$	$\sqrt{\frac{\pi}{a}}e^{-\omega^2/(4a)}$

Formula 4.26. Linear Systems and Matrix Exponential
The homogeneous linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

has solution

$$\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}(0).$$

If $\mathbf{A}$ is diagonalizable as $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$, then

$$e^{t\mathbf{A}} = \mathbf{S}e^{t\mathbf{D}}\mathbf{S}^{-1}, \qquad e^{t\mathbf{D}} = \operatorname{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t}).$$

Strategy. For a constant-coefficient linear system, the eigenvalues of $\mathbf{A}$ control the qualitative behavior. Negative real parts indicate decay, positive real parts indicate growth, and nonzero imaginary parts indicate rotation or oscillation.

Remark. The Laplace transform turns many differential equations into algebraic equations. This is its main computational advantage.

Definition 4.27. Heat Equation
The heat equation is the partial differential equation

$$u_t = \alpha u_{xx}, \qquad \alpha > 0.$$

It models diffusion, especially the flow of heat in a one-dimensional medium.

Definition 4.28. Wave Equation
The one-dimensional wave equation is

$$u_{tt} = c^2 u_{xx}.$$

It models waves traveling with speed c.

Theorem 4.29. d'Alembert's Formula
Let $c > 0$, let $f \in C^2(\mathbb{R})$, and let $g \in C^1(\mathbb{R})$. The classical solution of

$$u_{tt} = c^2 u_{xx}, \qquad u(x,0) = f(x), \qquad u_t(x,0) = g(x),$$

is

$$u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,\mathrm{d}s.$$

Remark. d'Alembert's formula shows that waves move along the characteristic lines $x - ct =$ constant and $x + ct =$ constant.

Definition 4.30. Laplace Equation
Laplace's equation is

$$\Delta u = 0.$$

In two variables, this means

$$u_{xx} + u_{yy} = 0.$$

A function satisfying Laplace's equation is called harmonic.

Definition 4.31. Poisson Equation
Poisson's equation is

$$\Delta u = f.$$

It is an inhomogeneous version of Laplace's equation.

Remark. The heat equation, wave equation, and Laplace equation are three fundamental partial differential equations with distinct physical interpretations.

Chapter 5

Combinatorics and Discrete Mathematics

Related POSTECH Courses: (MATH261) Discrete Mathematics
Related Books: Introductory Combinatorics (Richard A. Brualdi)

Formula 5.1. Binomial Theorem; Newton's Binomial Formula
For every nonnegative integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

In particular,

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

Remark. The coefficients $\binom{n}{k}$ are the entries of Pascal's triangle. This is one of the most recognizable bridges between algebra and counting.

Strategy. Bijection, Double Counting, and Complementary Counting. Before expanding a complicated sum, try to count the same objects in a second way. Bijections, double counting, and counting the complement are among the fastest and most reusable contest methods.

Formula 5.2. Binomial Inversion
The relations

$$b_n = \sum_{k=0}^{n} \binom{n}{k} a_k$$

and

$$a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k$$

are equivalent.

Formula 5.3. Pascal's Identity and Vandermonde's Identity
Pascal's identity is

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Vandermonde's identity is

$$\sum_{k=0}^{r} \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}.$$

Formula 5.4. Multinomial Theorem
For nonnegative integers $k_1, \ldots, k_m$ with $k_1 + \cdots + k_m = n$,

$$(x_1 + \cdots + x_m)^n = \sum_{k_1+\cdots+k_m=n} \frac{n!}{k_1! \cdots k_m!} x_1^{k_1} \cdots x_m^{k_m}.$$

The coefficient

$$\frac{n!}{k_1! \cdots k_m!}$$

is called a multinomial coefficient.

Formula 5.5. Stars and Bars
The number of nonnegative integer solutions of

$$x_1 + x_2 + \cdots + x_n = k$$

is

$$\binom{n+k-1}{k} = \left(\!\!\binom{n}{k}\!\!\right).$$

Equivalently, the expansion of $(x_1 + x_2 + \cdots + x_n)^k$ has

$$\binom{n+k-1}{k}$$

distinct monomial terms.

Formula 5.6. Lattice Path Counting
The number of shortest lattice paths from $(0, 0)$ to (m, n) using only unit steps to the right and upward is

$$\binom{m+n}{m} = \binom{m+n}{n}.$$

Theorem 5.7. Reflection Principle and Ballot-Path Formula
The reflection principle converts a path that first crosses a boundary into an unrestricted reflected path. In particular, for integers $a \geq b \geq 0$, the number of paths from $(0, 0)$ to (a, b) using right and up steps and never entering the region $y > x$ is

$$\binom{a+b}{b} - \binom{a+b}{b-1} = \frac{a-b+1}{a+1}\binom{a+b}{b}.$$

When $a = b = n$, this becomes the Catalan number

$$\frac{1}{n+1}\binom{2n}{n}.$$

Strategy. Tail Switching for Intersecting Paths. When two directed lattice paths meet, exchange their tails after the first intersection. This produces a bijection with a pair of paths whose endpoints have been crossed. The method is a standard way to count or cancel intersecting path pairs.

Formula 5.8. Ordinary Generating Function Table
For a sequence $(a_n)_{n\geq 0}$, its ordinary generating function is

$$A(x) = \sum_{n=0}^{\infty} a_n x^n.$$

The basic examples are

Sequence a_n	Ordinary generating function $A(x)$	Conditions
1	$\frac{1}{1-x}$	$\lvert x\rvert < 1$
r^n	$\frac{1}{1-rx}$	$\lvert rx\rvert < 1$
n	$\frac{x}{(1-x)^2}$	$\lvert x\rvert < 1$
n^2	$\frac{x(1+x)}{(1-x)^3}$	$\lvert x\rvert < 1$
$\binom{n}{k}$	$\frac{x^k}{(1-x)^{k+1}}$	$k \in \mathbb{N}_0,\ \lvert x\rvert < 1$
$\binom{n+r}{r}$	$\frac{1}{(1-x)^{r+1}}$	$r \in \mathbb{N}_0,\ \lvert x\rvert < 1$
F_n, where $F_0 = 0$ and $F_1 = 1$	$\frac{x}{1-x-x^2}$	$\lvert x\rvert < \frac{\sqrt{5}-1}{2}$
$\frac{1}{n+1}\binom{2n}{n}$	$\frac{1-\sqrt{1-4x}}{2x}$	$\lvert x\rvert < \frac{1}{4}$

Tactic. Generating functions are useful when a recurrence or counting problem has a repeated structure; a known generating function can replace a long derivation.

Formula 5.9. Stirling Numbers and Bell Numbers
The Stirling number of the second kind

$$\left\{ {n \atop k} \right\}$$

counts partitions of an n-element set into k nonempty blocks and satisfies

$$\left\{ {n \atop k} \right\} = k\left\{ {n-1 \atop k} \right\} + \left\{ {n-1 \atop k-1} \right\}.$$

The Bell number

$$B_n = \sum_{k=0}^{n} \left\{ {n \atop k} \right\}$$

counts all set partitions of an n-element set.

Formula 5.10. Fibonacci Numbers; Binet's Formula
If $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$, then

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

In particular,

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \varphi.$$

Formula 5.11. Roots-of-Unity Filter
Let $\omega_m = e^{2\pi i/m}$. Then

$$\frac{1}{m}\sum_{j=0}^{m-1} \omega_m^{j(n-r)} = \begin{cases} 1, & n \equiv r \pmod{m}, \\ 0, & n \not\equiv r \pmod{m}. \end{cases}$$

Consequently, if $A(x) = \sum_{n\geq 0} a_n x^n$, then the terms with indices congruent to r modulo m are extracted by

$$\sum_{n\equiv r \,(\mathrm{mod}\, m)} a_n x^n = \frac{1}{m}\sum_{j=0}^{m-1} \omega_m^{-rj} A(\omega_m^j x).$$

Formula 5.12. Tower of Hanoi
Let T_n be the minimum number of moves needed to move n disks in the Tower of Hanoi puzzle. Then

$$T_n = 2T_{n-1} + 1, \qquad T_1 = 1,$$

and hence

$$T_n = 2^n - 1.$$

Remark. The recurrence comes from moving the top $n-1$ disks away, moving the largest disk once, and then moving the $n-1$ disks back on top of it.

Strategy. State Compression and Transfer Matrices. When a counting process depends only on finitely many recent states, place the transition counts in a matrix $\mathbf{T}$. Counts of length n then have the form

$$\mathbf{u}^{\mathsf{T}}\mathbf{T}^n\mathbf{v}.$$

This is effective for forbidden substrings, tilings, walks, and finite automata.

Theorem 5.13. Pigeonhole Principle; Dirichlet's Box Principle
If $n+1$ objects are placed into n boxes, then at least one box contains at least two objects. More generally, if N objects are placed into k boxes, then at least one box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

Strategy. The Pigeonhole Principle is often hidden in problems asking us to prove that two objects must share a property. The hard part is usually choosing the right "boxes."

Theorem 5.14. Consecutive-Sum Divisibility Lemma
For any integers $a_1, \dots, a_n$, there exists a nonempty consecutive block whose sum is divisible by n. Define prefix sums

$$S_0 = 0, \qquad S_k = \sum_{i=1}^{k} a_i.$$

If some $S_k \equiv 0 \pmod{n}$, use the first k terms. Otherwise, two of $S_1, \dots, S_n$ have the same residue modulo n, and their difference is the required block sum.

Strategy. Coloring, Parity, and Bipartite Obstructions. Colorings convert geometric or movement constraints into parity data. On a bipartite graph, every path alternates colors. Thus a Hamiltonian path can exist only if the two color classes differ in size by at most 1, and a Hamiltonian cycle requires the two classes to have equal size. Checkerboard colorings are especially effective for tiling and movement problems.

Theorem 5.15. Inclusion–Exclusion Principle; Sieve Formula
For finite sets $A_1, \dots, A_n$,

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1}|A_1 \cap \cdots \cap A_n|.$$

Remark. Inclusion–Exclusion is the standard correction mechanism for overcounting. It is also historically connected to sieve methods.

Formula 5.16. Derangement Formula
Let D_n be the number of derangements of n objects, that is, permutations with no fixed points. Then

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Moreover,

$$D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

Remark. The approximation $D_n \approx n!/e$ is very memorable. In probability language, a random permutation has no fixed point with probability approximately $1/e$.

Formula 5.17. Catalan Numbers
The nth Catalan number is

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

It begins as

$$1,\ 1,\ 2,\ 5,\ 14,\ 42,\ 132, \ldots.$$

Remark. Catalan numbers count many different objects, including correct parenthesizations, Dyck paths, triangulations of a polygon, and rooted full binary trees. The number 42 is a standard recognition item.

Theorem 5.18. Cayley's Formula
The number of labeled trees on n vertices is

$$n^{n-2}.$$

Remark. Cayley's formula gives a compact and surprising enumeration of labeled trees.

Theorem 5.19. Burnside's Lemma; Cauchy–Frobenius–Burnside Lemma
Suppose a finite group G acts on a finite set X. The number of orbits is

$$\frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|,$$

where

$$\operatorname{Fix}(g) = \{x \in X \mid g \cdot x = x\}.$$

Idea. Burnside's Lemma says that the number of essentially different objects is the average number of objects fixed by a symmetry.

Lemma 5.20. Handshaking Lemma
For a finite undirected graph $G = (V, E)$,

$$\sum_{v \in V} \deg(v) = 2|E|.$$

In particular, the number of vertices of odd degree is even.

Remark. The name comes from the idea that every handshake contributes 2 to the total count of hands shaken.

Theorem 5.21. Euler's Formula for Planar Graphs
If a connected planar graph has V vertices, E edges, and F faces, then

$$V - E + F = 2.$$

Remark. This is the graph-theoretic version of Euler's polyhedron formula. It is one of the most famous formulas involving planar graphs.

Formula 5.22. Regions Cut by Hyperplanes in General Position
The maximum number of regions into which n hyperplanes in general position divide d-dimensional space is

$$R_d(n) = \sum_{k=0}^{d} \binom{n}{k}.$$

In particular, n lines divide the plane into at most

$$1 + n + \binom{n}{2}$$

regions, and n planes divide $\mathbb{R}^3$ into at most

$$1 + n + \binom{n}{2} + \binom{n}{3}$$

regions.

Formula 5.23. Chords and Diagonals in Convex Position
If n points lie on a circle and no three interior chords are concurrent, drawing every chord divides the disk into

$$1 + \binom{n}{2} + \binom{n}{4}$$

regions. A convex n-gon has at most

$$\binom{n}{4}$$

interior intersection points of diagonals.

Formula 5.24. Triangles in a Complete Diagonal Drawing
For n vertices in convex position with no three diagonals concurrent, the number of triangles whose sides lie on sides or diagonals of the complete drawing is

$$\binom{n}{3} + 4\binom{n}{4} + 5\binom{n}{5} + \binom{n}{6}.$$

Formula 5.25. Triangulation with Interior Vertices
Suppose a polygonal disk is triangulated using V vertices in total, of which b lie on the boundary. Then the number of edges and triangular faces are

$$E = 3V - b - 3, \qquad F_{\triangle} = 2V - b - 2.$$

These values follow from Euler's formula and double counting the triangle-edge incidences.

Strategy. Double Counting with Euler's Formula. In a planar subdivision, count edge-face incidences first and then use $V - E + F = 2$. Interior edges are counted twice and boundary edges once. This is often faster and safer than trying to enumerate faces directly.

Theorem 5.26. Eulerian Trail Criterion
A connected finite graph has an Eulerian circuit if and only if every vertex has even degree. It has an Eulerian trail but not an Eulerian circuit if and only if exactly two vertices have odd degree.

Remark. This theorem is historically connected to Euler's solution of the Seven Bridges of Königsberg problem, which marks the beginning of graph theory and topology. The historical city Königsberg is now Kaliningrad, Russia.

Theorem 5.27. Hall's Marriage Theorem
Let $G = (X \cup Y, E)$ be a bipartite graph. There is a matching that covers every vertex of X if and only if, for every subset $S \subseteq X$,

$$|N(S)| \geq |S|,$$

where $N(S)$ is the set of neighbors of vertices in S.

Strategy. Hall's Marriage Theorem is the standard named theorem for proving that a perfect assignment or matching exists.

Theorem 5.28. Kőnig's Theorem
In every finite bipartite graph, the size of a maximum matching equals the size of a minimum vertex cover.

Theorem 5.29. Max-Flow Min-Cut Theorem
In a finite capacitated network, the maximum value of a source–sink flow equals the minimum capacity of a source–sink cut.

Theorem 5.30. Menger's Theorem, Vertex Form
For nonadjacent vertices u and v, the maximum number of pairwise internally vertex-disjoint u–v paths equals the minimum number of vertices whose deletion separates u from v.

Theorem 5.31. Kirchhoff's Matrix–Tree Theorem
Let

$$\mathbf{L} = \mathbf{D} - \mathbf{A}$$

be the Laplacian matrix of a finite graph. The number of spanning trees equals any cofactor of $\mathbf{L}$ *obtained by deleting the same row and column.*

Theorem 5.32. Sperner's Theorem
The largest family of subsets of $[n]$ in which no member contains another has size

$$\binom{n}{\lfloor n/2 \rfloor}.$$

Theorem 5.33. Dilworth's Theorem
In a finite partially ordered set, the maximum size of an antichain equals the minimum number of chains covering the poset.

Theorem 5.34. Turán's Theorem
Among all n-vertex graphs containing no K_{r+1}, the maximum number of edges is attained by the complete r-partite graph whose part sizes differ by at most 1.

Theorem 5.35. Dirac's and Ore's Hamiltonian Criteria
Let G be a simple graph on $n \geq 3$ vertices. If every vertex has degree at least $n/2$, then G is Hamiltonian. More generally, it is enough that

$$\deg(u) + \deg(v) \geq n$$

for every pair of nonadjacent vertices u, v.

Theorem 5.36. Ramsey's Theorem
For any positive integers r and s, there exists an integer $R(r, s)$ such that every red-blue coloring of the edges of a sufficiently large complete graph contains either a red K_r or a blue K_s.

Formula 5.37. The Party Problem
The classical Ramsey number

$$R(3, 3) = 6$$

says that among any six people, there are either three mutual acquaintances or three mutual strangers.

Remark. Ramsey theory is the mathematics of unavoidable structure. Its slogan is that complete disorder is impossible in a large enough system.

Theorem 5.38. Erdős–Szekeres Theorem
Any sequence of

$$(r-1)(s-1)+1$$

distinct real numbers contains either an increasing subsequence of length r or a decreasing subsequence of length s.

Remark. This is a beautiful pigeonhole-style result. It is often remembered as the monotone subsequence theorem.

Chapter 6

Probability and Statistics

Related POSTECH Courses: (MATH230) Probability & Statistics

Theorem 6.1. Law of Total Probability
If $B_1, \ldots, B_n$ form a partition of the sample space and $\mathbb{P}(B_i) > 0$ for all i, then

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Theorem 6.2. Law of Total Expectation; Tower Property
If X is integrable and Y is a random variable, then

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]\,.$$

When Y is discrete, this becomes

$$\mathbb{E}[X] = \sum_{y} \mathbb{E}[X \mid Y = y]\mathbb{P}(Y = y),$$

where the sum is taken over values y with positive probability.

Formula 6.3. Law of Total Variance

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y])\,.$$

Theorem 6.4. Bayes' Theorem
If $B_1, \ldots, B_n$ form a partition of the sample space and $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

In the two-event form,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Remark. Bayes' Theorem is the standard formula for reversing conditional probability. It updates a prior probability after observing evidence.

Definition 6.5. Independence
Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Equivalently, if $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Random variables X and Y are independent if events determined by X and events determined by Y are independent.

Tactic. "Disjoint" and "independent" are different. If two nonempty events are disjoint, then knowing that one occurred makes the other impossible; they are usually not independent.

Formula 6.6. Complement Rule and Inclusion–Exclusion for Probability
The complement rule is

$$\mathbb{P}(A^{\mathsf{c}}) = 1 - \mathbb{P}(A).$$

For two events,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proposition 6.7. Linearity of Expectation
If $X_1, \ldots, X_n$ are integrable random variables and $a_1, \ldots, a_n$ are scalars, then

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i].$$

Independence is not required.

Strategy. Linearity of expectation is often the most efficient method to count an expected number of objects. Introduce indicator random variables and add their expectations.

Proposition 6.8. Expectation of a Product of Independent Random Variables
If $X_1, \ldots, X_n$ are independent integrable random variables, then their product is integrable and

$$\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n].$$

In particular, if they are also identically distributed, the right-hand side is $(\mathbb{E}[X_1])^n$.

Formula 6.9. Common Probability Distributions
The most frequently used distributions are summarized below.

Distribution	Mass or density function	Mean	Variance
Bernoulli(p)	$\mathbb{P}(X = x) = p^x(1-p)^{1-x}$, $x \in \{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	$\mathbb{P}(X = k) = (1-p)^{k-1}p$, $k \geq 1$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$	λ	λ
Uniform(a, b)	$f(x) = \frac{1}{b-a}$, $a \leq x \leq b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$f(x) = \lambda e^{-\lambda x}$, $x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2

Strategy. Waiting-until questions usually start with a geometric distribution. If a running sum is also involved, combine the expected number of trials with the expected contribution per trial, or condition on the first step.

Formula 6.10. Variance and Covariance Identities
For random variables with finite second moments,

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

and

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y).$$

If X and Y are independent, then $\operatorname{Cov}(X, Y) = 0$, so their variances add. The converse need not hold.

Formula 6.11. Hypergeometric Distribution
Suppose a population has N objects, of which K are successes. If n objects are chosen without replacement and X is the number of successes chosen, then

$$\mathbb{P}(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

Its mean is

$$\mathbb{E}[X] = n\frac{K}{N}.$$

Theorem 6.12. Poisson Limit Theorem; Law of Small Numbers
Let $p_n \in [0,1]$ satisfy $np_n \to \lambda > 0$. For each fixed nonnegative integer k,

$$\binom{n}{k} p_n^k (1-p_n)^{n-k} \longrightarrow e^{-\lambda}\frac{\lambda^k}{k!}.$$

Thus $\mathrm{Bin}(n, p_n)$ *converges in distribution to* $\mathrm{Poi}(\lambda)$.

Remark. This theorem explains why the Poisson distribution models rare events: many trials, small success probability, and finite expected count.

Formula 6.13. Empirical Rule; 68–95–99.7 Rule
For a normal distribution, approximately

$$68\%$$

of the mass lies within one standard deviation of the mean,

$$95\%$$

lies within two standard deviations, and

$$99.7\%$$

lies within three standard deviations.

Definition 6.14. Modes of Convergence for Random Variables
Let X_n and X be random variables on the same probability space. The principal modes of convergence used in probability are:

(i) $X_n \xrightarrow{\text{a.s.}} X$ if
$$\mathbb{P}\big(\{\omega \mid X_n(\omega) \to X(\omega)\}\big) = 1;$$

(ii) $X_n \xrightarrow{\mathbb{P}} X$ if
$$\forall \varepsilon > 0, \qquad \mathbb{P}\big(|X_n - X| > \varepsilon\big) \to 0;$$

(iii) $X_n \xrightarrow{L^p} X$ for $p \geq 1$ if
$$\mathbb{E}\big[|X_n - X|^p\big] \to 0;$$

(iv) $X_n \xrightarrow{\text{d}} X$ if
$$F_{X_n}(x) \to F_X(x)$$
at every continuity point x of the distribution function F_X.

Almost-sure convergence implies convergence in probability, and convergence in probability implies convergence in distribution.

Theorem 6.15. Borel–Cantelli Lemmas
If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then $\mathbb{P}(A_n \textit{ infinitely often}) = 0$. *If the* A_n *are independent and the series diverges, then*

$$\mathbb{P}(A_n \textit{ infinitely often}) = 1.$$

Theorem 6.16. Weak Law of Large Numbers
Let $X_1, X_2, \ldots$ *be independent and identically distributed random variables with finite mean* μ *and finite variance* σ^2. *Then*

$$\frac{X_1 + \cdots + X_n}{n} \xrightarrow{\mathbb{P}} \mu.$$

This follows directly from Chebyshev's inequality because the variance of the sample mean is σ^2/n.

Idea. The Law of Large Numbers says that the sample average stabilizes near the true mean when the number of trials becomes large.

Theorem 6.17. Central Limit Theorem
Let $X_1, X_2, \ldots$ *be independent identically distributed random variables with mean* μ *and variance* $\sigma^2 > 0$. *Then*

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{\mathrm{d}} Z, \qquad Z \sim \mathrm{N}(0,1).$$

Thus the standardized sum converges in distribution to the standard normal distribution.

Remark. The Central Limit Theorem explains why the normal distribution appears so often in nature and statistics.

Theorem 6.18. Markov's Inequality
If X *is a nonnegative random variable and* $a > 0$, *then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 6.19. Chebyshev's Inequality
If X *has mean* μ *and variance* σ^2, *then for every* $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

Remark. Markov's inequality uses only the mean, while Chebyshev's inequality uses the variance. Both are probability bounds that work without assuming a normal distribution.

Theorem 6.20. Chernoff Bound for a Binomial Random Variable
If $X \sim \mathrm{Bin}(n,p)$ *and* $\mu = np$, *then for* $\delta > 0$,

$$\mathbb{P}(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}.$$

For $0 < \delta < 1$,

$$\mathbb{P}(|X - \mu| \geq \delta\mu) \leq 2e^{-\delta^2\mu/3}.$$

Theorem 6.21. Jensen's Inequality
If φ is convex, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

If φ is concave, the inequality is reversed.

Idea. Jensen's inequality says that, for a convex function, applying the function after averaging gives a value no larger than averaging after applying the function.

Formula 6.22. Exact Birthday Problem
Under the standard model of 365 equally likely birthdays and no leap day, the probability that at least two people among n share a birthday is

$$1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \qquad (0 \leq n \leq 365).$$

For $n \geq 366$, the probability is 1 by the pigeonhole principle.

Formula 6.23. Birthday Problem Approximation
In a group of n people, assuming 365 equally likely birthdays, the probability that at least two people share a birthday is approximately

$$1 - \exp\left(-\frac{n(n-1)}{730}\right).$$

For $n = 23$, this probability is already greater than $1/2$.

Remark. The birthday problem is famous because the answer is surprisingly small: only 23 people are needed for a better-than-even chance of a shared birthday.

Formula 6.24. Expected Number of Runs in Coin Tosses
In n tosses of a fair coin, a run is a maximal consecutive block of identical outcomes. If R_n is the number of runs, then

$$\mathbb{E}[R_n] = 1 + \frac{n-1}{2} = \frac{n+1}{2}.$$

Idea. The first toss starts one run. For each of the remaining $n - 1$ positions, a new run begins exactly when the current toss differs from the previous toss, which has probability $1/2$.

Formula 6.25. Monty Hall Problem
There are three doors, exactly one of which hides a prize. The contestant first chooses a door. A host who knows the prize location must then open one of the other doors that hides no prize; whenever two such doors are available, the host chooses between them by a rule independent of the prize location. The contestant is then offered the unopened door. Under this protocol, staying wins with probability $1/3$, whereas switching wins with probability $2/3$.

Remark. The host's action is not random information: the host always reveals a losing door. This is why the original $2/3$ probability of having chosen incorrectly transfers to the remaining unopened door.

Formula 6.26. Coupon Collector
If there are n equally likely coupon types, the expected number of trials needed to see all n types is

$$nH_n = n\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right).$$

Strategy. For a fair die, the expected number of rolls needed to see all six faces is $6H_6$. If a question asks about the order in which the first appearances occur, ignore repeated rolls and look only at the random ordering of the six faces.

Formula 6.27. Gambler's Ruin
Suppose a gambler starts with i dollars and stops when reaching either 0 dollars or N dollars. If the probability of winning each round is p and $q = 1 - p$, then the probability of reaching N before 0 is

$$\frac{1-(q/p)^i}{1-(q/p)^N} \qquad (p \neq q).$$

In the fair case $p = q = 1/2$, the probability is

$$\frac{i}{N}.$$

Remark. Gambler's ruin is a classic random walk result. It is a good example where the fair case has a very simple answer.

Chapter 7

Geometry

Synthetic Geometry

Formula 7.1. Basic Plane and Solid Geometry Table
The basic measurement formulas are

Figure	Perimeter or surface area	Area or volume
Triangle	$a+b+c$	$\frac{1}{2}bh$
Parallelogram	$2(a+b)$	bh
Trapezoid	$a+b+c+d$	$\frac{1}{2}(a+c)h$
Regular n-gon	na	$\frac{1}{2}(na)r = \frac{na^2}{4\tan(\pi/n)}$
Circle	$2\pi r$	πr^2
Circular sector	$r\theta$	$\frac{1}{2}r^2\theta$
Rectangular box	$2(ab+bc+ca)$	abc
Right circular cylinder	$2\pi r(r+h)$	$\pi r^2 h$
Right circular cone	$\pi r(r+\ell)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$

In the regular-polygon row, r is the apothem; in the cone row, $\ell = \sqrt{r^2+h^2}$ is the slant height. Sector angles are measured in radians.

Theorem 7.2. Triangle Angle Sum and Exterior Angle Theorem
For every triangle $\triangle ABC$,

$$\angle A + \angle B + \angle C = \pi.$$

An exterior angle equals the sum of the two nonadjacent interior angles.

Proposition 7.3. Triangle Congruence and Similarity Criteria
Two triangles are congruent under any of the criteria

$$\text{SSS}, \qquad \text{SAS}, \qquad \text{ASA}, \qquad \text{AAS}, \qquad \text{RHS}.$$

They are similar under any of the criteria

$$\text{AA}, \qquad \text{SAS}, \qquad \text{SSS},$$

where the latter two use proportional corresponding side lengths.

Theorem 7.4. Midpoint Theorem and Centroid Ratio
The segment joining the midpoints of two sides of a triangle is parallel to the third side and has half its length. The three medians are concurrent at the centroid G, and G divides every median in the ratio

$$2 : 1$$

measured from the vertex.

Strategy. Reverse Engineering in Geometric Constructions. First draw a completed configuration satisfying the desired conditions. Then identify which points, perpendiculars, parallels, angle bisectors, circles, or reflections in the completed picture can be constructed from the original data. Familiar configurations involving triangle centers, similar triangles, and power of a point are especially useful targets.

Theorem 7.5. Happy Ending Problem, Five-Point Case
Among any five points in the plane, no three collinear, there are four points that are the vertices of a convex quadrilateral.

Idea. Examine the convex hull. If it has at least four vertices, choose four of them. If it is a triangle, use the line through the two interior points and choose two hull vertices lying on the same side of that line.

Definition 7.6. Classical Triangle Centers
For a triangle $\triangle ABC$, the most important classical centers are:

$$O = \text{circumcenter}, \qquad I = \text{incenter}, \qquad H = \text{orthocenter}, \qquad G = \text{centroid}.$$

Here O is the center of the circumcircle, I is the center of the incircle, H is the intersection of the altitudes, and G is the intersection of the medians.

Theorem 7.7. Gergonne Point and Nagel Point
The cevians joining the vertices of a triangle to the contact points of its incircle with the opposite sides are concurrent at the Gergonne point. The cevians joining the vertices to the contact points of the opposite excircles are concurrent at the Nagel point.

Theorem 7.8. Symmedian Lemma and Lemoine Point
Let the A-symmedian of $\triangle ABC$ meet BC at K. Then

$$\frac{BK}{KC} = \frac{AB^2}{AC^2}.$$

The three symmedians are concurrent at the Lemoine point, also called the symmedian point.

Theorem 7.9. Fermat Point; Torricelli Point
If every angle of $\triangle ABC$ is less than 120°, there is a unique point P minimizing

$$PA + PB + PC.$$

It satisfies

$$\angle APB = \angle BPC = \angle CPA = 120^\circ.$$

If one angle is at least 120°, the minimizing point is that vertex.

Formula 7.10. Extended Law of Sines
For a triangle $\triangle ABC$ with side lengths

$$a = BC, \qquad b = CA, \qquad c = AB,$$

and circumradius R, we have

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Formula 7.11. Law of Cosines
For a triangle $\triangle ABC$ with side lengths

$$a = BC, \qquad b = CA, \qquad c = AB,$$

we have

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

The analogous formulas hold after cyclic permutation of a, b, c.

Formula 7.12. Heron's Formula
Let $\triangle ABC$ have side lengths a, b, c and semiperimeter

$$s = \frac{a+b+c}{2}.$$

Then its area K is

$$K = \sqrt{s(s-a)(s-b)(s-c)}.$$

Also,

$$K = rs \qquad \text{and} \qquad K = \frac{abc}{4R},$$

where r is the inradius and R is the circumradius.

Strategy. The formulas

$$K = rs, \qquad K = \frac{abc}{4R}$$

are often more useful than Heron's formula when incircles or circumcircles appear.

Formula 7.13. Triangle Area Formulas
For a triangle with side lengths a, b, c, angles A, B, C, circumradius R, and inradius r,

$$[ABC] = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \frac{1}{2}ab\sin C,$$

and

$$[ABC] = rs = \frac{abc}{4R}.$$

Theorem 7.14. Weitzenböck's Inequality
For a triangle with side lengths a, b, c and area Δ,

$$a^2 + b^2 + c^2 \geq 4\sqrt{3}\,\Delta,$$

with equality if and only if the triangle is equilateral.

Formula 7.15. Inradius and Exradii
For a triangle of area K and semiperimeter s, the inradius is

$$r = \frac{K}{s}.$$

If r_a, r_b, r_c are the exradii opposite A, B, C, respectively, then

$$r_a = \frac{K}{s-a}, \qquad r_b = \frac{K}{s-b}, \qquad r_c = \frac{K}{s-c}.$$

Formula 7.16. Median Length Formula; Apollonius' Theorem
If m_a is the median from A to side $a = BC$, then

$$m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}.$$

Equivalently, if M is the midpoint of BC, then

$$AB^2 + AC^2 = 2(AM^2 + BM^2).$$

Theorem 7.17. Stewart's Theorem
Let D lie on side BC of a triangle $\triangle ABC$. Write

$$a = BC, \qquad b = CA, \qquad c = AB, \qquad m = BD, \qquad n = DC, \qquad d = AD,$$

so that $a = m + n$. Then

$$b^2 m + c^2 n = a(d^2 + mn).$$

Equivalently,

$$d^2 = \frac{b^2 m + c^2 n}{a} - mn.$$

Remark. Stewart's Theorem is the standard general formula for the length of a cevian. The median-length formula is the special case $m = n = a/2$.

Formula 7.18. Internal Angle-Bisector Length
Let the internal angle bisector from A meet BC at D, and denote its length by $\ell_a = AD$. For

$$a = BC, \qquad b = CA, \qquad c = AB,$$

we have

$$\ell_a^2 = bc\left(1 - \frac{a^2}{(b+c)^2}\right) = \frac{4bc\, s(s-a)}{(b+c)^2},$$

where $s = (a + b + c)/2$.

Theorem 7.19. Angle Bisector Theorem
In a triangle $\triangle ABC$, suppose the internal angle bisector of $\angle A$ meets BC at D. Then

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

Theorem 7.20. Ceva's Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively. Then the cevians AD, BE, CF are concurrent if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1,$$

using directed lengths.

Remark. Ceva's Theorem is the standard theorem for proving concurrence of three lines in a triangle.

Theorem 7.21. Trigonometric Form of Ceva's Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively. The cevians AD, BE, CF are concurrent if and only if

$$\frac{\sin\angle BAD}{\sin\angle DAC} \cdot \frac{\sin\angle CBE}{\sin\angle EBA} \cdot \frac{\sin\angle ACF}{\sin\angle FCB} = 1,$$

with the usual directed-angle convention when points lie on extended sides.

Strategy. Mass Points. In a cevian configuration, assign masses to vertices so that a point dividing a segment balances the endpoint masses inversely to the segment lengths. For example, if $D \in BC$, then

$$\frac{BD}{DC} = \frac{m_C}{m_B}.$$

Consistent masses convert several ratio computations into weighted averages and often recover Ceva-type relations immediately.

Theorem 7.22. Menelaus' Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively. Then the points D, E, F are collinear if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1,$$

using directed lengths.

Remark. Ceva is for concurrence, while Menelaus is for collinearity. This pair is one of the most important toolkits in olympiad geometry.

Theorem 7.23. Euler Line Theorem
In any non-equilateral triangle, the circumcenter O, centroid G, and orthocenter H are collinear. Moreover,

$$OG : GH = 1 : 2.$$

The line passing through these points is called the Euler line.

Theorem 7.24. Nine-Point Circle Theorem; Euler Circle
In any triangle, the following nine points lie on one circle: the three side midpoints, the three feet of the altitudes, and the three midpoints between the orthocenter and the vertices. This circle is called the nine-point circle.

Remark. The center of the nine-point circle is the midpoint of the segment joining the circumcenter O and the orthocenter H.

Theorem 7.25. Feuerbach's Theorem
The nine-point circle of a triangle is internally tangent to the incircle and externally tangent to each of the three excircles.

Theorem 7.26. Euler's Formula for the Incenter and Circumcenter
For a triangle with circumradius R, inradius r, circumcenter O, and incenter I, we have

$$OI^2 = R^2 - 2Rr.$$

In particular,

$$R \geq 2r.$$

Remark. The inequality $R \geq 2r$ is called Euler's inequality for triangles. Equality holds exactly for an equilateral triangle.

Strategy. Spiral Similarity. When two segments must be compared through both an angle and a ratio, look for a center of spiral similarity. It often reveals the required similar triangles immediately.

Theorem 7.27. Inscribed Angle Theorem
An inscribed angle equals half the central angle subtending the same arc. Consequently, angles subtending the same chord are equal.

Theorem 7.28. Cyclic Quadrilateral Criteria
Let A, B, C, D be four distinct points with no three collinear. Using directed angles modulo π, the points are concyclic if and only if

$$\angle BAC \equiv \angle BDC \pmod{\pi}.$$

If $ABCD$ is a convex quadrilateral in cyclic order, this is equivalent to

$$\angle ABC + \angle ADC = \pi.$$

Theorem 7.29. Tangent–Chord Theorem; Alternate Segment Theorem
The angle between a tangent to a circle and a chord through the point of tangency equals the inscribed angle subtending that chord in the opposite arc.

Theorem 7.30. Power of a Point Theorem
Let P be a point and let a line through P meet a circle at A and B. The product

$$PA \cdot PB$$

is independent of the choice of the line through P. It is called the power of P with respect to the circle.

Formula 7.31. Tangent–Secant Theorem
If PT is tangent to a circle at T and a secant through P meets the circle at A and B, then

$$PT^2 = PA \cdot PB.$$

Theorem 7.32. Radical Axis Theorem
For two nonconcentric circles, the set of points having equal powers with respect to the two circles is a line. This line is called the radical axis of the two circles.

Remark. Radical axes are powerful because they turn circle configurations into collinearity statements.

Theorem 7.33. Ptolemy's Theorem
Let A, B, C, D be the vertices of a convex cyclic quadrilateral in this cyclic order. Then

$$AC \cdot BD = AB \cdot CD + BC \cdot DA.$$

Conversely, equality in Ptolemy's inequality for a convex quadrilateral characterizes cyclicity.

Theorem 7.34. Ptolemy's Inequality
For any four points A, B, C, D in the Euclidean plane,

$$AC \cdot BD \le AB \cdot CD + BC \cdot DA.$$

Equality holds in the cyclic convex case and in the corresponding degenerate collinear case.

Formula 7.35. Brahmagupta's Formula
If a cyclic quadrilateral has side lengths a, b, c, d and semiperimeter

$$s = \frac{a+b+c+d}{2},$$

then its area K is

$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}.$$

Heron's formula is the limiting triangular case.

Theorem 7.36. Varignon's Theorem
The midpoints of the four sides of any quadrilateral form a parallelogram. Its area is half the area of the original quadrilateral.

Theorem 7.37. Pitot's Theorem
If a convex quadrilateral with consecutive side lengths a, b, c, d has an incircle tangent to all four sides, then

$$a + c = b + d.$$

Theorem 7.38. Euler's Quadrilateral Theorem
If M and N are the midpoints of the diagonals of a quadrilateral with side lengths a, b, c, d and diagonal lengths p, q, then

$$a^2 + b^2 + c^2 + d^2 = p^2 + q^2 + 4MN^2.$$

The parallelogram law is the case $M = N$.

Theorem 7.39. Simson Line Theorem
Let P be a point on the circumcircle of $\triangle ABC$. The feet of the perpendiculars from P to the three lines AB, BC, CA are collinear. This line is called the Simson line of P with respect to $\triangle ABC$.

Theorem 7.40. Miquel's Theorem
In a complete quadrilateral, the circumcircles of the four triangles formed by choosing three of the four lines pass through a common point. This point is called the Miquel point.

Theorem 7.41. Butterfly Theorem
Let M be the midpoint of a chord PQ of a circle. Through M, draw two other chords AB and CD, and let AD and BC meet PQ at X and Y, respectively. Then M is also the midpoint of XY.

Theorem 7.42. Morley's Trisector Theorem
In any triangle, the adjacent trisectors of the three angles intersect in three points that form an equilateral triangle.

Theorem 7.43. Napoleon's Theorem
Construct equilateral triangles externally on the three sides of any triangle. Then the centers of these three equilateral triangles form an equilateral triangle.

Theorem 7.44. British Flag Theorem
For any point P in the plane of a rectangle $ABCD$,

$$PA^2 + PC^2 = PB^2 + PD^2.$$

Theorem 7.45. Van Aubel's Theorem
Construct squares externally on the four sides of a quadrilateral. The segments joining the centers of opposite squares are equal in length and perpendicular.

Theorem 7.46. Viviani's Theorem
For any point P inside an equilateral triangle of altitude h, the sum of the perpendicular distances from P to the three sides is h.

Theorem 7.47. Desargues' Theorem
If two triangles are perspective from a point, then the three intersections of corresponding sides are collinear. Conversely, in the projective plane, collinearity of those intersections implies concurrence of the three lines joining corresponding vertices.

Theorem 7.48. Pappus' Hexagon Theorem
Let A, B, C lie on one line and A', B', C' lie on another. Then the three points

$$AB' \cap A'B, \qquad AC' \cap A'C, \qquad BC' \cap B'C$$

are collinear.

Theorem 7.49. Homothety Centers of Two Circles
For two nonconcentric circles, the two external common tangents meet at the external homothety center, and the two internal common tangents meet at the internal homothety center, when these intersections exist. Each homothety center lies on the line joining the two circle centers.

Formula 7.50. Circle Inversion
Under inversion with center O and radius ρ, a point $P \neq O$ is sent to the point P' on the ray OP satisfying

$$OP \cdot OP' = \rho^2.$$

Inversion preserves angles in magnitude, sends circles through O to lines not through O, and sends such lines back to circles through O.

Formula 7.51. Regular Polygon Tiling Angle Condition
If q regular p-gons meet edge-to-edge at each vertex of a Euclidean tiling, then

$$q\left(1 - \frac{2}{p}\right)\pi = 2\pi,$$

equivalently,

$$(p-2)(q-2) = 4.$$

Thus the only tilings by one type of regular polygon are

$$(p, q) = (3, 6), (4, 4), (6, 3).$$

Formula 7.52. Obtuse Triangles from Equally Spaced Points
Let N equally spaced points lie on a circle. The number of obtuse triangles determined by three of the points is

$$N\binom{k-1}{2} \quad \text{if } N = 2k,$$

and

$$N\binom{k}{2} \quad \text{if } N = 2k+1.$$

For even N, triangles containing a diameter are right rather than obtuse.

Fact 7.53. Alternating-Area Invariant in a Regular Even-Gon
Let $V_1, \dots, V_{2m}$ be the vertices of a regular $2m$-gon in cyclic order, and let P be any point inside it. Then

$$[PV_1V_2] + [PV_3V_4] + \cdots + [PV_{2m-1}V_{2m}]$$

equals

$$[PV_2V_3] + [PV_4V_5] + \cdots + [PV_{2m}V_1].$$

The equality follows from the cancellation of alternating directed edge vectors.

Theorem 7.54. Classification of Platonic Solids
There are exactly five Platonic solids:

tetrahedron, cube, octahedron, dodecahedron, icosahedron.

Analytic Geometry

Formula 7.55. Coordinate Geometry Table
For points $A = (x_1, y_1)$ and $B = (x_2, y_2)$,

Quantity	Formula
Distance	$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Quantity	Formula
Internal division $AP : PB = m : n$	$\left(\frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n}\right)$
Slope	$\frac{y_2 - y_1}{x_2 - x_1}$, when $x_1 \neq x_2$
Point-slope line	$y - y_1 = m(x - x_1)$
General line	$ax + by + c = 0$
Distance from (x_0, y_0) to $ax + by + c = 0$	$\frac{\lvert ax_0 + by_0 + c\rvert}{\sqrt{a^2 + b^2}}$

Formula 7.56. Triangle Centers in Coordinates
For $A = (x_1, y_1)$, $B = (x_2, y_2)$, and $C = (x_3, y_3)$, the centroid is

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right).$$

If the side lengths opposite A, B, C are a, b, c, respectively, then the incenter is

$$I = \frac{aA + bB + cC}{a + b + c}.$$

Formula 7.57. Barycentric Coordinates
Let A, B, C be noncollinear. Every point P in their affine plane has unique coordinates (α, β, γ) satisfying

$$\alpha + \beta + \gamma = 1, \qquad \mathbf{p} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}.$$

For P inside $\triangle ABC$,

$$\alpha = \frac{[PBC]}{[ABC]}, \qquad \beta = \frac{[PCA]}{[ABC]}, \qquad \gamma = \frac{[PAB]}{[ABC]}.$$

Formula 7.58. Shoelace Formula; Gauss's Area Formula
For a polygon with vertices

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

in cyclic order, its area is

$$\frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1} y_i)\right|,$$

where

$$(x_{n+1}, y_{n+1}) = (x_1, y_1).$$

Remark. The Shoelace Formula is one of the cleanest examples of analytic geometry: a geometric area becomes a determinant-like coordinate calculation.

Formula 7.59. Oriented Area and Collinearity Test
For points $A = (x_1, y_1)$, $B = (x_2, y_2)$, and $C = (x_3, y_3)$, the oriented double area is

$$2[ABC]_{\mathrm{or}} = \det\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}.$$

Hence A, B, C are collinear if and only if this determinant is 0.

Formula 7.60. Plane Rotation
Counterclockwise rotation through angle θ about the origin sends

$$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

In complex notation, the same transformation is simply

$$z \longmapsto e^{i\theta} z.$$

Formula 7.61. Reflection across a Line through the Origin
If a line through the origin makes angle θ with the positive x-axis, reflection across that line is

$$z \longmapsto e^{2i\theta}\overline{z}$$

in complex notation.

Strategy. Affine Normalization. Affine transformations preserve collinearity, parallelism, ratios along one line, concurrence, and ratios of areas. Therefore, when a problem depends only on these properties, one may send a triangle to convenient coordinates such as

$$A = (0, 0), \qquad B = (1, 0), \qquad C = (0, 1).$$

Angles, lengths, and circles are not generally preserved.

Theorem 7.62. Cavalieri's Principle
If two plane regions lie between the same pair of parallel lines and every line parallel to them cuts the regions in segments whose lengths have a constant ratio c, then their areas have ratio c. The three-dimensional analogue replaces segment lengths by cross-sectional areas and concludes the same ratio for volumes.

Strategy. Unfolding Method for Reflection Problems. A broken path reflecting from a line or face becomes a straight segment after reflecting the region across each boundary in succession. Thus a shortest reflected path is found by measuring a straight-line distance in the unfolded picture and then folding the path back.

Formula 7.63. Lattice Segment and Cell Counts
A segment from $(0, 0)$ to (m, n), where $m, n \in \mathbb{N}$, passes through

$$m + n - \gcd(m, n)$$

unit squares. The number of lattice points on the segment, including its endpoints, is

$$\gcd(m, n) + 1.$$

In three dimensions, a segment from $(0, 0, 0)$ to (a, b, c) passes through

$$a + b + c - \gcd(a, b) - \gcd(a, c) - \gcd(b, c) + \gcd(a, b, c)$$

unit cubes.

Formula 7.64. Vector Test for Parallel Segments
Directed segments AB and CD are parallel if and only if

$$\mathbf{b} - \mathbf{a} = \lambda(\mathbf{d} - \mathbf{c})$$

for some scalar λ. Equality with $\lambda = 1$ identifies equal directed segments and is often the most efficient method to detect a hidden parallelogram.

Formula 7.65. Vector Projection
If $\mathbf{u} \neq \mathbf{0}$, then the projection of $\mathbf{v}$ onto the line spanned by $\mathbf{u}$ is

$$\operatorname{proj}_{\mathbf{u}} \mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}.$$

The scalar component of $\mathbf{v}$ in the direction of $\mathbf{u}$ is

$$\frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{u}\|}.$$

Formula 7.66. Vector Equations of Lines and Planes
A line through a point $\mathbf{p}$ with direction vector $\mathbf{v} \neq \mathbf{0}$ is

$$\mathbf{x} = \mathbf{p} + t\mathbf{v} \qquad (t \in \mathbb{R}).$$

A plane through a point $\mathbf{p}$ with normal vector $\mathbf{n} \neq \mathbf{0}$ is

$$(\mathbf{x} - \mathbf{p}) \cdot \mathbf{n} = 0.$$

Formula 7.67. Distance from a Point to a Line
In $\mathbb{R}^3$, the distance from a point $\mathbf{p}_0$ to the line

$$\mathbf{x} = \mathbf{p} + t\mathbf{v}$$

is

$$\frac{\|(\mathbf{p}_0 - \mathbf{p}) \times \mathbf{v}\|}{\|\mathbf{v}\|}.$$

In $\mathbb{R}^2$, the distance from (x_0, y_0) to the line $ax + by + c = 0$ is

$$\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.$$

Formula 7.68. Distance from a Point to a Plane
The distance from a point $\mathbf{p}_0 = (x_0, y_0, z_0)$ to the plane

$$ax + by + cz + d = 0$$

is

$$\frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Formula 7.69. Sphere Equation
A sphere with center $\mathbf{c} = (a, b, c)$ and radius r has equation

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2.$$

Equivalently,

$$\|\mathbf{x} - \mathbf{c}\| = r.$$

Strategy. Line–Circle Intersection by a Discriminant. Substitute a parametric line into a circle equation. The resulting quadratic equation has two, one, or no real solutions according as its discriminant is positive, zero, or negative. The zero-discriminant case is the tangency condition.

Formula 7.70. Circle Equation
The circle with center (a, b) and radius r has equation

$$(x - a)^2 + (y - b)^2 = r^2.$$

The general equation

$$x^2 + y^2 + Dx + Ey + F = 0$$

can be converted to center-radius form by completing the square.

Theorem 7.71. Three Perpendiculars Theorem
Let Π be a plane, let $P \notin \Pi$, and let H be the orthogonal projection of P onto Π. Let $\ell \subseteq \Pi$ pass through a point Q, and suppose that $HQ \perp \ell$. Then

$$PQ \perp \ell.$$

Equivalently, a line in a plane that is perpendicular to the orthogonal projection of an oblique line is perpendicular to the oblique line itself.

Remark. In coordinate geometry, the three perpendiculars theorem is often replaced by dot products and projections. Still, it is a standard spatial-geometry fact from the classical school curriculum.

Definition 7.72. Conic Section
A conic section is a curve obtained by intersecting a plane with a double cone. The nondegenerate conics are:

ellipse, parabola, hyperbola.

Equivalently, a conic can be described using a focus, a directrix, and an eccentricity e.

Formula 7.73. Focus–Directrix Definition of Conics
A conic is the set of points P satisfying

$$\frac{\text{dist}(P, \text{focus})}{\text{dist}(P, \text{directrix})} = e.$$

It is an ellipse if

$$0 < e < 1,$$

a parabola if

$$e = 1,$$

and a hyperbola if

$$e > 1.$$

Formula 7.74. Standard Forms of Conics
The standard form of an ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

The standard form of a hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

The standard form of a parabola opening to the right is

$$y^2 = 4px.$$

Formula 7.75. Eccentricity of an Ellipse and a Hyperbola
For an ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad (a \geq b > 0),$$

the eccentricity is

$$e = \frac{c}{a}, \qquad c^2 = a^2 - b^2.$$

For a hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

the eccentricity is

$$e = \frac{c}{a}, \qquad c^2 = a^2 + b^2.$$

Formula 7.76. Tangent Lines to Standard Conics
At a point (x_1, y_1) on the indicated conic, the tangent line is

$$xx_1 + yy_1 = r^2 \qquad \text{for } x^2 + y^2 = r^2,$$

$$yy_1 = 2a(x + x_1) \qquad \text{for } y^2 = 4ax,$$

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \qquad \text{for } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

and

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \qquad \text{for } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Theorem 7.77. Reflection Property of Conics
At a point of a nondegenerate conic, the tangent makes equal angles with the two relevant focal directions. Consequently, a ray emitted from one focus of an ellipse reflects toward the other focus; a ray parallel to the axis of a parabola reflects through its focus; and a ray directed toward one focus of a hyperbola reflects along a ray whose backward extension passes through the other focus.

Remark. The reflection property gives a geometric reason why conics appear in optics, satellite dishes, whispering galleries, and orbital mechanics.

Theorem 7.78. Quadratic Classification of Conics
Consider a real quadratic equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

The discriminant

$$B^2 - 4AC$$

determines the type of the nondegenerate conic:

$$B^2 - 4AC < 0 \quad \Rightarrow \quad \textit{ellipse-type},$$

$$B^2 - 4AC = 0 \quad \Rightarrow \quad \textit{parabola-type},$$

and

$$B^2 - 4AC > 0 \quad \Rightarrow \quad \textit{hyperbola-type}.$$

Remark. This is the analytic-geometry counterpart of classifying conic sections by their shapes.

Formula 7.79. Polar Equation of a Conic
With a focus at the pole, a conic has polar equation

$$r = \frac{\ell}{1 + e\cos\theta}$$

after choosing a suitable axis. Here e is the eccentricity and ℓ is the semi-latus rectum.

Remark. The same equation describes ellipses, parabolas, and hyperbolas depending on whether $e < 1$, $e = 1$, or $e > 1$.

Definition 7.80. Apollonius Circle
Given two fixed points A and B and a positive constant $k \neq 1$, the locus of points P satisfying

$$\frac{PA}{PB} = k$$

is a circle. This circle is called an Apollonius circle.

Remark. Apollonius circles turn a ratio of distances into a concrete circle, which is a common hidden structure in geometry problems.

Theorem 7.81. Descartes' Circle Theorem; Soddy Circle Theorem
Suppose four circles are mutually tangent, and let their curvatures be

$$k_1, k_2, k_3, k_4,$$

where curvature means the reciprocal of the radius, with sign convention for an enclosing circle. Then

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2).$$

Remark. Descartes' Circle Theorem is a striking formula: tangency of four circles becomes one quadratic equation in their curvatures.

Formula 7.82. Boundary Lattice Points on a Segment
If a lattice segment has displacement $(\Delta x, \Delta y)$, then the number of lattice points on it, including both endpoints, is

$$\gcd(|\Delta x|, |\Delta y|) + 1.$$

Summing the gcd contributions over polygon edges gives the number B of boundary lattice points without double-counting vertices.

Theorem 7.83. Pick's Theorem
Let P be a simple polygon whose vertices lie on lattice points in the plane. If I is the number of lattice points strictly inside P and B is the number of lattice points on the boundary of P, then

$$\text{area}(P) = I + \frac{B}{2} - 1.$$

Remark. Pick's Theorem is a beautiful bridge between geometry and arithmetic: area is determined by counting lattice points.

Chapter 8

Number Theory

Related POSTECH Courses: (MATH304) Introduction to Number Theory
Related Books: Introduction to Number Theory (William W. Adams, Larry Joel Goldstein)

Theorem 8.1. Fundamental Theorem of Arithmetic
Every integer $n > 1$ can be written as a product of prime numbers. Moreover, this factorization is unique up to the order of the factors. Equivalently,

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where $p_1, \ldots, p_k$ are distinct primes and $\alpha_1, \ldots, \alpha_k$ are positive integers, and this expression is unique up to reordering.

Remark. This theorem is often called unique prime factorization. It is the reason prime numbers are treated as the basic building blocks of integers.

Theorem 8.2. Euclid's Theorem; Infinitude of Primes
There are infinitely many prime numbers.

Idea. If $p_1, \ldots, p_n$ are finitely many primes, consider

$$p_1 p_2 \cdots p_n + 1.$$

This number is not divisible by any prime on the list.

Theorem 8.3. Bézout's Identity; Bézout's Lemma
Let $a, b \in \mathbb{Z}$, not both zero. Then there exist integers $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b).$$

In particular, if $\gcd(a, b) = 1$, then there exist integers $x, y \in \mathbb{Z}$ such that

$$ax + by = 1.$$

Remark. Bézout's identity is the algebraic heart of the Euclidean algorithm and modular inverses.

Algorithm 8.4. Euclidean Algorithm
To compute $\gcd(a, b)$ for integers $a, b \in \mathbb{Z}$, repeatedly divide the larger number by the smaller number and replace the pair by

$$(b, r),$$

where r is the remainder. Continue until the remainder becomes 0. The last nonzero remainder is $\gcd(a, b)$.

Proposition 8.5. Basic Rules of Congruences
If

$$a \equiv b \pmod{n}, \qquad c \equiv d \pmod{n},$$

then

$$a + c \equiv b + d \pmod{n}, \qquad ac \equiv bd \pmod{n}.$$

If $\gcd(c, n) = 1$, *then*

$$ac \equiv bc \pmod{n} \implies a \equiv b \pmod{n}.$$

Theorem 8.6. Linear Congruence Criterion
The congruence

$$ax \equiv b \pmod{n}$$

has a solution if and only if

$$\gcd(a, n) \mid b.$$

If $d = \gcd(a, n)$ *and* $d \mid b$, *then there are exactly* d *solutions modulo* n.

Formula 8.7. Least Common Multiple from Prime Factorization
If

$$a = \prod_p p^{\alpha_p}, \qquad b = \prod_p p^{\beta_p},$$

then

$$\gcd(a, b) = \prod_p p^{\min(\alpha_p, \beta_p)}, \qquad \operatorname{lcm}(a, b) = \prod_p p^{\max(\alpha_p, \beta_p)}.$$

More generally, the least common multiple of several integers is found by taking the largest exponent of each prime appearing in the list.

Formula 8.8. Basic Divisibility Tests
For a decimal integer N, the following tests are useful.

- Divisibility by 2, 4, and 8 depends on the last 1, 2, and 3 digits, respectively.
- Divisibility by 3 and 9 depends on the digit sum.
- Divisibility by 5 depends on the last digit.
- Divisibility by 11 depends on the alternating sum of the digits.

Strategy. Digit problems often combine an lcm condition, a digit-sum condition, and a last-digits condition. First minimize the number of digits, then arrange the final few digits to satisfy the strongest divisibility conditions.

Definition 8.9. Complete Residue System
A set of n integers is a complete residue system modulo n if its elements represent every residue class modulo n exactly once. To prove that $a_0, \ldots, a_{n-1}$ form such a system, it is enough to prove

$$a_i \equiv a_j \pmod{n} \implies i = j.$$

Strategy. Modular Casework and Base Representation. When a problem repeats division by N, rewrite the integer in base N. The quotient deletes the final digit and the remainder reveals it. For divisibility or existence problems, first split integers into residue classes and look for a complete residue system rather than checking values individually.

Strategy. Infinite Descent. Assume a positive-integer solution exists and choose one minimizing a natural positive quantity. Construct a strictly smaller solution of the same type, contradicting well-ordering.

Strategy. Vieta Jumping. If an integer equation is quadratic in one variable, regard that variable as one root of a monic quadratic. Viète's formulas give the other integral root. Prove that it yields a smaller positive solution and apply infinite descent.

Theorem 8.10. Frobenius Coin Theorem for Two Denominations
If a and b are coprime positive integers, the largest integer not representable as

$$ax + by, \qquad x, y \in \mathbb{N}_0,$$

is $ab - a - b$. Exactly

$$\frac{(a-1)(b-1)}{2}$$

positive integers are not representable.

Theorem 8.11. Chinese Remainder Theorem; CRT
Let $m_1, \ldots, m_k$ be pairwise relatively prime positive integers. Then the system

$$x \equiv a_1 \pmod{m_1}, \qquad x \equiv a_2 \pmod{m_2}, \qquad \ldots, \qquad x \equiv a_k \pmod{m_k}$$

has a solution. Moreover, the solution is unique modulo

$$M = m_1 m_2 \cdots m_k.$$

Strategy. CRT is the standard theorem for combining several congruence conditions into one congruence.

Formula 8.12. Euler's Phi Function Formula
If

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

is the prime factorization of n, then

$$\varphi(n) = n \prod_{i=1}^{k} \left(1 - \frac{1}{p_i}\right).$$

In particular, for a prime power,

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1).$$

Formula 8.13. Sum-of-Divisors Function
If

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$

then the number of positive divisors of n is

$$\tau(n) = (\alpha_1 + 1) \cdots (\alpha_k + 1),$$

and the sum of positive divisors of n is

$$\sigma(n) = \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}.$$

Formula 8.14. p-Adic Valuation of a Factorial; Legendre's Formula
For a prime p and a positive integer n,

$$\nu_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor.$$

Only finitely many terms are nonzero. This formula gives the exponent of p in $n!$ and is especially useful for trailing-zero and divisibility questions.

Theorem 8.15. Möbius Inversion Formula
If arithmetic functions f and g satisfy

$$g(n) = \sum_{d|n} f(d),$$

then

$$f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right).$$

Theorem 8.16. Fermat's Little Theorem
Let p be a prime number. If $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for every integer a,

$$a^p \equiv a \pmod{p}.$$

Theorem 8.17. Euler's Theorem; Euler–Fermat Theorem
If $\gcd(a, n) = 1$*, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Remark. Fermat's Little Theorem is the special case of Euler's Theorem when $n = p$ is prime, since $\varphi(p) = p - 1$.

Proposition 8.18. Multiplicative Order
If $\gcd(a, n) = 1$*, then $[a]_n$ lies in the finite group $(\mathbb{Z}/n\mathbb{Z})^\times$. Its multiplicative order* $\mathrm{ord}_n(a)$ *divides*

$$|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n).$$

Consequently, for every positive integer k,

$$a^k \equiv 1 \pmod{n} \iff \mathrm{ord}_n(a) \mid k.$$

Theorem 8.19. Primitive Root Theorem; Gauss's Theorem on Primitive Roots
The multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic if and only if

$$n \in \{1, 2, 4, p^k, 2p^k\},$$

where p is an odd prime and $k \geq 1$.

Theorem 8.20. Existence and Uniqueness of Finite Fields
Every finite field has $q = p^n$ elements for some prime p and positive integer n. Conversely, for each prime power $q = p^n$, there exists a field $\mathbb{F}_q$ with q elements, unique up to isomorphism. Its multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.

Algorithm 8.21. Modular Exponentiation by Period Reduction
To find the last digits of a^n, compute a^n modulo a suitable power of 10. Look for a short period in the powers of a modulo that modulus. For example,

$$7^4 = 2401 \equiv 1 \pmod{100},$$

so exponents of 7 may be reduced modulo 4 when finding the last two digits.

Strategy. Last-digit and last-two-digit questions are modular arithmetic problems. Work modulo 10 or 100, not with the full number.

Theorem 8.22. Wilson's Theorem
A positive integer $p > 1$ is prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

Remark. Wilson's theorem is famous because it gives a clean factorial characterization of primes, even though it is not efficient as a primality test.

Theorem 8.23. Lucas's Theorem
Let p be prime and write

$$n = \sum_{i=0}^{r} n_i p^i, \qquad k = \sum_{i=0}^{r} k_i p^i$$

in base p. Then

$$\binom{n}{k} \equiv \prod_{i=0}^{r} \binom{n_i}{k_i} \pmod{p}.$$

Theorem 8.24. Kummer's Theorem
The valuation

$$\nu_p\left(\binom{n}{k}\right)$$

equals the number of carries when k and $n-k$ are added in base p.

Theorem 8.25. Lifting the Exponent Lemma; LTE
Let p be an odd prime, and suppose that $p \mid x-y$ while $p \nmid xy$. Then, for every positive integer n,

$$\nu_p(x^n - y^n) = \nu_p(x-y) + \nu_p(n).$$

A standard companion form is: if n is odd, $p \mid x+y$, and $p \nmid xy$, then

$$\nu_p(x^n + y^n) = \nu_p(x+y) + \nu_p(n).$$

Fact 8.26. Quadratic Residues Modulo Small Moduli
Squares have very restricted residues modulo small integers:

$$x^2 \equiv 0, 1 \pmod{4}, \qquad x^2 \equiv 0, 1, 4 \pmod{8}.$$

In particular, an integer congruent to 2 or 3 modulo 4 cannot be a square, and an integer congruent to $2, 3, 5, 6, 7$ modulo 8 cannot be a square.

Tactic. For Diophantine equations, first reduce modulo 4, 8, 3, 5, or 9. Many impossible equations are eliminated by the small list of possible square residues.

Definition 8.27. Legendre Symbol
Let p be an odd prime. The Legendre symbol is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1, & \text{if } a \text{ is a quadratic nonresidue modulo } p, \\ 0, & \text{if } p \mid a. \end{cases}$$

Theorem 8.28. Euler's Criterion
Let p be an odd prime. For any integer a,

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Equivalently, if $p \nmid a$, then

$$a^{(p-1)/2} \equiv \begin{cases} 1 \pmod{p}, & \textit{if } a \textit{ is a quadratic residue modulo } p, \\ -1 \pmod{p}, & \textit{if } a \textit{ is a quadratic nonresidue modulo } p. \end{cases}$$

Theorem 8.29. Law of Quadratic Reciprocity; Gauss's Theorem of Quadratic Reciprocity
Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$$

unless both p and q are congruent to 3 modulo 4.

Remark. Quadratic reciprocity is one of the crown jewels of elementary number theory. Gauss called it the golden theorem.

Theorem 8.30. Fermat's Two-Square Theorem; Fermat's Theorem on Sums of Two Squares
An odd prime p can be written as a sum of two squares,

$$p = a^2 + b^2$$

for some integers $a, b \in \mathbb{Z}$, if and only if

$$p \equiv 1 \pmod{4}.$$

Also,

$$2 = 1^2 + 1^2.$$

Theorem 8.31. Sum of Two Squares Criterion
A positive integer

$$n = 2^a \prod_{p\equiv 1 \pmod{4}} p^{\alpha_p} \prod_{q\equiv 3 \pmod{4}} q^{\beta_q}$$

is representable as $n = x^2 + y^2$ with integers x, y if and only if every exponent β_q is even.

Formula 8.32. Brahmagupta–Fibonacci Identity

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2 = (ac + bd)^2 + (ad - bc)^2.$$

Formula 8.33. Sophie Germain's Identity

$$a^4 + 4b^4 = (a^2 - 2ab + 2b^2)(a^2 + 2ab + 2b^2).$$

Theorem 8.34. Lagrange's Four-Square Theorem
Every nonnegative integer can be written as a sum of four integer squares. That is, for every $n \in \mathbb{N}_0$, there exist integers $a, b, c, d \in \mathbb{Z}$ such that

$$n = a^2 + b^2 + c^2 + d^2.$$

Remark. The two-square theorem has a congruence condition. The four-square theorem has no exception at all.

Theorem 8.35. Continued-Fraction Best Approximation Theorem
Let p_k/q_k be the convergents of the simple continued fraction of an irrational number α. Then

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k q_{k+1}}.$$

Moreover, the convergents are best approximations of the second kind: if $p, q \in \mathbb{Z}$, $0 < q < q_{k+1}$, and $p/q \neq p_k/q_k$, then

$$|q_k\alpha - p_k| < |q\alpha - p|.$$

Strategy. Problems asking for an unusually accurate fraction with the smallest possible denominator often signal a continued-fraction expansion.

Definition 8.36. Pell Equation
A Pell equation has the form
$$x^2 - Dy^2 = 1,$$
where D is a positive nonsquare integer. If (x_1, y_1) is the fundamental positive solution, then all positive solutions satisfy
$$x_n + y_n\sqrt{D} = (x_1 + y_1\sqrt{D})^n.$$
Continued fractions of $\sqrt{D}$ produce the fundamental solution.

Theorem 8.37. Dirichlet's Theorem on Arithmetic Progressions
If a and d are relatively prime positive integers, then the arithmetic progression
$$a,\ a+d,\ a+2d,\ a+3d, \dots$$
contains infinitely many primes.

Remark. Dirichlet's theorem is a major result connecting primes and arithmetic progressions. The condition $\gcd(a, d) = 1$ is necessary.

Theorem 8.38. Bertrand's Postulate; Chebyshev's Theorem
For every integer $n > 1$, there exists a prime p such that
$$n < p < 2n.$$

Theorem 8.39. Prime Number Theorem
Let $\pi(x)$ be the number of primes less than or equal to x. Then
$$\pi(x) \sim \frac{x}{\ln x} \qquad \text{as } x \to \infty.$$

Idea. The Prime Number Theorem says that the "probability" that a large integer near x is prime is roughly $1/\ln x$. This is a mental model, not a literal probability statement.

Definition 8.40. Riemann Zeta Function
For $\mathrm{Re}(s) > 1$, the Riemann zeta function is defined by
$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$
It also has the Euler product
$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

Remark. The Euler product is the bridge between the zeta function and prime numbers. The Riemann Hypothesis asserts that the nontrivial zeros of $\zeta(s)$ all have real part $1/2$.

Theorem 8.41. Euclid–Euler Theorem on Perfect Numbers
An even positive integer N is perfect if and only if
$$N = 2^{p-1}(2^p - 1),$$
where $2^p - 1$ is a Mersenne prime.

Remark. A perfect number is a positive integer equal to the sum of its positive proper divisors. For example,
$$6 = 1 + 2 + 3.$$

Theorem 8.42. Fermat's Last Theorem
For every integer $n \geq 3$, there do not exist positive integers a, b, c such that

$$a^n + b^n = c^n.$$

Remark. Fermat's Last Theorem was proved by Andrew Wiles. It is one of the most famous theorems in the history of mathematics.

Chapter 9

Algebra

Related POSTECH Courses: (MATH301) Modern Algebra I, (MATH302) Modern Algebra II
Related Books: Abstract Algebra (David S. Dummit, Richard M. Foote)

Definition 9.1. Group
A group is a set G with a binary operation $* : G \times G \to G$ s.t.

$$\forall a, b, c \in G, \qquad (a * b) * c = a * (b * c),$$

$$\exists e \in G \text{ such that } \quad \forall a \in G, \qquad e * a = a * e = a,$$

and

$$\forall a \in G, \qquad \exists a^{-1} \in G \text{ such that } \quad a * a^{-1} = a^{-1} * a = e.$$

The group is abelian if

$$\forall a, b \in G, \qquad a * b = b * a.$$

Theorem 9.2. Disjoint-Cycle Decomposition of a Permutation
Every permutation $\sigma \in S_n$ can be written as a product of disjoint cycles, uniquely up to the order of the cycles and cyclic rotation within each cycle. If $c(\sigma)$ is the number of cycles, including fixed points, then the minimum number of transpositions needed to express σ is

$$n - c(\sigma).$$

Formula 9.3. Inversions and the Sign of a Permutation
The inversion number is

$$\mathrm{inv}(\sigma) = |\{(i, j) \mid i < j,\ \sigma(i) > \sigma(j)\}|.$$

The sign satisfies

$$\mathrm{sgn}(\sigma) = (-1)^{\mathrm{inv}(\sigma)} = (-1)^{n - c(\sigma)}.$$

Thus any transposition reverses parity.

Fact 9.4. Dihedral Group
The symmetries of a regular n-gon form the dihedral group

$$D_{2n} = \langle r, s \mid r^n = e,\ s^2 = e,\ srs = r^{-1} \rangle.$$

Its elements are

$$e, r, r^2, \ldots, r^{n-1}, \qquad s, rs, r^2 s, \ldots, r^{n-1} s.$$

This presentation is the standard algebraic model for rotation and reflection counting with Burnside's Lemma.

Definition 9.5. Cyclic Group
A group G is cyclic if there exists an element $g \in G$ such that every element of G is a power of g. In this case, we write

$$G = \langle g \rangle.$$

Theorem 9.6. Lagrange's Theorem
If G is a finite group and $H \leq G$, then

$$|H| \mid |G|.$$

Equivalently, the order of every subgroup divides the order of the group.

Corollary 9.7. Order of an Element
If G is a finite group and $g \in G$, then the order of g divides $|G|$. In particular,

$$g^{|G|} = e.$$

Theorem 9.8. Cauchy's Theorem
If G is a finite group and a prime p divides $|G|$, then G contains an element of order p.

Theorem 9.9. First Isomorphism Theorem for Groups
If $\varphi : G \to H$ is a group homomorphism, then

$$G/\ker\varphi \cong \operatorname{im}\varphi.$$

Theorem 9.10. Cayley's Theorem
Every group is isomorphic to a subgroup of a symmetric group. In particular, every finite group of order n is isomorphic to a subgroup of S_n.

Theorem 9.11. Orbit–Stabilizer Theorem
If a finite group G acts on a set X and $x \in X$, then

$$|G \cdot x| = [G : G_x] = \frac{|G|}{|G_x|},$$

where $G \cdot x$ is the orbit of x and G_x is its stabilizer.

Theorem 9.12. Class Equation
Let G be a finite group, and choose one representative x_i from each conjugacy class not contained in the center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)].$$

Theorem 9.13. Sylow Theorems
Let G be a finite group and suppose

$$|G| = p^k m, \qquad p \nmid m,$$

where p is prime. Then G has a subgroup of order p^k, called a Sylow p-subgroup. Moreover, all Sylow p-subgroups are conjugate, and the number n_p of Sylow p-subgroups satisfies

$$n_p \equiv 1 \pmod{p}, \qquad n_p \mid m.$$

Strategy. For a finite group, subgroup orders and element orders must divide the group order. When a prime divides $|G|$, Cauchy's theorem guarantees an element of that prime order.

Fact 9.14. Quaternion Group
The quaternion group is

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

with

$$i^2 = j^2 = k^2 = ijk = -1.$$

It is nonabelian, but every subgroup of Q_8 is normal.

Remark. The quaternion group is a standard small group with behavior that is less intuitive than cyclic or dihedral groups, so it is a natural source of quick algebra questions.

Theorem 9.15. Fundamental Theorem of Finite Abelian Groups
Every finite abelian group is isomorphic to a direct product of cyclic groups of prime-power order:

$$G \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_r^{a_r}}.$$

The multiset of prime powers is uniquely determined by G, up to reordering. Equivalently, G admits a unique invariant-factor decomposition

$$G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}, \qquad n_1 \mid n_2 \mid \cdots \mid n_k,$$

up to isomorphism.

Definition 9.16. Ring and Field
A ring is a set R with operations $+$ and $\cdot$ such that $(R, +)$ is an abelian group, multiplication is associative, and multiplication distributes over addition on both sides. Throughout this book, rings are assumed to have a multiplicative identity 1_R unless stated otherwise. A field is a commutative ring in which $1_R \neq 0_R$ and every nonzero element has a multiplicative inverse.

Definition 9.17. Integral Domain
An integral domain is a commutative ring with identity in which there are no zero divisors. In other words, if $ab = 0$, then $a = 0$ or $b = 0$.

Formula 9.18. Basic Factorization Identities
The most frequently used polynomial factorizations include

$$a^2 - b^2 = (a-b)(a+b),$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2), \qquad a^3 + b^3 = (a+b)(a^2 - ab + b^2).$$

More generally,

$$a^n - b^n = (a-b) \sum_{k=0}^{n-1} a^{n-1-k} b^k,$$

and, when n is odd,

$$a^n + b^n = (a+b) \sum_{k=0}^{n-1} (-1)^k a^{n-1-k} b^k.$$

Formula 9.19. Cubic Factorization Identity
For all a, b, c,

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca),$$

or equivalently,

$$a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)\left((a-b)^2 + (b-c)^2 + (c-a)^2\right).$$

In particular, if $a + b + c = 0$, then

$$a^3 + b^3 + c^3 = 3abc.$$

Algorithm 9.20. Polynomial Division Algorithm
Let F be a field. If $f(x), g(x) \in F[x]$ and $g(x) \neq 0$, then there exist unique polynomials $q(x), r(x) \in F[x]$ such that

$$f(x) = q(x)g(x) + r(x), \qquad r(x) = 0 \vee \deg r < \deg g.$$

Theorem 9.21. Factor Theorem
For a polynomial $f(x) \in F[x]$ and an element $a \in F$,

$$f(a) = 0 \iff x - a \text{ divides } f(x).$$

Proposition 9.22. Root-Counting Principle for Polynomials
A nonzero polynomial of degree n over a field has at most n roots. Hence two polynomials of degree at most n that agree at $n+1$ distinct points are identical.

Formula 9.23. Lagrange Interpolation Formula
For distinct $x_0, \dots, x_n$, the unique polynomial of degree at most n satisfying $P(x_i) = y_i$ is

$$P(x) = \sum_{i=0}^{n} y_i \prod_{\substack{0 \le j \le n \\ j \neq i}} \frac{x - x_j}{x_i - x_j}.$$

Formula 9.24. Viète's Formulas
If

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = a_n \prod_{i=1}^{n} (x - r_i),$$

where the roots are counted with multiplicity, then

$$\sum_{i=1}^{n} r_i = -\frac{a_{n-1}}{a_n}, \qquad \prod_{i=1}^{n} r_i = (-1)^n \frac{a_0}{a_n}.$$

More generally, the elementary symmetric sums of the roots are the coefficients with alternating signs, divided by a_n.

Formula 9.25. Newton's Identities; Newton–Girard Formulas
Let e_k be the elementary symmetric polynomials in roots $r_1, \dots, r_n$, and let

$$p_k = r_1^k + \cdots + r_n^k.$$

With $e_0 = 1$, for $1 \le k \le n$,

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k e_k = 0.$$

Formula 9.26. Polynomial Discriminant
If

$$f(x) = a_n \prod_{i=1}^{n} (x - r_i),$$

then

$$\operatorname{Disc}(f) = a_n^{2n-2} \prod_{1 \le i < j \le n} (r_i - r_j)^2.$$

It vanishes if and only if f has a repeated root.

Theorem 9.27. Rational Root Theorem
Let

$$f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x].$$

If a rational number p/q in lowest terms is a root of f, then

$$p \mid a_0, \qquad q \mid a_n.$$

Theorem 9.28. Eisenstein's Irreducibility Criterion
Let

$$f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x].$$

If there exists a prime p such that

$$p \nmid a_n, \qquad p \mid a_i \quad (0 \le i < n), \qquad p^2 \nmid a_0,$$

then f is irreducible over $\mathbb{Q}$.

Theorem 9.29. Fundamental Theorem of Symmetric Polynomials
Every symmetric polynomial can be expressed uniquely as a polynomial in the elementary symmetric polynomials.

Theorem 9.30. Combinatorial Nullstellensatz, Coefficient Form
Let $P \in F[x_1, \ldots, x_n]$ have total degree $t_1 + \cdots + t_n$, and suppose the coefficient of $x_1^{t_1} \cdots x_n^{t_n}$ is nonzero. If

$$|S_i| > t_i$$

for every finite set $S_i \subseteq F$, then some $(s_1, \ldots, s_n) \in S_1 \times \cdots \times S_n$ satisfies

$$P(s_1, \ldots, s_n) \neq 0.$$

Strategy. Polynomial Method. Encode a forbidden configuration as zeros of a polynomial, then use a degree bound, interpolation, factorization, or a nonzero coefficient to prove that the polynomial cannot vanish everywhere.

Theorem 9.31. Fundamental Theorem of Algebra
Every nonconstant complex polynomial has at least one complex root. Equivalently, every polynomial of degree $n \ge 1$ over $\mathbb{C}$ splits into n linear factors over $\mathbb{C}$, counted with multiplicity.

Definition 9.32. Algebraically Closed Field
A field F is algebraically closed if every nonconstant polynomial in $F[x]$ has a root in F.

Theorem 9.33. Fundamental Theorem of Galois Theory; Galois Correspondence
For a finite Galois extension L/K, there is an inclusion-reversing correspondence between intermediate fields $K \subseteq E \subseteq L$ and subgroups $H \le \mathrm{Gal}(L/K)$, given by

$$E \longmapsto \mathrm{Gal}(L/E), \qquad H \longmapsto L^H.$$

Theorem 9.34. Abel–Ruffini Theorem
There is no formula for the roots of a general polynomial of degree 5 or higher using only the field operations and radicals.

Remark. The Abel–Ruffini theorem does not say that every quintic equation is unsolvable. It says that no universal radical formula exists for the general quintic.

Chapter 10

Topology

Related POSTECH Courses: (MATH321) General Topology
Related Books: Topology (James R. Munkres)

Definition 10.1. Topology
A topology on a set X is a collection $\mathcal{T}$ of subsets of X whose elements are called open sets, satisfying:

(i) $\varnothing, X \in \mathcal{T}$;

(ii) arbitrary unions of sets in $\mathcal{T}$ belong to $\mathcal{T}$;

(iii) finite intersections of sets in $\mathcal{T}$ belong to $\mathcal{T}$.

Definition 10.2. Open and Closed Sets
If $(X, \mathcal{T})$ is a topological space, a subset $U \subseteq X$ is open if $U \in \mathcal{T}$. A subset $F \subseteq X$ is closed if $X \setminus F \in \mathcal{T}$.

Definition 10.3. Compactness
A topological space X is compact if every open cover has a finite subcover. Equivalently, for every family $\mathcal{U} \subseteq \mathcal{T}$,

$$X \subseteq \bigcup_{U \in \mathcal{U}} U \implies \exists U_1, \ldots, U_n \in \mathcal{U} \text{ such that } X \subseteq U_1 \cup \cdots \cup U_n.$$

Definition 10.4. Connectedness
A topological space X is connected if it cannot be written as the union of two disjoint nonempty open sets. Equivalently, the only subsets of X that are both open and closed are $\varnothing$ and X.

Theorem 10.5. Jordan Curve Theorem
Every simple closed curve in the plane separates the plane into exactly two connected components, an interior and an exterior, and is the common boundary of both.

Definition 10.6. Continuity in Topological Spaces
A map $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is continuous if the inverse image of every open set is open:

$$\forall V \in \mathcal{T}_Y, \qquad f^{-1}(V) \in \mathcal{T}_X.$$

Definition 10.7. Homeomorphism
A homeomorphism is a bijective continuous map whose inverse is also continuous. Equivalently, $f : X \xrightarrow{\sim} Y$ is a homeomorphism if f and f^{-1} are continuous. Two spaces are homeomorphic, written $X \cong Y$, if such a map exists.

Theorem 10.8. Brouwer Fixed-Point Theorem
Every continuous map from a closed ball in $\mathbb{R}^n$ to itself has at least one fixed point.

Theorem 10.9. Borsuk–Ulam Theorem
Every continuous map

$$f : S^n \to \mathbb{R}^n$$

takes some pair of antipodal points to the same value.

Corollary 10.10. Ham Sandwich Theorem
Any n bounded measurable subsets of $\mathbb{R}^n$ can be bisected simultaneously by one hyperplane.

Theorem 10.11. Hairy Ball Theorem
Every continuous tangent vector field on an even-dimensional sphere S^{2m} vanishes at some point.

Theorem 10.12. Invariance of Domain
If $U \subseteq \mathbb{R}^n$ is open and $f : U \to \mathbb{R}^n$ is continuous and injective, then $f(U)$ is open and f is a homeomorphism from U onto $f(U)$.

Definition 10.13. Manifold
An n-dimensional topological manifold is a space that locally looks like $\mathbb{R}^n$. More precisely, each point has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$.

Definition 10.14. Fundamental Group
The fundamental group $\pi_1(X, x_0)$ records loops in X based at x_0, up to continuous deformation. It is one of the basic algebraic invariants of a topological space.

Fact 10.15. Fundamental Groups of Familiar Spaces
The following basic examples are worth recognizing:

$$\pi_1(S^1) \cong \mathbb{Z}, \qquad \pi_1(S^n) \cong \{e\} \quad (n \geq 2), \qquad \pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2.$$

Strategy. For introductory topology questions, the most common quick checks are: compactness means every open cover has a finite subcover, connectedness means the space cannot be separated into two disjoint nonempty open sets, and a homeomorphism is a continuous bijection with continuous inverse.

Formula 10.16. Euler Characteristic of a Surface
For a triangulated closed surface, the Euler characteristic is

$$\chi = V - E + F.$$

For an orientable closed surface of genus g,

$$\chi = 2 - 2g.$$

Theorem 10.17. Classification of Closed Surfaces
Every connected closed surface is homeomorphic to either an orientable surface of genus $g \geq 0$ or a nonorientable surface obtained as a connected sum of $k \geq 1$ projective planes.

Theorem 10.18. Poincaré Conjecture
Every closed simply connected 3-manifold is homeomorphic to the 3-sphere S^3.

Remark. The Poincaré conjecture was proved by Grigori Perelman using Ricci flow. The fundamental correspondence is

$$\text{Poincaré conjecture} \longleftrightarrow \text{Perelman} \longleftrightarrow \text{Ricci flow}.$$

Theorem 10.19. Thurston Geometrization Theorem
Every closed orientable 3-manifold first decomposes as a connected sum of prime 3-manifolds. After cutting each irreducible prime summand along a canonical family of incompressible tori, every resulting piece admits a complete locally homogeneous metric modeled on one of the eight Thurston geometries:

$$S^3,\ E^3,\ H^3,\ S^2 \times \mathbb{R},\ H^2 \times \mathbb{R},\ \widetilde{\mathrm{SL}}_2(\mathbb{R}),\ \mathrm{Nil},\ \mathrm{Sol}.$$

Fact 10.20. The Eight Thurston Geometries
The eight model geometries are

$$S^3, \quad \mathrm{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}}_2(\mathbb{R}), \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Chapter 11

Miscellaneous Topics

Strategy. Iterated Knowledge Elimination. Let Ω_0 be a finite set of possible states. For each agent i and state $\omega \in \Omega$, let

$$\mathcal{I}_i^{\Omega}(\omega) = \{\omega' \in \Omega \mid \omega' \text{ gives agent } i \text{ the same information as } \omega\}.$$

Relative to the current state space Ω, agent i knows the state precisely when

$$|\mathcal{I}_i^{\Omega}(\omega)| = 1.$$

A truthful public announcement restricts Ω to exactly the states at which that announcement is true. Repeating this restriction models successive statements about knowledge or ignorance. In particular, the statement "I knew that Q" means

$$\forall \omega' \in \mathcal{I}_i^{\Omega}(\omega), \qquad Q(\omega').$$

Strategy. Transform the Given Objects. Replace the original configuration by an equivalent one whose structure is easier to see: partition a region, introduce prefix sums, encode a movement by vectors, pass to residues, diagonalize a matrix, or complete a partial object. The transformed problem should preserve the quantity or property being asked about.

Strategy. Invariant, Monovariant, and Extremal Principles. An invariant remains unchanged under every allowed move. A monovariant moves strictly in one direction and therefore prevents cycles or proves termination. The extremal principle chooses a largest, smallest, first, or last object and exploits the extra restrictions forced by extremality. These are among the most common contest ideas hidden behind elementary statements.

Strategy. Complete the Configuration and Work Backward. First imagine a completed solution, then determine which parts are forced and which choices remain free. This is effective in geometric constructions, permutation completions, lattice paths, tilings, and assignment problems. Fixing one object under a symmetry often removes redundant cases before the remaining count begins.

Strategy. Rotate, Reflect, and Superimpose. A rotation, reflection, or translated copy can convert a broken path into a straight one, expose equal vectors, pair terms, or turn a local pattern into a global invariant. When two configurations overlap, compare what cancels and what remains rather than recomputing each separately.

Strategy. Homogenization and Normalization. If an inequality is homogeneous, scale the variables to impose a simple condition such as $a + b + c = 1$, $abc = 1$, or $x^2 + y^2 = 1$.

Strategy. Equality Cases and Smoothing. Identify the expected equality case before selecting an inequality. For a symmetric expression, replacing two variables by their average often reveals the extremal configuration.

Theorem 11.1. Triangle and Reverse Triangle Inequalities
For real or complex numbers x, y,

$$\big||x| - |y|\big| \leq |x - y| \leq |x| + |y|.$$

More generally, every norm satisfies

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

Formula 11.2. Basic Sum-of-Squares Identities
For real a, b, c,

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}\Big((a-b)^2 + (b-c)^2 + (c-a)^2\Big) \geq 0.$$

Consequently,

$$a^2 + b^2 + c^2 \geq ab + bc + ca$$

and

$$(a + b + c)^2 \leq 3(a^2 + b^2 + c^2).$$

Formula 11.3. Frequently Used Two-Variable Inequalities
For real x, y,

$$x^2 + y^2 \geq 2xy.$$

For positive x, y,

$$\frac{x}{y} + \frac{y}{x} \geq 2, \qquad x + \frac{1}{x} \geq 2.$$

Equality holds exactly when the two compared positive quantities are equal.

Theorem 11.4. Lagrange's Identity
For real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$,

$$\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) - \left(\sum_{i=1}^{n} a_i b_i\right)^2 = \sum_{1 \leq i < j \leq n} (a_i b_j - a_j b_i)^2.$$

The nonnegativity of the right-hand side gives the Cauchy–Schwarz inequality immediately.

Theorem 11.5. QM–AM–GM–HM Inequalities
For positive $x_1, \ldots, x_n$,

$$\sqrt{\frac{x_1^2 + \cdots + x_n^2}{n}} \geq \frac{x_1 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 \cdots x_n} \geq \frac{n}{\frac{1}{x_1} + \cdots + \frac{1}{x_n}}.$$

Equality holds if and only if all variables are equal.

Theorem 11.6. Weighted AM–GM Inequality
If $w_i \geq 0$, $\sum_{i=1}^{n} w_i = 1$, and $x_i > 0$, then

$$\sum_{i=1}^{n} w_i x_i \geq \prod_{i=1}^{n} x_i^{w_i}.$$

Equality holds when all x_i with positive weight are equal.

Theorem 11.7. Power Mean Inequality
For positive $x_1, \ldots, x_n$ and real $r \neq 0$, define

$$M_r = \left(\frac{x_1^r + \cdots + x_n^r}{n} \right)^{1/r},$$

and define $M_0 = (x_1 \cdots x_n)^{1/n}$. If $r < s$, then

$$M_r \leq M_s.$$

The harmonic, geometric, arithmetic, and quadratic means correspond to $r = -1, 0, 1, 2$, respectively.

Theorem 11.8. Young's Inequality
If $x, y \geq 0$, $p, q > 1$, and

$$\frac{1}{p} + \frac{1}{q} = 1,$$

then

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}.$$

Theorem 11.9. Cauchy–Schwarz Inequality, Engel Form
For real a_i and positive b_i,

$$\sum_{i=1}^{n} \frac{a_i^2}{b_i} \geq \frac{(a_1 + \cdots + a_n)^2}{b_1 + \cdots + b_n}.$$

Theorem 11.10. Hölder's Inequality
If $p, q > 1$ and $1/p + 1/q = 1$, then

$$\sum_{i=1}^{n} |a_i b_i| \leq \left(\sum_{i=1}^{n} |a_i|^p \right)^{1/p} \left(\sum_{i=1}^{n} |b_i|^q \right)^{1/q}.$$

Theorem 11.11. Minkowski's Inequality
For $p \geq 1$,

$$\left(\sum_{i=1}^{n} |a_i + b_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^{n} |a_i|^p \right)^{1/p} + \left(\sum_{i=1}^{n} |b_i|^p \right)^{1/p}.$$

Theorem 11.12. Rearrangement Inequality
If

$$a_1 \leq \cdots \leq a_n, \qquad b_1 \leq \cdots \leq b_n,$$

then for every permutation $\sigma \in S_n$,

$$\sum_{i=1}^{n} a_i b_{n+1-i} \leq \sum_{i=1}^{n} a_i b_{\sigma(i)} \leq \sum_{i=1}^{n} a_i b_i.$$

Theorem 11.13. Chebyshev's Sum Inequality
If $a_1 \leq \cdots \leq a_n$ and $b_1 \leq \cdots \leq b_n$, then

$$\frac{1}{n} \sum_{i=1}^{n} a_i b_i \geq \left(\frac{1}{n} \sum_{i=1}^{n} a_i \right) \left(\frac{1}{n} \sum_{i=1}^{n} b_i \right).$$

Theorem 11.14. Schur's Inequality
For $a, b, c \geq 0$ and $r \geq 0$,

$$\sum_{\text{cyc}} a^r (a-b)(a-c) \geq 0.$$

Formula 11.15. Nesbitt's Inequality
For positive a, b, c,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Theorem 11.16. Bernoulli's Inequality
If $x \geq -1$ and $r \geq 1$, then

$$(1+x)^r \geq 1 + rx.$$

Strategy. Comparing Powers by Logarithms. To compare two positive numbers such as a^b and b^a, take logarithms:

$$a^b > b^a \quad \Longleftrightarrow \quad b \ln a > a \ln b \quad \Longleftrightarrow \quad \frac{\ln a}{a} > \frac{\ln b}{b}.$$

The function $\ln x/x$ is decreasing for $x > e$. This quickly handles comparisons such as 11^9 versus 9^{11} or e^π versus π^e.

Formula 11.17. More Useful Series Values
The following values are common in fast calculation problems:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1, \qquad \sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4},$$

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}, \qquad \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}.$$

Strategy. These values often appear after a simple transformation. For example,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right) = \frac{\ln 2}{2}.$$

Part II

Facts to Memorize

Chapter 12

Mathematical Prizes and Congresses

Science Quiz often asks for the short association between a prize, a congress, a laureate, and a mathematical achievement. This chapter records the facts that should be recognized quickly: not full biographies, but the names, years, anchors, and historical landmarks most likely to become quiz answers.

Fact 12.1. Science Quiz Prize Triad
The three mathematics prizes that should be recognized first are

$$\text{Fields Medal}, \qquad \text{Abel Prize}, \qquad \text{Wolf Prize in Mathematics}.$$

Recognize each prize through the association

$$\text{prize} \quad \longleftrightarrow \quad \text{laureate} \quad \longleftrightarrow \quad \text{achievement or area}.$$

Strategy. History questions often ask for a short association rather than a full biography. Current prominence, a recent award, or a connection to the POSTECH–KAIST context makes such an association especially relevant.

Prize	Origin and first award	Namesake	Administration and cycle
Fields Medal	Proposed at ICM Toronto in 1924; first awarded in 1936.	John Charles Fields, Canadian mathematician and organizer of ICM 1924, who donated funds for the medals.	Awarded by the IMU at the ICM, normally every four years; subject to the age condition.
Abel Prize	Established by the Norwegian Parliament in 2002; first awarded in 2003.	Niels Henrik Abel, Norwegian mathematician known for algebra, analysis, and the impossibility of solving the general quintic by radicals.	Awarded annually by the Norwegian Academy of Science and Letters; presented in Oslo; no age restriction.
Wolf Prize in Mathematics	The Wolf Foundation was established in 1975; the prizes were first awarded in 1978.	Ricardo Wolf, a German-born Cuban inventor, diplomat, and philanthropist; he was not a mathematician.	Awarded by the Wolf Foundation in Israel; normally annual, although mathematics is not necessarily awarded every year; no age restriction.

Fields Medal and ICM

The International Congress of Mathematicians, commonly abbreviated as ICM, is the most important international congress in mathematics. The Fields Medal is awarded by the International Mathematical Union at the ICM every four years. At ICM Toronto in 1924, a resolution proposed two gold medals for outstanding mathematical achievement. John Charles Fields later donated funds for the medals that bear his name. A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress. Since 1966, up to four medals may be awarded at an ICM.

Year	ICM location	Medalists	Quiz anchors
1936	Oslo, Norway	Lars Ahlfors; Jesse Douglas	Complex analysis, Riemann surfaces; minimal surfaces, Plateau problem.
1950	Cambridge, USA	Laurent Schwartz; Atle Selberg	Distribution theory; analytic number theory, sieve methods, zeta functions.
1954	Amsterdam, Netherlands	Kunihiko Kodaira; Jean-Pierre Serre	Complex geometry, algebraic geometry; algebraic topology, sheaf theory.
1958	Edinburgh, UK	Klaus Roth; René Thom	Diophantine approximation; cobordism, differential topology.
1962	Stockholm, Sweden	Lars Hörmander; John Milnor	Partial differential equations; differential topology, exotic spheres.
1966	Moscow, USSR	Michael Atiyah; Paul Cohen; Alexander Grothendieck; Stephen Smale	K-theory and index theorem; forcing and continuum hypothesis; modern algebraic geometry; generalized Poincaré conjecture.
1970	Nice, France	Alan Baker; Heisuke Hironaka; Serge Novikov; John G. Thompson	Transcendental number theory; resolution of singularities; topology and geometry; finite group theory.
1974	Vancouver, Canada	Enrico Bombieri; David Mumford	Number theory, complex analysis, minimal surfaces; moduli spaces and algebraic geometry.
1978	Helsinki, Finland	Pierre Deligne; Charles Fefferman; Grigory Margulis; Daniel Quillen	Weil conjectures; several complex variables; Lie groups and ergodic theory; algebraic K-theory.
1982	Warsaw, Poland (held in 1983)	Alain Connes; William Thurston; Shing-Tung Yau	Noncommutative geometry; 3-manifolds and low-dimensional topology; Calabi conjecture and geometric analysis.
1986	Berkeley, USA	Simon Donaldson; Gerd Faltings; Michael Freedman	Gauge theory and 4-manifolds; Mordell conjecture and arithmetic geometry; topological 4-dimensional Poincaré conjecture.
1990	Kyoto, Japan	Vladimir Drinfeld; Vaughan Jones; Shigefumi Mori; Edward Witten	Quantum groups and Langlands program; Jones polynomial; minimal model program; mathematical physics and quantum field theory.
1994	Zurich, Switzerland	Jean Bourgain; Pierre-Louis Lions; Jean-Christophe Yoccoz; Efim Zelmanov	Analysis and Banach spaces; partial differential equations; dynamical systems; restricted Burnside problem.

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Year	ICM location	Medalists	Quiz anchors
1998	Berlin, Germany	Richard Borcherds; W. Timothy Gowers; Maxim Kontsevich; Curtis T. McMullen	Moonshine and automorphic forms; functional analysis and combinatorics; algebraic geometry and mathematical physics; complex dynamics.
2002	Beijing, China	Laurent Lafforgue; Vladimir Voevodsky	Langlands correspondence over function fields; motivic cohomology and $\mathbb{A}^1$-homotopy theory.
2006	Madrid, Spain	Andrei Okounkov; Grigori Perelman; Terence Tao; Wendelin Werner	Probability, representation theory, algebraic geometry; Ricci flow; PDE, harmonic analysis, additive combinatorics; SLE and Brownian motion.
2010	Hyderabad, India	Elon Lindenstrauss; Ngo Bao Chau; Stanislav Smirnov; Cédric Villani	Ergodic theory and number theory; Fundamental Lemma; conformal invariance; Boltzmann equation and optimal transport.
2014	Seoul, South Korea	Artur Avila; Manjul Bhargava; Martin Hairer; Maryam Mirzakhani	Dynamical systems; geometry of numbers; stochastic PDE; Riemann surfaces and moduli spaces.
2018	Rio de Janeiro, Brazil	Caucher Birkar; Alessio Figalli; Peter Scholze; Akshay Venkatesh	Fano varieties and minimal model program; optimal transport; p-adic geometry; number theory and homogeneous dynamics.
2022	Originally Saint Petersburg, Russia; virtual congress; awards in Helsinki, Finland	Hugo Duminil-Copin; June Huh; James Maynard; Maryna Viazovska	Phase transitions in statistical physics; Hodge theory in combinatorics and matroids; prime numbers; sphere packing and the E_8 lattice.
2026	Philadelphia, USA	Yu Deng; John Pardon; Jacob Tsimerman; Hong Wang	PDE and kinetic theory; symplectic geometry and topology; arithmetic geometry and o-minimality; harmonic analysis, geometric measure theory, and the Kakeya problem.

Fact 12.2. ICM and Fields Medal landmarks
The following associations are especially useful.

- ICM stands for International Congress of Mathematicians.
- IMU stands for International Mathematical Union.
- The Fields Medal was first awarded in 1936 at ICM Oslo.
- The first two Fields Medalists were Lars Ahlfors and Jesse Douglas.
- The first year with four Fields Medalists was 1966.
- Jean-Pierre Serre is the youngest Fields Medalist.
- Edward Witten was the first physicist to receive the Fields Medal; as an undergraduate, he majored in history rather than mathematics.
- Andrew Wiles received a special IMU silver plaque in 1998 for his proof of Fermat's Last Theorem.

- Grigori Perelman declined the Fields Medal in 2006.
- Maryam Mirzakhani was the first woman Fields Medalist.
- Maryna Viazovska was the second woman Fields Medalist.
- ICM 2014 was held in Seoul, South Korea.
- ICM 2022 was originally scheduled for Saint Petersburg, Russia. It was held as a virtual congress, and the prize ceremony took place in Helsinki, Finland.
- ICM 2026: Philadelphia, USA, July 23–30, 2026.
- There is no ICM scheduled for 2027: the regular cycle proceeds from 2026 to 2030.
- ICM 2030 is scheduled for Glasgow, UK; the associated IMU General Assembly is scheduled for Dublin, Ireland.

Fact 12.3. Fields Medal design
The obverse of the Fields Medal bears a profile of Archimedes and the Latin inscription

TRANSIRE SUUM PECTUS MUNDOQUE POTIRI.

The reverse shows a sphere inscribed in a cylinder, recalling the result that Archimedes requested for his tomb. For the corresponding sphere and circumscribed cylinder,

$$V_{\text{cylinder}} : V_{\text{sphere}} = 3 : 2,$$

equivalently

$$\frac{V_{\text{sphere}}}{V_{\text{cylinder}}} = \frac{2}{3}.$$

Abel Prize

The Abel Prize is awarded by the Norwegian Academy of Science and Letters. It is named after Niels Henrik Abel. The Norwegian Parliament established the modern prize in 2002, the bicentenary of Abel's birth, and the first award was made in 2003. Unlike the Fields Medal, the Abel Prize has no age restriction. For Science Quiz, the Abel Prize is often asked through the pattern

$$\text{year} \quad \longleftrightarrow \quad \text{laureate} \quad \longleftrightarrow \quad \text{famous theorem or area.}$$

Year	Laureate(s)	Quiz anchors
2003	Jean-Pierre Serre	Algebraic topology, algebraic geometry, number theory; first Abel Prize laureate.
2004	Michael Atiyah; Isadore Singer	Atiyah–Singer index theorem.
2005	Peter D. Lax	Partial differential equations and applied mathematics.
2006	Lennart Carleson	Harmonic analysis; convergence of Fourier series.
2007	Srinivasa S. R. Varadhan	Probability theory; large deviations.
2008	John G. Thompson; Jacques Tits	Finite groups, classification of finite simple groups; buildings and group theory.
2009	Mikhail Gromov	Geometry, geometric group theory, symplectic geometry.

Continued on the next page.

Year	Laureate(s)	Quiz anchors
2010	John Tate	Number theory and arithmetic geometry.
2011	John Milnor	Topology, geometry, dynamics; exotic spheres.
2012	Endre Szemerédi	Discrete mathematics and theoretical computer science; Szemerédi's theorem.
2013	Pierre Deligne	Algebraic geometry; Weil conjectures.
2014	Yakov Sinai	Dynamical systems, ergodic theory, mathematical physics.
2015	John F. Nash Jr.; Louis Nirenberg	Nonlinear PDE, geometry, Nash embedding theorem.
2016	Andrew Wiles	Fermat's Last Theorem; modularity of elliptic curves.
2017	Yves Meyer	Wavelet theory; applications from signal compression to gravitational-wave analysis.
2018	Robert P. Langlands	Langlands program; representation theory and number theory.
2019	Karen Uhlenbeck	Geometric analysis and gauge theory; first woman Abel Prize laureate.
2020	Hillel Furstenberg; Gregory Margulis	Probability, dynamics, ergodic theory, Lie groups.
2021	László Lovász; Avi Wigderson	Discrete mathematics, theoretical computer science, complexity theory.
2022	Dennis P. Sullivan	Topology, geometric topology, dynamical systems.
2023	Luis A. Caffarelli	Nonlinear partial differential equations, free-boundary problems, Monge–Ampère equation.
2024	Michel Talagrand	Probability theory, functional analysis, spin glasses, concentration inequalities.
2025	Masaki Kashiwara	Algebraic analysis, D-modules, representation theory, crystal bases.
2026	Gerd Faltings	Arithmetic geometry; Mordell conjecture, Lang's conjectures, Diophantine geometry.

Fact 12.4. Abel Prize landmarks
The following associations summarize the prize's main quiz anchors.

- Abel Prize ↔ Norwegian Academy of Science and Letters.
- Abel Prize ↔ Niels Henrik Abel.
- First Abel Prize laureate ↔ Jean-Pierre Serre.
- First woman Abel Prize laureate ↔ Karen Uhlenbeck.
- Since 2019, the Abel Prize has carried a monetary award of NOK 7.5 million, on the order of one million US dollars.
- John Milnor ↔ Fields Medal 1962 and Abel Prize 2011.
- Andrew Wiles ↔ Abel Prize 2016 ↔ Fermat's Last Theorem.
- Yves Meyer ↔ Abel Prize 2017 ↔ wavelets.
- Robert Langlands ↔ Abel Prize 2018 ↔ Langlands program.
- Gerd Faltings ↔ Fields Medal 1986 and Abel Prize 2026.

Wolf Prize in Mathematics

The Wolf Prize is awarded by the Wolf Foundation, established in Israel in 1975 by Ricardo Wolf. Wolf was a German-born Cuban inventor, diplomat, and philanthropist, not a mathematician. The prizes were first awarded in 1978. In the sciences, the Wolf Prize is awarded in Medicine, Agriculture, Mathematics, Chemistry, and Physics. It is especially important historically because many Wolf Prize in Mathematics laureates later received, or had already received, other major mathematical prizes. Unlike the Fields Medal, it is not restricted by age, and mathematics need not be awarded every year.

Year	Laureate(s)	Quiz anchors
1978	Israel Gelfand; Carl L. Siegel	Representation theory, functional analysis; number theory, several complex variables.
1979	Jean Leray; André Weil	PDE and algebraic topology; algebraic geometry and number theory.
1980	Henri Cartan; Andrey Kolmogorov	Several complex variables and algebraic topology; probability theory and dynamical systems.
1981	Lars Ahlfors; Oscar Zariski	Complex analysis and Riemann surfaces; algebraic geometry.
1982	Mark Krein; Hassler Whitney	Functional analysis; differential topology and Whitney embedding.
1983	Shiing-Shen Chern	Global differential geometry, Chern classes.
1984	Paul Erdős	Combinatorics, number theory, probabilistic method.
1985	Kunihiko Kodaira; Hans Lewy	Complex geometry; partial differential equations.
1986	Samuel Eilenberg; Atle Selberg	Algebraic topology and category theory; analytic number theory and Selberg trace formula.
1987	Kiyoshi Itô; Peter D. Lax	Stochastic calculus; partial differential equations and applied mathematics.
1988	Lars Hörmander; Friedrich Hirzebruch	Partial differential equations; topology, characteristic classes, Hirzebruch–Riemann–Roch.
1989	Alberto Calderón; John Milnor	Harmonic analysis and singular integrals; topology and exotic spheres.
1990	Ennio De Giorgi; Ilya Piatetski-Shapiro	Minimal surfaces and regularity theory; automorphic forms.
1992	Lennart Carleson; John G. Thompson	Harmonic analysis; finite group theory.
1993	Mikhail Gromov; Jacques Tits	Geometry and geometric group theory; buildings and group theory.
1995/96	Robert Langlands; Andrew Wiles	Langlands program; Fermat's Last Theorem.
1996/97	Joseph Keller; Yakov Sinai	Applied mathematics and waves; dynamical systems and mathematical physics.
1999	László Lovász; Elias M. Stein	Discrete mathematics and algorithms; harmonic analysis.
2000	Raoul Bott; Jean-Pierre Serre	Topology and geometry; algebraic topology, algebraic geometry, number theory.
2001	Vladimir Arnold; Saharon Shelah	Dynamical systems and singularity theory; mathematical logic and set theory.

Continued on the next page.

Year	Laureate(s)	Quiz anchors
2002/03	Mikio Sato; John Tate	Algebraic analysis and hyperfunctions; arithmetic geometry.
2005	Gregory Margulis; Serge Novikov	Lie groups, ergodic theory, discrete subgroups; topology and geometry.
2006/07	Stephen Smale; Hillel Furstenberg	Differential topology and dynamical systems; ergodic theory and probability.
2008/09	Pierre Deligne; Phillip Griffiths; David Mumford	Algebraic geometry, Hodge theory, moduli spaces.
2010	Shing-Tung Yau; Dennis Sullivan	Geometric analysis; topology and dynamics.
2012	Michael Aschbacher; Luis Caffarelli	Finite groups; nonlinear PDE and free-boundary problems.
2013	George Mostow; Michael Artin	Rigidity, geometry, Lie groups; algebraic geometry.
2014	Peter Sarnak	Number theory, analysis, geometry, automorphic forms.
2015	James Arthur	Trace formula, automorphic representations.
2017	Richard Schoen; Charles Fefferman	Geometric analysis; analysis and partial differential equations.
2018	Alexander Beilinson; Vladimir Drinfeld	Geometric representation theory, mathematical physics, Langlands program.
2019	Jean-François Le Gall; Gregory Lawler	Probability theory, random walks, Brownian motion, SLE.
2020	Simon Donaldson; Yakov Eliashberg	Differential geometry, topology, gauge theory; symplectic and contact topology.
2022	George Lusztig	Representation theory and related areas.
2023	Ingrid Daubechies	Wavelet theory, time-frequency analysis, signal processing.
2024	Adi Shamir; Noga Alon	Mathematical cryptography; combinatorics, graph theory, theoretical computer science.

Fact 12.5. Wolf Prize landmarks
The following associations summarize the prize's main quiz anchors.

- Wolf Prize ↔ Wolf Foundation.
- The name honors Ricardo Wolf; it does not refer to a mathematician named Wolf.
- Wolf Prize in Mathematics ↔ no Fields-style age restriction.
- Paul Erdős ↔ Wolf Prize 1984 ↔ combinatorics and probabilistic method.
- John Milnor ↔ Wolf Prize 1989; together with the Fields Medal and Abel Prize, this makes him a standard example connected to all three major mathematics prizes.
- Andrew Wiles and Robert Langlands shared the 1995/96 Wolf Prize in Mathematics.
- Ingrid Daubechies ↔ Wolf Prize 2023 ↔ wavelets.
- Adi Shamir and Noga Alon ↔ Wolf Prize 2024 ↔ cryptography and combinatorics.

Fact 12.6. Prize recognition
For concise review, remember the following map first.

- Fields Medal ↔ ICM ↔ IMU ↔ every four years ↔ age condition.

- Abel Prize $\leftrightarrow$ Norwegian Academy of Science and Letters $\leftrightarrow$ no age restriction.
- Wolf Prize $\leftrightarrow$ Wolf Foundation $\leftrightarrow$ no Fields-style age restriction.
- Fields Medal questions often ask for the year, ICM location, laureate, and keyword.
- Abel and Wolf questions often ask for the laureate–area association.

Chapter 13

Mathematicians, Institutions, History, and Anecdotes

Science Quiz repeatedly asks for a mathematician from a birth year, a named theorem, a field, a book, or a short biographical clue. The most efficient preparation is therefore chronological: recognize the person, place the person in the correct period, and attach one or two unmistakable achievements.

Mathematicians and Mathematical Institutions

The following table is ordered by birth. Ancient dates are approximate, and "fl." indicates the period in which a person is known to have worked. Living mathematicians are written with an open-ended life span, such as "1980–". The third column distinguishes birthplace or historical origin, nationality or citizenship when it is stated, and later residence or major place of work. An arrow records chronological movement only; it does not by itself assert a change of citizenship. When a modern national label would be anachronistic, the historical polity is given together with its present geographical location. Botong Wang is included without a birth year because no reliable public birth year was identified.

Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Thales of Miletus	c. 624–c. 546 BCE	Ancient Greece (Miletus; now Türkiye)	Geometry	Early deductive geometry; Thales' theorem; a right angle in a semicircle.
Pythagoras	c. 570–c. 495 BCE	Ancient Greece (Samos)	Geometry, number theory	Pythagorean theorem; Pythagorean school; numerical ratios in music.
Euclid	fl. c. 300 BCE	Hellenistic Egypt	Geometry, foundations	*Elements*; axiomatic method; Euclidean algorithm; infinitely many primes.
Archimedes	c. 287–212 BCE	Greek Sicily (now Italy)	Geometry, mechanics	Method of exhaustion; bounds for π; lever; buoyancy; sphere and circumscribed cylinder.
Apollonius of Perga	c. 240–c. 190 BCE	Ancient Greece (Perga; now Türkiye)	Geometry	*Conics*; systematic study of ellipses, parabolas, and hyperbolas.
Diophantus	c. 200–c. 284	Roman Egypt	Number theory, algebra	*Arithmetica*; Diophantine equations.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Liu Hui	c. 225–c. 295	China	Arithmetic, geometry	Commentary on *The Nine Chapters*; polygon method for π; elimination procedures.
Hypatia	c. 355–415	Roman Egypt	Geometry, astronomy	Alexandrian teacher and commentator; associated with editions of Diophantus and Apollonius.
Zu Chongzhi	429–500	China	Numerical mathematics, astronomy	$3.1415926 < \pi < 3.1415927$; the approximation $\pi \approx 355/113$.
Aryabhata	476–550	India	Arithmetic, trigonometry, astronomy	*Aryabhatiya*; sine tables; place-value computation and astronomical algorithms.
Brahmagupta	598–c. 668	India	Algebra, number theory	Rules for zero and negative numbers; Brahmagupta's formula; quadratic equations.
Muhammad al-Khwarizmi	c. 780–c. 850	Khwarazm (now Uzbekistan)	Algebra, arithmetic	Systematic algebra; the words *algebra* and *algorithm*.
Omar Khayyam	1048–1131	Persia (now Iran)	Algebra, geometry	Classification of cubic equations and geometric solutions by conic sections.
Bhāskara II	1114–1185	India	Algebra, number theory, astronomy	*Līlāvatī*; indeterminate equations; cyclic methods related to Pell equations.
Fibonacci	c. 1170–after 1240	Italy	Arithmetic, number theory	*Liber Abaci*; spread of Hindu–Arabic numerals in Europe; Fibonacci sequence.
Madhava of Sangamagrama	c. 1340–c. 1425	India	Infinite series, trigonometry	Kerala school; series for π, sine, and cosine.
Jamshid al-Kashi	c. 1380–1429	Persia (now Iran)	Numerical mathematics, astronomy	Decimal fractions; high-precision computation of π; law of cosines.
Niccolò Tartaglia	c. 1499–1557	Republic of Venice (now Italy)	Algebra	A method for depressed cubic equations; the Cardano–Tartaglia priority dispute.
Gerolamo Cardano	1501–1576	Duchy of Milan (now Italy)	Algebra, probability	*Ars Magna*; cubic and quartic equations; early probability.
Robert Recorde	c. 1512–1558	Wales	Arithmetic, algebra	Introduced the equality sign $=$ in *The Whetstone of Witte* (1557).
Lodovico Ferrari	1522–1565	Italy	Algebra	General solution of the quartic equation, published in Cardano's *Ars Magna*.
Rafael Bombelli	1526–1572	Italy	Algebra	*L'Algebra*; systematic rules for complex numbers in solutions of cubic equations.
François Viète	1540–1603	France	Algebra, trigonometry	Systematic symbolic algebra; Viète's formulas.
Simon Stevin	1548–1620	Flanders (now Belgium)	Arithmetic, mechanics	Promoted decimal fractions in Europe; work on statics and hydrostatics.
John Napier	1550–1617	Scotland	Computation, logarithms	Introduced logarithms in print in 1614; Napier's bones.
Thomas Harriot	c. 1560–1621	England	Algebra, astronomy	Introduced the inequality signs $<$ and $>$ in work published posthumously in 1631.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Galileo Galilei	1564–1642	Italy	Mathematical physics	Mathematical description of motion; telescopic astronomy.
Johannes Kepler	1571–1630	Holy Roman Empire (now Germany)	Geometry, astronomy	Laws of planetary motion; Kepler conjecture on sphere packing.
Marin Mersenne	1588–1648	France	Number theory, acoustics	Mersenne primes $2^p - 1$; scientific correspondence network in seventeenth-century Europe.
Girard Desargues	1591–1661	France	Projective geometry	Desargues' theorem; foundational ideas of projective geometry.
René Descartes	1596–1650	France (birthplace or historical origin) → Dutch Republic (later residence/work)	Analytic geometry, algebra	Cartesian coordinates; *La Géométrie*; rule of signs.
Bonaventura Cavalieri	1598–1647	Italy	Geometry	Method of indivisibles, an important precursor to integral calculus.
Pierre de Fermat	c. 1607–1665	France	Number theory, geometry, probability	Fermat's Last Theorem; little theorem; analytic geometry; probability correspondence with Pascal.
John Wallis	1616–1703	England	Analysis, geometry	Introduced the symbol ∞; Wallis product for π; infinitesimal methods.
Blaise Pascal	1623–1662	France	Probability, geometry	Pascal's triangle; foundations of probability with Fermat; projective geometry.
Christiaan Huygens	1629–1695	Dutch Republic (Netherlands)	Probability, mechanics	First printed treatise on probability; pendulum and wave theory.
Isaac Barrow	1630–1677	England	Geometry, analysis	Geometric form of the fundamental theorem of calculus; Newton's teacher at Cambridge.
Isaac Newton	1643–1727	England	Calculus, mechanics	Fluxions; fundamental theorem of calculus; generalized binomial theorem; laws of motion and gravitation.
Gottfried Wilhelm Leibniz	1646–1716	Holy Roman Empire (now Germany)	Calculus, logic	Independent development of calculus; dx and $\int$ notation; binary arithmetic.
Jakob Bernoulli	1655–1705	Switzerland	Probability, analysis	Bernoulli numbers; law of large numbers; *Ars Conjectandi.*
Guillaume de l'Hôpital	1661–1704	France	Analysis	First calculus textbook; l'Hôpital's rule, developed from lessons by Johann Bernoulli.
Abraham de Moivre	1667–1754	France (birthplace or historical origin) → England (later residence/work)	Probability, complex numbers	De Moivre's formula; normal approximation to the binomial distribution.
Johann Bernoulli	1667–1748	Switzerland	Analysis, mechanics	Brachistochrone problem; early calculus; teacher of Euler and l'Hôpital.
William Jones	1675–1749	Wales	Mathematics, navigation	Introduced the symbol π for the circle ratio in 1706; Euler later popularized it.
Brook Taylor	1685–1731	England	Analysis	Taylor's theorem and Taylor series; finite differences.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Christian Goldbach	1690–1764	Prussia (now Russia) (birthplace or historical origin) → Russia (later residence/work)	Number theory	Goldbach conjecture, proposed in correspondence with Euler.
James Stirling	1692–1770	Scotland	Analysis, combinatorics	Stirling's approximation; Stirling numbers.
Colin Maclaurin	1698–1746	Scotland	Analysis, geometry	Maclaurin series; *Treatise of Fluxions*.
Daniel Bernoulli	1700–1782	Netherlands (birthplace or historical origin) → Switzerland (later residence/work)	Probability, fluid mechanics	Bernoulli principle; expected utility and the St. Petersburg paradox.
Thomas Bayes	c. 1701–1761	England	Probability	Bayes' theorem; posthumous 1763 publication.
Leonhard Euler	1707–1783	Switzerland (birthplace or historical origin) → Russia and Prussia (later residence/work)	Analysis, number theory, geometry	Basel problem; Euler formula and identity; graph theory; polyhedron formula; standardized much modern notation.
Jean le Rond d'Alembert	1717–1783	France	Analysis, mechanics	Wave equation; d'Alembert's principle; work on the fundamental theorem of algebra.
Maria Gaetana Agnesi	1718–1799	Italy	Analysis	*Instituzioni analitiche*; one of the earliest comprehensive calculus textbooks; the witch of Agnesi curve.
Joseph-Louis Lagrange	1736–1813	Sardinia (now Italy) (birthplace or historical origin) → Prussia and France (later residence/work)	Analysis, mechanics, number theory	Lagrangian mechanics; multipliers; four-square theorem.
Gaspard Monge	1746–1818	France	Geometry	Descriptive geometry; Monge formulation of optimal transport.
Pierre-Simon Laplace	1749–1827	France	Probability, celestial mechanics	Laplace transform; Laplacian; analytic theory of probability.
Adrien-Marie Legendre	1752–1833	France	Number theory, analysis	Legendre symbol; elliptic integrals; least squares.
Joseph Fourier	1768–1830	France	Analysis, PDE	Fourier series and transform; heat equation.
Sophie Germain	1776–1831	France	Number theory, elasticity	Work on Fermat's Last Theorem and vibrating plates; used the pseudonym M. LeBlanc.
Carl Friedrich Gauss	1777–1855	Brunswick (now Germany)	Number theory, geometry, statistics	*Disquisitiones Arithmeticae*; quadratic reciprocity; Gaussian distribution; least squares; curvature.
Siméon Denis Poisson	1781–1840	France	Analysis, probability, physics	Poisson distribution and equation; mathematical physics and potential theory.
Bernard Bolzano	1781–1848	Bohemia (now Czechia)	Analysis, logic	Early rigorous analysis; intermediate value theorem; Bolzano– Weierstrass theorem.
Augustin-Louis Cauchy	1789–1857	France	Analysis	Rigorous limits and convergence; Cauchy sequences; integral theorem; determinant theory.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
August Ferdinand Möbius	1790–1868	Saxony (now Germany)	Geometry, number theory	Möbius strip; Möbius transformations and inversion function.
Charles Babbage	1791–1871	England	Computation, numerical mathematics	Difference Engine and Analytical Engine.
Nikolai Lobachevsky	1792–1856	Russia	Geometry	Independent creation of hyperbolic, non-Euclidean geometry.
George Green	1793–1841	England	Analysis, mathematical physics	Green's theorem, Green's identities, and Green functions.
Niels Henrik Abel	1802–1829	Norway	Algebra, analysis	Impossibility of a radical formula for the general quintic; elliptic functions; Abel Prize namesake.
János Bolyai	1802–1860	Transylvania (now Romania); Hungarian	Geometry	Independent creation of hyperbolic geometry.
Carl Gustav Jacobi	1804–1851	Prussia (now Germany)	Analysis, mechanics, number theory	Jacobian determinant; elliptic functions; Hamilton–Jacobi theory.
Peter Gustav Lejeune Dirichlet	1805–1859	France (now Germany); German	Number theory, analysis	Primes in arithmetic progressions; Dirichlet characters; modern function concept.
William Rowan Hamilton	1805–1865	Ireland	Algebra, mechanics	Quaternions; Hamiltonian mechanics; Hamilton–Jacobi equation.
Augustus De Morgan	1806–1871	British India (birthplace or historical origin) → UK (later residence/work)	Logic, algebra	De Morgan's laws; formal logic; mathematical induction terminology.
Joseph Liouville	1809–1882	France	Analysis, number theory	First proof that transcendental numbers exist; Liouville's theorem.
Hermann Grassmann	1809–1877	Prussia (now Poland); German	Algebra, geometry	*Ausdehnungslehre*; vector spaces and exterior algebra.
Ernst Kummer	1810–1893	Prussia (now Poland); German	Number theory, algebra	Ideal numbers; regular primes; major progress on Fermat's Last Theorem.
Évariste Galois	1811–1832	France	Algebra	Galois theory; solvability by radicals through groups.
James Joseph Sylvester	1814–1897	England	Algebra, number theory	Matrix and invariant theory; coined the term "matrix"; Sylvester's law of inertia.
Karl Weierstrass	1815–1897	Prussia (now Germany)	Analysis	ε–δ rigor; uniform convergence; nowhere-differentiable function.
George Boole	1815–1864	England (birthplace or historical origin) → Ireland (later residence/work)	Logic, algebra	Boolean algebra; algebraic treatment of logic.
Ada Lovelace	1815–1852	England	Computation	Notes on Babbage's Analytical Engine; an algorithm for Bernoulli numbers often described as the first published computer program.
George Gabriel Stokes	1819–1903	Ireland (birthplace or historical origin) → UK (later residence/work)	Analysis, mathematical physics	Stokes' theorem; fluid equations; optics.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Pafnuty Chebyshev	1821–1894	Russia	Number theory, approximation, probability	Chebyshev polynomials and inequality; work on prime distribution.
Arthur Cayley	1821–1895	England	Algebra, geometry	Matrix algebra; Cayley's theorem; Cayley–Hamilton theorem.
Charles Hermite	1822–1901	France	Analysis, number theory	Proved e transcendental in 1873; Hermite polynomials and normal forms.
Leopold Kronecker	1823–1891	Prussia (now Poland); German	Algebra, number theory	Kronecker product and delta; arithmetic approach to algebra.
Bernhard Riemann	1826–1866	Kingdom of Hanover (now Germany)	Analysis, geometry, number theory	Riemann surfaces and integral; Riemannian geometry; zeta function and hypothesis.
Richard Dedekind	1831–1916	Brunswick (now Germany)	Algebra, foundations, number theory	Dedekind cuts; ideals; rigorous construction of the real numbers.
Camille Jordan	1838–1922	France	Algebra, analysis	Jordan normal form; Jordan curve theorem; permutation groups.
Sophus Lie	1842–1899	Norway	Geometry, differential equations	Lie groups and Lie algebras; continuous symmetries.
Georg Cantor	1845–1918	Russian Empire (birthplace or historical origin) → Germany (later residence/work)	Set theory	Founder of set theory; diagonal argument; cardinalities; Cantor set.
Wilhelm Killing	1847–1923	Prussia (now Germany)	Lie theory, geometry	Classification program for simple Lie algebras; Killing form.
Gottlob Frege	1848–1925	Germany	Logic, foundations	Predicate logic; *Begriffsschrift*; logicist foundations of arithmetic.
Felix Klein	1849–1925	Prussia (now Germany)	Geometry, group theory	Erlangen program; Klein bottle; Klein four-group.
Ferdinand Georg Frobenius	1849–1917	Prussia (now Germany)	Algebra, analysis	Frobenius theorem and endomorphism; character theory of finite groups.
Sofia Kovalevskaya	1850–1891	Russia (birthplace or historical origin) → Sweden (later residence/work)	Analysis, PDE	Cauchy–Kovalevskaya theorem; rigid-body dynamics.
Ferdinand von Lindemann	1852–1939	Germany	Number theory, analysis	Proved π transcendental in 1882, establishing the impossibility of squaring the circle.
Henri Poincaré	1854–1912	France	Topology, dynamics, geometry	Poincaré conjecture; qualitative dynamics; foundational topology.
Andrey Markov	1856–1922	Russia	Probability, number theory	Markov chains; Markov inequality; stochastic dependence.
Giuseppe Peano	1858–1932	Italy	Logic, foundations	Peano axioms; symbolic logic; space-filling Peano curve.
David Hilbert	1862–1943	Prussia (Königsberg; now Russia); German	Foundations, algebra, geometry	Twenty-three problems; Hilbert spaces and basis theorem; axiomatic geometry.
John Charles Fields	1863–1932	Canada	Complex analysis	Canadian mathematician who proposed and endowed the Fields Medal.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Hermann Minkowski	1864–1909	Russian Empire (now Lithuania) (birthplace or historical origin) → Germany (later residence/work)	Geometry, number theory	Geometry of numbers; Minkowski spacetime; convex body theorem.
Jacques Hadamard	8 Dec 1865–17 Oct 1963	France	Analysis, number theory	Prime number theorem; Cauchy–Hadamard theorem; Hadamard matrices.
Charles-Jean de la Vallée Poussin	1866–1962	Belgium	Analysis, number theory	Independent proof of the prime number theorem in 1896.
Felix Hausdorff	1868–1942	Germany	Topology, set theory	Hausdorff spaces; Hausdorff measure and dimension.
Élie Cartan	1869–1951	France	Differential geometry, Lie theory	Lie groups and algebras; differential forms; moving frames.
Ernst Zermelo	1871–1953	Germany	Set theory, foundations	Axiomatization of set theory; well-ordering theorem; Zermelo–Fraenkel set theory.
Bertrand Russell	1872–1970	Wales, UK	Logic, foundations	Russell's paradox; *Principia Mathematica* with Whitehead.
Alfred Young	1873–1940	England	Algebra, representation theory	Young diagrams and Young tableaux for symmetric groups.
Henri Lebesgue	1875–1941	France	Analysis, measure theory	Lebesgue measure and integral; dominated convergence theorem.
G. H. Hardy	1877–1947	England	Analysis, number theory	Hardy–Weinberg principle; Hardy–Littlewood circle method; collaboration with Ramanujan.
Maurice Fréchet	1878–1973	France	Analysis, topology	Metric spaces; Fréchet derivative; abstract functional spaces.
Frigyes Riesz	1880–1956	Austria-Hungary (now Hungary)	Functional analysis	Riesz representation theorem; L^p spaces and operator theory.
L. E. J. Brouwer	1881–1966	Netherlands	Topology, foundations	Brouwer fixed-point theorem; invariance of domain; intuitionism.
Wacław Sierpiński	1882–1969	Russian Empire (now Poland); Polish	Set theory, topology	Sierpiński triangle and carpet; continuum and descriptive set theory.
Emmy Noether	1882–1935	Germany (birthplace or historical origin) → USA (later residence/work)	Algebra, mathematical physics	Noetherian rings and modules; symmetry–conservation theorem.
J. E. Littlewood	1885–1977	England	Analysis, number theory	Hardy–Littlewood circle method and conjectures; analytic number theory.
Hermann Weyl	1885–1955	Germany (birthplace or historical origin) → Switzerland and USA (later residence/work)	Geometry, representation theory, physics	Weyl groups and character formula; gauge ideas; foundations.
Hugo Steinhaus	1887–1972	Austria-Hungary (now Ukraine); Polish	Analysis, probability	Lwów school; Steinhaus theorem; co-founder of *Studia Mathematica*.
George Pólya	1887–1985	Austria-Hungary (now Hungary) (birthplace or historical origin) → USA (later residence/work)	Combinatorics, probability	Pólya enumeration theorem; random walks; *How to Solve It*.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Srinivasa Ramanujan	1887–1920	British India	Number theory, analysis	Partitions, modular forms, mock theta functions; 1729.
Richard Courant	1888–1972	Prussia (now Poland) (birthplace or historical origin) → Germany and USA (later residence/work)	Analysis, PDE	Courant Institute namesake; finite elements; Courant–Friedrichs–Lewy condition.
Ludwig Wittgenstein	1889–1951	Austria (birthplace or historical origin) → UK (later residence/work)	Logic, philosophy	*Tractatus*; language games; major influence on logical positivism and philosophy of language.
Stefan Banach	1892–1945	Austria-Hungary (now Poland); Polish	Functional analysis	Banach spaces; contraction mapping theorem; Lwów school and *The Scottish Book*.
Norbert Wiener	1894–1964	USA	Analysis, probability	Wiener process and measure; harmonic analysis; founder of cybernetics.
Emil Artin	1898–1962	Austria-Hungary (now Czechia) (birthplace or historical origin) → Germany and USA (later residence/work)	Algebra, number theory	Artin reciprocity; braid groups; modern algebraic number theory.
Oscar Zariski	1899–1986	Russian Empire (now Belarus) (birthplace or historical origin) → Italy and USA (later residence/work)	Algebraic geometry	Zariski topology; foundations of modern algebraic geometry.
Juliusz Schauder	1899–1943	Austria-Hungary (now Ukraine); Polish	Functional analysis, PDE	Schauder fixed-point theorem and Schauder estimates.
Antoni Zygmund	1900–1992	Russian Empire (now Poland) (birthplace or historical origin) → USA (later residence/work)	Harmonic analysis	*Trigonometric Series*; Chicago school of harmonic analysis.
Alfred Tarski	1901–1983	Russian Empire (now Poland) (birthplace or historical origin) → USA (later residence/work)	Logic, foundations	Semantic definition of truth; Tarski's theorem; Banach–Tarski paradox.
Andrey Kolmogorov	1903–1987	Russia / USSR	Probability, dynamics	Axioms of probability; Kolmogorov complexity; turbulence and KAM theory.
Alonzo Church	1903–1995	USA	Logic, computation	Lambda calculus; Church's theorem; Church–Turing thesis.
John von Neumann	1903–1957	Austria-Hungary (now Hungary) (birthplace or historical origin) → USA (later residence/work)	Analysis, computation, game theory	Operator algebras; quantum mechanics; minimax theorem; stored-program architecture.
Stanisław Mazur	1905–1981	Austria-Hungary (now Ukraine); Polish	Functional analysis	Lwów school; Banach–Mazur theorem; Scottish Book basis problem and its live-goose prize.
Kurt Gödel	1906–1978	Austria-Hungary (now Czechia) (birthplace or historical origin) → USA (later residence/work)	Logic, foundations	Completeness and incompleteness theorems; consistency of choice and CH with ZF.
André Weil	1906–1998	France (birthplace or historical origin) → USA (later residence/work)	Number theory, algebraic geometry	Weil conjectures; adeles; founding member of Bourbaki.
Hans Fitting	1906–1938	Germany	Group theory, algebra	Fitting subgroup, lemma, decomposition, and ideals.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Lars Ahlfors	1907–1996	Finland (birthplace or historical origin) → USA (later residence/work)	Complex analysis	Riemann surfaces; Ahlfors theory; one of the first two Fields Medalists.
Claude Chevalley	1909–1984	South Africa; French (birthplace or historical origin) → USA (later residence/work)	Algebra, number theory	Chevalley groups; class field theory; founding member of Bourbaki.
Stanisław Ulam	1909–1984	Austria-Hungary (now Ukraine); Polish (birthplace or historical origin) → USA (later residence/work)	Set theory, probability	Monte Carlo method; Ulam spiral; Scottish Book and Manhattan Project.
Saunders Mac Lane	1909–2005	USA	Algebra, topology	Co-founder of category theory; Eilenberg–Mac Lane spaces.
Shiing-Shen Chern	1911–2004	China (birthplace or historical origin) → USA (later residence/work)	Differential geometry, topology	Chern classes; Chern–Gauss–Bonnet theorem; Chern Medal namesake.
Alan Turing	1912–1954	England	Logic, computation	Turing machine; computability; codebreaking; Turing test.
Paul Erdős	1913–1996	Austria-Hungary (now Hungary); Hungarian	Number theory, combinatorics	Probabilistic method; Erdős number; prolific collaboration.
Israel Gelfand	1913–2009	Russian Empire (now Ukraine) (birthplace or historical origin) → USSR and USA (later residence/work)	Functional analysis, representation theory	Gelfand representation; representation theory; influential seminar.
Samuel Eilenberg	1913–1998	Russian Empire (now Poland) (birthplace or historical origin) → USA (later residence/work)	Algebraic topology	Co-founder of category theory and homological algebra; Eilenberg–Mac Lane spaces.
Mark Kac	1914–1984	Russian Empire (now Ukraine); Polish (birthplace or historical origin) → USA (later residence/work)	Probability, mathematical physics	Feynman–Kac formula; "Can one hear the shape of a drum?"
George Dantzig	1914–2005	USA	Optimization, statistics	Linear programming; simplex method; the mistaken-homework anecdote.
Laurent Schwartz	1915–2002	France	Analysis	Theory of distributions; Fields Medal 1950.
Claude Shannon	1916–2001	USA	Information theory	Entropy, channel capacity, and the mathematical theory of communication.
Atle Selberg	1917–2007	Norway (birthplace or historical origin) → USA (later residence/work)	Analytic number theory	Selberg sieve and trace formula; elementary proof of the prime number theorem with Erdős.
Benoit Mandelbrot	1924–2010	Poland (birthplace or historical origin) → France and USA (later residence/work)	Geometry, probability	Fractal geometry; Mandelbrot set; popularized the term "fractal."
Louis Nirenberg	1925–2020	Canada (birthplace or historical origin) → USA (later residence/work)	Analysis, PDE	Fundamental estimates for elliptic PDE; Abel Prize 2015 with John Nash.
Peter D. Lax	1926–	Hungary (birthplace or historical origin) → USA (later residence/work)	Analysis, PDE	Lax pair and Lax equivalence theorem; numerical analysis; Abel Prize 2005.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Jean-Pierre Serre	1926–	France	Algebraic topology, geometry, number theory	Youngest Fields Medalist; first Abel laureate; Serre duality and spectral sequences.
Alexander Grothendieck	1928–2014	Germany; stateless, later French	Algebraic geometry	Schemes, topoi, étale cohomology; transformed modern algebraic geometry.
John Nash	1928–2015	USA	Geometry, game theory, PDE	Nash equilibrium; Nash embedding theorem; Abel Prize 2015.
Michael Atiyah	1929–2019	England; British	Geometry, topology	Atiyah–Singer index theorem; K-theory; Fields and Abel prizes.
Stephen Smale	1930–	USA	Topology, dynamics	Higher-dimensional Poincaré conjecture; Smale horseshoe; Smale's problems.
Lars Hörmander	1931–2012	Sweden	Analysis, PDE	Linear partial differential operators; microlocal analysis; Fields Medal 1962.
John Milnor	1931–	USA	Topology, geometry, dynamics	Exotic 7-spheres; Milnor fibration; Fields, Wolf, and Abel prizes.
Paul Cohen	1934–2007	USA	Logic, set theory	Forcing; independence of the continuum hypothesis and axiom of choice.
Yakov Sinai	1935–	USSR (now Russia) (birthplace or historical origin) → USA (later residence/work)	Dynamical systems, probability	Sinai billiards; Kolmogorov–Sinai entropy; Abel Prize 2014.
Robert Langlands	1936–	Canada (birthplace or historical origin) → USA (later residence/work)	Number theory, representation theory	Langlands program; reciprocity and functoriality; Abel Prize 2018.
Vladimir Arnold	1937–2010	USSR (now Ukraine); Russian	Dynamical systems, geometry	KAM theory; Arnold conjecture; singularity theory.
John Horton Conway	1937–2020	England (birthplace or historical origin) → USA (later residence/work)	Algebra, combinatorics	Surreal numbers; cellular automaton *Life*; Conway groups and knot notation.
Donald Knuth	1938–	USA	Algorithms, combinatorics	*The Art of Computer Programming*; TeX; analysis of algorithms.
Alan Baker	1939–2018	England	Number theory	Linear forms in logarithms; transcendence theory; Fields Medal 1970.
Dennis Sullivan	1941–	USA	Topology, dynamical systems	Surgery theory, rational homotopy theory, and conformal dynamics; Abel Prize 2022.
Karen Uhlenbeck	1942–	USA	Geometric analysis	Gauge theory and harmonic maps; first woman Abel laureate.
Mikhail Gromov	1943–	USSR (now Russia) (birthplace or historical origin) → France (later residence/work)	Geometry, topology	Gromov–Hausdorff distance; geometric group theory; Abel Prize 2009.
Per Enflo	1944–	Sweden	Functional analysis	Solved Mazur's Scottish Book basis problem and received the promised live goose.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Pierre Deligne	1944–	Belgium	Algebraic geometry	Proof of the Weil conjectures; Fields, Wolf, and Abel prizes.
Grigory Margulis	1946–	USSR (now Russia) (birthplace or historical origin) → USA (later residence/work)	Lie groups, dynamics	Superrigidity, arithmeticity, and homogeneous dynamics; Fields and Abel prizes.
William Thurston	1946–2012	USA	Topology, geometry	Geometrization conjecture; eight geometries; low-dimensional topology.
Masaki Kashiwara	1947–	Japan	Algebraic analysis, representation theory	D-modules, microlocal analysis, and crystal bases; Abel Prize 2025.
Luis Caffarelli	1948–	Argentina (birthplace or historical origin) → USA (later residence/work)	Nonlinear PDE	Regularity theory, free-boundary problems, and the Monge–Ampère equation; Abel Prize 2023.
László Lovász	1948–	Hungary (birthplace or historical origin) → USA and Hungary (later residence/work)	Combinatorics, theoretical computer science	Lovász local lemma, graph limits, and algorithms; Abel Prize 2021.
Shing-Tung Yau	1949–	China (birthplace or historical origin) → Hong Kong and USA (later residence/work)	Geometric analysis	Calabi conjecture; Ricci-flat metrics; Fields Medal 1982.
Edward Witten	1951–	USA	Mathematical physics	First physicist to receive the Fields Medal; quantum field theory, topology, and string theory.
Michel Talagrand	1952–	France	Probability, functional analysis	Concentration inequalities, stochastic processes, and spin glasses; Abel Prize 2024.
Adi Shamir	1952–	Israel	Cryptography, theoretical computer science	Co-inventor of RSA and Shamir secret sharing; Turing Award and Wolf Prize 2024.
Andrew Wiles	1953–	England (birthplace or historical origin) → USA and UK (later residence/work)	Number theory	Fermat's Last Theorem via modularity; special IMU plaque and Abel Prize.
Gerd Faltings	1954–	West Germany (now Germany)	Arithmetic geometry	Mordell conjecture; Fields Medal 1986; Abel Prize 2026.
Ingrid Daubechies	1954–	Belgium (birthplace or historical origin) → USA (later residence/work)	Analysis, applied mathematics	Orthogonal wavelets; Daubechies wavelets; Wolf Prize 2023.
Efim Zelmanov	1955–	USSR (now Russia) (birthplace or historical origin) → USA (later residence/work)	Algebra, group theory	Restricted Burnside problem; Fields Medal 1994.
Yitang Zhang	1955–	China (birthplace or historical origin) → USA (later residence/work)	Analytic number theory	Proved the first bounded-gap theorem for infinitely many pairs of primes, initiating rapid progress on prime gaps.
Avi Wigderson	1956–	Israel (birthplace or historical origin) → USA (later residence/work)	Theoretical computer science	Complexity theory, randomness, and pseudorandomness; Abel Prize 2021 and Turing Award.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Noga Alon	1956–	Israel (birthplace or historical origin) → USA (later residence/work)	Combinatorics, graph theory	Probabilistic method, algebraic combinatorics, and theoretical computer science; Wolf Prize 2024.
Simon Donaldson	1957–	England	Geometry, topology	Gauge theory and smooth 4-manifolds; Fields Medal 1986.
Maxim Kontsevich	1964–	USSR (now Russia) (birthplace or historical origin) → France (later residence/work)	Geometry, mathematical physics	Deformation quantization; mirror symmetry; Fields Medal 1998.
Grigori Perelman	1966–	USSR (now Russia)	Geometry, topology	Poincaré and geometrization conjectures via Ricci flow; declined the Fields Medal and Millennium Prize.
Elon Lindenstrauss	1970–	Israel	Ergodic theory, number theory	Measure rigidity and applications to number theory; Fields Medal 2010.
Cédric Villani	1973–	France	PDE, optimal transport	Boltzmann equation and Landau damping; Fields Medal 2010; *Théorème vivant*.
Manjul Bhargava	1974–	Canada; Canadian–American	Number theory	Higher composition laws; arithmetic statistics; Fields Medal 2014.
Terence Tao	1975–	Australia (birthplace or historical origin) → USA (later residence/work)	Analysis, PDE, combinatorics	Green–Tao theorem; harmonic analysis; Fields Medal 2006.
Martin Hairer	1975–	Switzerland; Austrian (birthplace or historical origin) → UK (later residence/work)	Probability, stochastic PDE	Theory of regularity structures; Fields Medal 2014.
Maryam Mirzakhani	1977–2017	Iran (birthplace or historical origin) → USA (later residence/work)	Geometry, dynamics	Moduli spaces and Riemann surfaces; first woman Fields Medalist.
Caucher Birkar	1978–	Iran; Kurdish (birthplace or historical origin) → UK (later residence/work)	Algebraic geometry	Birational geometry and boundedness of Fano varieties; Fields Medal 2018.
Akshay Venkatesh	1981–	India (birthplace or historical origin) → Australia and USA (later residence/work)	Number theory, dynamics	Number theory and homogeneous dynamics; Fields Medal 2018.
June Huh	1983–	USA (birthplace or historical origin) → South Korea and USA (later residence/work)	Combinatorics, algebraic geometry	Hodge theory for matroids; log-concavity conjectures; Fields Medal 2022.
Alessio Figalli	1984–	Italy (birthplace or historical origin) → Switzerland (later residence/work)	Analysis, optimal transport	Optimal transport and PDE; Fields Medal 2018.
Maryna Viazovska	1984–	Ukraine (birthplace or historical origin) → Switzerland (later residence/work)	Number theory, discrete geometry	Optimal sphere packing in dimensions 8 and 24; Fields Medal 2022.
Hugo Duminil-Copin	1985–	France (birthplace or historical origin) → Switzerland (later residence/work)	Probability, mathematical physics	Phase transitions and percolation; Fields Medal 2022.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
James Maynard	1987–	England	Number theory	Small gaps between primes and Diophantine approximation; Fields Medal 2022.
Peter Scholze	1987–	East Germany (now Germany)	Arithmetic geometry	Perfectoid spaces and p-adic geometry; Fields Medal 2018.
Jacob Tsimerman	1988–	USSR (now Russia) (birthplace or historical origin) → Israel and Canada (later residence/work)	Arithmetic geometry	O-minimality in arithmetic and complex algebraic geometry; André–Oort and related conjectures; Fields Medal 2026.
Yu Deng	1989–	China (birthplace or historical origin) → USA (later residence/work)	PDE, kinetic theory	Boltzmann equation from hard-sphere dynamics, wave kinetic equations, and probabilistic nonlinear Schrödinger equations; Fields Medal 2026.
John Pardon	1989–	USA	Symplectic geometry, topology	Virtual fundamental cycles, Fukaya categories, group actions on 3-manifolds, and knot theory; Fields Medal 2026.
Hong Wang	1991–	China (birthplace or historical origin) → France and USA (later residence/work)	Harmonic analysis, geometric measure theory	Fourier restriction, local smoothing, and the three-dimensional Kakeya conjecture; third woman Fields Medalist, 2026.
Botong Wang	birth year not publicly listed	China (birthplace) → United States (later residence/work)	Algebraic geometry, topology	Topology of algebraic varieties and interactions with combinatorics and applied mathematics.

Mathematical Institutions

The institutions below include learned societies, international unions, higher-education institutions, and research institutes. A research institute may host scholars and seminars without awarding degrees.

Institution	Founded	Location	Role and distinctive facts
Collège de France	1530	Paris, France	Public research chairs and lectures; admits listeners without enrollment and does not grant degrees.
École normale supérieure (ENS)	1794	Paris, France	Selective higher-education and research institution; unlike IAS, IHES, and SLMath, it has degree-granting programs.
London Mathematical Society (LMS)	1865	London, UK	Learned society that promotes mathematics through journals, meetings, grants, and prizes.
Société Mathématique de France (SMF)	1872	Paris, France	French learned society for mathematical research, publications, conferences, and professional exchange.

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Institution	Founded	Location	Role and distinctive facts
American Mathematical Society (AMS)	1888	Providence, USA	Began as the New York Mathematical Society and adopted its present name in 1894; publishes *Mathematical Reviews* and MathSciNet.
Deutsche Mathematiker-Vereinigung (DMV)	1890	Germany	German mathematical society; supports research, communication, education, and public engagement.
Institut Mittag-Leffler	1916	Djursholm, Sweden	International research center of the Royal Swedish Academy of Sciences; associated with *Acta Mathematica*.
International Mathematical Union (IMU)	1920; reconstituted 1951	Secretariat in Berlin, Germany	International union that supports global cooperation, organizes the ICM, and awards the Fields Medal and other IMU prizes.
Institut Henri Poincaré (IHP)	1928	Paris, France	International center for mathematics and theoretical physics; thematic programs, seminars, library, and outreach.
Institute for Advanced Study (IAS)	1930	Princeton, USA	Independent institute for fundamental research; home to a School of Mathematics; has no degree programs.
Mathematisches Forschungsinstitut Oberwolfach (MFO)	1944	Oberwolfach, Germany	International workshop and research center famous for small, intensive meetings and the Oberwolfach Research Institute for Mathematics.
Tata Institute of Fundamental Research (TIFR)	1945	Mumbai, India	National center for fundamental research in mathematics and the sciences; central to the development of modern mathematics in India.
Institut des Hautes Études Scientifiques (IHES)	1958	Bures-sur-Yvette, France	Private advanced research institute in mathematics and theoretical physics; hosts a small permanent faculty and many visitors; does not grant degrees.
Research Institute for Mathematical Sciences (RIMS)	1963	Kyoto, Japan	Kyoto University institute for mathematical research and collaboration; publishes *Publications of RIMS*.
Max Planck Institute for Mathematics (MPIM)	1980	Bonn, Germany	Max Planck Society institute for pure mathematics; long-term and visitor research programs.
Simons Laufer Mathematical Sciences Institute (SLMath)	1982	Berkeley, USA	Founded as MSRI and renamed in 2022; collaborative research programs, workshops, and outreach; no degree programs.
Fields Institute for Research in Mathematical Sciences	1992	Toronto, Canada	Collaborative research institute named after John Charles Fields; distinct from the Fields Medal.
Isaac Newton Institute for Mathematical Sciences	1992	Cambridge, UK	UK national and international visitor institute for long thematic programs in the mathematical sciences.

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Institution	Founded	Location	Role and distinctive facts
Korea Institute for Advanced Study (KIAS)	1996	Seoul, South Korea	Government-founded institute for basic research in mathematics, physics, and computational sciences; no formal curriculum or degree programs.
Clay Mathematics Institute (CMI)	1998	Denver, USA	Private foundation supporting research, fellowships, and publications; announced the seven Millennium Prize Problems in 2000.
Institute for Pure and Applied Mathematics (IPAM)	1999	Los Angeles, USA	UCLA-based institute connecting mathematics with science, engineering, technology, and society through long programs and workshops.
Banff International Research Station (BIRS)	2003	Banff, Canada	International mathematical research station for focused workshops, collaborative teams, and training programs.
National Institute for Mathematical Sciences (NIMS)	2005	Daejeon, South Korea	Government research institute for mathematical research, applications, computation, and public engagement.
IBS Center for Geometry and Physics (IBS-CGP)	2012	POSTECH, Pohang, South Korea	Institute for Basic Science center founded for geometry, mathematical physics, symplectic topology, and international collaboration.

Mathematical History

The following chronology records changes that altered mathematical language, method, or subject matter. It is intentionally written as short recall statements rather than as a continuous historical essay.

- **c. 1800–1600 BCE.** Babylonian tablets already contained sophisticated arithmetic, quadratic procedures, and examples of Pythagorean triples.
- **c. 1650 BCE.** The Rhind Mathematical Papyrus, copied by the scribe Ahmes from an older source, recorded Egyptian arithmetic and geometric problems.
- **c. 300 BCE.** Euclid organized Greek geometry and number theory axiomatically in the thirteen books of *Elements*.
- **3rd century BCE.** Archimedes used the method of exhaustion to obtain rigorous area and volume results and proved
$$\frac{223}{71} < \pi < \frac{22}{7}.$$
- **263.** Liu Hui completed his influential commentary on *The Nine Chapters on the Mathematical Art* and refined polygon methods for approximating π.
- **5th century.** Zu Chongzhi obtained $3.1415926 < \pi < 3.1415927$ and the approximation $355/113$.
- **628.** Brahmagupta's *Brahmasphutasiddhanta* gave systematic arithmetic rules involving zero and negative numbers.

- **c. 825.** Al-Khwarizmi's treatise on *al-jabr* gave algebra its name; the Latin form of his name produced the word *algorithm*.
- **1202.** Fibonacci's *Liber Abaci* promoted Hindu–Arabic numerals and place-value arithmetic in Latin Europe.
- **14th–16th centuries.** Madhava and the Kerala school developed infinite series for trigonometric functions and π, centuries before the European calculus.
- **1545.** Cardano published *Ars Magna*, containing methods for general cubic and quartic equations, based in part on work of del Ferro, Tartaglia, and Ferrari.
- **1557.** Robert Recorde introduced the equality sign $=$ in *The Whetstone of Witte*.
- **1614.** John Napier published logarithms, transforming difficult multiplication and astronomical computation into addition.
- **1631.** Thomas Harriot's work, published posthumously, introduced the inequality signs $<$ and $>$.
- **1637.** Descartes' *La Géométrie* joined algebra and geometry through coordinates and equations.
- **1654.** The Fermat–Pascal correspondence on games of chance helped establish mathematical probability.
- **1655.** John Wallis introduced the infinity symbol ∞ in *De sectionibus conicis*.
- **1665–1666.** During the plague years, Newton developed major parts of his calculus of fluxions, series methods, mechanics, and gravitation.
- **1675.** Leibniz first used the elongated-S integral sign $\int$ in a manuscript; his differential notation $\mathrm{d}x$ and $\mathrm{d}y$ became standard.
- **1684.** Leibniz published the first paper on differential calculus. Newton and Leibniz are now credited with independent developments of calculus.
- **1687.** Newton published the *Philosophiæ Naturalis Principia Mathematica*, placing mechanics and gravitation in a mathematical framework.
- **1706.** William Jones used the symbol π for the ratio of circumference to diameter; Euler later popularized it.
- **1713 and 1738.** Nicolaus Bernoulli proposed the St. Petersburg paradox, and Daniel Bernoulli analyzed it using expected utility.
- **1715.** Brook Taylor published the general expansion now called Taylor's theorem.
- **1730s.** Euler adopted e for the base of natural logarithms and $f(x)$ for a function; these notations became standard through his work.
- **1734–1735.** Euler solved the Basel problem,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

- **1736.** Euler's Königsberg bridges paper founded graph theory and helped initiate topology.

- **1742.** Christian Goldbach wrote to Euler about the conjecture now usually stated as: every even integer greater than 2 is a sum of two primes.
- **1755.** Euler used capital sigma Σ for summation in *Institutiones calculi differentialis*.
- **1763.** Bayes' essay on inverse probability was published posthumously by Richard Price.
- **1777.** Euler used i for $\sqrt{-1}$ in a manuscript published in 1794; Gauss later helped make the notation standard.
- **1799.** Gauss's doctoral dissertation supplied the first substantially complete proof of the fundamental theorem of algebra.
- **1801.** Gauss published *Disquisitiones Arithmeticae*, reorganizing number theory; in the same year his least-squares calculations helped recover the orbit of Ceres.
- **1821.** Cauchy's *Cours d'Analyse* made limits, convergence, and continuity central to rigorous analysis.
- **1824.** Abel proved that the general quintic equation cannot be solved by radicals.
- **1829–1832.** Lobachevsky and Bolyai published independent constructions of hyperbolic geometry, breaking the exclusive hold of Euclidean geometry.
- **1830s–1846.** Galois connected polynomial equations with permutation groups; Liouville published Galois' manuscripts posthumously in 1846.
- **1843.** Hamilton discovered the quaternions, with
$$i^2 = j^2 = k^2 = ijk = -1.$$
- **1844.** Liouville proved that transcendental numbers exist, providing the first explicit examples.
- **1847.** Boole published *The Mathematical Analysis of Logic*, treating logic by algebraic operations.
- **1854.** Riemann's habilitation lecture introduced a broad theory of manifolds and curvature, transforming geometry.
- **1858.** August Ferdinand Möbius and Johann Benedict Listing independently described the one-sided surface now called the Möbius strip.
- **1872.** Dedekind published Dedekind cuts and Cantor published a construction of the real numbers, helping complete the arithmetization of analysis.
- **1873.** Charles Hermite proved that e is transcendental.
- **1874.** Cantor proved that the real algebraic numbers are countable and that transcendental numbers are uncountable.
- **1882.** Ferdinand von Lindemann proved that π is transcendental. Consequently, squaring the circle with straightedge and compass is impossible.
- **1889.** Peano published axioms for the natural numbers in a symbolic logical language.
- **1895.** Poincaré published *Analysis Situs*, a foundational work of algebraic topology.

- **1896.** Hadamard and de la Vallée Poussin independently proved the prime number theorem,
$$\pi(x) \sim \frac{x}{\ln x}.$$
- **1899.** Hilbert's *Foundations of Geometry* gave a modern axiomatic treatment of Euclidean geometry.
- **1900.** At the second ICM in Paris, Hilbert presented ten of the twenty-three problems in his published list; the list guided much of twentieth-century mathematics.
- **1901.** Russell discovered the paradox of the set of all sets that are not members of themselves, exposing a crisis in naive set theory.
- **1904–1908.** Zermelo used the axiom of choice to prove the well-ordering theorem and then proposed an axiomatization of set theory, later refined into ZF and ZFC.
- **1910–1913.** Whitehead and Russell published the three volumes of *Principia Mathematica*.
- **1918.** Emmy Noether proved that continuous symmetries of an action correspond to conservation laws.
- **1922.** Banach's work established complete normed spaces as central objects of functional analysis; his contraction principle became a standard existence theorem.
- **1930.** The Institute for Advanced Study was founded in Princeton as an independent research institution without degree programs.
- **1931.** Gödel proved his incompleteness theorems: every sufficiently strong consistent effectively axiomatized formal system is incomplete and cannot prove its own consistency.
- **1933.** Kolmogorov axiomatized probability using measure theory.
- **1935.** The Bourbaki group began publishing under the fictional name Nicolas Bourbaki; mathematicians of the Lwów school began recording problems in *The Scottish Book*.
- **1936.** Turing formalized computation through the Turing machine and proved the undecidability of the halting problem. The first Fields Medals were awarded in Oslo.
- **1940 and 1963.** Gödel proved that CH is consistent with ZF if ZF is consistent; Cohen introduced forcing and proved that CH is independent of ZFC, subject to consistency.
- **1947.** George Dantzig introduced the simplex method for linear programming.
- **1948.** Shannon founded information theory by defining entropy and channel capacity mathematically.
- **1950.** Nash introduced the equilibrium concept now called Nash equilibrium.
- **1956.** Milnor discovered exotic smooth structures on the 7-sphere.
- **1958.** The Institut des Hautes Études Scientifiques was founded near Paris as a research institute without degree programs.
- **1963.** The Atiyah–Singer index theorem connected the analytic index of an elliptic operator with topological data.
- **1970.** Matiyasevich completed earlier work of Davis, Putnam, and Robinson, proving that Hilbert's tenth problem has no algorithmic solution.

- **1976.** Appel and Haken proved the Four Color Theorem with extensive computer assistance, creating a landmark debate about computer-assisted proof.
- **1982.** The Mathematical Sciences Research Institute was founded in Berkeley; it was renamed the Simons Laufer Mathematical Sciences Institute in 2022.
- **1983.** Faltings proved the Mordell conjecture: a smooth projective curve of genus greater than 1 over a number field has only finitely many rational points.
- **1985.** Louis de Branges proved the Bieberbach conjecture on coefficients of univalent functions.
- **1994–1995.** Wiles, with the crucial repair developed with Richard Taylor, proved Fermat's Last Theorem through the modularity of semistable elliptic curves.
- **1998 and 2014.** Hales announced a computer-assisted proof of the Kepler conjecture; the Flyspeck project later completed a formal verification.
- **2002–2003.** Perelman posted papers completing Hamilton's Ricci-flow program and proving the Poincaré and geometrization conjectures.
- **2004.** Green and Tao proved that the prime numbers contain arbitrarily long arithmetic progressions.
- **2013.** Yitang Zhang proved the first finite bound on gaps between infinitely many pairs of primes; Maynard, Tao, and the Polymath project rapidly improved bounded-gap methods.
- **2014.** ICM Seoul was held in South Korea, and Maryam Mirzakhani became the first woman to receive the Fields Medal.
- **2016.** Maryna Viazovska proved the optimality of the E_8 lattice sphere packing in dimension 8; joint work soon settled dimension 24 using the Leech lattice.
- **2022.** June Huh received the Fields Medal for work joining combinatorics, algebraic geometry, and Hodge theory; Maryna Viazovska became the second woman Fields Medalist.
- **2025.** Hong Wang and Joshua Zahl proved the three-dimensional Kakeya conjecture: a subset of $\mathbb{R}^3$ that contains a unit line segment in every direction has full Hausdorff and Minkowski dimension 3.

Fact 13.1. Königsberg and Kaliningrad
Königsberg, home of the Seven Bridges problem and birthplace of Hilbert, became Kaliningrad after World War II. Kaliningrad is now part of Russia.

Mathematical Anecdotes

Some famous stories are documented, while others survive as traditional legends. A story whose details are uncertain is explicitly identified below; the uncertainty is part of the fact to remember.

- **c. fifth century BCE: Pythagorean irrationality legend.** A famous but historically uncertain legend says that Hippasus was drowned after revealing the irrationality of $\sqrt{2}$. The discovery itself shattered the belief that every magnitude is a ratio of integers.
- **c. 300 BCE: No royal road.** Proclus reports that when Ptolemy asked for an easier route to geometry, Euclid replied that there is no royal road to geometry.

- **third century BCE: Archimedes and the crown.** Vitruvius tells the traditional story that Archimedes recognized how displacement could test a crown's purity and ran out crying "Eureka!" The precise details are not independently verified.
- **212 BCE: Archimedes' circles.** Ancient accounts say that Archimedes was killed during the Roman capture of Syracuse while absorbed in a geometric diagram. The familiar wording "Do not disturb my circles" is traditional.
- **75 BCE: Archimedes' tomb.** Cicero reported rediscovering the tomb of Archimedes by recognizing the carved sphere and cylinder that commemorated Archimedes' favorite result.
- **1539–1545: Cardano and Tartaglia.** Tartaglia disclosed his cubic method to Cardano under a promise of secrecy. Cardano later published it in *Ars Magna* after learning that del Ferro had found the method earlier, beginning a celebrated priority dispute.
- **c. 1637: Fermat's margin.** Fermat wrote that he had a marvelous proof of his last theorem but that the margin was too small to contain it. No valid general proof by Fermat is known.
- **c. 1666: Newton's apple.** Newton later told William Stukeley that seeing an apple fall prompted thoughts about gravity. The story does not require that an apple struck Newton's head.
- **late seventeenth–early eighteenth centuries: The calculus priority dispute.** Newton and Leibniz developed calculus independently, but a bitter priority dispute divided British and continental mathematics for decades.
- **1766 onward: Euler's eyesight.** Euler continued producing mathematics at an extraordinary rate after losing almost all of his sight, dictating long calculations from memory.
- **c. 1780s: Gauss and the arithmetic sum.** A traditional schoolroom story says that young Gauss immediately summed $1 + 2 + \cdots + 100$ by pairing terms to obtain 5050. The details vary among later accounts.
- **1790s: Sophie Germain's pseudonym.** Because women were excluded from the École Polytechnique, Germain studied its lecture notes and submitted work under the name "M. LeBlanc." Lagrange became a supporter after learning her identity.
- **1832: Galois' last night.** Galois wrote urgent mathematical notes before the duel that killed him at age 20. He did not create all of Galois theory in one night; the notes summarized and revised ideas developed earlier.
- **1843: Hamilton's bridge.** On October 16, 1843, Hamilton carved the quaternion relations
$$i^2 = j^2 = k^2 = ijk = -1$$
into Brougham Bridge in Dublin after the key idea came to him during a walk.
- **1895: No Nobel Prize in Mathematics.** The popular story that Nobel excluded mathematics because of a romantic rivalry is unsupported. Nobel's will simply named physics, chemistry, physiology or medicine, literature, and peace.
- **1900: Hilbert's problems.** Hilbert's Paris address did not orally present all twenty-three problems; ten were delivered in the lecture, while the complete list appeared in print.

- **1915: Hilbert and Noether.** When objections were raised to Noether teaching at Göttingen because she was a woman, Hilbert reportedly replied that the university was not a bathhouse.
- **1918: The taxi-cab number.** Hardy told Ramanujan that taxi number 1729 seemed dull. Ramanujan immediately answered that it is the smallest number expressible as a sum of two positive cubes in two ways:

$$1729 = 1^3 + 12^3 = 9^3 + 10^3.$$

- **1935 onward: Nicolas Bourbaki.** Bourbaki is not one mathematician but the collective pseudonym of a group that sought a unified, axiomatic presentation of modern mathematics.
- **1935–1941: *The Scottish Book*.** Mathematicians of the Lwów school wrote problems in a notebook kept at the Scottish Café. Prizes ranged from coffee or beer to a live goose.
- **1936 and 1972: Mazur's goose.** Stanisław Mazur offered a live goose for Problem 153 in 1936. Per Enflo solved the basis problem in 1972, and Mazur presented the promised goose in a televised ceremony.
- **1939: Dantzig's homework.** George Dantzig arrived late to Jerzy Neyman's statistics class, copied two unsolved problems from the board believing they were homework, and later submitted solutions to both.
- **1947: Gödel's citizenship hearing.** Before his 1947 US citizenship examination, Gödel found what he believed was a logical route to dictatorship in the Constitution. Einstein and Oskar Morgenstern accompanied him and helped prevent the hearing from becoming a constitutional debate.
- **1947: Hilbert's hotel.** The infinite-hotel thought experiment, popularized by George Gamow, illustrates that a countably infinite full hotel can still accommodate more guests by moving each guest to another numbered room.
- **twentieth century: Erdős' language.** Erdős called children "epsilons" and imagined an ideal book in which God keeps the most elegant proofs. The Erdős number measures collaboration distance from him.
- **1954: Turing's apple.** Turing died from cyanide poisoning in 1954, and a bitten apple was found nearby. The popular claim that the apple was deliberately poisoned, or that it inspired the Apple logo, is not established.
- **1956: Milnor's exotic spheres.** Milnor discovered that a topological 7-sphere can carry differentiable structures not equivalent to the standard one. The group of oriented homotopy 7-spheres has order 28.
- **1976: The computer proof controversy.** The 1976 proof of the Four Color Theorem required computer checking of many cases, forcing mathematicians to debate what it means for a proof to be humanly inspectable.
- **1993–1994: Wiles' secret work.** Wiles worked privately for years on Fermat's Last Theorem. After announcing a proof in 1993, a gap was found; Wiles and Richard Taylor repaired it in 1994.
- **2006 and 2010: Perelman's refusals.** Perelman declined the 2006 Fields Medal and the Clay Millennium Prize for the Poincaré conjecture.

- **2010s: Villani's public image.** Cédric Villani became recognizable for a cravat and spider brooch. His book *Théorème vivant* recounts work with Clément Mouhot on nonlinear Landau damping.
- **2018: Birkar's stolen medal.** Caucher Birkar's Fields Medal was stolen shortly after the 2018 ceremony in Rio de Janeiro; the IMU later presented him with a replacement.

Chapter 14

Famous Problems and Conjectures

Famous problems are usually not tested by asking for complete proofs. They are tested through the following information: the standard statement, the person or group attached to the problem, whether it is solved, and the main idea or keyword attached to the solution. The goal of this chapter is to make those associations immediate.

Millennium Prize Problems

The Clay Mathematics Institute announced the seven Millennium Prize Problems in 2000. Each problem carries a prize of one million US dollars. Among the seven, only the Poincaré conjecture has been solved.

Problem	Standard mathematical form	Status	Quiz anchors
Poincaré Conjecture	Every closed simply connected 3-manifold is homeomorphic to S^3.	Solved.	Poincaré; Perelman; Ricci flow.
Riemann Hypothesis	Every nontrivial zero of $\zeta(s)$ has real part 1/2.	Open.	Riemann; zeta function; prime numbers.
P versus NP	Is $P = NP$?	Open.	Cook, Karp; computational complexity.
Birch and Swinnerton-Dyer Conjecture	For an elliptic curve E, $\operatorname{rk} E(\mathbb{Q})$ equals the order of vanishing of $L(E, s)$ at $s = 1$.	Open.	Elliptic curves; L-functions.
Hodge Conjecture	Rational Hodge classes on smooth projective complex varieties should come from algebraic cycles.	Open.	Hodge theory; algebraic geometry.
Navier–Stokes Existence and Smoothness	Smooth initial data for the 3-dimensional incompressible Navier–Stokes equations should have global smooth solutions, or one must find a breakdown.	Open.	Fluid mechanics; PDE; turbulence.
Yang–Mills Existence and Mass Gap	Construct quantum Yang–Mills theory on $\mathbb{R}^4$ for compact simple gauge groups and prove a positive mass gap.	Open.	Gauge theory; quantum field theory; mass gap.

Fact 14.1. Millennium Prize status check

Among the seven Millennium Prize Problems, the Poincaré conjecture is the only one that has been solved. The remaining six are the Riemann Hypothesis, P versus NP, the Birch and Swinnerton-Dyer conjecture, the Hodge conjecture, Navier–Stokes existence and smoothness, and Yang–Mills existence and mass gap.

Theorem 14.2. Poincaré Conjecture
Every closed simply connected 3*-manifold is homeomorphic to the* 3*-sphere* S^3*.*

Remark. Henri Poincaré posed the problem in the early twentieth century. Grigori Perelman proved it using Ricci flow and ideas from Hamilton's program. The principal Quiz anchor is

$$\text{Poincaré conjecture} \quad \longleftrightarrow \quad \text{Perelman} \quad \longleftrightarrow \quad \text{Ricci flow.}$$

Hypothesis 14.3. Riemann Hypothesis
Let

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

be the Riemann zeta function, initially defined for $\mathrm{Re}(s) > 1$ *and continued meromorphically. The Riemann Hypothesis states that every nontrivial zero of* $\zeta(s)$ *has real part* $1/2$*.*

Idea. The zeta function is connected to prime numbers by Euler's product

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Thus RH is a statement about the hidden regularity of the distribution of prime numbers.

Open Problem 14.4. P versus NP
Determine whether

$$P = NP.$$

Here P is the class of decision problems solvable in polynomial time, and NP is the class of decision problems whose proposed solutions can be verified in polynomial time.

Remark. The usual slogan is: can every solution that can be checked quickly also be found quickly? The anchors are Cook, Karp, NP-completeness, and theoretical computer science.

Conjecture 14.5. Birch and Swinnerton-Dyer Conjecture
For an elliptic curve E *over* $\mathbb{Q}$*,* $\operatorname{rk} E(\mathbb{Q})$ *equals the order of vanishing of its* L*-function* $L(E, s)$ *at* $s = 1$*.*

Remark. The problem connects rational points on elliptic curves with analytic behavior of L-functions. The phrase "rk equals order of vanishing" is the main quiz memory.

Conjecture 14.6. Hodge Conjecture
On a smooth projective complex variety, every rational cohomology class of type (p, p) *should be a rational linear combination of classes of algebraic cycles.*

Remark. This is one of the deepest bridges between topology and algebraic geometry. The principal Quiz anchor is

$$\text{Hodge classes} \quad \longleftrightarrow \quad \text{algebraic cycles.}$$

Open Problem 14.7. Navier–Stokes Existence and Smoothness
For the 3-dimensional incompressible Navier–Stokes equations, determine whether smooth initial data always lead to global smooth solutions, or whether finite-time singularities can occur.

Remark. This is a PDE problem motivated by fluid mechanics. The Quiz anchors are incompressible fluid, global regularity, singularity, and turbulence.

Open Problem 14.8. Yang–Mills Existence and Mass Gap
Construct a mathematically rigorous quantum Yang–Mills theory on $\mathbb{R}^4$ for every compact simple gauge group and prove that it has a positive mass gap.

Remark. The problem comes from quantum field theory. In quiz form, the Quiz anchors are gauge theory, quantum field theory, and mass gap.

Hilbert's Problems

David Hilbert published a list of 23 problems after his address at the 1900 ICM in Paris. He presented ten of them orally. The list became a program for twentieth-century mathematics, but its items are not all yes–no questions; several are broad research programs whose status cannot be reduced to a single word.

No.	Subject	Modern status and principal names
1	Continuum hypothesis and well-ordering	CH is independent of ZFC, assuming consistency: Gödel proved relative consistency and Cohen proved independence using forcing.
2	Consistency of arithmetic	Gödel's incompleteness theorems rule out the strongest form of Hilbert's program; Gentzen later gave a relative consistency proof using transfinite induction.
3	Scissors congruence of polyhedra	Solved negatively by Max Dehn through the Dehn invariant: equal volume does not imply equidecomposability.
4	Geometries with straight-line geodesics	A broad program rather than one fixed proposition; substantially developed through metric and convex geometry.
5	Continuous groups and Lie groups	Solved affirmatively by Gleason and by Montgomery–Zippin: locally Euclidean topological groups are Lie groups.
6	Axiomatization of physics	An open-ended program involving probability, mechanics, and physical theories; no single final solution.
7	Transcendence of a^b	Solved by the Gelfond–Schneider theorem when a and b are algebraic, $a \neq 0, 1$, and b is irrational.
8	Prime-number problems	Includes the Riemann Hypothesis and Goldbach-type questions; central parts remain open.
9	General reciprocity laws	Largely answered by Artin reciprocity and class field theory.
10	Algorithm for Diophantine equations	Solved negatively by the Davis–Putnam–Robinson–Matiyasevich theorem: no such general algorithm exists over $\mathbb{Z}$.
11	Quadratic forms over algebraic number fields	Essentially solved through Hasse's local–global theory and later arithmetic developments.
12	Explicit class field theory	Solved for important special cases, but the general *Kronecker Jugendtraum* remains open.
13	Seventh-degree equations and functions	The continuous-function interpretation was solved by Kolmogorov and Arnold; stricter algebraic variants remain distinct.
14	Finite generation of certain algebras	Solved negatively in general by Nagata's counterexample.

Continued on the next page.

No.	Subject	Modern status and principal names
15	Rigorous foundation of Schubert calculus	Fulfilled through modern intersection theory and algebraic geometry.
16	Real algebraic curves and limit cycles	Both parts contain major open questions; the limit-cycle problem is open even in low degrees.
17	Nonnegative rational functions as sums of squares	Solved affirmatively by Emil Artin using sums of squares of rational functions.
18	Space groups, tilings, and sphere packing	Major parts were solved by Bieberbach and, for the three-dimensional Kepler conjecture, by Hales; it is a multi-part problem.
19	Regularity of variational solutions	Solved through work of De Giorgi, Nash, Moser, and others.
20	General boundary-value problems	A broad program developed through modern functional analysis and PDE; not one universally closed proposition.
21	Riemann–Hilbert problem	The unrestricted original existence claim is false by Bolibrukh's counterexample; important variants are solved.
22	Uniformization by automorphic functions	Solved by Koebe and Poincaré through the uniformization theorem.
23	Further development of the calculus of variations	A research program that stimulated extensive modern theory rather than a single proposition with one completion date.

Fact 14.9. Hilbert problem anchors
Hilbert's first problem concerns the continuum hypothesis; the second concerns consistency; the eighth includes the Riemann Hypothesis; the tenth concerns an algorithm for Diophantine equations; and the sixteenth includes the number and arrangement of limit cycles.

Landau's Problems

At the 1912 ICM in Cambridge, Edmund Landau identified four elementary prime-number problems that he regarded as exceptionally difficult. All four remain open.

Problem	Statement	Status and anchor
Goldbach Conjecture	Every even integer greater than 2 is a sum of two primes.	Open; additive number theory.
Twin Prime Conjecture	There are infinitely many primes p for which $p+2$ is prime.	Open; bounded prime gaps are known.
Legendre Conjecture	For every positive integer n, there is a prime between n^2 and $(n+1)^2$.	Open; primes in short intervals.
Primes of the form n^2+1	There are infinitely many positive integers n for which n^2+1 is prime.	Open; polynomial prime values.

Other Famous Problems and Conjectures

The following problems are not all open problems. Some are solved theorems, but they remain famous because of their history, statements, or solution methods. They are collected together rather than divided into mathematical fields.

Problem	Standard form	People and status	Quiz anchor
Basel Problem	Evaluate $\sum_{n=1}^{\infty} 1/n^2$.	Solved by Euler: $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$.	Zeta value $\zeta(2)$; Euler.
Fermat's Last Theorem	$x^n + y^n = z^n$ has no positive integer solutions for $n \geq 3$.	Fermat; proved by Wiles with Taylor–Wiles.	Elliptic curves and modular forms.
Four Color Theorem	Every planar map can be colored with at most four colors.	Appel–Haken; later Robertson–Sanders–Seymour–Thomas.	First famous computer-assisted proof.
Kepler Conjecture	The densest packing of congruent spheres in $\mathbb{R}^3$ is the face-centered cubic packing.	Kepler; proved by Hales.	Sphere packing; Flyspeck project.
Thurston Geometrization Conjecture	Closed orientable 3-manifolds decompose into pieces modeled on eight geometries.	Thurston; completed through Perelman's work.	Eight Thurston geometries.
Continuum Hypothesis	There is no cardinality strictly between $\|\mathbb{N}\|$ and $\|\mathbb{R}\|$.	Cantor; independent of ZFC by Gödel and Cohen.	Forcing; independence.
Collatz Conjecture	Iterating $n \mapsto n/2$ for even n and $n \mapsto 3n+1$ for odd n eventually reaches 1.	Collatz; open.	Elementary iteration; nonlinear dynamics.
abc Conjecture	If $a + b = c$ and a, b, c are coprime, then c is controlled by the radical of abc, up to ε.	Masser–Oesterlé; open in the standard sense.	Radical, Diophantine equations.

Theorem 14.10. Basel Problem
Euler evaluated the reciprocal-square series:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \zeta(2) = \frac{\pi^2}{6}.$$

Remark. The problem was first posed by Pietro Mengoli in 1644 and became associated with the Bernoulli family in Basel. Euler's solution was a landmark because an infinite sum over integers produced a value involving π.

Theorem 14.11. Fermat's Last Theorem
For every integer $n \geq 3$, there do not exist positive integers x, y, z such that

$$x^n + y^n = z^n.$$

Remark. Fermat wrote that he had a marvelous proof, but no proof appeared in his margin. Andrew Wiles proved the theorem through the modularity of semistable elliptic curves. The principal Quiz anchor is

$$\text{Fermat's Last Theorem} \quad \longleftrightarrow \quad \text{Wiles} \quad \longleftrightarrow \quad \text{elliptic curves and modular forms.}$$

Theorem 14.12. Four Color Theorem
Every planar map can be colored with at most four colors so that any two adjacent regions have different colors.

Remark. The theorem was proved by Kenneth Appel and Wolfgang Haken in 1976 with extensive computer assistance. This made it one of the most famous early examples of a computer-assisted proof.

Theorem 14.13. Kepler Conjecture
The maximum density of a packing of congruent spheres in $\mathbb{R}^3$ *is attained by the face-centered cubic packing and equals*

$$\frac{\pi}{3\sqrt{2}}.$$

Remark. Thomas Hales proved the Kepler conjecture using a large computer-assisted argument. The Flyspeck project later produced a formal verification of the proof. In higher-dimensional sphere packing, Maryna Viazovska proved the optimality of the E_8 lattice in dimension 8.

Theorem 14.14. Thurston Geometrization Theorem
Every closed orientable 3*-manifold can be decomposed into pieces that admit one of the eight Thurston geometries:*

$$S^3,\ E^3,\ H^3,\ S^2\times\mathbb{R},\ H^2\times\mathbb{R},\ \widetilde{\mathrm{SL}}_2(\mathbb{R}),\ \mathrm{Nil},\ \mathrm{Sol}.$$

Remark. The Poincaré conjecture is a special case of the geometrization program. The 2018 Science Quiz asked for the number of Thurston geometries; the answer is 8.

Hypothesis 14.15. Continuum Hypothesis
There is no set whose cardinality is strictly between the cardinality of the natural numbers and the cardinality of the real numbers. Equivalently,

$$2^{\aleph_0} = \aleph_1.$$

Remark. Kurt Gödel proved that CH cannot be disproved from ZFC if ZFC is consistent. Paul Cohen proved that CH cannot be proved from ZFC if ZFC is consistent. Cohen introduced forcing and received the Fields Medal in 1966.

Open Problem 14.16. Goldbach Conjecture
Every even integer greater than 2 is a sum of two prime numbers.

Remark. Goldbach is one of the easiest famous open problems to state. It belongs to additive number theory. The weak Goldbach conjecture, concerning sums of three odd primes, was proved by Harald Helfgott.

Open Problem 14.17. Twin Prime Conjecture
There are infinitely many primes p such that

$$p + 2$$

is also prime.

Remark. The modern Quiz anchors are bounded gaps between primes, Yitang Zhang, James Maynard, Terence Tao, and the Polymath project. These results do not prove the twin prime conjecture, but they prove that infinitely many prime pairs occur within some fixed bounded distance.

Open Problem 14.18. Collatz Conjecture
Start with a positive integer n. If n is even, replace it by $n/2$. If n is odd, replace it by $3n + 1$. The Collatz conjecture states that repeated iteration always eventually reaches 1.

Remark. This problem is famous because the rule is elementary but the global behavior is still unknown. It is also called the $3n + 1$ problem.

Conjecture 14.19. *abc* **Conjecture**
For every $\varepsilon > 0$*, there exists a constant* K_ε *such that, whenever* a, b, c *are pairwise coprime positive integers satisfying* $a + b = c$*, we have*

$$c < K_\varepsilon \,\mathrm{rad}(abc)^{1+\varepsilon},$$

where $\mathrm{rad}(n)$ *is the product of the distinct prime divisors of* n*.*

Remark. The conjecture is powerful because it predicts that the additive relation $a + b = c$ strongly restricts the repeated prime factors of abc. The keyword is "radical." Shinichi Mochizuki announced a claimed proof in 2012, and the papers were later published, but no broadly accepted proof has been established; the conjecture is therefore recorded here as open.

Fact 14.20. June Huh and combinatorial log-concavity
June Huh's work connects combinatorics with algebraic geometry and Hodge theory. The central associations are:

- Read's conjecture $\leftrightarrow$ chromatic polynomials.
- Brylawski's and Okounkov's conjectures $\leftrightarrow$ matroid characteristic polynomials and log-concavity.
- Rota's conjecture $\leftrightarrow$ log-concavity for matroids.
- Mason's conjecture $\leftrightarrow$ independent sets of matroids.
- Hodge theory $\leftrightarrow$ the geometric method behind these combinatorial log-concavity results.

In a quiz, if June Huh appears with a list of conjectures, the odd one out is often a famous conjecture from a different field, such as the Poincaré conjecture.

Conjecture 14.21. Hadwiger Conjecture
Every finite graph G with chromatic number $\chi(G) = k$ contains the complete graph K_k as a minor.

Remark. The conjecture is known for $k \leq 6$ and remains open in general. The case $k = 5$ is equivalent to the Four Color Theorem.

Conjecture 14.22. Hadamard Conjecture
For every positive integer n, there exists a $4n \times 4n$ matrix $\mathbf{H}$ with entries in $\{-1, 1\}$ such that

$$\mathbf{H}\mathbf{H}^{\mathsf{T}} = 4n\mathbf{I}_{4n}.$$

Remark. A matrix satisfying this orthogonality condition is a Hadamard matrix. The necessary divisibility condition is known, but existence for every positive multiple of 4 remains open.

Conjecture 14.23. Union-Closed Sets Conjecture
Let $\mathcal{F}$ be a finite nonempty family of finite sets such that

$$A, B \in \mathcal{F} \implies A \cup B \in \mathcal{F}.$$

Then some element belongs to at least half of the sets in $\mathcal{F}$.

Remark. The conjecture was proposed by Péter Frankl in 1979. It is also called Frankl's union-closed sets conjecture and remains open.

Conjecture 14.24. Lonely Runner Conjecture
Suppose that $n \geq 2$ runners move at distinct constant speeds around a unit circle. For each runner, there is a time at which that runner is at circular distance at least

$$\frac{1}{n}$$

from every other runner.

Remark. After subtracting the chosen runner's speed from every speed, the problem becomes one of simultaneous Diophantine approximation. It remains open in general.

Open Problem 14.25. Odd Perfect Number Problem
Determine whether there exists an odd positive integer n satisfying

$$\sigma(n) = 2n,$$

where $\sigma(n)$ is the sum of the positive divisors of n.

Remark. Euclid and Euler characterized the even perfect numbers through Mersenne primes. No odd perfect number is known, and nonexistence has not been proved.

Conjecture 14.26. Legendre Conjecture
For every positive integer n, there is a prime p satisfying

$$n^2 < p < (n+1)^2.$$

Remark. This is stronger than what follows from the prime number theorem and remains open.

Conjecture 14.27. Erdős–Straus Conjecture
For every integer $n \geq 2$, there exist positive integers x, y, z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Remark. The conjecture concerns Egyptian-fraction decompositions and is verified computationally for enormous ranges, but no general proof is known.

Conjecture 14.28. Beal Conjecture
If positive integers satisfy

$$A^x + B^y = C^z, \qquad x, y, z > 2,$$

then A, B, and C have a common prime factor.

Remark. Fermat's Last Theorem is the equal-exponent case with pairwise coprime bases. The Beal conjecture remains open.

Conjecture 14.29. Schanuel Conjecture
If $z_1, \ldots, z_n \in \mathbb{C}$ are linearly independent over $\mathbb{Q}$, then

$$\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(z_1, \ldots, z_n, e^{z_1}, \ldots, e^{z_n}) \geq n.$$

Remark. Schanuel's conjecture is a central organizing conjecture in transcendental number theory. It would imply many famous results and conjectures about algebraic independence.

Open Problem 14.30. Infinitely Many Mersenne Primes
Determine whether there are infinitely many primes of the form

$$2^p - 1,$$

where p is prime.

Remark. Every even perfect number has the Euclid–Euler form

$$2^{p-1}(2^p - 1)$$

with $2^p - 1$ prime. Thus the infinitude of even perfect numbers is equivalent to the infinitude of Mersenne primes.

Open Problem 14.31. Perfect Cuboid Problem
Determine whether there exists a rectangular box with integer edge lengths a, b, c for which all three face diagonals and the space diagonal are integers:

$$a^2 + b^2, \quad b^2 + c^2, \quad c^2 + a^2, \quad a^2 + b^2 + c^2$$

are all perfect squares.

Remark. Boxes with integer edges and integer face diagonals are known as Euler bricks. Requiring the space diagonal to be integral gives the still-open perfect cuboid problem.

Theorem 14.32. Catalan–Mihăilescu Theorem
The only solution in integers $x, y, a, b > 1$ of

$$x^a - y^b = 1$$

is

$$3^2 - 2^3 = 1.$$

Remark. Catalan posed the conjecture in 1844, and Preda Mihăilescu proved it in 2002. Thus 8 and 9 are the only consecutive positive perfect powers greater than 1.

Open Problem 14.33. Smooth Four-Dimensional Poincaré Conjecture
Determine whether every smooth closed 4-manifold that is homotopy equivalent to S^4 is diffeomorphic to S^4.

Remark. The topological 4-dimensional Poincaré conjecture was proved by Freedman. The smooth 4-dimensional case remains open, while the other dimensions are settled.

Theorem 14.34. Kakeya Conjecture in Three Dimensions
If a set $K \subseteq \mathbb{R}^3$ contains a unit line segment in every direction, then

$$\dim_{\mathrm{H}} K = 3,$$

and its Minkowski dimension is also 3.

Remark. Besicovitch showed that Kakeya sets may have Lebesgue measure 0. Hong Wang and Joshua Zahl proved the full dimension conjecture in $\mathbb{R}^3$ in 2025. The analogous conjecture remains open in dimensions $n \geq 4$.

Open Problem 14.35. Inscribed Square Problem
Determine whether every Jordan curve in the plane contains four points that are the vertices of a square.

Remark. Also called the Toeplitz square-peg problem, it is known for many regular classes of curves but remains open for arbitrary Jordan curves.

Fact 14.36. Kissing Numbers
The kissing number τ_n is the maximum number of nonoverlapping unit spheres that can touch one unit sphere in $\mathbb{R}^n$. Famous exact values include

$$\tau_2 = 6, \qquad \tau_3 = 12, \qquad \tau_4 = 24, \qquad \tau_8 = 240, \qquad \tau_{24} = 196560.$$

Exact values are unknown in most dimensions.

Open Problem 14.37. Hilbert–Smith Conjecture
Every locally compact topological group that acts faithfully and continuously on a connected manifold should be a Lie group.

Remark. The problem asks whether manifolds admit genuinely non-Lie continuous symmetry groups. It remains open in full generality.

Conjecture 14.38. Jacobian Conjecture
Let

$$F = (F_1, \ldots, F_n) : \mathbb{C}^n \to \mathbb{C}^n$$

be a polynomial map. If

$$\det\left(\frac{\partial F_i}{\partial x_j}\right)$$

is a nonzero constant, then F has a polynomial inverse.

Remark. Keller posed the conjecture in 1939. It is open for every $n \geq 2$, even though many reductions are known.

Open Problem 14.39. Inverse Galois Problem
Determine whether every finite group G occurs as

$$\mathrm{Gal}(K/\mathbb{Q})$$

for some finite Galois extension K of $\mathbb{Q}$.

Remark. The problem is solved for many classes of groups but remains open in general.

Open Problem 14.40. Hilbert's Tenth Problem over $\mathbb{Q}$
Determine whether there is an algorithm that decides, for every polynomial

$$f \in \mathbb{Z}[x_1, \dots, x_n],$$

whether the equation $f = 0$ has a solution in $\mathbb{Q}^n$.

Remark. Matiyasevich's theorem gives a negative answer over $\mathbb{Z}$. The analogous decision problem over $\mathbb{Q}$ remains open.

Theorem 14.41. Classification of Finite Simple Groups
Every finite simple group is one of the following:

- *a cyclic group of prime order;*
- *an alternating group A_n with $n \geq 5$;*
- *a finite simple group of Lie type;*
- *one of the* 26 *sporadic simple groups.*

Remark. The proof is distributed across thousands of pages by many authors. The largest sporadic group is the Monster group.

Theorem 14.42. Restricted Burnside Problem
For fixed positive integers d and n, there are only finitely many finite d-generated groups of exponent n, up to isomorphism.

Remark. Efim Zelmanov solved the restricted Burnside problem using the theory of Lie algebras and received the 1994 Fields Medal. The unrestricted Burnside problem has a negative answer in general.

Fact 14.43. Langlands Program
The Langlands program predicts deep correspondences between arithmetic objects such as Galois representations and analytic objects such as automorphic representations. It is not one conjecture but a network of conjectures linking number theory, representation theory, and geometry.

Open Problem 14.44. Three-Dimensional Euler Singularity Problem
For the incompressible Euler equations in $\mathbb{R}^3$, determine whether smooth finite-energy initial data can develop a singularity in finite time.

Remark. The Euler equations model an ideal fluid without viscosity. Their global regularity problem is distinct from the Navier–Stokes Millennium problem.

Conjecture 14.45. Falconer Distance Conjecture
If a compact set $E \subseteq \mathbb{R}^n$ satisfies

$$\dim_{\mathrm{H}} E > \frac{n}{2},$$

then its distance set

$$\Delta(E) = \{\|\mathbf{x} - \mathbf{y}\| : \mathbf{x}, \mathbf{y} \in E\}$$

has positive Lebesgue measure.

Remark. The conjecture connects harmonic analysis, geometric measure theory, and discrete distance problems. It remains open in general.

Open Problem 14.46. Hilbert's Sixteenth Problem
The second part asks for a uniform upper bound, in terms of the degree n, on the number and arrangement of limit cycles of a planar polynomial vector field of degree n.

Remark. Even the quadratic case is not completely understood. The first part of Hilbert's sixteenth problem concerns the topology of real algebraic curves.

Theorem 14.47. Bieberbach Conjecture
If

$$f(z) = z + \sum_{n=2}^{\infty} a_n z^n$$

is analytic and injective on the unit disk, then

$$|a_n| \leq n \qquad (n \geq 2).$$

Remark. Ludwig Bieberbach conjectured the bound in 1916, and Louis de Branges proved it in 1985.

Theorem 14.48. Kadison–Singer Problem
Every pure state on the diagonal maximal abelian subalgebra of bounded operators on ℓ^2 has a unique extension to a state on all bounded operators.

Remark. Kadison and Singer posed the problem in 1959. Marcus, Spielman, and Srivastava solved it in 2013 through an equivalent finite-dimensional partition problem and the method of interlacing polynomials.

Part III

Mathematics in 2026

Chapter 15

Mathematics in 2026

This chapter records mathematical events, awards, and recent developments relevant to the 2026 Science Quiz. Confirmed entries and topics to monitor are listed separately.

Confirmed Current Entries

- **2025 Shaw Prize in Mathematical Sciences.** Kenji Fukaya was awarded the 2025 Shaw Prize in Mathematical Sciences for his work in symplectic geometry, the Fukaya category, symplectic topology, mirror symmetry, and gauge theory.
- **2025 AI and the IMO.** Systems reported by Google DeepMind and OpenAI reached gold-medal-level scores on the 2025 International Mathematical Olympiad problems. Google DeepMind's performance was evaluated through the IMO process, whereas OpenAI reported an independent evaluation. Relevant Quiz anchors include mathematical reasoning, olympiad problems, natural-language proofs, formal verification, and AI-assisted mathematics.
- **2026 ICM and IMU General Assembly.** The International Congress of Mathematicians is scheduled for Philadelphia, USA, from July 23 to July 30, 2026. The 20th IMU General Assembly is scheduled for New York City from July 20 to July 21, 2026.
- **2026 IMU prizes at ICM.** The Fields Medals, IMU Abacus Medal, Carl Friedrich Gauss Prize, and Chern Medal Award are associated with the opening ceremony. The Leelavati Prize is associated with the closing ceremony.
- **2026 Abel Prize.** Gerd Faltings received the 2026 Abel Prize for introducing powerful tools in arithmetic geometry and resolving long-standing Diophantine conjectures of Mordell and Lang. Relevant Quiz anchors include arithmetic geometry, the Mordell conjecture, Faltings' theorem, the Mordell–Lang conjecture, and Diophantine geometry.
- **2026 Breakthrough Prize in Mathematics.** Frank Merle received the 2026 Breakthrough Prize in Mathematics for work on nonlinear evolution equations, including stability, singularity formation, and resolution into solitons.
- **2026 Clay Research Awards.** The awards recognized three groups: Tuomas Orponen, Pablo Shmerkin, Hong Wang, and Joshua Zahl; Robert Burklund, Jeremy Hahn, Ishan Levy, and Tomer Schlank; and Yu Deng and Zaher Hani. Relevant Quiz anchors include the Furstenberg set conjecture, the Kakeya conjecture, the Telescope Conjecture, and the Boltzmann equation.
- **Moving sofa problem.** Jineon Baek posted the preprint *Optimality of Gerver's Sofa*, claiming a resolution of the moving sofa problem by proving that Gerver's sofa is optimal.

Its status as a preprint claim must be distinguished from publication and independent verification.

Topics to Monitor

- **ICM 2026 prize results.** The 2026 Fields Medalists, IMU Abacus Medalist, Gauss Prize laureate, Chern Medal laureate, and Leelavati Prize laureate, together with the principal mathematical keywords attached to their work.
- **ICM 2026 mathematical themes.** Major fields, theorems, and research programs represented in the plenary and invited lectures.
- **IMO 2026.** Host city, participating countries, team results, medal cutoffs, perfect scores, and especially memorable solutions, with human competition results distinguished from AI evaluations.
- **AI theorem proving and formalization.** Lean and other proof assistants, autoformalization, verified proof search, and human–AI collaboration, with olympiad benchmark performance distinguished from formal machine-checked proofs and research-level progress.
- **Geometric Langlands program.** Dennis Gaitsgory, the 2025 Breakthrough Prize, and the precise scope and publication status of the large collaborative proof project.
- **2026 Shaw Prize and Wolf Prize.** Official laureates and the principal mathematical areas recognized by the two foundations.
- **Korea-related mathematics news.** Major prizes, appointments, deaths, young mathematicians, KIAS and IBS activity, and work connected to POSTECH, KAIST, and Professor June Huh, especially when a clear theorem, problem, or named object is involved.
- **Open-problem progress.** Bounded gaps between primes, the Navier–Stokes problem, geometric analysis, arithmetic geometry, and other major research areas, with numerical evidence, preprint claims, and completed proofs clearly distinguished.
- **2026 POSTECH–KAIST Science War.** Season-specific trends from practice problems, team notes, and the official match.

Part IV

Past Problems

Problem 2010.1. Determine whether there exist positive integers x and y satisfying

$$x^2 + y^2 = 963.$$

Solution. No such positive integers exist. By the quadratic-residue criterion modulo 4 from Mathematical Concepts, for any integer a, we have

$$(a^2 \equiv 0 \pmod 4) \vee (a^2 \equiv 1 \pmod 4).$$

Hence the sum of two integer squares can be congruent only to

$$0, \quad 1, \quad \text{or} \quad 2 \pmod 4.$$

However,

$$963 \equiv 3 \pmod 4.$$

Thus 963 cannot be written as the sum of two integer squares.

Therefore, the answer is $\boxed{\text{no}}$.

□

Problem 2010.2. Identify the mathematician who became famous for refusing the Fields Medal after solving the Poincaré conjecture.

Solution. Grigori Perelman proved the Poincaré conjecture using Ricci flow. He was awarded the Fields Medal in 2006, but declined it.

Therefore, the answer is $\boxed{\text{Grigori Perelman}}$.

□

Problem 2010.3. Determine the country of origin of John Charles Fields, after whom the Fields Medal is named.

Solution. John Charles Fields was a Canadian mathematician. The Fields Medal is named after him.

Therefore, the answer is $\boxed{\text{Canada}}$.

□

Problem 2010.4. Determine the maximum possible number of distinct eigenvalues of a 5×5 matrix.

Solution. The eigenvalues of a 5×5 matrix $\mathbf{A}$ are the roots of its characteristic polynomial

$$p_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I}_5 - \mathbf{A}).$$

This polynomial has degree 5. Therefore it can have at most 5 distinct roots, and hence $\mathbf{A}$ can have at most 5 distinct eigenvalues.

Therefore, the answer is $\boxed{5}$.

□

Problem 2010.5. In 1742, a Prussian mathematician wrote to Euler and proposed a conjecture. Euler later stated it in the following form:

Every even integer greater than 2 can be expressed as the sum of two primes.

This is an old unsolved problem in number theory. It has been verified for very large numbers, but it has not been proved in general. Identify this conjecture.

Solution. The statement that every even integer greater than 2 can be expressed as the sum of two primes is known as Goldbach's conjecture.

Therefore, the answer is $\boxed{\text{Goldbach's conjecture}}$.

$\square$

Problem 2010.6. Brownian motion is modeled by a stochastic process named after the American mathematician who developed its mathematical theory and later founded cybernetics. For this process, increments over disjoint time intervals are independent, and an increment over an interval of length t is Gaussian with mean 0 and variance t. Identify the mathematician.

Solution. The process described is the Wiener process, the standard mathematical model of Brownian motion. It is named after Norbert Wiener.

Hence the mathematician is $\boxed{\text{Norbert Wiener}}$.

$\square$

Problem 2010.7. This set is a closed uncountable set and is also perfect. Although it is uncountable, it has Lebesgue measure 0. It is self-similar and is often used in the construction of fractals. It is obtained from the closed interval $[0, 1]$ by repeatedly removing the open middle third, then doing the same to the remaining intervals. It is named after the mathematician widely known as the founder of set theory. Identify this set.

Solution. The set obtained by repeatedly removing the open middle third from $[0, 1]$ is the Cantor set. It is closed, perfect, uncountable, self-similar, and has Lebesgue measure 0.

Therefore, the answer is $\boxed{\text{Cantor set}}$.

$\square$

Problem 2010.8. Determine the year in which Terence Tao received the Fields Medal.

Solution. Terence Tao received the Fields Medal at the 2006 International Congress of Mathematicians.

Therefore, the answer is $\boxed{2006}$.

$\square$

Problem 2011.1. Identify the international mathematical congress, held in South Korea in 2014, at which the Fields Medal is awarded.

Solution. The Fields Medal is awarded at the International Congress of Mathematicians, commonly abbreviated as ICM. The 2014 ICM was held in Seoul, South Korea.

Therefore, the answer is $\boxed{\text{International Congress of Mathematicians, or ICM}}$.

$\square$

Problem 2011.2. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. If

$$\mathbf{AB} - \mathbf{I}_n = \mathbf{0},$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix. Determine

$$\mathbf{BA} - \mathbf{I}_n.$$

Solution. The condition

$$\mathbf{AB} - \mathbf{I}_n = \mathbf{0}$$

means that

$$\mathbf{AB} = \mathbf{I}_n.$$

Since $\mathbf{A}$ and $\mathbf{B}$ are square matrices, this implies that $\mathbf{B}$ is the inverse of $\mathbf{A}$. Hence

$$\mathbf{BA} = \mathbf{I}_n.$$

Therefore,

$$\mathbf{BA} - \mathbf{I}_n = \mathbf{I}_n - \mathbf{I}_n = \mathbf{0}.$$

Therefore, the answer is $\boxed{\mathbf{0}}$.

□

Problem 2011.3. Differentiate $(1 + x)^{10}$ three times, substitute $x = 0$, and determine the resulting value.

Solution. We compute the third derivative:

$$\frac{\mathrm{d}^3}{\mathrm{d}x^3}(1 + x)^{10} = 10 \cdot 9 \cdot 8(1 + x)^7.$$

Substituting $x = 0$, we get

$$10 \cdot 9 \cdot 8(1 + 0)^7 = 720.$$

Therefore, the answer is $\boxed{720}$.

□

Problem 2011.4. Give at least two real-valued functions defined on $\mathbb{R}$ that remain unchanged after being differentiated twice. In other words, give at least two functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f'' = f.$$

Solution. Two simple examples are

$$f(x) = e^x \qquad \text{and} \qquad g(x) = e^{-x}.$$

Indeed,

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} e^x = e^x$$

and

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} e^{-x} = e^{-x}.$$

Thus both functions remain unchanged after being differentiated twice.

Therefore, two valid examples are $\boxed{e^x \text{ and } e^{-x}}$.

□

Problem 2011.5. Let $\mathbf{A}$ be a 5×5 matrix such that every entry in the ith row is equal to i. Find the sum of the five eigenvalues of $\mathbf{A}$, counted with multiplicity.

Solution. By the trace–eigenvalue identity in Mathematical Concepts, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to $\mathrm{tr}(\mathbf{A})$.

In this matrix, the diagonal entries are

$$1, \quad 2, \quad 3, \quad 4, \quad 5.$$

Therefore,

$$\mathrm{tr}(\mathbf{A}) = 1 + 2 + 3 + 4 + 5 = 15.$$

Hence the sum of the five eigenvalues of $\mathbf{A}$, counted with multiplicity, is $\boxed{15}$.

□

Problem 2011.6. A train takes 22 seconds to completely cross an 82 m bridge. When its speed is doubled, it takes 50 seconds to completely cross a 706 m bridge. Determine the length of the train.

Solution. Let L be the length of the train, and let v be its original speed in meters per second.

To completely cross a bridge, the train must travel the length of the bridge plus its own length. Hence

$$\frac{L+82}{v} = 22.$$

When the speed is doubled, the train takes 50 seconds to completely cross a 706 m bridge, so

$$\frac{L+706}{2v} = 50.$$

Equivalently,

$$L + 82 = 22v, \qquad L + 706 = 100v.$$

Subtracting the first equation from the second gives

$$624 = 78v,$$

so

$$v = 8.$$

Therefore,

$$L + 82 = 22 \cdot 8 = 176,$$

and hence

$$L = 94.$$

Thus the length of the train is $\boxed{94\,\mathrm{m}}$.

□

Problem 2011.7. Form an integer using the digits

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Each of these digits must be used at least once, and digits may be repeated. Determine the smallest such integer divisible by all of

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Solution. The number must be divisible by

$$\operatorname{lcm}(1, 2, 3, 4, 6, 7, 8, 9) = 504.$$

Thus it must be divisible by 7, 8, and 9.

If each digit is used exactly once, then the sum of the digits is

$$1 + 2 + 3 + 4 + 6 + 7 + 8 + 9 = 40,$$

which is not divisible by 9. Hence no 8-digit number works.

If we use exactly one extra digit d, then the digit sum becomes $40 + d$. For divisibility by 9, we need

$$40 + d \equiv 0 \pmod 9,$$

so

$$d \equiv 5 \pmod 9.$$

But the digit 5 is not allowed. Hence no 9-digit number works.

Therefore, the answer must have at least 10 digits. For a 10-digit number, we need to add two extra digits. Their sum must be congruent to 5 modulo 9. Among the allowed digits, the possible extra pairs are

$$(1, 4), \qquad (2, 3), \qquad (6, 8), \qquad (7, 7).$$

The lexicographically smallest admissible multiset is

$$1, 1, 2, 3, 4, 4, 6, 7, 8, 9.$$

With these digits, the smallest possible beginning is

$$112344.$$

It remains to arrange the last four digits $6, 7, 8, 9$ so that the whole number is divisible by 504.

Since the digit sum is already divisible by 9, it remains to check divisibility by 7 and 8. The last three digits must be divisible by 8. Among the permutations of three of the digits $6, 7, 8, 9$, the possible last three digits divisible by 8 are

$$768, \qquad 896, \qquad 968, \qquad 976.$$

Hence the only candidates with prefix 112344 are

$$1123449768, \qquad 1123447896, \qquad 1123447968, \qquad 1123448976.$$

Checking divisibility by 7, only

$$1123449768$$

is divisible by 7. Indeed,

$$1123449768 = 504 \cdot 2229067.$$

Therefore it is divisible by all of

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Thus the smallest possible integer is $\boxed{1123449768}$.

□

Problem 2011.8. Determine the greatest integer less than or equal to

$$-\frac{1}{\ln\left(\frac{365}{366}\right)}.$$

Hint. Use the Taylor approximation of $\ln(1 - x)$.

Solution. We write

$$\frac{365}{366} = 1 - \frac{1}{366}.$$

Using

$$\ln(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots,$$

with $x = \dfrac{1}{366}$, we get

$$-\ln\left(\frac{365}{366}\right) = \frac{1}{366} + \frac{1}{2\cdot 366^2} + \frac{1}{3\cdot 366^3} + \cdots.$$

Hence

$$\frac{1}{366} < -\ln\left(\frac{365}{366}\right) < \frac{1}{365}.$$

Taking reciprocals gives

$$365 < -\frac{1}{\ln\left(\frac{365}{366}\right)} < 366.$$

Therefore, the greatest integer less than or equal to the given number is $\boxed{365}$.

□

Problem 2011.9. Evaluate the integral

$$\int_0^{\pi/2} \frac{1}{1+\tan^{\sqrt{2}} t}\,\mathrm{d}t.$$

Hint. Use the substitution $x = \dfrac{\pi}{2} - t$.

Solution. Let

$$I = \int_0^{\pi/2} \frac{1}{1+\tan^{\sqrt{2}} t}\,\mathrm{d}t.$$

Use the substitution

$$x = \frac{\pi}{2} - t.$$

Then

$$\tan t = \tan\left(\frac{\pi}{2} - x\right) = \cot x = \frac{1}{\tan x}.$$

Hence

$$I = \int_0^{\pi/2} \frac{1}{1+\cot^{\sqrt{2}} x}\,\mathrm{d}x = \int_0^{\pi/2} \frac{\tan^{\sqrt{2}} x}{1+\tan^{\sqrt{2}} x}\,\mathrm{d}x.$$

Adding this expression to the original expression for I, we obtain

$$2I = \int_0^{\pi/2} \left(\frac{1}{1+\tan^{\sqrt{2}} x} + \frac{\tan^{\sqrt{2}} x}{1+\tan^{\sqrt{2}} x}\right)\mathrm{d}x = \int_0^{\pi/2} 1\,\mathrm{d}x = \frac{\pi}{2}.$$

Therefore,

$$I = \frac{\pi}{4}.$$

Therefore, the answer is $\boxed{\dfrac{\pi}{4}}$.

□

Problem 2011.10. This French mathematician was born on December 8, 1865, and died on October 17, 1963, living to the age of 97. He is well known for his proof of the prime number theorem, which states that the number of primes less than or equal to x is asymptotically approximated by $\dfrac{x}{\ln x}$ for large x. He was also personally connected to the Dreyfus affair and became a strong supporter of Zionism.

Identify this mathematician.

Hint. The following clues are useful:

(a) The Cauchy–__________ theorem gives the radius of convergence of a power series.

(b) A __________ matrix is a square matrix whose entries are all $+1$ or -1, and such matrices are also used in error-correcting codes.

Solution. The mathematician described in the problem is Jacques Hadamard.

The first hint refers to the Cauchy–Hadamard theorem, which gives the radius of convergence of a power series. The second hint refers to a Hadamard matrix, a square matrix whose entries are all $+1$ or -1 and whose rows are mutually orthogonal.

Hadamard is also famous for proving the prime number theorem in 1896, independently of Charles Jean de la Vallée Poussin. Therefore, the answer is $\boxed{\text{Jacques Hadamard}}$.

□

Problem 2012.1. This British mathematical genius died of cyanide poisoning shortly before his 42nd birthday. A famous account says that he died after eating an apple laced with cyanide. Identify this mathematician.

Solution. The mathematician described in the problem is Alan Turing.

Turing was a British mathematician, logician, and computer scientist. He made fundamental contributions to computability theory through the concept of the Turing machine, and he also played a crucial role in codebreaking during World War II.

Turing died in 1954 from cyanide poisoning, shortly before his 42nd birthday. The story involving an apple laced with cyanide is widely known, although some details of the event remain historically uncertain.

Therefore, the answer is $\boxed{\text{Alan Turing}}$.

□

Problem 2012.2. Evaluate the infinite series

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots .$$

Solution. The given series is

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

This is the famous Basel problem. Euler proved that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the value of the series is $\boxed{\dfrac{\pi^2}{6}}$.

□

Problem 2012.3. Identify the mathematician who first proved the law of quadratic reciprocity and discovered more than seven proofs of it.

Solution. The law of quadratic reciprocity is one of the central theorems in elementary number theory. It describes a deep symmetry between the solvability of

$$x^2 \equiv p \pmod{q} \qquad \text{and} \qquad x^2 \equiv q \pmod{p}$$

for odd primes p and q.

Carl Friedrich Gauss gave the first complete proof of the law of quadratic reciprocity. He regarded it as one of the jewels of number theory and found several different proofs of the theorem.

Therefore, the answer is $\boxed{\text{Carl Friedrich Gauss}}$.

□

Problem 2012.4. Let $\mathbf{A}$ be a 10×10 matrix such that every entry in the kth row is equal to k. Find the sum of the eigenvalues of $\mathbf{A}$, counted with multiplicity.

Solution. By the trace–eigenvalue identity in Mathematical Concepts, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to $\operatorname{tr}(\mathbf{A})$.

In this matrix, the diagonal entries are

$$1, \quad 2, \quad 3, \quad \ldots, \quad 10.$$

Therefore,

$$\operatorname{tr}(\mathbf{A}) = 1 + 2 + 3 + \cdots + 10 = \frac{10 \cdot 11}{2} = 55.$$

Hence the sum of the eigenvalues of $\mathbf{A}$, counted with multiplicity, is $\boxed{55}$.

□

Problem 2012.5. This philosopher was born as the youngest son of a wealthy Austrian steel magnate who supported many artists. He showed talent in several fields, especially mathematics and philosophy. His work had a major influence on logical positivism and on the philosophy of language, and it also influenced the humanities, social sciences, and the arts. His later philosophy is especially known for emphasizing ordinary language use.

Identify this person.

Solution. The person described in the problem is Ludwig Wittgenstein.

Wittgenstein was born into a wealthy Austrian family and studied logic, mathematics, and philosophy. His early work, especially the *Tractatus Logico-Philosophicus*, strongly influenced logical positivism and the Vienna Circle.

Later, in works such as *Philosophical Investigations*, Wittgenstein emphasized the importance of ordinary language and argued that the meaning of a word is closely connected to its use in language.

Therefore, the answer is $\boxed{\text{Ludwig Wittgenstein}}$.

□

Problem 2013.1. A complex number is called *algebraic* if it is a root of a nonzero polynomial with integer coefficients; otherwise, it is called *transcendental*. Joseph Liouville first proved the existence of transcendental numbers in 1844. Hilbert's seventh problem, one of the 23 problems presented at the 1900 International Congress of Mathematicians in Paris, also concerns transcendental numbers. It was solved independently by Aleksandr Gelfond and Theodor Schneider in the 1930s, and Alan Baker received the 1970 Fields Medal in part for major advances in this area.

Which mathematician proved in 1873 that e, the base of the natural logarithm, is transcendental? Ferdinand von Lindemann later generalized this mathematician's method and proved in 1882 that π is transcendental.

Solution. The mathematician was Charles Hermite. In 1873, Hermite proved that

$$e$$

is transcendental. Lindemann subsequently developed related ideas to prove the transcendence of π in 1882.

Therefore, the answer is $\boxed{\text{Charles Hermite}}$.

$\square$

Problem 2013.2. Beginning around 1935, mathematicians associated with the Lwów school gathered in cafés near the university and recorded problems in a notebook. The names appearing in it include Banach, Mazur, Ulam, Kac, Zygmund, Schauder, Erdős, Dugundji, Steinhaus, and Ahlfors. Some problems carried prizes. In particular, on November 6, 1936, Stanisław Mazur offered a live goose for Problem 153, later associated with the basis problem. Per Enflo solved the problem in 1972, and Mazur presented him with the promised goose in a televised ceremony.

Determine the title by which this famous notebook is known.

Solution. The notebook is known as *The Scottish Book*, named after the Scottish Café in Lwów, where members of the Lwów school of mathematics frequently met and discussed problems.

Therefore, the answer is $\boxed{\textit{The Scottish Book}}$.

$\square$

Problem 2013.3. Using exactly three occurrences of the digit 9, together with standard arithmetic operations, parentheses, powers, and square roots, construct each of the numbers

$$1, \qquad 6, \qquad 8.$$

Solution. One set of constructions is

$$\left(\frac{9}{9}\right)^9 = 1, \qquad \frac{9+9}{\sqrt{9}} = 6, \qquad 9 - \frac{9}{9} = 8.$$

Each expression uses exactly three occurrences of the digit 9. Alternative constructions include

$$9^{9-9} = 1, \qquad \sqrt{9\cdot 9} - \sqrt{9} = 6, \qquad 9 - \frac{\sqrt{9}}{\sqrt{9}} = 8.$$

$\square$

Problem 2014.1. This twentieth-century German mathematician is well known for a theorem stating that symmetries of Hamiltonian dynamical systems correspond to conservation laws. Identify this mathematician.

Solution. The theorem described in the problem is Noether's theorem. It states, roughly speaking, that continuous symmetries of a physical system correspond to conserved quantities. For example, time-translation symmetry corresponds to conservation of energy, and spatial-translation symmetry corresponds to conservation of momentum.

The mathematician associated with this theorem is Emmy Noether, one of the most important algebraists of the twentieth century.

Therefore, the answer is $\boxed{\text{Emmy Noether}}$.

$\square$

Problem 2014.2. In the famous Chinese mathematical classic *The Nine Chapters on the Mathematical Art*, often compared with Euclid's *Elements* as a foundational mathematical classic, the value of π appears as 3. Later, Liu Hui, who wrote a commentary on *The Nine Chapters*, obtained the approximation 3.14 by refining the method of inscribed polygons. Identify the ancient Chinese mathematician who further refined this method and obtained the approximations

$$\frac{355}{113} \quad \text{and} \quad \frac{22}{7}$$

for π.

Solution. The mathematician described in the problem is Zu Chongzhi.

Liu Hui improved the approximation of π by using polygons with many sides. Zu Chongzhi later obtained the very accurate bounds

$$3.1415926 < \pi < 3.1415927,$$

and gave two famous rational approximations:

$$\frac{22}{7} \qquad \text{and} \qquad \frac{355}{113}.$$

The approximation $\dfrac{355}{113}$ is especially accurate and is traditionally known as the Milü, or "accurate ratio."

Therefore, the answer is $\boxed{\text{Zu Chongzhi}}$.

□

Problem 2014.3. Consider the 2014×2014 matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

Solution. Let $\mathbf{A}$ be the given matrix, and let $\mathbf{J}_{2014}$ be the 2014×2014 matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J}_{2014} - \mathbf{I}_{2014}.$$

The matrix $\mathbf{J}_{2014}$ has eigenvalue 2014 for the vector

$$\mathbf{1} = (1, 1, \ldots, 1)^{\mathsf{T}},$$

since

$$\mathbf{J}_{2014}\mathbf{1} = 2014\mathbf{1}.$$

Therefore,

$$\mathbf{A1} = (\mathbf{J}_{2014} - \mathbf{I}_{2014})\mathbf{1} = 2014\mathbf{1} - \mathbf{1} = 2013\mathbf{1}.$$

Hence 2013 is an eigenvalue of $\mathbf{A}$.

Now take any vector $\mathbf{v} = (v_1, \ldots, v_{2014})^{\mathsf{T}}$ such that

$$v_1 + \cdots + v_{2014} = 0.$$

Then $\mathbf{J}_{2014}\mathbf{v} = \mathbf{0}$, so

$$\mathbf{Av} = (\mathbf{J}_{2014} - \mathbf{I}_{2014})\mathbf{v} = -\mathbf{v}.$$

Thus -1 is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2013, so the multiplicity of -1 is 2013.

Therefore, the eigenvalues are

$$\boxed{2013 \text{ with multiplicity } 1, \qquad -1 \text{ with multiplicity } 2013}.$$

□

Problem 2014.4. Let $m \geq 3$ and $n \geq 3$. Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq \mathbb{R}^m, \qquad \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} \subseteq \mathbb{R}^n.$$

Assume that

$$V = \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$$

is a 2-dimensional subspace of $\mathbb{R}^m$, and that

$$W = \operatorname{span}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3)$$

is a 3-dimensional subspace of $\mathbb{R}^n$. Determine $\operatorname{rk}(\mathbf{A})$ for the $m \times n$ matrix

$$\mathbf{A} = \mathbf{v}_1\mathbf{w}_1^\mathsf{T} + \mathbf{v}_2\mathbf{w}_2^\mathsf{T} + \mathbf{v}_3\mathbf{w}_3^\mathsf{T}.$$

Solution. Let

$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{pmatrix}, \qquad \mathbf{Q} = \begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 \end{pmatrix}.$$

Then

$$\mathbf{A} = \mathbf{P}\mathbf{Q}^\mathsf{T}.$$

Since $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ span a 2-dimensional subspace, we have

$$\operatorname{rk}(\mathbf{P}) = 2.$$

Since $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ span a 3-dimensional subspace, they are linearly independent, and hence

$$\operatorname{rk}(\mathbf{Q}) = 3.$$

Therefore, the linear map $\mathbf{Q}^\mathsf{T} : \mathbb{R}^n \to \mathbb{R}^3$ is surjective.

It follows that the image of $\mathbf{A} = \mathbf{P}\mathbf{Q}^\mathsf{T}$ is

$$\operatorname{im}(\mathbf{A}) = \mathbf{P}(\operatorname{im}(\mathbf{Q}^\mathsf{T})) = \mathbf{P}(\mathbb{R}^3) = \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3).$$

This space has dimension 2. Hence

$$\operatorname{rk}(\mathbf{A}) = 2.$$

Therefore, the answer is $\boxed{2}$.

□

Problem 2015.1. Read the following passage and fill in the blank with the title of the mathematical classic, either in English or in Korean.

> For 2000 years, __________ was not only the core of mathematical education but also at the heart of Western culture. The most glowing tributes to __________ do not, in fact, come from mathematicians, but from philosophers, politicians, and others.
>
> Here are some examples:
>
> (a) He studied and nearly mastered the six books of __________ since he was a member of Congress. He regrets his want of education, and does what he can to supply the want.
>
> Abraham Lincoln, *Short Autobiography*
>
> (b) He studied __________ until he could demonstrate with ease all the propositions in the six books.
>
> *Herndon's Life of Lincoln*
>
> (c) At the age of eleven, I began __________. This was one of the great events of my life, as dazzling as first love. I had not imagined there was anything so delicious in the world.
>
> Bertrand Russell, *Autobiography*, Vol. 1

Solution. The passage is referring to Euclid's *Elements*, one of the most influential works in the history of mathematics.

For centuries, Euclid's *Elements* served as a standard model of rigorous mathematical reasoning and was central to mathematical education. The quotations from Abraham Lincoln and Bertrand Russell emphasize how deeply the book influenced not only mathematicians, but also philosophers, politicians, and intellectual culture more broadly.

Therefore, the answer is $\boxed{\textit{The Elements}, \text{ or Euclid's } \textit{Elements}}$.

□

Problem 2015.2. Let $\mathbf{A}$ be an invertible square matrix. Suppose that, for a suitable permutation matrix $\mathbf{P}$, Gaussian elimination gives an LU-decomposition

$$\mathbf{PA} = \mathbf{LU}.$$

It may also happen that another permutation matrix $\widetilde{\mathbf{P}}$ gives another LU-decomposition

$$\widetilde{\mathbf{P}}\mathbf{A} = \widetilde{\mathbf{L}}\widetilde{\mathbf{U}}.$$

Show that the sets of absolute values of the pivots can be different. In other words, show by example that

$$\{|u_{kk}| \mid 1 \le k \le n\} \ne \{|\widetilde{u}_{kk}| \mid 1 \le k \le n\}.$$

Hint. Look for a counterexample in the 2×2 case.

Solution. Consider the invertible matrix

$$\mathbf{A} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{3} & 1 \end{pmatrix}.$$

With no row exchange, we have the LU-decomposition

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{5}{6} \end{pmatrix}.$$

Hence the absolute values of the pivots are

$$\left\{1, \frac{5}{6}\right\}.$$

Now let $\widetilde{\mathbf{P}}$ be the permutation matrix that swaps the two rows. Then

$$\widetilde{\mathbf{P}}\mathbf{A} = \begin{pmatrix} \frac{1}{3} & 1 \\ 1 & \frac{1}{2} \end{pmatrix}.$$

This matrix has the LU-decomposition

$$\widetilde{\mathbf{P}}\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 1 \\ 0 & -\frac{5}{2} \end{pmatrix}.$$

Hence the absolute values of the pivots are

$$\left\{\frac{1}{3}, \frac{5}{2}\right\}.$$

Therefore,

$$\left\{1, \frac{5}{6}\right\} \ne \left\{\frac{1}{3}, \frac{5}{2}\right\}.$$

This shows that the set of absolute values of the pivots can depend on the chosen row permutation.

□

Problem 2015.3. In $\mathbb{R}^2$, the so-called Mercedes-Benz system is defined by

$$\{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2\}, \qquad \mathbf{v}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \end{pmatrix} \qquad (k = 0, 1, 2).$$

Find vectors $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2 \in \mathbb{R}^3$ such that they have the same length, are mutually orthogonal, have positive z-coordinates, and satisfy

$$\mathbf{v}_k = P_{xy}(\mathbf{w}_k) \qquad (k = 0, 1, 2),$$

where P_{xy} denotes the projection onto the xy-plane. Here $\mathbb{R}^2$ is regarded as the subspace of $\mathbb{R}^3$ with z-coordinate equal to 0.

Solution. Since $P_{xy}(\mathbf{w}_k) = \mathbf{v}_k$, each $\mathbf{w}_k$ must have the form

$$\mathbf{w}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \\ c \end{pmatrix}$$

for some positive constant c.

For $i \neq j$, the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$ is 120°, so

$$\mathbf{v}_i \cdot \mathbf{v}_j = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}.$$

Therefore,

$$\mathbf{w}_i \cdot \mathbf{w}_j = \mathbf{v}_i \cdot \mathbf{v}_j + c^2 = -\frac{1}{2} + c^2.$$

To make the vectors mutually orthogonal, we need

$$-\frac{1}{2} + c^2 = 0,$$

so

$$c = \frac{1}{\sqrt{2}}.$$

Hence one possible choice is

$$\boxed{\mathbf{w}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \\ \frac{1}{\sqrt{2}} \end{pmatrix} \qquad (k = 0, 1, 2)}.$$

These vectors all have squared length

$$1 + \frac{1}{2} = \frac{3}{2},$$

have positive z-coordinates, project to the given vectors $\mathbf{v}_k$, and are mutually orthogonal.

□

Problem 2015.4. This French mathematician was born in 1973 and received the Fields Medal in 2010 at the age of 36. On his website, he described himself as “Mathématicien, Professeur de l'Université de Lyon, Directeur de l'Institut Henri Poincaré." He is also famous for his stylish clothing.

Together with Clément Mouhot, he completed a theorem that played a central role in his Fields Medal-winning work. He later wrote a book about the process of developing this theorem, and the title of the book is also one of his nicknames.

Write the name of this mathematician, using the correct French spelling, and the title of the book.

Solution. The mathematician described in the problem is Cédric Villani.

Villani received the Fields Medal in 2010 for his work on nonlinear Landau damping and the Boltzmann equation. He is also widely recognized for his distinctive style, often including a cravat and a spider brooch.

The book mentioned in the problem is *Théorème vivant*, which may be translated as *Living Theorem*. It describes the process through which Villani and Clément Mouhot developed the theorem that contributed to Villani's Fields Medal-winning work.

Therefore, the answer is $\boxed{\text{Cédric Villani, } \textit{Théorème vivant}}$.

□

Problem 2015.5. On the front side of the Fields Medal, there is a profile of Archimedes together with the phrase by Marcus Manilius:

"TRANSIRE SUUM PECTUS MUNDOQUE POTIRI."

On the back side, there is another inscription together with the two solid figures that were engraved on Archimedes' tomb.

Identify the two solid figures and determine their volume ratio.

Solution. The two solid figures are a sphere and its circumscribed cylinder.

Suppose the sphere has radius R. Then the volume of the sphere is

$$V_S = \frac{4}{3}\pi R^3.$$

The circumscribed cylinder has radius R and height $2R$, so its volume is

$$V_C = \pi R^2 \cdot 2R = 2\pi R^3.$$

Therefore,

$$V_S : V_C = \frac{4}{3}\pi R^3 : 2\pi R^3 = 2 : 3.$$

Thus the two figures are a sphere and its circumscribed cylinder, and the volume ratio is $\boxed{2 : 3}$.

□

Problem 2015.6. Let $f : \mathbb{R} \to \mathbb{R}$ be a periodic function with period 2π. For $x \in (0, 2\pi]$, define

$$f(x) = \lceil \max(3, |\pi - x|) - 3 \rceil,$$

and extend f periodically, so that

$$f(x + 2\pi) = f(x)$$

for all $x \in \mathbb{R}$. Determine whether the following limit converges:

$$\lim_{n\to\infty} f(n).$$

Hint. Use $3.14 < \pi < 3.145$.

Solution. For $x \in (0, 2\pi]$, the function f takes only the values 0 and 1. More precisely,

$$f(x) = 1$$

when

$$0 < x < \pi - 3 \qquad \text{or} \qquad \pi + 3 < x \leq 2\pi,$$

and

$$f(x) = 0$$

when

$$\pi - 3 \leq x \leq \pi + 3.$$

Thus f is a 2π-periodic function which is 1 near multiples of 2π, and 0 on the large middle interval.

Since $1/(2\pi)$ is irrational, the fractional parts

$$\left\{ \frac{n}{2\pi} \right\}, \qquad n = 1, 2, 3, \ldots,$$

are dense in $[0, 1]$. Equivalently, the representatives of the integers modulo 2π are dense in $(0, 2\pi]$. Hence there are infinitely many positive integers n whose representatives modulo 2π lie in

$$(0, \pi - 3) \cup (\pi + 3, 2\pi],$$

and for these integers we have

$$f(n) = 1.$$

There are also infinitely many positive integers n whose residues modulo 2π lie in

$$(\pi - 3, \pi + 3),$$

and for these integers we have

$$f(n) = 0.$$

The sequence $f(n)$ has a subsequence equal to 1 and another subsequence equal to 0, so it does not converge as $n \to \infty$.

Therefore, the answer is $\boxed{\text{the limit does not exist}}$.

□

Problem 2015.7. This person is believed to have lived approximately from 624 BCE to 546 BCE. He is often regarded as one of the earliest figures worthy of being called a mathematician. He introduced the method of proof to geometric facts that had been known empirically through land surveying, helping to lay the foundations of geometry that were later systematized in Euclid's *Elements*.

The theorems attributed to him include:

(a) Vertical angles are equal.

(b) The base angles of an isosceles triangle are equal.

(c) Triangle congruence criteria such as side-angle-side and angle-side-angle.

(d) An angle inscribed in a semicircle is a right angle.

Identify this person.

Solution. The person described in the problem is Thales of Miletus.

Thales is traditionally regarded as one of the earliest Greek mathematicians. He is especially important because he is associated not merely with observing geometric facts, but with proving them. Several elementary geometric results are attributed to him, including the theorem that an angle inscribed in a semicircle is a right angle.

Therefore, the answer is $\boxed{\text{Thales of Miletus}}$.

□

Problem 2015.8. Determine which of the following mathematicians was born earliest.

(a) Henri Poincaré
(b) Karl Weierstrass
(c) Julius Wilhelm Richard Dedekind
(d) Andrei Andreyevich Markov
(e) Georg Cantor

Solution. We compare their years of birth:

Karl Weierstrass	1815,
Julius Wilhelm Richard Dedekind	1831,
Georg Cantor	1845,
Henri Poincaré	1854,
Andrei Andreyevich Markov	1856.

Among these mathematicians, the one born earliest is Karl Weierstrass.

Therefore, the answer is $\boxed{\text{Karl Weierstrass}}$.

□

Problem 2015.9. The first International Congress of Mathematicians at which the Fields Medal was awarded was held in 1936. Determine the country in which it was held.

Solution. The Fields Medal was first awarded at the 1936 International Congress of Mathematicians. The 1936 ICM was held in Oslo, Norway.

Therefore, the answer is $\boxed{\text{Norway}}$.

□

Problem 2015.10. Find the sum of the distinct prime numbers appearing in the prime factorization of 15!.

Solution. The prime numbers appearing in the prime factorization of 15! are exactly the primes less than or equal to 15:

$$2, \quad 3, \quad 5, \quad 7, \quad 11, \quad 13.$$

Therefore, their sum is

$$2 + 3 + 5 + 7 + 11 + 13 = 41.$$

Therefore, the answer is $\boxed{41}$.

□

Problem 2015.11. In three-dimensional space, suppose that the lengths of the orthogonal projections of a line segment ℓ onto the yz-plane, the zx-plane, and the xy-plane are 5, 5, and 6, respectively. Determine the length of ℓ.

Solution. Let the displacement vector of the line segment be

$$\mathbf{v} = (a, b, c).$$

Then the length of ℓ is

$$\|\mathbf{v}\| = \sqrt{a^2 + b^2 + c^2}.$$

The projection of $\mathbf{v}$ onto the yz-plane has length

$$\sqrt{b^2 + c^2} = 5.$$

Similarly, the projections onto the zx-plane and the xy-plane give

$$\sqrt{c^2 + a^2} = 5, \qquad \sqrt{a^2 + b^2} = 6.$$

Squaring and adding these three equations, we get

$$(b^2 + c^2) + (c^2 + a^2) + (a^2 + b^2) = 5^2 + 5^2 + 6^2.$$

Hence

$$2(a^2 + b^2 + c^2) = 25 + 25 + 36 = 86,$$

so

$$a^2 + b^2 + c^2 = 43.$$

Therefore, the length of ℓ is

$$\|\mathbf{v}\| = \sqrt{43}.$$

Therefore, the answer is $\boxed{\sqrt{43}}$.

□

Problem 2015.12. In three-dimensional space, consider the matrix obtained by rotating space by $30°$ around the line passing through the origin and the point $(1, 1, 2)$. Determine the number of real eigenvalues of this matrix.

Solution. A rotation in $\mathbb{R}^3$ around an axis fixes every vector lying on the rotation axis. Therefore, the direction vector

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

is an eigenvector with eigenvalue 1.

On the plane perpendicular to the rotation axis, the matrix acts as a $30°$ rotation. A nontrivial rotation in a two-dimensional plane has no real eigenvalues unless the rotation angle is $0°$ or $180°$. Since the angle here is $30°$, the two remaining eigenvalues are non-real complex conjugates.

Hence the only real eigenvalue is 1. Therefore, the number of real eigenvalues is $\boxed{1}$.

□

Problem 2015.13. A right circular cone has base radius 3 and height 6. Determine the maximum possible volume of a cylinder inscribed in this cone.

Solution. Let x be the radius of the inscribed cylinder, and let h be its height. By similarity of triangles in the cone, the radius of the horizontal cross-section decreases linearly from 3 to 0 as the height increases from 0 to 6. Hence

$$x = 3 - \frac{h}{2}.$$

Equivalently,
$$h = 6 - 2x.$$
Therefore, the volume of the cylinder is
$$V(x) = \pi x^2 h = \pi x^2(6 - 2x) = 6\pi x^2 - 2\pi x^3,$$
where $0 \leq x \leq 3$.

Differentiating, we get
$$V'(x) = 12\pi x - 6\pi x^2 = 6\pi x(2 - x).$$
Thus the critical point in the interval is
$$x = 2.$$
At this point,
$$h = 6 - 2 \cdot 2 = 2,$$
so the maximum volume is
$$V(2) = \pi \cdot 2^2 \cdot 2 = 8\pi.$$
Therefore, the maximum possible volume is $\boxed{8\pi}$.

□

Problem 2015.14. Suppose that a polynomial equation of degree 7 with real coefficients has three non-real complex roots whose absolute values are all different. Determine the number of real roots of the equation.

Solution. For a polynomial with real coefficients, every non-real complex root appears together with its complex conjugate.

Suppose the three given non-real roots are
$$\alpha_1, \quad \alpha_2, \quad \alpha_3,$$
and their absolute values are all different. Then their complex conjugates
$$\overline{\alpha_1}, \quad \overline{\alpha_2}, \quad \overline{\alpha_3}$$
are also roots. Since
$$|\alpha_i| = |\overline{\alpha_i}|,$$
and the three given roots have distinct absolute values, these conjugate roots give three additional non-real roots.

Thus the polynomial has at least 6 non-real roots. Since the degree is 7, there is exactly one remaining root, and it must be real.

Therefore, the equation has $\boxed{1}$ real root.

□

Problem 2015.15. This mathematician was born in 1882 and died in 1935. When she was being considered for a professorship at the University of Göttingen, there was strong opposition. In response, Hilbert famously argued, "I do not see that the sex of the candidate is an argument against her admission as a Privatdozent. After all, we are a university, not a bathhouse."

She made remarkable contributions to several areas of mathematics, especially modern algebra, and much of twentieth-century abstract algebra was deeply influenced by her work. Identify this mathematician.

Solution. The mathematician described in the problem is Emmy Noether.

Noether made fundamental contributions to modern algebra, especially through her work on rings, ideals, and modules. She is also famous for Noether's theorem, which connects symmetries and conservation laws in physics.

The biographical details in the problem, including her connection with Göttingen and Hilbert's defense of her academic appointment, point to Emmy Noether.

Therefore, the answer is $\boxed{\text{Emmy Noether}}$.

$\square$

Problem 2015.16. Determine the number of distinct terms in the expansion of

$$(x_1 + x_2 + \cdots + x_7)^4.$$

Solution. Each term in the expansion has the form

$$x_1^{k_1} x_2^{k_2} \cdots x_7^{k_7},$$

where

$$k_1 + k_2 + \cdots + k_7 = 4$$

and each k_i is a nonnegative integer.

Therefore, the number of distinct terms is the number of nonnegative integer solutions to

$$k_1 + k_2 + \cdots + k_7 = 4.$$

By the stars and bars formula, this number is

$$\binom{4+7-1}{7-1} = \binom{10}{6} = 210.$$

Therefore, the answer is $\boxed{210}$.

$\square$

Problem 2015.17. Let

$$\mathbf{v} = \mathbf{w} = (1, 2, 3, 4)$$

be row vectors. Consider the 4×4 matrix obtained as the product

$$\mathbf{v}^\mathsf{T}\mathbf{w}.$$

Find the sum and the product of the eigenvalues of this matrix.

Solution. The matrix is

$$\mathbf{v}^\mathsf{T}\mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \end{pmatrix}.$$

By the trace–eigenvalue identity in Mathematical Concepts, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to its trace. Therefore,

$$\sum_{i=1}^{4} \lambda_i = \operatorname{tr}(\mathbf{v}^\mathsf{T}\mathbf{w}) = 1^2 + 2^2 + 3^2 + 4^2 = 30.$$

The product of the eigenvalues is equal to the determinant. Since $\mathbf{v}^\mathsf{T}\mathbf{w}$ is an outer product of two vectors, we have $\operatorname{rk}(\mathbf{v}^\mathsf{T}\mathbf{w}) = 1$. In particular, as a 4×4 matrix, it is not invertible, so

$$\det(\mathbf{v}^\mathsf{T}\mathbf{w}) = 0.$$

Hence the product of the eigenvalues is

$$\prod_{i=1}^{4} \lambda_i = 0.$$

Therefore, the sum and product of the eigenvalues are $\boxed{30 \text{ and } 0}$, respectively.

□

Problem 2015.18. Evaluate the infinite series

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}.$$

Hint. Use

$$\frac{e^x - 1}{x} = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!},$$

then differentiate and substitute $x = 1$.

Solution. We use a simple telescoping identity:

$$\frac{n}{(n+1)!} = \frac{1}{n!} - \frac{1}{(n+1)!}.$$

Therefore,

$$\sum_{n=1}^{N} \frac{n}{(n+1)!} = \sum_{n=1}^{N} \left(\frac{1}{n!} - \frac{1}{(n+1)!}\right) = 1 - \frac{1}{(N+1)!}.$$

Taking $N \to \infty$, we obtain

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = 1.$$

Therefore, the answer is $\boxed{1}$.

□

Problem 2017.1. Determine all possible positive integers that can occur as the number of faces of a regular polyhedron, that is, a Platonic solid.

Solution. There are exactly five Platonic solids:

Solid	Number of faces
Tetrahedron	4
Cube	6
Octahedron	8
Dodecahedron	12
Icosahedron	20

Therefore, the possible numbers of faces of a regular polyhedron are

$$\boxed{4,\ 6,\ 8,\ 12,\ 20}.$$

□

Problem 2017.2. Let (F_n) be the Fibonacci sequence defined by

$$F_0 = 0, \qquad F_1 = 1, \qquad F_{n+1} = F_n + F_{n-1}.$$

Determine
$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n}.$$

Solution. By the linear-recurrence method developed in Mathematical Concepts, Binet's formula is
$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$
Since $|\psi/\varphi| < 1$,
$$\frac{F_{n+1}}{F_n} = \varphi \, \frac{1-(\psi/\varphi)^{n+1}}{1-(\psi/\varphi)^n} \longrightarrow \varphi.$$
Therefore,
$$\boxed{\frac{1+\sqrt{5}}{2}}.$$

□

Problem 2017.3. For a positive integer n, define the $n \times n$ matrix $\mathbf{C}$ by
$$c_{ij} = \binom{i+j-2}{i-1} \qquad (1 \le i, j \le n).$$
Now define another $n \times n$ matrix $\mathbf{B}$ by
$$b_{ij} = \begin{cases} c_{ij}, & (i,j) \ne (n,n), \\ c_{ij} + 2016, & (i,j) = (n,n). \end{cases}$$
Determine $\det(\mathbf{B})$.

Solution. We first compute the determinant of $\mathbf{C}$. By Vandermonde's identity,
$$\binom{i+j-2}{i-1} = \sum_{k=1}^{\min(i,j)} \binom{i-1}{k-1}\binom{j-1}{k-1}.$$
Hence
$$\mathbf{C} = \mathbf{L}\mathbf{L}^{\mathsf{T}},$$
where
$$l_{ik} = \begin{cases} \binom{i-1}{k-1}, & k \le i, \\ 0, & k > i. \end{cases}$$
The matrix $\mathbf{L}$ is lower triangular and all of its diagonal entries are 1. Therefore,
$$\det(\mathbf{L}) = 1,$$
and so
$$\det(\mathbf{C}) = \det(\mathbf{L}\mathbf{L}^{\mathsf{T}}) = (\det(\mathbf{L}))^2 = 1.$$

The matrix $\mathbf{B}$ differs from $\mathbf{C}$ only in the (n,n)-entry. Since the determinant is linear in each entry, we have
$$\det(\mathbf{B}) = \det(\mathbf{C}) + 2016 \cdot M_{nn},$$
where M_{nn} is the (n,n)-minor of $\mathbf{C}$.

This minor is the determinant of the upper-left $(n-1)\times(n-1)$ submatrix of $\mathbf{C}$, which has the same form as $\mathbf{C}$ with size $n-1$. Therefore,

$$M_{nn} = 1.$$

Hence

$$\det(\mathbf{B}) = 1 + 2016 \cdot 1 = 2017.$$

Therefore, the answer is $\boxed{2017}$.

□

Problem 2017.4. In 2017, this prize was awarded to the French mathematician Yves Meyer. The Norwegian Academy of Science and Letters announced on March 21 that Meyer would receive the prize, worth six million Norwegian kroner, in recognition of his decisive contributions to the development of wavelet theory, which is used in areas ranging from gravitational wave detection to movie file compression.

Identify this prize, which is often called the Nobel Prize of mathematics.

Solution. The prize described in the problem is the Abel Prize.

The Abel Prize is awarded by the Norwegian Academy of Science and Letters and is often described as one of the most prestigious prizes in mathematics. In 2017, it was awarded to Yves Meyer for his fundamental contributions to the mathematical theory of wavelets.

Therefore, the answer is $\boxed{\text{Abel Prize}}$.

□

Problem 2017.5. Express the integral

$$\int_0^1 x^m(1-x)^n \,\mathrm{d}x$$

in terms of a binomial coefficient closely related to the positive integers m and n.

Solution. This integral is a beta integral. For positive integers m and n, we have

$$\int_0^1 x^m(1-x)^n \,\mathrm{d}x = \frac{m!n!}{(m+n+1)!}.$$

This is the factorial form of the answer.

To express it using a binomial coefficient, recall that

$$\binom{m+n}{m} = \frac{(m+n)!}{m!n!}.$$

Hence

$$\frac{m!n!}{(m+n+1)!} = \frac{1}{m+n+1} \cdot \frac{m!n!}{(m+n)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Therefore,

$$\int_0^1 x^m(1-x)^n \,\mathrm{d}x = \frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Therefore, the answer is $\boxed{\frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}}$.

□

Problem 2018.1. The following people are this year's Fields Medalists. Excluding choice (5), identify the second-oldest medalist among them.

(1) Caucher Birkar
(2) Alessio Figalli
(3) Peter Scholze
(4) Akshay Venkatesh
(5) The person who stole Birkar's medal

Solution. The 2018 Fields Medalists were

(1)	Caucher Birkar	born in 1978,
(2)	Alessio Figalli	born in 1984,
(3)	Peter Scholze	born in 1987,
(4)	Akshay Venkatesh	born in 1981.

Among these four, the second oldest is Akshay Venkatesh. Therefore, the answer is

$$\boxed{(4)}.$$

Choice (5) is a joke referring to the incident in which Caucher Birkar's Fields Medal was stolen shortly after the 2018 Fields Medal ceremony. Birkar later received a replacement medal. □

Problem 2018.2. According to William Thurston's geometrization conjecture, a 3-dimensional manifold can be cut along spheres and tori so that each resulting piece has one of n possible geometric structures. Determine n.

Solution. The 3-dimensional geometric structures appearing in Thurston's geometrization conjecture are the eight Thurston geometries:

$$S^3, \quad \mathrm{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}}_2(\mathbb{R}), \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Hence $n = 8$, so the answer is

$$\boxed{8}.$$

□

Problem 2018.3. Find the sum of the following infinite series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots .$$

Solution. The given series is one half of the alternating harmonic series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right).$$

Recall the Taylor expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots .$$

Substituting $x = 1$, we obtain

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots .$$

Therefore,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{\ln 2}{2}.$$

Therefore, the answer is $\boxed{\dfrac{\ln 2}{2}}$.

□

Problem 2018.4. Determine which of the following five Fields Medalists does not have a Korean doctoral student.

(1) William Thurston
(2) Maxim Kontsevich
(3) Grigory Margulis
(4) Terence Tao
(5) Efim Zelmanov

Solution. According to the original answer key, the exceptional mathematician among the five listed Fields Medalists is

$$\boxed{\text{(2) Maxim Kontsevich}}.$$

□

Problem 2019.1. Consider the 2019×2019 matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

Solution. Let $\mathbf{A}$ be the given matrix, and let $\mathbf{J}_{2019}$ be the 2019×2019 matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J}_{2019} - \mathbf{I}_{2019}.$$

First, consider the vector

$$\mathbf{1} = (1, 1, \ldots, 1)^{\mathsf{T}}.$$

Since

$$\mathbf{J}_{2019}\mathbf{1} = 2019\mathbf{1},$$

we get

$$\mathbf{A}\mathbf{1} = (\mathbf{J}_{2019} - \mathbf{I}_{2019})\mathbf{1} = 2019\mathbf{1} - \mathbf{1} = 2018\mathbf{1}.$$

Hence 2018 is an eigenvalue.

Now take any vector $\mathbf{v} = (v_1, \ldots, v_{2019})^{\mathsf{T}}$ satisfying

$$v_1 + \cdots + v_{2019} = 0.$$

Then

$$\mathbf{J}_{2019}\mathbf{v} = \mathbf{0},$$

and therefore

$$\mathbf{A}\mathbf{v} = (\mathbf{J}_{2019} - \mathbf{I}_{2019})\mathbf{v} = -\mathbf{v}.$$

Thus -1 is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2018, so the multiplicity of -1 is 2018.

Therefore, the eigenvalues are

$$\boxed{2018 \text{ with multiplicity } 1, \qquad -1 \text{ with multiplicity } 2018}.$$

□

Problem 2019.2. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2}\,\mathrm{d}x.$$

Solution. Let

$$u = 2x.$$

Then

$$\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u.$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2}\,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{8}{1+u^2}\cdot\frac{1}{2}\,\mathrm{d}u = 4\int_{-\infty}^{\infty} \frac{1}{1+u^2}\,\mathrm{d}u.$$

Since

$$\int_{-\infty}^{\infty} \frac{1}{1+u^2}\,\mathrm{d}u = \pi,$$

we obtain

$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2}\,\mathrm{d}x = 4\pi.$$

Therefore, the answer is $\boxed{4\pi}$.

□

Problem 2019.3. Compute the value of $\pi + e$ up to five decimal places.

Solution. Using the approximations

$$\pi \approx 3.14159265, \qquad e \approx 2.71828183,$$

we get

$$\pi + e \approx 3.14159265 + 2.71828183 = 5.85987448.$$

Therefore, up to five decimal places,

$$\pi + e \approx 5.85987.$$

Therefore, the answer is $\boxed{5.85987}$.

□

Problem 2020.1. The International Congress of Mathematicians, abbreviated as ICM, is a congress for mathematicians from around the world. It is organized every four years by the International Mathematical Union and may be thought of as a kind of Olympic Games for mathematicians. The 2014 ICM was held in South Korea, and the 2018 ICM was held in Brazil. As of 2019, determine the country in which the 2022 ICM was scheduled to be held.

Solution. ICM stands for the International Congress of Mathematicians. It is the main international congress in mathematics and is organized every four years by the International Mathematical Union.

The 2014 ICM was held in Seoul, South Korea, and the 2018 ICM was held in Rio de Janeiro, Brazil. At the reference date specified in the problem, the 2022 ICM was scheduled to be held in Saint Petersburg, Russia. Therefore, the answer is $\boxed{\text{Russia}}$.

The congress was later changed to a virtual event, with the 2022 awards ceremony held in Helsinki, Finland. The answer above therefore records the planned host country at the date specified in the original problem.

□

Problem 2020.2. A fair six-sided die has the numbers $-2, 0, 2, 3, 4, 5$ written on its faces. The die is rolled 10 times, and X is defined to be the product of the 10 numbers obtained. Determine $\mathbb{E}[X]$.

Solution. Let Y be the number obtained from one roll of the die. Since all six faces are equally likely,

$$\mathbb{E}[Y] = \frac{-2+0+2+3+4+5}{6} = 2.$$

Now let $Y_1, Y_2, \ldots, Y_{10}$ be the numbers obtained from the 10 rolls. Then

$$X = Y_1 Y_2 \cdots Y_{10}.$$

Since the rolls are independent, the random variables $Y_1, Y_2, \ldots, Y_{10}$ are independent. Therefore,

$$\mathbb{E}[X] = \mathbb{E}[Y_1 Y_2 \cdots Y_{10}] = \mathbb{E}[Y_1]\mathbb{E}[Y_2] \cdots \mathbb{E}[Y_{10}].$$

All the Y_i have the same distribution as Y, so

$$\mathbb{E}[X] = (\mathbb{E}[Y])^{10} = 2^{10} = 1024.$$

Therefore, the answer is $\boxed{1024}$.

□

Problem 2020.3. Determine which of the following Fields Medalists was not serving as a professor at the time.

(1) Cédric Villani
(2) Elon Lindenstrauss
(3) Peter Scholze
(4) Martin Hairer
(5) Terence Tao

Solution. According to the original answer key and the temporal context of the problem, Cédric Villani was then serving in French politics rather than in a professorial position. The other choices were serving as professors.

Therefore, the answer is $\boxed{\text{(1) Cédric Villani}}$.

□

Problem 2020.4. Which of the following statements is correct?

1. On a simply connected open subset of $\mathbb{R}^3$, every continuously differentiable vector field with zero curl is a gradient field.
2. Every differentiable function is twice differentiable.
3. Every square matrix is diagonalizable.
4. Every bounded sequence converges.

Solution. Statement (1) is the conservative-field criterion from Mathematical Concepts. The assumptions of continuous differentiability and simple connectedness are essential. The other statements are false in general: differentiability need not iterate, matrices may be defective, and bounded sequences need not converge.

Hence the correct choice is $\boxed{(1)}$.

□

Problem 2020.5. Determine which of the following mathematics-related academic institutions is a degree-granting school.

(1) Institut des Hautes Études Scientifiques, IHES
(2) École normale supérieure
(3) Mathematical Sciences Research Institute, MSRI
(4) Institute for Advanced Study, IAS
(5) Collège de France

Solution. École normale supérieure is one of France's most prestigious higher education institutions. It operates educational programs and is naturally regarded as a degree-granting school.

By contrast, IHES, MSRI, and IAS are research institutes rather than degree-granting schools. Collège de France is also a famous academic institution that offers public lectures, but it does not grant degrees.

Therefore, the answer is $\boxed{\text{(2) École normale supérieure}}$.

□

Problem 2021.1. Determine which of A and B is larger, where

$$A = 1 + \cfrac{1}{1 + \frac{1}{1 - \frac{1}{1+4}}}, \qquad B = 1 + \cfrac{1}{1 + \frac{1}{1 + \frac{1}{1+3}}}.$$

Solution. First, we compute A:

$$A = 1 + \cfrac{1}{1 + \frac{1}{1 - \frac{1}{5}}} = 1 + \cfrac{1}{1 + \frac{1}{\frac{4}{5}}} = 1 + \cfrac{1}{1 + \frac{5}{4}} = 1 + \cfrac{1}{\frac{9}{4}} = 1 + \frac{4}{9} = \frac{13}{9}.$$

Next, we compute B:

$$B = 1 + \cfrac{1}{1 + \frac{1}{1 + \frac{1}{4}}} = 1 + \cfrac{1}{1 + \frac{1}{\frac{5}{4}}} = 1 + \cfrac{1}{1 + \frac{4}{5}} = 1 + \cfrac{1}{\frac{9}{5}} = 1 + \frac{5}{9} = \frac{14}{9}.$$

Since

$$\frac{13}{9} < \frac{14}{9},$$

we have $A < B$. Therefore, the answer is $\boxed{B}$.

□

Problem 2021.2. Determine the nearest integer to

$$\frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^8.$$

Solution. Binet's formula gives

$$F_8 = \frac{\varphi^8 - \psi^8}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

Thus

$$\frac{\varphi^8}{\sqrt{5}} = F_8 + \frac{\psi^8}{\sqrt{5}}.$$

Since $|\psi| < 1$,

$$0 < \frac{|\psi|^8}{\sqrt{5}} < \frac{1}{2}.$$

Therefore the nearest integer is $F_8 = 21$, so the answer is

$$\boxed{21}.$$

□

Problem 2021.3. Choose all groups among the following that are not finitely generated:

(1) The fundamental group of an open unit disk with finitely many punctures.
(2) $\mathrm{SL}_2(\mathbb{Z})$, the group of 2×2 integer matrices with determinant 1.
(3) $\mathbb{C}^\times$, the group of nonzero complex numbers under multiplication.
(4) The Monster group, the largest sporadic finite simple group.

Solution. First, consider an open unit disk with finitely many punctures. If the removed points are $p_1, \ldots, p_n$, then the fundamental group is isomorphic to the free group on n generators:

$$\pi_1(D \setminus \{p_1, \ldots, p_n\}) \cong F_n.$$

Hence the group in choice (1) is finitely generated.

The group $\mathrm{SL}_2(\mathbb{Z})$ is also finitely generated. For example, it is generated by the two matrices

$$\mathbf{S} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{T} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Thus choice (2) is finitely generated.

On the other hand, $\mathbb{C}^\times$ is an uncountable abelian group. Every finitely generated group is countable, so $\mathbb{C}^\times$ cannot be finitely generated. Therefore, choice (3) is not finitely generated.

Finally, the Monster group is a finite group. Every finite group is finitely generated, since one may simply take all of its elements as generators. Hence choice (4) is finitely generated.

Therefore, the only group in the list that is not finitely generated is $\boxed{(3)\ \mathbb{C}^\times}$.

□

Problem 2021.4. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Find all eigenvalues of $\mathbf{A}$ together with their multiplicities. Also determine whether $\mathbf{A}$ is diagonalizable.

Solution. We can write $\mathbf{A}$ as

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}.$$

Thus $\mathrm{rk}(\mathbf{A}) = 1$.

First, let

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{A}\mathbf{v} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix} = 6\mathbf{v}.$$

Hence 6 is an eigenvalue of $\mathbf{A}$.

Since $\operatorname{rk}(\mathbf{A}) = 1$, its nullity is 2. Therefore, the eigenspace corresponding to the eigenvalue 0 has dimension 2. Indeed,

$$\mathbf{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y + 3z \\ x + 2y + 3z \\ x + 2y + 3z \end{pmatrix},$$

so

$$E_0 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y + 3z = 0 \right\}.$$

Thus the geometric multiplicity of 0 is 2.

Now

$$\operatorname{tr}(\mathbf{A}) = 1 + 2 + 3 = 6.$$

Since the sum of the eigenvalues, counted with algebraic multiplicity, is equal to $\operatorname{tr}(\mathbf{A})$, the eigenvalues are

$$6, \quad 0, \quad 0.$$

Hence the algebraic multiplicities are

$$6 : 1, \qquad 0 : 2.$$

Finally, the eigenspace for 6 has dimension 1, and the eigenspace for 0 has dimension 2. The total dimension of the eigenspaces is therefore

$$1 + 2 = 3.$$

Hence there exists a basis of $\mathbb{R}^3$ consisting of eigenvectors of $\mathbf{A}$. Therefore, $\mathbf{A}$ is diagonalizable.

Thus the eigenvalues are $\boxed{6, 0, 0}$, with algebraic multiplicities 1 and 2, respectively, and $\boxed{\mathbf{A} \text{ is diagonalizable}}$.

$\square$

Problem 2022.1. Determine the largest of the following numbers.

$$\begin{array}{ll} (1) & e^\pi \\ (2) & \pi^e \\ (3) & 7\pi \\ (4) & \binom{7}{2} \\ (5) & 4! \end{array}$$

Solution. We compare the values directly:

$$e^\pi \approx 23.1407, \qquad \pi^e \approx 22.4592, \qquad 7\pi \approx 21.9911.$$

Also,

$$\binom{7}{2} = \frac{7 \cdot 6}{2} = 21, \qquad 4! = 24.$$

Therefore, the largest number among the choices is $\boxed{(5)\ 4!}$.

$\square$

Problem 2022.2. Three people store a jointly owned treasure in a safe and want to lock it. They want no single person to be able to open the safe alone, but any two or more people should always be able to open it together. This can be done by putting n locks on the safe, making m copies of the key for each lock, and distributing the keys appropriately among the three people. Find n and m so that the number of locks n is minimized.

Solution. Let the three people be A, B, and C.

Since any two people should be able to open the safe, the key to each lock must be given to at least two people. If a certain lock had its key given to only one person, then the other two people would not be able to open that lock together.

Thus, for each lock, exactly one of the three people should not receive the key. On the other hand, no single person should be able to open the safe alone. Therefore, each person must be missing the key to at least one lock.

Hence we need at least three locks: one lock whose key is not given to A, one lock whose key is not given to B, and one lock whose key is not given to C. This lower bound is achievable by distributing the keys as follows:

	A	B	C
L_A	X	O	O
L_B	O	X	O
L_C	O	O	X

That is, the key to L_A is given to B and C, the key to L_B is given to A and C, and the key to L_C is given to A and B.

Then each person is unable to open one of the locks, so no one can open the safe alone. However, any two people together can open all three locks. Therefore, the minimum is achieved by $\boxed{n = 3,\ m = 2}$.

□

Problem 2022.3. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2+7}\,\mathrm{d}x.$$

Solution. Let

$$u = x - 2.$$

Then the limits of integration remain $-\infty$ and ∞, so

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2+7}\,\mathrm{d}x = 3\int_{-\infty}^{\infty} \frac{1}{u^2+7}\,\mathrm{d}u.$$

We use the standard formula

$$\int_{-\infty}^{\infty} \frac{1}{u^2+a^2}\,\mathrm{d}u = \frac{\pi}{a}.$$

Here $a = \sqrt{7}$, and therefore

$$3\int_{-\infty}^{\infty} \frac{1}{u^2+7}\,\mathrm{d}u = 3\cdot\frac{\pi}{\sqrt{7}} = \frac{3\pi}{\sqrt{7}}.$$

Therefore, the answer is $\boxed{\dfrac{3\pi}{\sqrt{7}}}$.

□

Problem 2022.4. A couple has two children.

(a) Given that at least one child is a daughter, determine the probability that both children are daughters.

(b) Given that a specified child is a daughter named Youngmi, determine the probability that the other child is also a daughter.

Solution. Let G denote a girl and B denote a boy. If we distinguish the two children by birth order, then the possible gender combinations are

$$GG, \quad GB, \quad BG, \quad BB.$$

For part (a), the condition that one child is a daughter means that at least one child is a girl. Thus the possible cases are

$$GG, \quad GB, \quad BG.$$

Among these three cases, only GG has the property that the other child is also a daughter. Therefore,

$$\mathbb{P}(\text{both are daughters} \mid \text{at least one is a daughter}) = \frac{1}{3}.$$

For part (b), the information that one child is a daughter named Youngmi is more specific. In the intended interpretation of the problem, this information identifies one particular child. Then the gender of the other child is still equally likely to be G or B. Hence

$$\mathbb{P}(\text{the other child is a daughter}) = \frac{1}{2}.$$

Therefore, the answers are $\boxed{\text{(a) } \frac{1}{3}, \text{ (b) } \frac{1}{2}}$.

□

Problem 2022.5. Determine which of the following conjectures was not solved by Professor June Huh.

(1) Read's conjecture
(2) Brylawski's conjecture
(3) Poincaré conjecture
(4) Rota's conjecture
(5) Okounkov's conjecture

Solution. Professor June Huh is known for connecting combinatorics with algebraic geometry and using these ideas to solve several major conjectures in combinatorics. His work is related to conjectures such as Read's conjecture, Brylawski's conjecture, Rota's conjecture, and Okounkov's conjecture.

On the other hand, the Poincaré conjecture is a famous problem in 3-dimensional topology. It was proved by Grigori Perelman using Ricci flow, not by June Huh.

Therefore, the conjecture in the list that was not solved by June Huh is

$$\boxed{\text{(3) Poincaré conjecture}}.$$

□

Problem 2022.6. This mathematician, often called the father of linear programming, is famous for solving difficult problems while he was a doctoral student. One day, while studying at UC Berkeley, he arrived late to a class. The statistics professor Jerzy Neyman had written two famous unsolved problems on the blackboard, but the student thought they were homework problems. He took them home, solved them, and submitted complete solutions a few days later, thinking only that he had turned in the homework late. Identify this mathematician.

Solution. The mathematician described in the problem is George Dantzig.

George Dantzig is known as one of the founders of linear programming and as the creator of the simplex method. For this reason, he is often called the father of linear programming.

The anecdote in the problem is also a famous story about Dantzig. While he was a doctoral student at UC Berkeley, he arrived late to Jerzy Neyman's statistics class and copied down two problems written on the blackboard, believing that they were homework problems. He later submitted solutions, not realizing that the problems were actually open problems in statistics.

Therefore, the answer is $\boxed{\text{George Dantzig}}$.

□

Problem 2022.7. Two fair coins are tossed independently. Determine the probability that at least one head appears.

Solution. The complement event is that both coins show tails. By independence,

$$\mathbb{P}(\text{both tails}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Hence

$$\mathbb{P}(\text{at least one head}) = 1 - \frac{1}{4} = \boxed{\frac{3}{4}}.$$

□

Problem 2023.1. Determine the year in which the Fields Medal, received by Professor June Huh, was first awarded.

Solution. The Fields Medal is one of the most prestigious awards in mathematics, and it is awarded by the International Mathematical Union. Professor June Huh received the Fields Medal in 2022.

The Fields Medal was first awarded in 1936. The first two recipients were Lars Ahlfors and Jesse Douglas.

Therefore, the answer is $\boxed{1936}$.

□

Problem 2023.2. Consider the 6×6 square grid with vertices $(0,0)$, $(6,0)$, $(0,6)$, and $(6,6)$. A path starts at $(0,0)$ and ends at $(6,6)$, moving only one unit to the right or one unit upward at each step. Determine the number of such paths that never go above the line $x = y$. Equivalently, determine the number of such paths that stay in the region

$$\{(x,y) \in \mathbb{R}^2 \mid y \leq x\}.$$

Solution. To go from $(0,0)$ to $(6,6)$, a path must consist of 6 right steps and 6 upward steps. Without the condition $y \leq x$, the total number of such paths is

$$\binom{12}{6}.$$

We now count the paths that do go above the line $y = x$. By the reflection principle stated in Mathematical Concepts, the number of paths from $(0,0)$ to $(6,6)$ that at some point satisfy $y > x$ is

$$\binom{12}{5}.$$

Therefore, the number of paths that always stay in the region $y \le x$ is

$$\binom{12}{6} - \binom{12}{5}.$$

Computing this gives

$$\binom{12}{6} - \binom{12}{5} = 924 - 792 = 132.$$

Therefore, the answer is $\boxed{132}$.

□

Problem 2023.3. A point on the surface of the Earth has the following property: starting from that point, one moves 5 km south, then 5 km east, and then 5 km north, and returns to the original point. Determine the number of such points on the surface of the Earth.

Assume that the surface of the Earth is a perfect two-dimensional sphere, and that the North Pole and the South Pole are antipodal points. At the North Pole, the directions north, east, and west are not defined, and at the South Pole, the directions south, east, and west are not defined. Starting points for which one of these undefined directions occurs during the movement are excluded.

Solution. The North Pole is one such point. If we start at the North Pole, move 5 km south, then move 5 km east along a latitude circle, and finally move 5 km north, we return to the North Pole.

There are also many such points near the South Pole. Let Q be the point reached after moving 5 km south from the starting point. Suppose that the latitude circle passing through Q has circumference

$$\frac{5}{n}\,\text{km}$$

for some positive integer n. Then moving 5 km east from Q goes exactly n full times around that latitude circle and returns to Q. After that, moving 5 km north returns to the original starting point.

Near the South Pole, the circumferences of latitude circles approach 0. Hence, for every positive integer n, there is a latitude circle near the South Pole whose circumference is exactly $\frac{5}{n}$ km. For each such latitude circle, every point lying 5 km north of it gives a valid starting point.

Therefore, there are not just one but infinitely many such points on the Earth's surface. In fact, this construction gives entire latitude circles of valid starting points, so there are uncountably many such points. In particular, the number of such points is

$$\boxed{\text{infinitely many}}.$$

□

Problem 2023.4. Let $f : (-1, 1) \to \mathbb{R}$ be a continuous function. For each $\lambda > 0$, define

$$f_\lambda(x) := \lambda f\left(\frac{x}{\lambda}\right),$$

where f_λ is defined for $x \in (-\lambda, \lambda)$.

Suppose that for every sequence $\lambda_i \to \infty$, the functions f_{λ_i} converge locally uniformly to 0 on $\mathbb{R}$. Determine whether the following statement is true or false:

$$(f \text{ is differentiable at } 0) \wedge (f'(0) = 0).$$

Solution. We prove that the statement is true.

First, evaluate at $x = 0$. For every sequence $\lambda_i \to \infty$, we have

$$f_{\lambda_i}(0) = \lambda_i f(0).$$

Since f_{λ_i} converges locally uniformly to 0, in particular $f_{\lambda_i}(0) \to 0$ at the fixed point $x = 0$. Hence

$$\lambda_i f(0) \to 0.$$

This is possible for all sequences $\lambda_i \to \infty$ only if

$$f(0) = 0.$$

Now we show that $f'(0) = 0$. Let $h_i \to 0$ be any sequence with $h_i \neq 0$. For all sufficiently large i, we have $h_i \in (-1, 1)$. Define

$$\lambda_i = \frac{1}{|h_i|}.$$

Then $\lambda_i \to \infty$, and

$$h_i = \frac{\operatorname{sgn}(h_i)}{\lambda_i}.$$

By the assumption, $f_{\lambda_i} \to 0$ locally uniformly on $\mathbb{R}$. In particular, this convergence is uniform on the compact set $\{-1, 1\}$. Therefore,

$$f_{\lambda_i}(\operatorname{sgn}(h_i)) \to 0.$$

But

$$f_{\lambda_i}(\operatorname{sgn}(h_i)) = \lambda_i f\left(\frac{\operatorname{sgn}(h_i)}{\lambda_i}\right) = \lambda_i f(h_i).$$

Hence

$$\lambda_i f(h_i) \to 0.$$

Since $f(0) = 0$, we get

$$\left|\frac{f(h_i) - f(0)}{h_i}\right| = \left|\frac{f(h_i)}{h_i}\right| = \lambda_i |f(h_i)| \to 0.$$

Thus

$$\frac{f(h_i) - f(0)}{h_i} \to 0$$

for every sequence $h_i \to 0$ with $h_i \neq 0$. Therefore, f is differentiable at 0 and

$$f'(0) = 0.$$

Therefore, the answer is $\boxed{\text{True}}$.

□

Problem 2024.1. Determine the number of occurrences of the digit 0 when all natural numbers from 100 to 999 are written consecutively.

Solution. Every natural number from 100 to 999 is a three-digit number.

In a three-digit number, the hundreds digit cannot be 0. Therefore, the digit 0 can appear only in the tens digit or the units digit.

First, count the numbers whose tens digit is 0. The hundreds digit can be one of $1, 2, \ldots, 9$, and the units digit can be one of $0, 1, \ldots, 9$. Hence there are

$$9 \cdot 10 = 90$$

such numbers.

Next, count the numbers whose units digit is 0. The hundreds digit can be one of $1, 2, \ldots, 9$, and the tens digit can be one of $0, 1, \ldots, 9$. Hence there are again

$$9 \cdot 10 = 90$$

such numbers.

Therefore, the total number of appearances of the digit 0 is

$$90 + 90 = 180.$$

Therefore, the answer is $\boxed{180}$.

□

Problem 2024.2. Determine which of the following people was born latest.

(1) John von Neumann
(2) Alan Turing
(3) John Nash
(4) Pierre de Fermat

Solution. We compare their years of birth:

(1) John von Neumann: born in 1903,
(2) Alan Turing: born in 1912,
(3) John Nash: born in 1928,
(4) Pierre de Fermat: born in 1607.

Among these four, the person born the latest is John Nash.

Therefore, the answer is $\boxed{\text{(3) John Nash}}$.

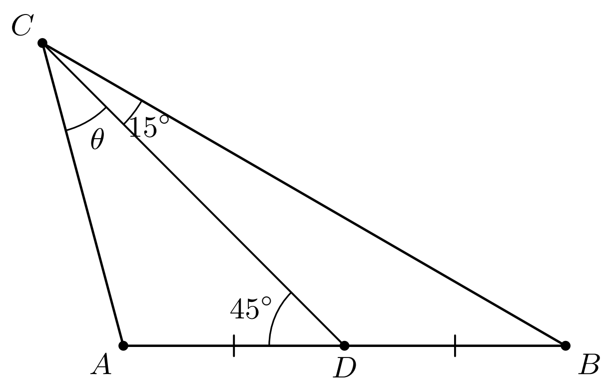
□

Problem 2024.3. In triangle ABC, the point D is the midpoint of $\overline{AB}$. Suppose that

$$\angle BCD = 15^\circ \qquad \text{and} \qquad \angle ADC = 45^\circ.$$

Determine

$$\theta = \angle ACD.$$



Solution. Since A, D, B are collinear,

$$\angle BDC = 180^\circ - \angle ADC = 135^\circ.$$

Therefore, in $\triangle BCD$,

$$\angle CBD = 180^\circ - 135^\circ - 15^\circ = 30^\circ.$$

By the Sine Rule,

$$\frac{DB}{CD} = \frac{\sin 15^\circ}{\sin 30^\circ} = 2\sin 15^\circ.$$

Let $\theta = \angle ACD$. In $\triangle ACD$,

$$\angle CAD = 180^\circ - 45^\circ - \theta = 135^\circ - \theta.$$

Again by the Sine Rule,

$$\frac{AD}{CD} = \frac{\sin\theta}{\sin(135^\circ - \theta)}.$$

Since $AD = DB$, we obtain

$$\frac{\sin\theta}{\sin(135^\circ - \theta)} = 2\sin 15^\circ.$$

Expanding the denominator gives

$$\sin\theta = \sqrt{2}\sin 15^\circ(\sin\theta + \cos\theta).$$

Since

$$\sqrt{2}\sin 15^\circ = \frac{\sqrt{3}-1}{2},$$

it follows that

$$\tan\theta = \frac{1}{\sqrt{3}}.$$

As θ is an interior angle of the triangle, we conclude that

$$\boxed{\theta = 30^\circ}.$$

□

Problem 2024.4. Let C be a convex subset of $\mathbb{R}^3$. Here convex means that for any two points $\mathbf{p}, \mathbf{q} \in C$, the line segment joining $\mathbf{p}$ and $\mathbf{q}$ is also contained in C.

Suppose that the following two facts are known:

(i) The intersection of C with the xy-plane is

$$C \cap \{z = 0\} = \{(x, y, 0) \mid x^2 + 4y^2 \le 1\}.$$

(ii) The set C contains the z-axis.

Determine the volume of C in the region $-1 \le z \le 1$, or determine that the given information is insufficient.

(1) This cannot be determined.
(2) π
(3) 2π
(4) 4π

Solution. Let

$$E = \{(x, y) \in \mathbb{R}^2 \mid x^2 + 4y^2 \le 1\}.$$

By condition (i),

$$C \cap \{z = 0\} = E \times \{0\}.$$

The ellipse E has semiaxes 1 and $\frac{1}{2}$, so its area is

$$\operatorname{area}(E) = \pi \cdot 1 \cdot \frac{1}{2} = \frac{\pi}{2}.$$

For each height z, define the horizontal cross-section

$$C_z = \{(x, y) \in \mathbb{R}^2 \mid (x, y, z) \in C\}.$$

We claim that C_z has the same area as E for every z.

First, we show that $C_z \subseteq E$. Suppose that $(x, y, z) \in C$ and

$$x^2 + 4y^2 > 1.$$

Since C contains the z-axis, the point $(0, 0, t)$ belongs to C for every $t \in \mathbb{R}$. By convexity, the line segment joining (x, y, z) and $(0, 0, t)$ is contained in C.

For any $\alpha \in (0, 1)$, we may choose

$$t = -\frac{\alpha z}{1 - \alpha}$$

so that this segment intersects the xy-plane at the point

$$(\alpha x, \alpha y, 0).$$

Since α can be chosen sufficiently close to 1, we may arrange that

$$(\alpha x)^2 + 4(\alpha y)^2 > 1.$$

This would give a point of $C \cap \{z = 0\}$ lying outside $E \times \{0\}$, contradicting condition (i). Therefore,

$$C_z \subseteq E$$

for every z.

Conversely, we show that the interior of E is contained in every C_z. Take (x, y) with

$$x^2 + 4y^2 < 1.$$

Choose $\alpha \in (0, 1)$ sufficiently close to 1 so that

$$\left(\frac{x}{\alpha}\right)^2 + 4\left(\frac{y}{\alpha}\right)^2 \leq 1.$$

Then

$$\left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) \in C.$$

Also, since C contains the z-axis,

$$\left(0, 0, \frac{z}{1 - \alpha}\right) \in C.$$

By convexity, their convex combination also belongs to C, and

$$\alpha\left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) + (1 - \alpha)\left(0, 0, \frac{z}{1 - \alpha}\right) = (x, y, z).$$

Hence $(x, y) \in C_z$.

Thus, for every z,

$$\operatorname{int}(E) \subseteq C_z \subseteq E.$$

Since the boundary of the ellipse has area 0, we have

$$\text{area}(C_z) = \text{area}(E) = \frac{\pi}{2}$$

for every z. Therefore, the volume of C in the region $-1 \leq z \leq 1$ is

$$\int_{-1}^{1} \text{area}(C_z)\, \mathrm{d}z = \int_{-1}^{1} \frac{\pi}{2}\, \mathrm{d}z = \pi.$$

Therefore, the answer is $\boxed{(2)\ \pi}$.

□

Problem 2025.1. A fair coin is tossed repeatedly until either two heads in a row or two tails in a row appear. Determine the expected number of tosses before the process stops.

Solution. On the first toss, it is impossible to have two consecutive equal outcomes. After the first toss, the process stops exactly when the next toss gives the same result as the previous toss.

Since the coin is fair, from the second toss onward we have

$$\mathbb{P}(\text{the new toss matches the previous toss}) = \frac{1}{2}.$$

Therefore, after the first toss, the number of additional tosses needed until the process stops follows a geometric distribution with success probability $\frac{1}{2}$. Its expected value is

$$\frac{1}{1/2} = 2.$$

Since the first toss is always made, the expected total number of tosses is

$$1 + 2 = 3.$$

Therefore, the answer is $\boxed{3}$.

□

Problem 2025.2. The nineteenth-century mathematicians Cauchy and Weierstrass established rigorous definitions of limits. Identify the first mathematician to prove that the sum of the reciprocals of the squares of all positive integers is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

(1) David Hilbert
(2) Georg Cantor
(3) Bernhard Riemann
(4) Leonhard Euler

Solution. The problem of finding the exact value of the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is known as the Basel problem.

This problem had been known since the seventeenth century, and it was Leonhard Euler who first found and proved the famous identity

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the answer is $\boxed{\text{(4) Leonhard Euler}}$.

□

Problem 2025.3. Find the equation of the smallest sphere tangent to all four planes

$$x = 0, \qquad y = 0, \qquad z = 0, \qquad x + y + z = 3.$$

Solution. Since the sphere is tangent to the three coordinate planes

$$x = 0, \qquad y = 0, \qquad z = 0,$$

and we are looking for the smallest such sphere, its center must lie in the first octant and have the form

$$\mathbf{c} = (a, a, a)$$

by symmetry. The distance from $\mathbf{c}$ to each coordinate plane is a, so the radius of the sphere is

$$r = a.$$

The sphere must also be tangent to the plane

$$x + y + z = 3.$$

The distance from (a, a, a) to this plane is

$$\frac{|a + a + a - 3|}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{3 - 3a}{\sqrt{3}},$$

since the smallest sphere lies between the coordinate planes and the plane $x + y + z = 3$. This distance must equal the radius a, so

$$\frac{3 - 3a}{\sqrt{3}} = a.$$

Hence

$$3 - 3a = \sqrt{3}a,$$

and therefore

$$a = \frac{3}{3 + \sqrt{3}} = \frac{3 - \sqrt{3}}{2}.$$

Thus the center is

$$\left(\frac{3 - \sqrt{3}}{2}, \frac{3 - \sqrt{3}}{2}, \frac{3 - \sqrt{3}}{2} \right),$$

and the radius is

$$\frac{3 - \sqrt{3}}{2}.$$

Therefore, the equation of the sphere is

$$\boxed{\left(x - \frac{3 - \sqrt{3}}{2} \right)^2 + \left(y - \frac{3 - \sqrt{3}}{2} \right)^2 + \left(z - \frac{3 - \sqrt{3}}{2} \right)^2 = \left(\frac{3 - \sqrt{3}}{2} \right)^2}.$$

$\square$

Problem 2025.4. There are 200 cards numbered from 1 to 200, all initially facing up. For each integer from 1 to 200, we call out that integer and flip every card whose number is a multiple of it. Determine the number of cards facing up after all 200 calls.

(1) 142
(2) 153
(3) 164
(4) 175
(5) 186

Solution. A card numbered n is flipped once for each positive divisor of n. Hence it is flipped $\tau(n)$ times, where $\tau(n)$ is the number of positive divisors of n.

A positive integer has an odd number of positive divisors if and only if it is a perfect square. Therefore, exactly the cards numbered

$$1^2, \quad 2^2, \quad 3^2, \quad \dots, \quad 14^2$$

are flipped an odd number of times, since

$$14^2 = 196 \leq 200 < 225 = 15^2.$$

These 14 cards end up facing down. The remaining

$$200 - 14 = 186$$

cards are facing up.

Therefore, the answer is $\boxed{(5)\ 186}$.

$\square$

Problem 2025.5. Determine which of 11^9 and 9^{11} is larger.

Solution. Taking logarithms, it is enough to compare

$$9 \ln 11 \qquad \text{and} \qquad 11 \ln 9.$$

Equivalently, compare

$$\frac{\ln 11}{11} \qquad \text{and} \qquad \frac{\ln 9}{9}.$$

The function $f(x) = \ln x / x$ is decreasing for $x > e$. Since $9 < 11$ and both are greater than e, we have

$$\frac{\ln 9}{9} > \frac{\ln 11}{11}.$$

Hence

$$11 \ln 9 > 9 \ln 11,$$

so

$$9^{11} > 11^9.$$

Therefore, the larger number is $\boxed{9^{11}}$.

$\square$

Problem 2025.6. Roll a fair six-sided die repeatedly until a 6 appears. Determine the expected sum of all rolls, including the final 6.

Solution. Let E be the desired expected sum. On the first roll, if the result is 6, the process stops and contributes 6. If the result is one of $1, 2, 3, 4, 5$, then the process returns to the same situation after contributing that roll.

Therefore,

$$E = \frac{1}{6} \cdot 6 + \frac{1}{6} \sum_{k=1}^{5} (k + E).$$

Hence

$$E = 1 + \frac{15 + 5E}{6} = \frac{7}{2} + \frac{5}{6}E.$$

Thus

$$\frac{1}{6}E = \frac{7}{2}, \qquad E = 21.$$

Therefore, the answer is $\boxed{21}$.

□

Problem 2025.7. Determine the probability that at least two people in a group of 10 share the same birthday. Assume a 365-day year and that each day is equally likely.

Solution. It is easier to compute the complementary probability that all 10 birthdays are distinct. The first person may have any birthday, the second must avoid that birthday, the third must avoid the first two birthdays, and so on. Hence

$$\mathbb{P}(\text{all birthdays are distinct}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{356}{365}.$$

Therefore,

$$\mathbb{P}(\text{at least two people share a birthday}) = 1 - \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{356}{365}.$$

Therefore, the answer is

$$\boxed{1 - \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{356}{365}}.$$

□

Problem 2025.8. Find the largest of the following numbers:

(1) e^{π}
(2) π^{e}
(3) $\sqrt{500}$
(4) 7π

Solution. Using the standard approximations

$$e \approx 2.718, \qquad \pi \approx 3.142,$$

we get

$$e^{\pi} \approx 23.14, \qquad \pi^{e} \approx 22.46, \qquad \sqrt{500} \approx 22.36, \qquad 7\pi \approx 21.99.$$

Therefore, the answer is $\boxed{(1)\ e^{\pi}}$.

□

Problem 2025.9. In the Monty Hall problem, a contestant chooses one of three doors, exactly one of which hides a prize. A host who knows the prize location must open an unchosen door that hides no prize; when two such doors are available, the host follows a rule independent of the prize location. The contestant may then stay or switch to the only other unopened door. Which strategy has the greater winning probability?

Solution. The original choice contains the prize with probability 1/3 and is wrong with probability 2/3. Under the stated host protocol, whenever the original choice is wrong the only other unopened door contains the prize. Thus staying wins with probability 1/3, while switching wins with probability 2/3.

Therefore, $\boxed{\text{switching}}$ is optimal.

□

Problem 2025.11. Determine which of the following Fields Medalists majored in history rather than mathematics as an undergraduate.

(1) June Huh
(2) Caucher Birkar
(3) Edward Witten
(4) Simon Donaldson
(5) William Thurston

Solution. Edward Witten is famous as a mathematical physicist and was the first physicist to receive the Fields Medal. As an undergraduate, he majored in history rather than mathematics.

Therefore, the answer is $\boxed{\text{(3) Edward Witten}}$.

□

Problem 2025.12. Determine which of the following mathematicians was not alive during the twentieth century.

(1) Sophus Lie
(2) Wilhelm Killing
(3) Alfred Young
(4) Hans Fitting
(5) Botong Wang

Solution. Sophus Lie lived from 1842 to 1899, so he died before the twentieth century began. The other listed mathematicians were alive at some point during the twentieth century.

Therefore, the answer is $\boxed{\text{(1) Sophus Lie}}$.

□

Problem 2025.13. Königsberg was the capital of East Prussia and a major center of European scholarship for many years. It was also the hometown of Immanuel Kant, David Hilbert, and Jürgen Moser. The famous Seven Bridges of Königsberg problem, in which Leonhard Euler proved that no solution exists, marked the beginning of both topology and graph theory. Determine the present-day country containing the former city of Königsberg.

(1) Germany
(2) Poland
(3) Lithuania
(4) Russia
(5) Korea

Solution. The historical city Königsberg is now Kaliningrad. Kaliningrad is a city in Russia.

Therefore, the answer is $\boxed{\text{(4) Russia}}$.

□

Problem 2025.14. A fair coin is flipped 50 times, and each outcome is recorded as either heads (H) or tails (T). A run is a consecutive sequence of the same outcome. For example, if the result is $HHTTTH$, there are 3 runs: HH, TTT, and H. Determine the expected number of runs.

(1) 24.5
(2) 25
(3) 25.5
(4) 26

Solution. The first toss always starts one run. For each position from the second toss onward, a new run begins exactly when the current toss differs from the previous toss.

For a fair coin,

$$\mathbb{P}(\text{current toss differs from previous toss}) = \frac{1}{2}.$$

For $2 \le i \le 50$, let

$$I_i = \mathbb{1}_{\{\text{a new run starts at position } i\}}.$$

Then

$$R = 1 + \sum_{i=2}^{50} I_i.$$

By linearity of expectation,

$$\mathbb{E}[R] = 1 + \sum_{i=2}^{50} \mathbb{E}[I_i] = 1 + 49 \cdot \frac{1}{2} = 25.5.$$

Therefore, the answer is $\boxed{\text{(3) 25.5}}$.

□

Problem 2025.15. Determine which of the following theorems was published first.

(1) Bayes' Theorem
(2) Intermediate Value Theorem
(3) Cauchy–Schwarz Inequality
(4) Fundamental Theorem of Calculus

Solution. The Fundamental Theorem of Calculus is associated with Newton and Leibniz in the seventeenth century, while Bayes' Theorem, the Intermediate Value Theorem in its modern rigorous form, and the Cauchy–Schwarz Inequality were published later.

Therefore, the answer is $\boxed{\text{(4) Fundamental Theorem of Calculus}}$.

□

Problem 2025.16. Determine which of the following amounts is closest to the Abel Prize award.

(1) None
(2) About \$1,000,000
(3) About \$1,500,000
(4) About \$2,000,000

Solution. The Abel Prize is a major international mathematics prize awarded by the Norwegian Academy of Science and Letters. Among the listed options, the prize amount is closest to one million US dollars.

Therefore, the answer is $\boxed{\text{(2) About \$1,000,000}}$.

□

Problem 2025.17. Determine which of the following mathematicians has received both the Fields Medal and the Abel Prize.

(1) John Milnor
(2) Terence Tao
(3) June Huh
(4) No one

Solution. John Milnor received the Fields Medal in 1962 and the Abel Prize in 2011. Therefore he has received both prizes.

Therefore, the answer is $\boxed{\text{(1) John Milnor}}$.

□

Problem 2025.18. A fair six-sided die is rolled until all 6 faces have appeared. Determine the probability that no odd-numbered face appears before all even-numbered faces have appeared.

(1) 20%
(2) 15%
(3) 10%
(4) 5%

Solution. Ignore repeated rolls and look only at the order in which new faces appear. The condition is that the first three distinct faces to appear are exactly the even faces $2, 4, 6$ in some order.

The probability that the first new face is even is $3/6$. Given that one even face has appeared, the probability that the next new face is also even is $2/5$. Given that two even faces have appeared, the probability that the next new face is the remaining even face is $1/4$. Therefore the desired probability is

$$\frac{3}{6} \cdot \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{20} = 5\%.$$

Therefore, the answer is $\boxed{\text{(4) 5\%}}$.

□

Problem 2025.19. In the Tower of Hanoi puzzle, 5 disks are stacked on peg A from largest to smallest. The objective is to move all disks to peg C using auxiliary peg B, subject to the following rules:

- Only one disk may be moved at a time.
- A larger disk may not be placed on a smaller disk.
- Peg B may be used freely.

Determine the minimum number of moves required.

(1) 25
(2) 31
(3) 32
(4) 63
(5) 120

Solution. Let $T(n)$ be the minimum number of moves needed for n disks. To move n disks, first move the top $n-1$ disks to the auxiliary peg, then move the largest disk once, and finally move the $n-1$ disks onto the largest disk. Hence

$$T(n) = 2T(n-1) + 1, \qquad T(1) = 1.$$

Solving this recurrence gives

$$T(n) = 2^n - 1.$$

Therefore,

$$T(5) = 2^5 - 1 = 31.$$

Therefore, the answer is $\boxed{(2)\ 31}$.

□

Problem 2025.20. Determine the last two digits of 7^{123}, expressed as a two-digit number.

(1) 37
(2) 41
(3) 43
(4) 49

Solution. Work modulo 100. Since

$$7^4 = 2401 \equiv 1 \pmod{100},$$

and

$$123 = 4 \cdot 30 + 3,$$

we have

$$7^{123} = (7^4)^{30} 7^3 \equiv 1^{30} \cdot 343 \equiv 43 \pmod{100}.$$

Therefore, the answer is $\boxed{(3)\ 43}$.

□

Part V

Expected Problems

Standard Mathematical Problems

Problem 1. (1988 IMO, Vieta Jumping). Let a and b be positive integers such that $ab+1$ divides a^2+b^2. Show that
$$\frac{a^2+b^2}{ab+1}$$
is the square of an integer.

Solution. Let
$$k = \frac{a^2+b^2}{ab+1} \in \mathbb{N}.$$
Suppose, for contradiction, that k is not a perfect square. Among all positive integer solutions of
$$a^2+b^2 = k(ab+1),$$
choose one for which $a+b$ is minimal, and assume without loss of generality that $a \geq b$.

If $a=b$, then
$$k = \frac{2a^2}{a^2+1} < 2,$$
so $k=1$, a contradiction. Hence $a>b$. Regard the equation as a quadratic equation in a:
$$x^2 - kbx + b^2 - k = 0.$$
One root is a, so the other root is
$$c = kb - a \in \mathbb{Z}.$$
We claim that $c>0$. From
$$a(a-kb) = k - b^2,$$
if $a = kb$, then $k = b^2$, contrary to the assumption. If $a > kb$, then the left-hand side is at least $a \geq kb \geq k$, whereas the right-hand side is less than k, which is impossible. Therefore $a < kb$, and hence $c > 0$.

Since k is a positive integer that is not a perfect square, we have $k \geq 2$. Let
$$f(x) = x^2 - kbx + b^2 - k.$$
Since
$$f(b) = (2-k)b^2 - k < 0$$
and the two roots are c and a with $a > b$, it follows that
$$0 < c < b.$$
By Viète's formulas, c is the other root, and therefore
$$c^2 + b^2 = k(cb+1).$$
Thus (c,b) is another positive integer solution with
$$c + b < a + b,$$
contradicting the minimality of $a+b$. Hence k must be a perfect square.

□

Problem 2. (Sophomore's Dream). Prove the identity
$$\int_0^1 x^{-x}\,\mathrm{d}x = \sum_{n=1}^{\infty} \frac{1}{n^n}.$$

Solution. For $0 < x \le 1$, we have $-\ln x \ge 0$, and hence

$$x^{-x} = e^{-x\ln x} = \sum_{n=0}^{\infty} \frac{x^n(-\ln x)^n}{n!}.$$

Every term is nonnegative, so Tonelli's theorem from Mathematical Concepts allows us to integrate term by term:

$$\int_0^1 x^{-x}\,\mathrm{d}x = \sum_{n=0}^{\infty} \frac{1}{n!}\int_0^1 x^n(-\ln x)^n\,\mathrm{d}x.$$

In the nth integral, let $t = -(n+1)\ln x$. Then

$$\int_0^1 x^n(-\ln x)^n\,\mathrm{d}x = \frac{1}{(n+1)^{n+1}}\int_0^{\infty} t^n e^{-t}\,\mathrm{d}t = \frac{n!}{(n+1)^{n+1}}.$$

Therefore

$$\int_0^1 x^{-x}\,\mathrm{d}x = \sum_{n=0}^{\infty} \frac{1}{(n+1)^{n+1}} = \boxed{\sum_{n=1}^{\infty} \frac{1}{n^n}}.$$

Numerically, the common value is approximately

$$1.291285997\ldots.$$

$\square$

Problem 3. (1982 IMO). A function

$$f : \mathbb{Z}_{>0} \to \mathbb{Z}_{\ge 0}$$

satisfies

$$f(m+n) - f(m) - f(n) \in \{0, 1\}$$

for all positive integers m and n. Moreover,

$$f(2) = 0, \qquad f(3) > 0, \qquad f(9999) = 3333.$$

Determine $f(1982)$.

Solution. Repeatedly applying the given condition shows that, for all positive integers r and t,

$$rf(t) \le f(rt) \le rf(t) + r - 1.$$

Apply this estimate to

$$N = 9999 \cdot 1982$$

in two different ways. Using $t = 1982$ and then $t = 9999$ gives

$$9999f(1982) \le f(N) \le 9999f(1982) + 9998$$

and

$$1982 \cdot 3333 \le f(N) \le 1982 \cdot 3333 + 1981.$$

Consequently,

$$\frac{1982 \cdot 3333 - 9998}{9999} \le f(1982) \le \frac{1982 \cdot 3333 + 1981}{9999}.$$

The left endpoint lies strictly between 659 and 660, whereas the right endpoint lies strictly between 660 and 661. Since $f(1982)$ is an integer, it follows that

$$\boxed{f(1982) = 660}.$$

□

Problem 4. (Happy Ending Problem). Five points are given in the plane, no three of which are collinear. Prove that four of them are the vertices of a convex quadrilateral.

Solution. Consider the convex hull of the five points. If the convex hull has at least four vertices, any four of its vertices, taken in cyclic order, form a convex quadrilateral.

It remains to consider the case in which the convex hull is a triangle with vertices A, B, C. Let the remaining two points be P and Q; both lie in the interior of $\triangle ABC$. The line PQ contains none of A, B, C, so two of the three vertices lie in the same open half-plane determined by PQ. Relabel them as A and B.

The triangle $\triangle ABQ$ meets the line PQ only at Q, because A and B lie strictly on the same side of that line. Hence $P \notin \triangle ABQ$. Similarly, $Q \notin \triangle ABP$. Since A and B are vertices of the convex hull of all five points, neither lies in the triangle determined by the other three points. Consequently, all four points A, B, P, Q are vertices of their convex hull, which is a convex quadrilateral.

□

Problem 5. (Sylvester–Gallai Theorem). Let S be a finite set of points in the plane, not all collinear. Prove that there exists a line containing exactly two points of S.

Solution. Call a line determined by points of S *ordinary* if it contains exactly two points of S. Consider all pairs (P, ℓ) such that $P \in S$, the line ℓ is determined by at least two points of S, and $P \notin \ell$. Since S is finite and not collinear, the set of positive distances arising in this way is finite and nonempty. Choose (P, ℓ) so that

$$h = \operatorname{dist}(P, \ell) > 0$$

is as small as possible, and let H be the foot of the perpendicular from P to ℓ.

Suppose, for contradiction, that ℓ contains at least three points of S. We can choose distinct points $A, B \in S \cap \ell$ so that either $A = H$, or A and B lie on the same side of H with A closer to H. In either case,

$$AB < PB.$$

Indeed, if $A = H$, then $AB = HB < PB$; otherwise, $AB = HB - HA < HB < PB$.

Computing the area of $\triangle PAB$ with bases AB and PB gives

$$\frac{1}{2} AB\, h = [PAB] = \frac{1}{2} PB \operatorname{dist}(A, PB).$$

Hence

$$\operatorname{dist}(A, PB) = \frac{AB}{PB} h < h.$$

However, the line PB is determined by two points of S, and $A \notin PB$. This contradicts the minimality of h.

Therefore, ℓ contains exactly two points of S, so ℓ is the required ordinary line.

□

Problem 6. (Moser's Circle Problem). Place n points on a circle and join every pair of points by a chord. Assume that no three chords meet at a common point in the interior of the circle. Determine, with proof, the maximum number of regions into which the chords divide the circle.

Solution. There are

$$\binom{n}{2}$$

chords. Draw them one at a time. If a new chord contains r existing interior intersection points, those points divide it into $r+1$ segments, and each segment splits one existing region into two. Therefore the new chord creates $r+1$ new regions.

It follows that the final number of regions is

$$1 + \#\{\text{chords}\} + \#\{\text{interior intersections}\}.$$

Every interior intersection is determined by four boundary points: among the six chords joining those four points, exactly the two diagonals of their cyclic quadrilateral intersect in the interior. Conversely, every choice of four boundary points produces one such intersection. The assumption that no three chords are concurrent ensures that all these intersections are distinct. Hence there are

$$\binom{n}{4}$$

interior intersections.

Thus the maximum number of regions is

$$\boxed{1 + \binom{n}{2} + \binom{n}{4}}.$$

In particular, the first values are

$$1,\ 2,\ 4,\ 8,\ 16,\ 31, \dots,$$

so the apparent pattern 2^{n-1} fails when $n = 6$.

$\square$

Problem 7. (1976 IMO). Determine, with proof, the largest number that can be written as the product of positive integers whose sum is 1976. The number of factors is not prescribed.

Solution. A maximizing product contains no factor 1, since replacing a pair $1, a$ by $a+1$ preserves the sum and increases the product. It also contains no factor $k \geq 5$, because replacing k by 3 and $k-3$ preserves the sum and increases the product:

$$3(k-3) > k.$$

A factor 4 may be replaced by $2, 2$ without changing the product. Thus it suffices to use only 2 and 3.

Three factors equal to 2 should be replaced by two factors equal to 3, since

$$2+2+2 = 3+3 \qquad \text{and} \qquad 2^3 < 3^2.$$

Hence at most two factors equal to 2 occur. Since

$$1976 = 3 \cdot 658 + 2,$$

the only optimal remainder pattern is one factor equal to 2 and 658 factors equal to 3.

Therefore, the largest possible product is

$$\boxed{2 \cdot 3^{658}}.$$

$\square$

Problem 8. (2021 Putnam). A grasshopper starts at the origin in the coordinate plane and makes a sequence of hops. Each hop has length 5, and after every hop the grasshopper is at a point whose coordinates are both integers. Determine the smallest number of hops needed to reach

$$(2021, 2021).$$

Solution. If a hop has displacement vector (p, q), then

$$p^2 + q^2 = 25.$$

Thus the possible absolute coordinate pairs are $(5, 0)$ and $(4, 3)$, in either order. In particular, a single hop can increase the sum of the two coordinates by at most

$$4 + 3 = 7.$$

Since

$$2021 + 2021 = 4042 = 7 \cdot 577 + 3,$$

at least 578 hops are necessary.

This bound is attained because

$$(2021, 2021) = 288(3, 4) + 288(4, 3) + (5, 0) + (0, 5).$$

The right-hand side is a sum of

$$288 + 288 + 1 + 1 = 578$$

legal hop vectors. Therefore, the minimum number of hops is

$$\boxed{578}.$$

□

Direct Calculation Problems

Problem 1. Evaluate each of the following integrals. For indefinite integrals, the constant of integration may be omitted; for definite integrals, determine the exact value.

1.1. (2010 MIT Integration Bee). $\displaystyle\int_0^{\pi/2} \sin x \sin(2x) \sin(3x)\, \mathrm{d}x$

1.2. (2010 MIT Integration Bee). $\displaystyle\int_0^{1} \sin^2(\ln x)\, \mathrm{d}x$

1.3. (2010 MIT Integration Bee). $\displaystyle\int_1^{\infty} \frac{\mathrm{d}x}{x\sqrt{x^4 - 1}}$

1.4. (2010 MIT Integration Bee). $\displaystyle\int_0^{1} \frac{\ln(1 + x)}{1 + x^2}\, \mathrm{d}x$

1.5. (2010 MIT Integration Bee). $\displaystyle\int \frac{\cos x + x \sin x}{x(x + \cos x)}\, \mathrm{d}x$

1.6. (2010 MIT Integration Bee). $\displaystyle\int_0^{\pi/2} \frac{\mathrm{d}x}{\sin x + \sec x}$

1.7. (2010 MIT Integration Bee). $\displaystyle\int_0^{1} \sqrt{1 + x\sqrt{1 + x\sqrt{1 + \cdots}}}\, \mathrm{d}x$

1.8. (2010 MIT Integration Bee). $\displaystyle\int \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2}\right) \mathrm{d}x, \qquad x > 0, \quad x \neq 1$

1.9. (2010 MIT Integration Bee). $\displaystyle\int_1^2 \sqrt{x-1}\sqrt{2-x}\,\mathrm{d}x$

1.10. (2011 MIT Integration Bee). $\displaystyle\int \cos(\ln x)\,\mathrm{d}x, \qquad x > 0$

1.11. (2011 MIT Integration Bee). $\displaystyle\int_0^\pi \cos x\cos(3x)\cos(5x)\,\mathrm{d}x$

1.12. (2011 MIT Integration Bee). $\displaystyle\int \sin(101x)\sin^{99} x\,\mathrm{d}x$

1.13. (2011 MIT Integration Bee). $\displaystyle\int xe^{e^{x^2}+x^2}\,\mathrm{d}x$

1.14. (2011 MIT Integration Bee). $\displaystyle\int \sqrt{\frac{1-x}{1+x}}\,\mathrm{d}x, \qquad -1 < x < 1$

Solution 1.1. By the product-to-sum identities,

$$\sin x\sin(3x) = \frac{1}{2}(\cos(2x) - \cos(4x)),$$

and hence

$$\sin x\sin(2x)\sin(3x) = \frac{1}{4}(\sin(2x) + \sin(4x) - \sin(6x)).$$

Therefore,

$$\int_0^{\pi/2} \sin x\sin(2x)\sin(3x)\,\mathrm{d}x = \frac{1}{4}\left(1 - \frac{1}{3}\right) = \boxed{\frac{1}{6}}.$$

□

Solution 1.2. Let $t = -\ln x$. Then $x = e^{-t}$ and $\mathrm{d}x = -e^{-t}\,\mathrm{d}t$, so

$$\int_0^1 \sin^2(\ln x)\,\mathrm{d}x = \int_0^\infty e^{-t}\sin^2 t\,\mathrm{d}t.$$

Since $\sin^2 t = (1 - \cos(2t))/2$,

$$\int_0^\infty e^{-t}\sin^2 t\,\mathrm{d}t = \frac{1}{2}\left(1 - \frac{1}{5}\right) = \boxed{\frac{2}{5}}.$$

□

Solution 1.3. Let $u = x^2$. Then

$$\frac{\mathrm{d}x}{x} = \frac{\mathrm{d}u}{2u},$$

and therefore

$$\int_1^\infty \frac{\mathrm{d}x}{x\sqrt{x^4-1}} = \frac{1}{2}\int_1^\infty \frac{\mathrm{d}u}{u\sqrt{u^2-1}}.$$

With $u = \sec\theta$, this becomes

$$\frac{1}{2}\int_0^{\pi/2} \mathrm{d}\theta = \boxed{\frac{\pi}{4}}.$$

□

Solution 1.4. Let

$$I = \int_0^1 \frac{\ln(1+x)}{1+x^2}\,\mathrm{d}x.$$

Substituting $x = \tan t$ gives

$$I = \int_0^{\pi/4} \ln(1+\tan t)\,\mathrm{d}t.$$

Under $t \mapsto \pi/4 - t$,

$$1 + \tan\left(\frac{\pi}{4} - t\right) = \frac{2}{1+\tan t}.$$

Thus

$$I = \frac{\pi}{4}\ln 2 - I,$$

and consequently

$$\boxed{I = \frac{\pi}{8}\ln 2}.$$

□

Solution 1.5. We recognize the logarithmic derivative

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln|x| - \ln|x+\cos x|) = \frac{\cos x + x\sin x}{x(x+\cos x)}.$$

Hence

$$\boxed{\int \frac{\cos x + x\sin x}{x(x+\cos x)}\,\mathrm{d}x = \ln|x| - \ln|x+\cos x|}.$$

□

Solution 1.6. Write

$$I = \int_0^{\pi/2} \frac{\cos x}{1+\sin x\cos x}\,\mathrm{d}x.$$

Replacing x by $\pi/2 - x$ also gives

$$I = \int_0^{\pi/2} \frac{\sin x}{1+\sin x\cos x}\,\mathrm{d}x.$$

Adding the two expressions and setting $u = \sin x - \cos x$, we obtain

$$2I = \int_{-1}^{1} \frac{2}{3-u^2}\,\mathrm{d}u.$$

Therefore,

$$\boxed{I = \frac{1}{\sqrt{3}}\ln(2+\sqrt{3})}.$$

□

Solution 1.7. For $x \in [0,1]$, define the finite truncations

$$R_1(x) = 1, \qquad R_{n+1}(x) = \sqrt{1 + xR_n(x)}.$$

The map $y \mapsto \sqrt{1+xy}$ is increasing. Induction gives

$$1 \le R_n(x) \le \frac{1+\sqrt{5}}{2}$$

and $R_n(x) \le R_{n+1}(x)$. Hence $R_n(x)$ converges pointwise to a positive limit $R(x)$. Passing to the limit in the recurrence gives

$$R(x)^2 = 1 + xR(x),$$

and therefore

$$R(x) = \frac{x + \sqrt{x^2+4}}{2}.$$

The uniform bound above permits integration by dominated convergence, so

$$\int_0^1 R(x)\,\mathrm{d}x = \frac{1}{2}\int_0^1 x\,\mathrm{d}x + \frac{1}{2}\int_0^1 \sqrt{x^2+4}\,\mathrm{d}x.$$

Evaluating the elementary integral yields

$$\boxed{\frac{1+\sqrt{5}}{4} + \ln\left(\frac{1+\sqrt{5}}{2}\right)}.$$

□

Solution 1.8. On any interval contained in $(0,\infty)\setminus\{1\}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x}{\ln x}\right) = \frac{1}{\ln x} - \frac{1}{(\ln x)^2}.$$

Therefore,

$$\boxed{\int \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2}\right)\mathrm{d}x = \frac{x}{\ln x}}.$$

□

Solution 1.9. Let $u = x - 1$. Then

$$\int_1^2 \sqrt{x-1}\sqrt{2-x}\,\mathrm{d}x = \int_0^1 \sqrt{u(1-u)}\,\mathrm{d}u.$$

Setting $v = u - 1/2$ gives

$$\sqrt{u(1-u)} = \sqrt{\frac{1}{4} - v^2}.$$

Thus the integral is the area of a semicircle of radius 1/2:

$$\boxed{\frac{\pi}{8}}.$$

□

Solution 1.10. Let $t = \ln x$. Then $x = e^t$ and $\mathrm{d}x = e^t\,\mathrm{d}t$, so

$$\int \cos(\ln x)\,\mathrm{d}x = \int e^t \cos t\,\mathrm{d}t.$$

Therefore,

$$\boxed{\int \cos(\ln x)\,\mathrm{d}x = \frac{x}{2}(\cos(\ln x) + \sin(\ln x))}.$$

□

Solution 1.11. By repeated use of the product-to-sum identity,

$$\cos x \cos(3x) \cos(5x) = \frac{1}{4}(\cos x + \cos(3x) + \cos(7x) + \cos(9x)).$$

Each term has integral 0 on $[0, \pi]$. Hence

$$\boxed{\int_0^\pi \cos x \cos(3x) \cos(5x) \,\mathrm{d}x = 0}.$$

□

Solution 1.12. Differentiating a suitable product gives

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{100}\sin(100x)\sin^{100} x\right) &= \sin^{99} x(\sin x \cos(100x) + \cos x \sin(100x)) \\ &= \sin(101x)\sin^{99} x. \end{aligned}$$

Therefore,

$$\boxed{\int \sin(101x)\sin^{99} x \,\mathrm{d}x = \frac{1}{100}\sin(100x)\sin^{100} x}.$$

□

Solution 1.13. Since

$$e^{e^{x^2}+x^2} = e^{e^{x^2}} e^{x^2},$$

let $u = e^{x^2}$. Then $\mathrm{d}u = 2xe^{x^2}\,\mathrm{d}x$, and

$$\int xe^{e^{x^2}+x^2}\,\mathrm{d}x = \frac{1}{2}\int e^u \,\mathrm{d}u.$$

Hence

$$\boxed{\int xe^{e^{x^2}+x^2}\,\mathrm{d}x = \frac{1}{2}e^{e^{x^2}}}.$$

□

Solution 1.14. For $-1 < x < 1$,

$$\sqrt{\frac{1-x}{1+x}} = \frac{1-x}{\sqrt{1-x^2}}.$$

Therefore,

$$\begin{aligned} \int \sqrt{\frac{1-x}{1+x}}\,\mathrm{d}x &= \int \frac{\mathrm{d}x}{\sqrt{1-x^2}} - \int \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x \\ &= \boxed{\arcsin x + \sqrt{1-x^2}}. \end{aligned}$$

□

Problem 2. (Buffon's Needle). Infinitely many parallel lines are drawn in the plane, with distance d between consecutive lines. A needle of length ℓ, where

$$0 < \ell \leq d,$$

is dropped at random. Assume that its midpoint is uniformly distributed modulo one strip and, independently, that its direction is uniformly distributed. Determine the probability that the needle intersects one of the parallel lines.

Solution. Let X be the distance from the midpoint of the needle to the nearest parallel line, and let Θ be the acute angle between the needle and the lines. Then

$$0 \leq X \leq \frac{d}{2}, \qquad 0 \leq \Theta \leq \frac{\pi}{2},$$

and X and Θ are independent and uniform on these intervals.

For a fixed angle θ, the needle intersects a line exactly when its perpendicular half-length reaches the nearest line, that is, when

$$X \leq \frac{\ell}{2} \sin\theta.$$

Since $\ell \leq d$, the right-hand side never exceeds $d/2$. Therefore,

$$\begin{aligned} \mathbb{P}(\text{intersection}) &= \frac{2}{\pi} \int_0^{\pi/2} \mathbb{P}\left(X \leq \frac{\ell}{2}\sin\theta\right) \mathrm{d}\theta \\ &= \frac{2}{\pi} \int_0^{\pi/2} \frac{\ell \sin\theta}{d} \,\mathrm{d}\theta \\ &= \boxed{\frac{2\ell}{\pi d}}. \end{aligned}$$

□

Memorization Problems

Problem 1. (Yang–Mills Existence and Mass Gap). The Clay Mathematics Institute's Yang–Mills Existence and Mass Gap problem asks for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^d$ with a positive mass gap. Determine d, and state whether this case has been solved.

Solution. The Millennium Prize Problem is posed on four-dimensional Euclidean space:

$$\boxed{d = 4}.$$

As of 2026, the required rigorous construction and proof of a positive mass gap in this dimension remain open.

□

Problem 2. (Exotic Spheres). John Milnor first exhibited exotic spheres in dimension n: smooth manifolds homeomorphic, but not diffeomorphic, to the standard sphere S^n. Later work showed that the group Θ_n of oriented diffeomorphism classes of homotopy n-spheres has exactly k elements in this dimension. Determine

$$n + k.$$

Solution. Milnor's first exotic spheres were seven-dimensional, and Kervaire–Milnor theory gives

$$\Theta_7 \cong \mathbb{Z}/28\mathbb{Z}.$$

Thus, with orientation taken into account, there are 28 classes in total: one standard class and 27 exotic classes. Therefore $n = 7$ and $k = 28$, so

$$\boxed{n + k = 35}.$$

□

Reasoning and Puzzle Problems

Problem 1. (Sum-and-Product Puzzle). Two integers x and y satisfy

$$1 < x < y, \qquad x + y \leq 100.$$

The sum $x+y$ is disclosed privately to S, and the product xy is disclosed privately to P. The preceding information is common knowledge. The following conversation then takes place:

(i) P: "I do not know the pair."

(ii) S: "I knew that you did not know the pair."

(iii) P: "Now I know the pair."

(iv) S: "Now I also know the pair."

Determine (x, y).

Solution. Let

$$\Omega_0 = \{(x, y) \in \mathbb{N}^2 \mid (1 < x < y) \wedge (x + y \leq 100)\}.$$

For $\Omega \subseteq \Omega_0$ and $\omega = (x, y) \in \Omega$, define the information sets

$$\mathcal{I}_S^{\Omega}(\omega) = \{(u, v) \in \Omega \mid u + v = x + y\},$$

$$\mathcal{I}_P^{\Omega}(\omega) = \{(u, v) \in \Omega \mid uv = xy\}.$$

The four announcements successively restrict the state space as follows:

$$\Omega_1 = \{\omega \in \Omega_0 \mid |\mathcal{I}_P^{\Omega_0}(\omega)| > 1\},$$

$$\Omega_2 = \{\omega \in \Omega_1 \mid \mathcal{I}_S^{\Omega_0}(\omega) \subseteq \Omega_1\},$$

$$\Omega_3 = \{\omega \in \Omega_2 \mid |\mathcal{I}_P^{\Omega_2}(\omega)| = 1\},$$

and

$$\Omega_4 = \{\omega \in \Omega_3 \mid |\mathcal{I}_S^{\Omega_3}(\omega)| = 1\}.$$

The finite elimination is summarized by

stage	public information retained	number of states
0	$(1 < x < y) \wedge (x + y \leq 100)$	2352
1	P does not know the pair	1747
2	S knew that P did not know	145
3	P now knows the pair	86
4	S now knows the pair	1

At the second stage, the possible sums are exactly

$$11,\ 17,\ 23,\ 27,\ 29,\ 35,\ 37,\ 41,\ 47,\ 53.$$

These values follow by checking, for each admissible sum, whether every compatible factorization has at least two representations in Ω_0. After the third announcement, the only sum-information set of cardinality 1 is

$$\mathcal{I}_S^{\Omega_3}(4, 13) = \{(4, 13)\}.$$

Therefore,

$$\Omega_4 = \{(4, 13)\},$$

and the unique pair is

$$\boxed{(x, y) = (4, 13)}.$$

□

Problem 2. (Bertrand's Box Paradox). Three boxes contain coins as follows:

$$GG, \qquad GS, \qquad SS,$$

where G denotes a gold coin and S denotes a silver coin. A box is chosen uniformly at random, and then one of its two coins is chosen uniformly at random. The chosen coin is gold. Determine the conditional probability that the other coin in the same box is also gold.

Solution. Let B_{GG} be the event that the chosen box is GG, and let G be the event that the observed coin is gold. Then

$$\mathbb{P}(G \mid B_{GG}) = 1, \qquad \mathbb{P}(G \mid B_{GS}) = \frac{1}{2}, \qquad \mathbb{P}(G \mid B_{SS}) = 0.$$

Since the three boxes are initially equally likely, Bayes' theorem gives

$$\begin{aligned}\mathbb{P}(B_{GG} \mid G) &= \frac{\mathbb{P}(G \mid B_{GG})\mathbb{P}(B_{GG})}{\mathbb{P}(G \mid B_{GG})\mathbb{P}(B_{GG}) + \mathbb{P}(G \mid B_{GS})\mathbb{P}(B_{GS})} \\ &= \frac{1 \cdot (1/3)}{1 \cdot (1/3) + (1/2) \cdot (1/3)} = \boxed{\frac{2}{3}}.\end{aligned}$$

Equivalently, after observing gold, the selected coin is equally likely to be any of the three gold coins present among the boxes. Two of those three coins lie in the GG box.

□

Problem 3. (Josephus Problem). The people numbered

$$1, 2, \ldots, n$$

stand in a circle. Starting with person 1, count repeatedly around the circle and eliminate every second person. After each elimination, resume counting with the next remaining person. Determine the number of the last survivor.

Solution. Let $J(n)$ denote the number of the survivor. If $n = 2m$, the first circuit eliminates all even-numbered people. Relabeling the survivors

$$1, 3, \ldots, 2m - 1$$

as $1, 2, \ldots, m$ gives

$$J(2m) = 2J(m) - 1.$$

A similar relabeling when $n = 2m + 1$ gives

$$J(2m + 1) = 2J(m) + 1.$$

Write

$$n = 2^r + \ell, \qquad 0 \le \ell < 2^r.$$

Starting from $J(2^r) = 1$, a strong induction using these recurrences yields

$$\boxed{J(n) = 2\ell + 1}.$$

In binary notation this has a particularly simple interpretation. If

$$n = (1b_{r-1}b_{r-2}\cdots b_0)_2,$$

then

$$J(n) = (b_{r-1}b_{r-2}\cdots b_0 1)_2;$$

the leading binary digit of n is moved to the end.

□

Problem 4. (Conway's Soldiers). On an infinite checkerboard, every square on or below a horizontal line initially contains a soldier. A legal move jumps one soldier horizontally or vertically over an adjacent soldier into the next square, which must be empty; the jumped soldier is removed. Prove that no soldier can reach the fifth row above the initial line.

Solution. Let

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad q = \varphi^{-1},$$

so that $q + q^2 = 1$. Assign to a square the weight

$$q^d,$$

where d is its Manhattan distance from a fixed target square in the fifth row. A jump toward the target replaces weights

$$q^{d+2} + q^{d+1}$$

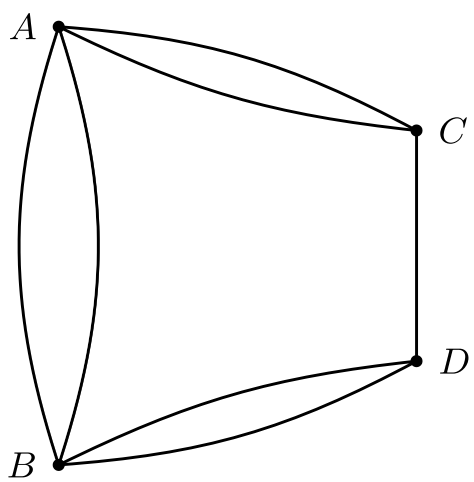
by q^d. Since $q + q^2 = 1$, this jump preserves total weight; every other legal jump does not increase it.

Summing the geometric series over the initially occupied half-plane gives total weight exactly 1 for a target in the fifth row. Any finite play moves only finitely many soldiers, so infinitely many initially occupied squares remain occupied and retain strictly positive total weight. If the target were occupied, it alone would contribute weight 1, and the untouched soldiers would make the total strictly greater than 1. This contradicts the nonincrease of total weight.

Therefore the fifth row cannot be reached.

□

Problem 5. (Seven Bridges of Königsberg). The city of Königsberg was divided by the Pregel River into four land masses, connected by seven bridges as shown below. Is it possible to start at one land mass, cross each bridge exactly once, and end at some land mass? Prove your answer.



Solution. Replace each land mass by a vertex and each bridge by an edge. Then the problem becomes the following graph-theoretic question: does the resulting connected multigraph have an Euler trail, that is, a trail using every edge exactly once?

In the graph above, the four vertex degrees are

$$3, \quad 3, \quad 3, \quad 5.$$

In particular, all four vertices have odd degree.

A connected graph has an Euler trail if and only if it has exactly zero or two vertices of odd degree. Since this graph has four odd vertices, an Euler trail does not exist.

Therefore it is impossible to cross each of the seven bridges exactly once.

□

Problem 6. (15 Puzzle). In the standard fifteen puzzle, a legal move consists of sliding one tile adjacent to the blank square into the blank square. Starting from the solved position on the left, determine whether one can reach the position on the right, in which only the tiles 14 and 15 have been interchanged.

solved position

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

14–15 challenge

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

Solution. Read the tiles row by row from left to right and top to bottom, omitting the blank square. Let I be the number of inversions in this resulting list, and let r be the row number of the blank square counted from the top. We claim that the parity of

$$I + r$$

is invariant under every legal move.

If the blank square moves horizontally, then the order of the numbered tiles in the row-by-row list does not change, so I is unchanged, and r is also unchanged. Hence the parity of $I + r$ is unchanged.

If the blank square moves vertically, then one tile moves up or down by one row. In the row-by-row list, that tile passes across exactly three other tiles, so the inversion number changes by 3, and therefore its parity changes. At the same time, the blank square moves to an adjacent row, so the parity of r also changes. Hence the parity of $I + r$ remains unchanged in this case as well.

In the solved position, the inversion number is

$$I = 0,$$

and the blank square lies in row 4, so

$$I + r \equiv 0 + 4 \equiv 0 \pmod{2}.$$

In the target position, the list differs only by the transposition of 14 and 15, so

$$I = 1,$$

while the blank square is still in row 4. Thus

$$I + r \equiv 1 + 4 \equiv 1 \pmod 2.$$

Since the parity of $I + r$ is invariant, the target position cannot be reached from the solved position.

Therefore the 14–15 challenge is impossible.

□

Problem 7. (Chomp). An $m \times n$ rectangular chocolate bar consists of unit-square pieces. The lower-left piece is poisoned.

On each turn, a player chooses one remaining piece and eats that piece together with every remaining piece weakly above it and weakly to its right. The player who eats the poisoned piece loses.

Prove that whenever the initial bar contains more than the poisoned piece alone, the first player has a winning strategy.

Solution. Since the game is finite and has no draws, one of the two players has a winning strategy. Suppose, for contradiction, that the second player has one.

Let the first player eat only the upper-right corner piece. According to the supposed winning strategy, the second player must have a non-poisoned response; otherwise the second player is forced to eat the poisoned piece and loses. Let that response be a remaining piece C. Eating C also removes the upper-right corner, because that corner lies above and to the right of C. Hence the position obtained after these two moves is exactly the position that would have resulted had the first player chosen C on the very first move.

After the original two-move sequence, this position is losing for the player whose turn it is, namely the first player. The first player could instead begin by choosing C and hand the identical losing position to the second player. This contradicts the assumption that the second player has a winning strategy.

Therefore the first player has a winning strategy. The argument proves its existence without identifying the winning first move in general.

□

Problem 8. (100 Prisoners and 100 Boxes). One hundred prisoners and one hundred boxes are labeled $1, 2, \ldots, 100$. The prisoners' labels are placed in the boxes according to a uniformly random permutation.

The prisoners enter the room one at a time. Each prisoner may open at most 50 boxes and must restore them before leaving. They may agree on a strategy beforehand but may not communicate after the process begins. They all survive if and only if every prisoner finds their own label.

Find a strategy whose probability of saving all the prisoners is greater than 30%.

Solution. Regard the contents of the boxes as a permutation π of $\{1, 2, \ldots, 100\}$, where box i contains the label $\pi(i)$. Prisoner i first opens box i. If it contains label j, the prisoner next opens box j, and continues following the labels in this way for at most 50 boxes.

This procedure follows the cycle of π containing i. Therefore every prisoner finds their own label if and only if every cycle of π has length at most 50.

For $51 \le k \le 100$, the probability that a uniformly random permutation contains a k-cycle is $1/k$. Moreover, two cycles of lengths greater than 50 cannot occur simultaneously. Hence the probability that some cycle is too long is

$$\sum_{k=51}^{100} \frac{1}{k}.$$

The probability that all prisoners survive is therefore

$$1 - \sum_{k=51}^{100} \frac{1}{k} = 1 - (H_{100} - H_{50}) \approx 0.3118278207.$$

Thus the cycle-following strategy saves all the prisoners with probability approximately

$$\boxed{31.18\%},$$

which is greater than 30%.

□

Problem 9. (St. Petersburg Paradox). A fair coin is tossed until heads appears for the first time. If the first head appears on the nth toss, the player receives

$$2^n$$

units of money. Compute the expected payoff. If decisions are based only on expected monetary value, determine how large an entry fee it would be rational to pay, and explain why the conclusion is regarded as paradoxical.

Solution. The probability that the first head appears on the nth toss is

$$\left(\frac{1}{2}\right)^n.$$

Therefore the expected payoff is

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} 2^n \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} 1 = \boxed{\infty}.$$

Under the rule of maximizing expected monetary value alone, subtracting any finite entry fee still leaves an infinite expected net payoff. Thus there is no finite upper bound on the fee that this rule recommends.

The conclusion conflicts sharply with ordinary judgment. For example, the payoff is at most 2 with probability 1/2 and at most 4 with probability 3/4, so most outcomes are small. The infinite expectation is driven by extremely rare but extremely large prizes. This mismatch shows that expected monetary value alone is not always an adequate decision criterion; finite resources, risk, and diminishing marginal utility of money can all change a rational valuation.

□

Mathematical Intuition Problems